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## X-Fed NeuroScan

Federated Learning with Explainable AI
for Medical Image Classification

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# X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification

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# Abstract

Medical image analysis plays a critical role in the early detection and diagnosis of various diseases. However, manual interpretation of radiographic images is time-consuming and subject to inter-observer variability. This project presents an end-to-end deep learning–based system for the automated detection of chest diseases and bone fractures from X-ray images. The proposed solution integrates multiple transfer learning models within a unified diagnostic framework, enabling efficient and scalable analysis of heterogeneous medical images.

The system architecture consists of a deep learning pipeline that includes image preprocessing, feature extraction using pre-trained convolutional neural networks, an intelligent medical image routing model, and task-specific diagnostic models. ResNet50 is employed as a core feature extractor due to its residual learning mechanism and suitability for medical image analysis, while YOLOv8 is utilized for bone fracture detection using a one-stage object detection approach. The chest disease detection task is addressed using a dedicated classification model optimized for thoracic imaging.

Preliminary experimental results demonstrate the feasibility of the proposed approach and indicate promising performance in detecting both bone fractures and chest-related conditions. The modular design of the system allows for extensibility and supports the integration of additional diagnostic tasks. This work highlights the potential of deep learning and transfer learning techniques to assist clinical decision-making and improve the efficiency of medical image diagnosis.

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# INTRODUCTION

# Chapter 1 Introduction

Medical imaging plays a critical role in modern healthcare, supporting clinicians in diagnosing a wide range of conditions, from bone fractures to chest tumors and other life-threatening emergencies. As imaging technologies become more advanced and widely available, the volume of medical scans produced each day continues to grow. This rapid increase has intensified the need for efficient, reliable, and scalable diagnostic support systems that can assist healthcare professionals in interpreting complex visual data.

Recent advances in computer vision have made automated medical image analysis a promising tool for improving diagnostic accuracy and reducing clinical workload. Deep learning models, in particular, have demonstrated exceptional performance in tasks such as classification, localization, and segmentation across many imaging modalities. However, despite their capabilities, traditional centralized learning approaches often face critical limitations—especially in sensitive domains like healthcare, where data privacy, institutional policies, and regulatory constraints restrict the sharing of patient information.

To address these challenges, new paradigms in machine learning and model transparency have emerged. Federated learning enables collaborative training across multiple medical institutions without requiring direct access to patient data, offering a pathway toward more diverse and robust models while preserving privacy. Complementing this, explainable AI techniques provide insights into model decisions, helping clinicians understand why a prediction was made and increasing trust in AI-assisted diagnostics.

This project aims to integrate these three powerful components—computer vision, explainable AI, and federated learning—into a unified system designed to support medical image analysis across a variety of diseases and emergency conditions. By leveraging decentralized learning and transparent decision-making, the system seeks to improve accuracy, reliability, and interpretability while respecting the privacy and variability inherent in real-world clinical environments.

## 1.1 Problem Definition

Accurate diagnosis, precise segmentation, and localized identification of multi-class thoracic pathologies from projectional radiography (X-ray) and cross-sectional computed tomography (CT) images remain a critical, life-saving bottleneck in contemporary emergency and clinical healthcare systems. In high-volume triage environments, intensive care units, and resource-constrained medical centers, clinicians and radiologists are subjected to immense cognitive fatigue, psychological stress, and severe time constraints. Under these taxing diagnostic conditions, many clinically critical anomalies can be visually subtle, transient, or easily obscured by surrounding dense anatomical architectures.

For instance, early-stage alveolar consolidations, hidden or subpleural tuberculous granulomas, micro-nodular viral patterns, or tiny, solitary tumor-like pulmonary lesions are inherently faint and frequently masked by overlying bone structures (such as the ribs, clavicles, and scapulae), complex central vascular configurations, or soft-tissue summation artifacts. Overlooking these fine radiological indicators can lead to severe diagnostic delays, inappropriate patient discharge, cross-infection risks in hospital wards, uncontrolled systemic disease progression, and significantly compromised long-term patient survival outcomes.

While modern deep learning architectures, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), have achieved state-of-the-art performance in isolated computer-aided diagnostic benchmarks, their meaningful clinical integration within hospital workflows remains profoundly limited by two overarching, systemic barriers:

1. **Data Siloing, Fragmentation, and Strict Privacy Barriers:** Training a deep learning model capable of high generalization across diverse clinical settings requires massive, multi-institutional, and highly heterogeneous datasets. However, medical data is heavily siloed due to strict ethical, legal, and regulatory compliance standards enforced worldwide, such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union. Institutions are strictly prohibited from aggregating, uploading, or sharing raw patient imaging data to centralized cloud repositories or external computing hubs. Consequently, models trained on isolated, single-institution datasets suffer from severe overfitting and fail to generalize when deployed across external diagnostic setups or varying patient demographics.
2. **The "Black-Box" Nature and Lack of Clinically Verifiable Interpretability:** Standard deep learning pipelines produce deterministic classification vectors or bounding coordinate values without generating clear, human-interpretable rationale for their decision-making. Medical practitioners cannot ethically, legally, or clinically trust automated diagnoses that lack transparent, physiological justification. If a model detects a malignancy or a highly infectious respiratory condition without visually mapping the corresponding lesion, patch, or parenchymal disruption, the clinician cannot easily verify, audit, or trust the output, limiting the artificial intelligence to a novelty tool rather than a viable diagnostic assistant.

Furthermore, medical data exhibits severe Non-Identically and Independently Distributed (Non-IID) characteristics across disparate hospitals. Variations in imaging hardware manufacturers (e.g., GE, Siemens, Philips), localized X-ray tube voltages, exposure settings, slice thicknesses in CT imaging, patient demographics, regional disease prevalence, and distinct institutional positioning protocols introduce massive data distribution shifts. When institutional datasets are highly heterogeneous, standard centralized optimization functions collapse, leading to catastrophic forgetting or severe performance degradation for minority disease classes or specific imaging modalities.

Given this multi-faceted medical and technical challenge, the central problem addressed by this research is formally stated as follows:

***How can we engineer an integrated, privacy-preserving, mathe-***

***matically robust, and highly explainable deep learning framework capable of simultaneously detecting and localizing four major thoracic disease classes (pneumonia, tuberculosis, COVID-19, and tumor-like chest masses/lung cancer) using heterogeneous X-ray and CT image streams, without transmitting sensitive patient data beyond institutional boundaries, and while generating clinically verifiable visual explanations that clinicians can trust?***

To formalize this problem within a unified multi-modal diagnostic pipeline, the target diseases, structural anomalies, and their corresponding diagnostic modalities are defined below:

- **Pneumonia (X-ray Modality):** An acute, inflammatory parenchymal lung infection causing fluid, pus, and cellular debris accumulation inside the alveoli. This condition manifests on chest radiographs as areas of increased opacity, visible as dense lobar consolidations, patchy bronchopneumonic infiltrates, or diffuse interstitial markings that obscure the underlying clear pulmonary margins.
- **Tuberculosis (TB) (X-ray/CT Modality):** A chronic, granulomatous infectious disease caused by *Mycobacterium tuberculosis*. Radiographically, it exhibits a highly diverse presentation including apical fibro-nodular infiltrates, thick-walled cavitary lesions resulting from caseous tissue necrosis, hilar or mediastinal lymphadenopathy, or disseminated miliary micro-nodules distributed throughout the lung fields.
- **COVID-19 (X-ray Modality):** A highly infectious viral pneumonia caused by SARS-CoV-2. It presents with characteristic subpleural, peripheral, and basal ground-glass opacities (GGOs) that can progress rapidly into confluent, bilateral consolidations with extensive pulmonary involvement, creating acute ventilation-perfusion mismatching.
- **Chest Masses, Lung Cancer, and Tumor-like Abnormalities (CT Modality):** Neoplastic, space-occupying cellular proliferations within the lung parenchyma, mediastinum, or bronchial tree. Evaluated via high-resolution chest CT scans, these present as solid, sub-solid, or ground-glass solitary pulmonary nodules or large masses, characterized by malignant morphological signatures such as spiculation, lobulation, and eccentric calcification patterns.
- **Normal / Baseline Cases (Multi-Modal):** Thoracic imaging datasets exhibiting entirely pristine, healthy lung fields, intact vascular markings, and uncompromised pleural spaces, serving as the baseline control class for the multi-class objective function to prevent false positive classifications.

Successfully executing an automated system to address these conditions within a single unified framework introduces several technical, mathematical, and algorithmic challenges:

- **Extreme Morphological Subtlety and Feature Overlap:** Many targeted pathologies exhibit overlapping visual characteristics. For instance, early-stage lung cancer can mimic a localized tuberculous granuloma on a CT scan, while viral pneumonia, atypical bacterial infections, and early COVID-19 can display identical ground-glass signatures on a chest radiograph. Models must capture fine-grained features to avoid cross-class misclassification.
- **Strict Regulatory, Legal, and Privacy Constraints:** The system must comply

with zero-trust data architectures. Raw clinical DICOM files cannot be pooled or aggregated centrally, necessitating decentralized training algorithms where only abstract model parameters, localized weights, or gradient updates are shared over secure networks.

- **Statistical Heterogeneity (Non-IID Data Profiles):** The random partitioning of data across different clinical sites yields highly unbalanced label distributions (feature and label skewness). The framework must remain resilient against local client drift, preventing individual hospital nodes from skewing the global aggregated model optimization path.
- **Explainable Artificial Intelligence (XAI) Alignment:** The model must tightly couple its predictive classification heads with pixel-level attribution maps (such as Gradient-weighted Class Activation Mapping or Attention Maps). These visual explanations must pinpoint the exact diseased parenchymal region or mass boundary, mapping directly to recognized radiological signs.
- **Communication Efficiency and Deployment Constraints:** Transmitting heavy deep-learning model parameters across decentralized institutional nodes introduces severe bandwidth, latency, and communication overhead. The framework must utilize highly optimized architectures to enable real-world federated deployment across medical centers with varying network conditions.

To systematically overcome these challenges, this work designs, implements, and evaluates an integrated, enterprise-grade deep framework: *X-Fed NeuroScan*. By synthesizing advanced edge-computed federated learning algorithms for privacy-preserving, collaborative multi-institutional model development with cutting-edge Explainable AI (XAI) pipelines, *X-Fed NeuroScan* delivers highly accurate, safe, and transparent multi-class diagnostic outputs. This framework preserves patient data privacy by design while offering visual justifications that allow clinicians to confidently verify and act upon the model's findings.

## 1.2 Challenges in Medical Image Analysis

Medical imaging plays a fundamental role in modern healthcare, providing clinicians with valuable information for diagnosing, monitoring, and treating a wide range of diseases. Imaging modalities such as chest X-rays and Computed Tomography (CT) scans are routinely used to assess abnormalities within the thoracic region, including pneumonia, tuberculosis, pulmonary edema, and lung cancer. Despite the significant advancements in imaging technologies, the interpretation and analysis of medical images remain highly challenging tasks.

Medical image analysis traditionally relies on the expertise of radiologists and physicians who must examine large volumes of imaging data and identify subtle patterns that may indicate disease. The complexity of anatomical structures, variability among patients, and limitations of human perception can make accurate diagnosis difficult. These challenges can affect diagnostic accuracy, increase workload, and potentially delay clinical decision-making.

Table 1.1: Systemic problem matrix: Mapping technical challenges to the corresponding structural elements of the framework for thoracic disease detection.

| Core Challenge Category | Radiological / Clinical Impact | Algorithmic Mitigations |
|---|---|---|
| **Visual Anomaly Subtlety** | Faint ground-glass patches or early masses are overlooked during clinical triage. | Advanced multi-scale feature extractors and high-resolution spatial processing pipelines. |
| **Data Siloing & Privacy** | Small institutional datasets limit model generalization and trigger model overfitting. | Decentralized federated learning protocol; raw patient data never leaves the local node. |
| **Statistical Non-IID Skew** | Variations in scanner brands and local demographics degrade centralized optimization. | Weighted model aggregation strategies designed to stabilize decentralized parameter updates. |
| **Opaque Decision-Making** | Clinicians reject AI classification due to an inability to audit or trust the underlying logic. | Deep XAI layer coupling that outputs spatial heatmaps corresponding to anatomical lesions. |

## Complex Anatomical Structures

The human thoracic cavity contains numerous overlapping anatomical structures, including the lungs, heart, ribs, blood vessels, diaphragm, and surrounding soft tissues. In chest X-ray images, these structures are projected onto a two-dimensional plane, causing different tissues to overlap and potentially obscure important pathological findings.

The presence of overlapping structures makes it difficult for clinicians to accurately distinguish between normal anatomical variations and disease-related abnormalities. Small lesions, nodules, or early-stage infections may be hidden behind ribs or other dense structures, increasing the likelihood of missed diagnoses.

## Subtle Appearance of Early-Stage Diseases

Many thoracic diseases initially manifest as subtle visual changes that are difficult to detect through routine examination. Early-stage lung cancer, for example, may appear as a small pulmonary nodule that is barely distinguishable from surrounding tissue. Similarly, mild pneumonia or early tuberculosis infections may produce only faint abnormalities within the lung fields.

Detecting these subtle findings requires significant expertise and experience. Even highly trained radiologists may encounter difficulties when abnormalities are small, poorly defined, or located in anatomically complex regions.

## Variability in Disease Presentation

The same disease can appear differently across patients due to factors such as age, disease stage, genetic characteristics, and the presence of other medical conditions. Furthermore, different diseases may exhibit similar radiological appearances, making differential diagnosis challenging.

For example, pulmonary infections, inflammatory conditions, and certain types of lung cancer can produce similar opacities in chest X-rays. Likewise, benign and malignant lung nodules may share similar characteristics in CT scans. This variability increases diagnostic complexity and may lead to uncertainty during image interpretation.

## Dependence on Radiologist Expertise

Accurate medical image interpretation requires extensive education, training, and clinical experience. Radiologists must possess detailed knowledge of anatomy, pathology, imaging physics, and disease progression in order to make reliable assessments.

However, diagnostic performance can vary among clinicians depending on their level of expertise and familiarity with specific conditions. Less experienced practitioners may overlook subtle findings or misinterpret complex cases. As a result, diagnostic consistency can differ between healthcare professionals and institutions.

## Inter-Observer and Intra-Observer Variability

One of the well-known challenges in medical imaging is the variability in interpretation among radiologists. Different specialists may provide different assessments when evaluating the same image, a phenomenon known as inter-observer variability.

Additionally, the same radiologist may occasionally reach different conclusions when reviewing an image at different times, which is referred to as intra-observer variability. Such inconsistencies can affect diagnostic reliability and complicate treatment planning, particularly in borderline or ambiguous cases.

## Large Volume of Imaging Data

Modern healthcare systems generate enormous quantities of medical images on a daily basis. Radiologists are often required to review hundreds of imaging studies during a single workday, creating significant workload pressures.

CT examinations present an additional challenge because a single scan may contain hundreds of image slices that must be carefully reviewed. Thorough examination of such large datasets is time-consuming and can contribute to fatigue, potentially increasing the risk of oversight or diagnostic errors.

## Time Constraints in Clinical Practice

Healthcare professionals frequently operate under strict time constraints, particularly in busy hospitals and emergency departments. Rapid diagnosis is often essential for initiating timely treatment and improving patient outcomes.

The need to balance speed with diagnostic accuracy can be challenging, especially when dealing with complex imaging studies. Under time pressure, subtle abnormalities may be overlooked, and difficult cases may require additional consultation or follow-up examinations.

## Image Quality Limitations

The diagnostic quality of medical images can be affected by several factors, including patient movement, improper positioning, low contrast, imaging noise, and equipment-related artifacts. Poor image quality may obscure important pathological findings and make interpretation more difficult.

In chest X-ray imaging, underexposure or overexposure can reduce visibility of anatomical details. In CT imaging, motion artifacts caused by breathing or patient movement can degrade image clarity. Such limitations may reduce diagnostic confidence and necessitate repeat imaging procedures.

## Diagnostic Errors and Missed Findings

Despite advances in medical imaging technology, diagnostic errors remain an unavoidable challenge. Errors may occur due to perceptual limitations, cognitive biases, fatigue, distraction, or the inherently complex nature of certain cases.

Missed lung nodules, overlooked fractures, and undetected early-stage diseases represent some of the most common diagnostic challenges in thoracic imaging. Because delayed diagnosis can significantly impact patient outcomes, minimizing such errors remains a major objective within radiology.

## Challenges in Lung Cancer Detection

Lung cancer detection presents particularly significant difficulties due to the diverse appearance of pulmonary nodules. Nodules may vary greatly in size, shape, texture, density, and location. Small malignant nodules can closely resemble benign lesions or normal anatomical structures.

Furthermore, certain tumors may be partially obscured by blood vessels, ribs, or surrounding tissues. Accurate identification and characterization of suspicious nodules often require careful examination across multiple CT slices and comparison with previous studies, making the diagnostic process both complex and time-intensive.

### Increasing Demand for Radiological Services

The global demand for medical imaging services continues to grow due to population growth, aging demographics, and increased utilization of diagnostic imaging technologies. However, many healthcare systems face shortages of trained radiologists and imaging specialists.

This imbalance between demand and available expertise can lead to increased workloads, longer reporting times, and reduced access to timely diagnoses. The challenge is particularly pronounced in developing regions and underserved healthcare environments.

## 1.3 Background and Motivation

The rapid growth of medical imaging data, coupled with increased clinical workloads, has made conventional diagnostic workflows increasingly insufficient. Although automated analysis using computer vision holds considerable promise in improving diagnostic accuracy and efficiency, practical implementation is hindered by challenges such as patient privacy restrictions, limited data set diversity, and substantial computational and data transfer demands. Therefore, a thorough understanding of both the potential benefits and inherent limitations of AI in medical imaging is essential for the design of systems that are accurate, interpretable, and scalable in heterogeneous clinical environments.

### Rise of Computer Vision in Automated Analysis

The rapid digitization of healthcare has led to a dramatic increase in medical imaging data, including X-ray, CT, MRI, and ultrasound scans. Radiologists now face the dual challenges of handling large patient loads and interpreting large volumes of images quickly and accurately. Computer vision has emerged as a crucial tool for addressing these demands by enhancing diagnostic efficiency and accuracy.

To understand why computer vision is now widely adopted in clinical workflows, consider the following advantages:

- **High sensitivity to subtle patterns**: AI can detect faint fractures or early-stage tumors.
- **Rapid processing**: Thousands of images can be analyzed within seconds.
- **Consistency**: Predictions remain stable regardless of fatigue or emotional state.
- **Adaptability**: Deep models can generalize to various imaging modalities.

The effectiveness of computer vision in medical imaging is reinforced by major achievements in the field. These advancements primarily arise from deep learning architectures such as CNNs and Vision Transformers. In practice, computer vision supports several core tasks:

1. Classification of abnormal vs. normal scans.

2. Localization of suspicious areas using object detection.

3. Segmentation of organs, lesions, and fracture regions.

4. Identification of patterns that deviate from healthy anatomy.

A concise summary of computer vision benefits in medical imaging is outlined in Table 1.2.

Table 1.2: Key Advantages of Computer Vision in Medical Image Automation

| **Computer Vision Capability** | **Role in Medical Imaging** |
|---|---|
| Feature extraction | Detects subtle abnormalities beyond human perception |
| Automation of routine tasks | Reduces workload and radiologist fatigue |
| High-speed inference | Enables rapid triage in emergencies |
| Generalization across cases | Handles variations in anatomy and imaging conditions |

## Limitations of Centralized Machine Learning

Despite the promise of AI-based diagnostic systems, models trained in a centralized manner face several constraints. These limitations arise mainly from restrictions on data availability, data diversity, and practical constraints in medical settings.

Centralized learning typically suffers from:

- **Restricted data sharing**: Patient data cannot be freely exchanged due to privacy laws.
- **Lack of representation**: Single-hospital datasets fail to capture global patient diversity.
- **High storage and transfer requirements**: Medical images are large and expensive to move.
- **Data imbalance**: Rare conditions remain underrepresented, harming model performance.

These issues do not simply degrade model accuracy—they introduce significant risks into clinical AI deployment. The consequences can be better understood through the following observations:

1. Models trained in isolation often misclassify images from new machines or demographics.

2. Performance can fluctuate between institutions due to imaging protocol differences.

3. Bias can emerge when the training set fails to include certain patient groups.

Table 1.3 summarizes these core limitations in a structured manner.

Table 1.3: Challenges of Centralized Machine Learning in Healthcare

| **Limitation** | **Impact** |
|---|---|
| Privacy restrictions | Prevent hospitals from sharing sensitive patient data |
| Limited dataset diversity | Reduces generalization to different populations or imaging devices |
| Large data transfer requirements | Increases cost and delays in model development |
| Imbalanced condition frequency | Degrades performance on rare diseases or subtle abnormalities |

Centralized machine learning, while powerful, therefore struggles to meet the scalability, diversity, and privacy needs of real-world medical AI deployment. These challenges motivate the development of more robust learning paradigms capable of overcoming data fragmentation and institutional isolation.

## 1.4 Suggested Solution

The challenges identified in the medical imaging domain—including subtle abnormalities, limited dataset diversity, privacy constraints, and variability in imaging standards—necessitate a more advanced diagnostic support system. The proposed project aims to address these limitations by integrating three complementary technologies: Computer Vision, Explainable AI, and Federated Learning. Together, these components form an intelligent system capable of assisting clinicians in identifying fractures, tumors, infections, and other abnormalities more reliably and transparently.

The core idea behind the project is to design a computer vision pipeline capable of analyzing X-ray and other medical imaging modalities with high accuracy. Deep learning models will detect patterns that may be difficult for clinicians to recognize quickly, especially in emergency or high-stress situations. By augmenting image interpretation with AI-based classification and region highlighting, the system serves as a powerful second-opinion tool, reducing oversight and improving diagnostic confidence.

To ensure that the system remains trustworthy and clinically interpretable, Explainable AI techniques are incorporated to reveal how the model arrives at its predictions. This transparency allows medical professionals to understand the rationale behind the AI's outputs, compare them with clinical knowledge, and detect any mistakes or biases early.

Moreover, Federated Learning is utilized to overcome the limitations of centralized training. Instead of pooling patient data into a single server—a process that is often legally and ethically restricted—hospitals collaboratively train the model without exposing sensitive information. This approach significantly increases dataset diversity while maintaining strict compliance with privacy regulations.

In summary, the proposed solution seeks to:

- Improve diagnostic accuracy through advanced computer vision algorithms.
- Increase clinician trust via transparent and interpretable model outputs.
- Overcome data-sharing barriers by enabling collaborative, privacy-preserving training.

These three components work together to form a robust, scalable, and clinically meaningful AI-based diagnostic support framework.

## Explainable Artificial Intelligence (XAI) in Thoracic Diagnostics

Explainable Artificial Intelligence (XAI) encompasses a robust suite of mathematical formulations, post-hoc algorithmic frameworks, and intrinsic architectural designs engineered to systematically dismantle the "black-box" paradigm of deep neural networks. By transforming opaque, multi-layered statistical representations into highly transparent, human-interpretable, and clinically auditable insights, XAI bridges the gap between advanced deep learning and clinical practice.

As deep learning models grow exponentially in depth and structural complexity—leveraging billions of abstract parameters across dense convolutional layers or self-attention blocks—their internal latent spaces become completely divorced from standard human logic. In safety-critical fields like thoracic oncology and pulmonology, where an algorithmic failure can lead to catastrophic diagnostic errors, this lack of transparency presents a significant barrier to clinical adoption. XAI methods resolve this issue by providing verifiable visual or textual justifications that explain how and why an algorithm arrived at a specific diagnostic conclusion.

Within the specific domain of multi-class thoracic imaging (encompassing complex pathologies like pneumonia, COVID-19, tuberculosis, and lung cancer), explainability shifts from a theoretical asset to a strict clinical, ethical, and legal necessity. Rather than operating as a binary oracle that simply outputs a classification vector or a bounding box coordinates, an advanced explainable system maps its final prediction to specific, localized pixel-level features. These visual justifications allow radiologists and treating physicians to evaluate the physiological validity of the model's outputs. This capability serves as a vital confirmatory layer during high-stress triage workflows and protects against catastrophic false positives or negatives.

XAI supports and enhances clinical decision-making through several key mechanisms:

- **High-Precision Spatial Localization:** Identifying and mapping the exact spatial coordinates, pixel boundaries, or macro-anatomical lung segments that drove the model's objective function toward a specific classification.

- **Spurious Feature Disconnection and Artifact Auditing:** Enabling developers and clinicians to verify that the network is optimization-grounded in genuine, peer-reviewed pathological markers (e.g., spiculation, ground-glass opacities, or cavitary walls) rather than over-indexing on irrelevant artifacts, such as digital scanner text, patient positioning markers, rib contours, or random background noise.
- **Algorithmic Error Detection and Bias Mitigation:** Facilitating the systematic detection of hidden model vulnerabilities, such as shortcut learning, distribution shifts, severe dataset bias, or systemic misinterpretations of normal structural variants (e.g., vascular bifurcations mistaken for micro-nodules).
- **Interactive Clinical Cross-Examination:** Providing a transparent visual baseline that empowers clinicians to confidently verify, critique, or override automated AI predictions by applying their extensive domain expertise and real-world experience.
- **Medico-Legal Compliance and Auditability:** Generating a clear, deterministic trail of visual evidence for every diagnostic output. This documentation satisfies rigorous regulatory standards, fits seamlessly into institutional electronic health records (EHR), and can be effectively communicated in patient-facing clinical reports.

A wide array of XAI methodologies are utilized across modern deep thoracic diagnostics, categorized primarily into gradient-based post-hoc attributions and intrinsic attention mechanisms. Linear and non-linear pixel-level saliency maps compute the direct gradients of the output score with respect to the input image, capturing highly granular sensitivity profiles. To provide cleaner, structurally localized visualizations, advanced frameworks leverage Class Activation Mapping (CAM) and Gradient-weighted Class Activation Mapping (Grad-CAM). These techniques tap into the final convolutional layers of a network to capture coarse semantic features and project them back onto the input image as heatmaps.

Furthermore, with the introduction of Vision Transformers (ViTs) in medical imaging, attention-based visualization frameworks (such as Rollout or Value-Decomposition Attention) directly extract the internal self-attention matrix weights. This generates highly accurate heatmaps that reflect the global context and long-range structural dependencies learned by the network. Beyond pure spatial heatmaps, modern XAI pipelines also incorporate quantitative feature attribution scores, concept-based explanations, and automated natural-language diagnostic text generation. These advanced techniques map directly onto standard radiological vocabularies, ensuring that the model's reasoning aligns with established clinical workflows. Figure 1.1 illustrates the conceptual integration of an explainability layer within an automated diagnostic pipeline, highlighting how opaque latent representations are translated into interpretable spatial heatmaps.

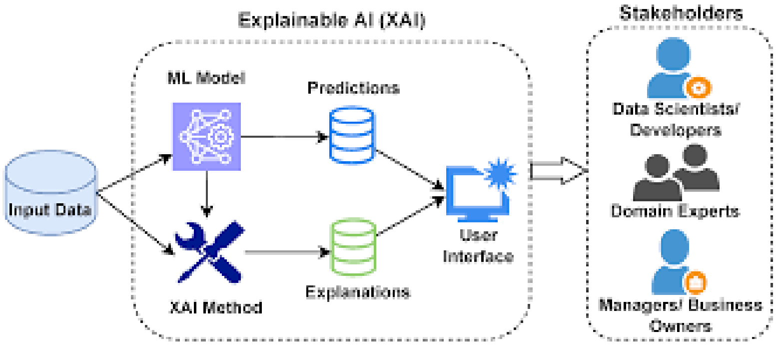



Figure 1.1: Explainable AI (XAI) layer integration within the automated diagnostic pipeline, illustrating the mapping of latent feature spaces to localized spatial heatmaps for clinical validation.

# Federated Learning Frameworks for Decentralized Thoracic Imaging

Federated Learning (FL) represents a transformative decentralized optimization paradigm designed to facilitate collaborative, multi-institutional machine learning model development while guaranteeing complete local data sovereignty. In classical centralized deep learning frameworks, large-scale medical imaging repositories must be aggregated from multiple disparate clinical sites into a singular, central cloud repository or a unified computing infrastructure. This centralized architectural model introduces massive single-point-of-failure vulnerabilities, extreme network bandwidth consumption, and significant regulatory hurdles.

Conversely, Federated Learning reverses this computational direction: rather than transmitting immense quantities of sensitive diagnostic imaging data over external networks to the centralized model, the abstract global model is transmitted directly to the decentralized institutional edge nodes. This foundational shift directly addresses the most critical hurdles in contemporary healthcare ecosystems, including strict patient privacy requirements, enterprise-grade data security protocols, legal liabilities, and the massive logistical burdens of cross-border medical dataset aggregation.

Within the high-stakes domain of clinical thoracic AI, datasets acquired across a multitude of independent hospitals, imaging centers, and regional health systems are fundamentally heterogeneous, geographically isolated, and legally protected[cite: 7, 10, 41]. Regulatory bodies enforce strict privacy statutes—such as the General Data Protection Regulation (GDPR) in the European Union and the Health Insurance Portability and Accountability Act (HIPAA) in the United States—which treat medical images and their associated metadata as highly sensitive Protected Health Information (PHI).

Federated Learning provides a mathematically sound solution to these constraints by ensuring that raw patient data remains isolated behind institutional firewall structures throughout the entire lifecycle of model optimization[cite: 46]. The global model undergoes iterative optimization cycles by distributing parameter weights to individual clients, executing localized stochastic gradient descent loops entirely on edge hardware, and transmitting only abstract weight deltas or gradient updates back to a central orchestrating server. This collaborative process enables the underlying network to learn from massive, multi-institutional datasets of diverse patient distributions and varying scanner hardware configurations, which is critical for building a highly generalizable model capable of detecting rare, subtle lung abnormalities.

Federated Learning enhances clinical artificial intelligence development through several mechanisms:

- **Complete Prevention of PHI Transmission:** Allowing diverse diagnostic centers to collaboratively train advanced deep networks without exposing, transmitting, or consolidating raw patient data.
- **Massive Global Generalizability and Scale:** Leveraging diverse datasets from distinct global populations, distinct geographic regions, and varying demographic slices, which directly prevents over-indexing on localized institutional distribution skews.
- **Significant Reduction of Data Breach Surfaces:** Mitigating systemic data breach risks and cyber-attacks by eliminating central honey-pot storage architectures, keeping sensitive information protected within local server environments.
- **Inherent Regulatory and Ethical Compliance:** Providing built-in compliance with evolving legal frameworks and data retention policies by minimizing data duplication and movement.
- **Continuous Edge-Driven Model Evolution:** Supporting lifelong and continuous learning patterns as edge institutions acquire new clinical imaging data during daily diagnostic workflows, without disrupting local institutional pipelines.

A multitude of aggregation strategies and architectural frameworks have been engineered to navigate the complex constraints of distributed medical imaging data. The foundational optimization baseline is *Federated Averaging (FedAvg)*, where the central server initializes global model weights, distributes them to selected clients, and then computes a coordinate-wise weighted average of the local parameters based on the specific volume of data hosted at each institutional node after local optimization rounds.

To stabilize convergence when dealing with severe non-IID client distributions, advanced aggregation algorithms like FedProx introduce proximal regularization terms to bound local parameter drift, while SCAFFOLD uses control variates to correct for localized client gradients. Furthermore, modern healthcare deployments implement security layers like Secure Aggregation (SecAgg) protocols based on cryptographic multi-party computation, alongside Differential Privacy (DP) techniques that add calibrated noise to local weight vectors to defend against reverse-engineering or model-inversion attacks. These privacy-preserving frameworks can also be paired with deep transfer learning, multi-task heads, and multi-modal feature fusion layers, allowing deep networks to optimize across distributed data streams while maintaining strong privacy guarantees. Figure 1.2 details the central-server approach to federated learning, illustrating the iterative parameter exchange loop between the global aggregator and the secure edge nodes.

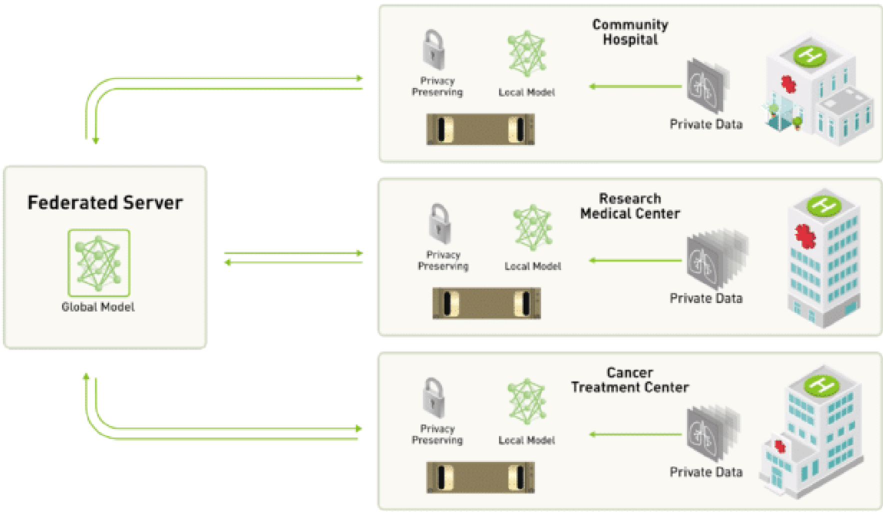



Figure 1.2: Centralized server architecture for federated learning, detailing the secure, iterative parameters exchange loop between the global orchestrator and localized institutional edge nodes.

## 1.5 Project Scope

The fundamental scope of this research and development initiative centers on engineering, validating, and establishing an enterprise-grade, decentralized, and highly explainable medical computer vision ecosystem dedicated exclusively to multi-class thoracic pathology diagnosis. This advanced platform is structurally and algorithmically optimized to detect, classify, and spatio-temporally localize four highly prevalent, clinically severe, and socio-economically impactful thoracic conditions: acute pneumonia, Coronavirus Disease 2019 (COVID-19), chronic pulmonary tuberculosis (TB), and malignant lung cancer/chest masses. These conditions were strategically selected due to their massive contribution to global respiratory morbidity and mortality, the high risk of catastrophic clinical consequences if their initial radiological presentation is overlooked, and the immense structural variability they present across projectional and cross-sectional medical imaging modalities. By synthesizing advanced deep neural architectures with zero-trust decentralized optimization frameworks and post-hoc visual attribution models, this project scope explicitly establishes a robust, privacy-preserving machine learning diagnostic engine that can operate seamlessly across multi-institutional hospital networks.

The core technical boundary of this project is strictly defined by the integration, training, and fine-tuning of multi-modal deep learning models engineered to process two distinct clinical imaging streams. For the detection and differentiation of acute inflammatory and viral respiratory conditions (pneumonia and COVID-19), the system utilizes high-resolution projectional Chest X-ray (CXR) DICOM datasets, capturing fine-grained variations in alveolar and interstitial attenuation. For chronic infectious diseases and neoplastic lesions (tuberculosis, dense pulmonary masses, and lung cancer), the analytical pipeline is expanded to ingest and process high-contrast, cross-sectional chest Computed Tomography (CT) scans, enabling the extraction of multi-planar spatial features, volumetric nodule boundaries, and subtle morphological variations like edge spiculation or irregular calcifications. The machine learning pipeline is mathematically optimized to balance the trade-offs between diagnostic sensitivity and specificity, guaranteeing high recall to minimize false negative outcomes in clinical triage while maintaining precision to prevent downstream diagnostic bottlenecks and patient anxiety.

To guarantee seamless real-world deployability, abstract technological feasibility, and practical utility within active clinical environments, the backend diagnostic engine is natively wrapped inside a secure, low-latency, and web-based enterprise application interface. This front-end medical console is architected to allow non-technical healthcare practitioners, including front-line triage nurses, general physicians, emergency room technicians, and specialized thoracic radiologists, to securely ingest imaging files, execute distributed inference pipelines, and interact dynamically with the localized diagnostic outputs without requiring prior knowledge of deep learning mechanics, server configurations, or parameter tuning. While this cross-platform interface serves as a critical component for simulating real-world medical deployment and validating clinical integration, the primary focus, computational allocation, and academic scope of this project remain rigorously fixed upon the underlying mathematical formulations, decentralized federated optimization trajectories, model convergence stability under non-IID conditions, and the quantitative reliability of the explainable AI visualizations.

The functional boundaries and high-level architectural mandates of the system are categorized across four distinct, integrated operational vectors:

- **Automated Multi-Class Thoracic Disease Identification:** Seamlessly processing heterogenous, multi-modal clinical image streams to perform differential classification across pneumonia, COVID-19, tuberculosis, and lung cancer/chest masses, effectively separating them from pristine normal baseline images.
- **Decentralized, Privacy-Preserving Collaborative Optimization:** Executing a secure, zero-trust federated learning protocol that orchestrates iterative parameter aggregation loops across distributed edge servers, strictly ensuring that protected health information (PHI) and raw patient images never exit local institutional firewalls.
- **Explainable Spatial Localization and Visual Justification:** Automatically coupling all deterministic classification outputs with high-fidelity, pixel-level attribution maps and self-attention weight heatmaps, explicitly highlighting the exact anatomical contours, parenchymal patches, or nodular boundaries that informed the network's predictive pathways.
- **Enterprise Web Deployment and Ingestion Architecture:** Constructing a secure web console featuring automated DICOM processing, smooth scaling across diverse hardware devices, and clean data visualizations tailored to match active clinical triage environments.

Crucially, the scope of this project is strictly defined by an assistive, clinician-in-the-loop operational philosophy and does *not*, under any circumstances, aim to replace or supersede the definitive diagnostic judgment, clinical intuition, or legal accountability of professional medical practitioners or board-certified radiologists. Instead, the framework is meticulously engineered to function as an advanced, high-throughput second-opinion mechanism and triage assistant capable of reducing diagnostic oversights, mitigating human cognitive fatigue, and accelerating clinical turnaround times in high-volume emergency environments.

Several technical boundaries are explicitly designated as out-of-scope for the current implementation phase to ensure absolute focus on the core machine learning and federated optimization objectives. The system is constrained to the four specified target conditions; the inclusion of other diffuse thoracic abnormalities (such as pneumothorax, chronic obstructive pulmonary disease, emphysema, or pleural effusions) falls entirely outside the project's current boundaries. Furthermore, while the federated learning layer is mathematically modeled and simulated to replicate a multi-hospital clinical network with severe statistical non-IID data distributions, the deployment scope is executed within distributed, multi-GPU simulated edge clusters rather than active, live-production hospital server backbones. Real-world institutional deployment across live physical hospital networks, real-time edge streaming, and expansion to handle non-thoracic disease modalities are deferred to future operational iterations, serving as logical continuations of this foundational architectural framework.

Table 1.4: Project scope boundary matrix: Delineating technical inclusions, explicit exclusions, and target clinical metrics.

| **Scope Category** | **Explicit Technical Inclusions** | **Explicit Technical Exclusions** |
|---|---|---|
| **Target Pathologies** | Pneumonia, COVID-19, Pulmonary Tuberculosis (TB), and Lung Cancer/Chest Masses. | Pneumothorax, Emphysema, Pleural Effusions, Scoliosis, and Bone Fractures. |
| **Imaging Modalities** | High-resolution projectional Chest X-rays (CXR) and cross-sectional chest CT scans. | Magnetic Resonance Imaging (MRI), Ultrasound, and Positron Emission Tomography (PET). |
| **Data Protocols** | Zero-trust Federated Learning (FedAvg) using simulated non-IID distributed edge clients. | Live central cloud data pooling or live physical network integration into active EHR backbones. |
| **Interpretability Layer** | Post-hoc visual attributions via Grad-CAM and self-attention rollout maps. | Automated clinical prescription generation or natural language therapy recommendations. |

## 1.6 Project Objectives

The objectives of this project are centered around creating a reliable, accurate, and accessible diagnostic support system for bone-fracture and chest tumor identification. The project aims to ensure that the system not only performs well in terms of machine learning metrics but also provides practical value to clinicians through usability, efficiency, and cost-effectiveness. Achieving clinical-level performance requires a well-defined set of goals that align both with technical constraints and real-world medical needs.

A primary objective is to develop a computer vision model capable of achieving high diagnostic performance. In particular, the system is designed to reach an accuracy above

90% across its targeted conditions while maintaining a recall rate that minimizes the risk of missed fractures or tumors. Since false negatives can lead to severe clinical consequences, optimizing recall is essential for ensuring the safety and reliability of the system.

In addition to predictive performance, another major objective is to deliver an intuitive and engaging web interface that enables medical personnel to interact with the system easily. The interface should allow users to upload images, view results, and interpret highlighted regions without requiring technical expertise. Ease of use is critical for facilitating adoption in real clinical workflows, where time efficiency and clarity are important.

Operational efficiency is also a key objective. The system must be designed in a manner that minimizes computational overhead and reduces ongoing costs. This includes optimizing model size, using lightweight inference strategies where possible, and ensuring that the web application remains responsive even with limited hardware resources.

To summarize, the project seeks to meet the following objectives:

- Achieve a diagnostic accuracy exceeding 90% on bone-fracture and chest tumor detection tasks.
- Maintain a high recall rate to reduce the likelihood of missed abnormalities.
- Provide a user-friendly, accessible, and engaging web interface suitable for non-technical clinical staff.
- Develop an efficient system that minimizes operational and computational costs.

Collectively, these objectives define the criteria by which the success of the project will be evaluated and guide the design decisions throughout development.

## 1.7 Project Challenges

Despite the remarkable progress achieved in recent years, medical image analysis remains a challenging field. The complexity of medical data, the critical nature of clinical decision-making, and the variability of imaging procedures introduce numerous obstacles that must be addressed to develop reliable and clinically applicable systems. The following subsections discuss the major challenges associated with medical image analysis, particularly in the context of chest disease detection using X-ray images and lung cancer detection using CT scans.

### Limited Availability of High-Quality Medical Data

One of the most significant challenges in medical image analysis is the limited availability of large, high-quality, and well-annotated datasets. Deep learning models generally require substantial amounts of labeled data to achieve high performance and robust generalization. However, obtaining medical images with accurate annotations is often difficult and expensive.

Medical image annotation requires the expertise of trained radiologists or medical specialists who must carefully examine each image and identify pathological findings. This process is time-consuming and resource-intensive. Furthermore, patient privacy regulations and ethical considerations often restrict the sharing of medical data between healthcare institutions, limiting the availability of publicly accessible datasets.

In the context of chest X-ray analysis, some diseases may occur infrequently, leading to severe class imbalance within datasets. Similarly, for lung cancer detection from CT scans, obtaining sufficiently large collections of confirmed malignant and benign cases can be challenging. These limitations can negatively affect model training and reduce diagnostic performance.

## Variability in Image Acquisition

Medical images are acquired using different imaging devices, protocols, and settings across hospitals and healthcare facilities. As a result, images may exhibit significant variations in quality, resolution, contrast, brightness, noise levels, and anatomical positioning.

Chest X-ray images obtained from different institutions may vary due to differences in machine manufacturers, patient positioning techniques, exposure settings, and image preprocessing procedures. Likewise, CT scans may differ in slice thickness, reconstruction algorithms, scanning protocols, and radiation dose levels.

Such variability can lead to distribution shifts between training and deployment environments. A model trained on data from one institution may experience performance degradation when applied to images acquired in another clinical setting. Ensuring robustness across diverse imaging conditions remains a major challenge for medical image analysis systems.

## Image Noise and Artifacts

Medical images frequently contain noise and artifacts that can obscure important anatomical structures and pathological findings. These imperfections may arise from patient movement, hardware limitations, acquisition errors, or environmental factors during image capture.

In chest X-ray imaging, artifacts such as medical devices, implants, external objects, and improper positioning can interfere with disease detection. CT scans may contain motion artifacts, beam-hardening effects, metal artifacts, and reconstruction-related distortions.

The presence of noise and artifacts can significantly reduce image quality and negatively affect the performance of automated detection systems. Consequently, effective preprocessing techniques, including image enhancement, denoising, and artifact removal, are often required before analysis.

## Complexity of Disease Manifestations

Many thoracic diseases exhibit highly variable visual characteristics, making automated detection particularly challenging. Diseases may present differently depending on their severity, progression stage, patient demographics, and coexisting medical conditions.

For example, pneumonia, tuberculosis, pulmonary edema, and COVID-19 may produce overlapping radiographic features in chest X-rays. Similarly, lung nodules observed in CT scans can vary considerably in size, shape, texture, density, and location. Some malignant nodules closely resemble benign abnormalities, increasing the difficulty of accurate classification.

Additionally, certain diseases may manifest as subtle changes that are difficult to detect even for experienced radiologists. Developing models capable of recognizing both obvious and subtle pathological patterns remains a critical research challenge.

## Class Imbalance in Medical Datasets

Medical datasets often suffer from class imbalance, where healthy cases significantly outnumber disease-positive cases or where some diseases occur much less frequently than others. This imbalance can lead machine learning models to favor majority classes during training.

For instance, in chest X-ray datasets, normal scans may constitute a large proportion of the available data, while rare diseases may only be represented by a small number of samples. Similarly, in lung cancer screening datasets, malignant nodules may be considerably less common than benign findings.

Class imbalance can result in poor sensitivity for detecting rare but clinically important conditions. Addressing this issue often requires specialized techniques such as data augmentation, oversampling, synthetic data generation, weighted loss functions, and balanced training strategies.

## Interpretability and Explainability

Healthcare professionals require transparency and trust when using artificial intelligence systems in clinical practice. Many state-of-the-art deep learning models function as black-box systems, producing predictions without providing clear explanations for their decisions.

In medical diagnosis, explainability is particularly important because clinicians must understand the reasoning behind automated recommendations before incorporating them into patient care decisions. A prediction without justification may not be sufficient in high-stakes medical environments.

To address this challenge, researchers employ explainable AI techniques such as saliency maps, Grad-CAM visualizations, attention mechanisms, and feature attribution methods. These approaches help identify image regions that contribute most strongly to model predictions, thereby improving transparency and clinical confidence.

## Generalization and Robustness

A medical image analysis system must perform reliably across different patient populations, imaging devices, healthcare institutions, and geographical regions. However, models trained on specific datasets may inadvertently learn dataset-specific characteristics rather than clinically meaningful features.

This issue can lead to reduced performance when models encounter unseen data during real-world deployment. Factors such as demographic differences, variations in disease prevalence, and differences in imaging protocols can all affect model generalization.

Ensuring robustness requires diverse training datasets, rigorous validation procedures, external testing across multiple institutions, and continuous performance monitoring after deployment.

## Three-Dimensional Nature of CT Data

Unlike chest X-rays, which provide two-dimensional projections, CT scans generate three-dimensional volumetric data. While this additional information improves diagnostic capabilities, it also introduces substantial computational and analytical challenges.

CT volumes may contain hundreds of slices for a single patient examination. Processing such large datasets requires considerable memory, storage capacity, and computational resources. Furthermore, accurately identifying lung nodules often requires analyzing spatial relationships across multiple slices.

Advanced techniques such as 3D convolutional neural networks, volumetric segmentation algorithms, and multi-view learning approaches are commonly employed to address these challenges. However, these methods generally demand greater computational complexity and longer training times.

## Accurate Segmentation of Anatomical Structures

Segmentation is a fundamental step in many medical image analysis pipelines. It involves identifying and isolating relevant anatomical structures such as lungs, airways, lesions, tumors, and nodules.

Accurate segmentation is particularly challenging when boundaries are unclear, tissues overlap, or pathological regions exhibit irregular shapes. In chest X-ray images, overlapping anatomical structures can complicate lung segmentation. In CT scans, tumors may have poorly defined margins or be attached to surrounding tissues.

Segmentation errors can propagate through subsequent analysis stages and adversely affect disease detection and classification performance. Therefore, robust segmentation algorithms remain an important area of ongoing research.

## Clinical Validation and Regulatory Requirements

Developing an accurate machine learning model is only one step toward clinical adoption. Medical image analysis systems must undergo extensive validation to demonstrate safety, reliability, and effectiveness in real-world healthcare environments.

Clinical validation typically involves testing models on large and diverse patient populations, comparing performance against expert radiologists, and assessing their impact on clinical workflows. Furthermore, healthcare technologies are subject to strict regulatory requirements and quality standards before deployment.

Meeting these regulatory and clinical validation requirements can be a lengthy and resource-intensive process, but it is essential for ensuring patient safety and maintaining trust in AI-assisted diagnostic systems.

## Ethical and Privacy Considerations

Medical image analysis systems operate on highly sensitive patient information. Protecting patient privacy and ensuring compliance with healthcare regulations are critical responsibilities throughout the development lifecycle.

Issues such as data security, informed consent, data ownership, and algorithmic bias must be carefully addressed. Models trained on non-representative datasets may exhibit reduced performance for certain demographic groups, potentially leading to inequitable healthcare outcomes.

Consequently, ethical AI development requires responsible data management practices, fairness assessments, transparency, and continuous evaluation to ensure that diagnostic systems benefit all patient populations.

## Computational Resource Requirements

Modern deep learning architectures used in medical image analysis often contain millions of trainable parameters and require substantial computational resources for training and inference. High-resolution chest X-ray images and volumetric CT scans further increase computational demands.

Training advanced models frequently requires powerful Graphics Processing Units (GPUs), large memory capacities, and extensive training times. These requirements may limit accessibility for smaller healthcare institutions and research groups with limited computational infrastructure.

Optimizing model efficiency while maintaining diagnostic accuracy remains an important challenge, particularly for deployment in resource-constrained clinical environments.

## 1.8 Summary

This chapter established the fundamental motivation, core medical context, and foundational technical architecture behind the development of an intelligent, decentralized, and highly transparent thoracic medical imaging platform. The rapid expansion of diagnostic imaging workloads within modern healthcare environments, combined with the extreme physiological subtlety of multi-class thoracic anomalies, has created an urgent, safety-critical requirement for enterprise-grade computer-aided diagnostic systems. Traditional radiographical interpretation processes are inherently vulnerable to human oversight caused by cognitive fatigue, subtle or overlapping pathological manifestations, and massive data distribution shifts induced by varying institutional imaging hardware and patient demographic skews.

To address these vulnerabilities, this chapter comprehensively explored the integration of advanced computer vision pipelines within clinical workflows, specifically targeting the detection and segmentation of highly infectious respiratory conditions and malignant neoplastic lesions. While deep learning architectures have demonstrated remarkable success in extracting high-level semantic abstractions from medical image grids, conventional centralized optimization paradigms face severe implementation barriers. Strict global data protection regulations and the risk of catastrophic data breaches legally restrict institutions from pooling sensitive patient records, which ultimately triggers model overfitting and prevents the development of generalized, clinically robust artificial intelligence models.

The rationale for the proposed platform directly resolves this technical and legal impasse by synthesizing state-of-the-art multi-scale deep neural networks with cutting-edge Explainable AI (XAI) pipelines and privacy-preserving Federated Learning (FL) aggregation strategies. This integrated design optimizes multi-class diagnostic sensitivity while providing pixel-level spatial attributions that map directly to peer-reviewed radiological criteria, allowing clinicians to confidently verify and audit the underlying decision-making logic. Simultaneously, the decentralized optimization framework enables secure, multi-institutional collaboration by transmitting only encrypted, abstract parameter updates over secure network layers, keeping raw patient records entirely localized and protected behind institutional firewalls.

Furthermore, the scope and technical objectives of this research were precisely defined to guide development toward real-world clinical viability. The architecture focuses on achieving maximum diagnostic sensitivity and localization accuracy for pneumonia, COVID-19, tuberculosis, and lung cancer across multi-modal image streams, while ensuring communication efficiency, robust edge-node security, and seamless clinical workflow integration. This chapter detailed the algorithmic and logistical challenges anticipated during development—such as severe statistical non-IID data skew, feature overlap, gradient drift, and adversarial inference risks—setting a clear benchmark for evaluating the platform's performance. In conclusion, this introductory analysis establishes the theoretical and practical foundation for an interpretable, privacy-preserving diagnostic platform, setting the stage for the detailed methodologies, experimental designs, and comparative performance evaluations presented in the subsequent chapters of this work.

# CHAPTER 2
# LITERATURE REVIEW

# Chapter 2 Literature Review

In this chapter, a literature review is provided in project-related areas. To build the framework (X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification), we need to get familiar with foundations of medical image analysis, deep learning-based classification techniques, and the challenges associated with handling sensitive clinical data. We also examine the principles and methodologies of Federated Learning (FL) as a privacy-preserving approach for multi-institutional model training, as well as Explainable AI (XAI) techniques that enhance model transparency and clinical trust. Additionally, this chapter reviews existing commercial medical imaging applications, related academic research, and publicly available datasets. Finally, we evaluate the tools, platforms, and frameworks that support the development of deep learning models, federated learning workflows, and explainable AI components.

The chapter is organized as follows:

- Section 2.1: Medical image history and challenges, with subsections on bone fractures and chest diseases.
- Section 2.2: Explainable AI (XAI) in medical image detection.
- Section 2.3: Federated Learning (FL) in medical image detection.
- Section 2.4: Combined approaches integrating detection with XAI and FL.
- Section 2.5: Summary of related papers.
- Section 2.6: Similar commercial applications.
- Section 2.7: Summary of the chapter.

## 2.1 Medical Image Diagnosis

**Medical imaging** is the cornerstone of modern diagnosis, providing non-invasive visualization of the body's interior using techniques like X-rays, Computed Tomography (CT), and Ultrasound. Due to the high volume and complexity of medical scans, interpretation is often time-consuming and subject to human variability. This process is increasingly enhanced by **Artificial Intelligence (AI)**, particularly **Computer Vision (CV)** and Deep Learning (DL) methods.

While traditional Deep Learning (DL) approaches have achieved high accuracy in radiological analysis for disease detection and bone fracture localization, their clinical deployment is significantly hindered. The primary issues stem from two critical needs: the strict requirement for **data privacy** and the **black-box nature** of AI decision-making. These barriers prevent the widespread, trustworthy adoption of powerful AI models in healthcare settings.

Consequently, the research focus has shifted toward specialized solutions: **Federated**

**Learning (FL)**, a privacy-preserving paradigm, and **Explainable AI (XAI)**, a necessary component for model transparency. This work provides a systematic review of the state-of-the-art in these domains. It begins by examining advanced deep learning architectures, analyzing the role of FL in data collaboration, and reviewing XAI techniques.

We will first detail the related work in chest X-ray diagnosis and bone fracture detection. Subsequently, it will explore the use of XAI and FL specifically within these two application areas, identifying the gap that this research aims to bridge.

### 2.1.1 Lung Cancer Detection

Lung cancer is one of the most prevalent and life-threatening malignancies worldwide and remains a leading cause of cancer-related mortality. According to global cancer statistics, lung cancer accounts for a significant proportion of newly diagnosed cancer cases each year and is responsible for more deaths than several other major cancers combined. The high mortality rate associated with lung cancer is largely attributed to the fact that the disease is frequently diagnosed at advanced stages, when treatment options become more limited and patient prognosis deteriorates considerably. Consequently, early detection has become one of the most important objectives in modern thoracic oncology, as timely diagnosis can substantially improve treatment outcomes and survival rates.

Lung cancer originates from the uncontrolled proliferation of abnormal cells within lung tissues. Over time, these abnormal cells can form tumors that interfere with normal respiratory function and may eventually spread to other parts of the body through a process known as metastasis. The disease can affect various regions of the lungs, including the bronchi, bronchioles, and alveolar structures. Depending on the cellular origin and pathological characteristics, lung cancer is generally classified into two major categories: Small Cell Lung Cancer (SCLC) and Non-Small Cell Lung Cancer (NSCLC).

Non-Small Cell Lung Cancer represents approximately 85% of all diagnosed lung cancer cases and includes several subtypes such as adenocarcinoma, squamous cell carcinoma, and large cell carcinoma. Adenocarcinoma is the most common subtype and frequently develops in the peripheral regions of the lungs. Squamous cell carcinoma often arises near the central airways and is strongly associated with smoking history. Large cell carcinoma is less common but tends to exhibit aggressive growth characteristics. Small Cell Lung Cancer, although less prevalent, is generally more aggressive and is characterized by rapid growth and early metastasis, making timely diagnosis particularly important.

The development of lung cancer is associated with multiple risk factors. Cigarette smoking remains the most significant risk factor and is responsible for the majority of lung cancer cases worldwide. Tobacco smoke contains numerous carcinogenic compounds capable of inducing genetic mutations that disrupt normal cellular regulation. However, lung cancer can also occur in non-smokers due to factors such as exposure to secondhand smoke, environmental pollutants, occupational hazards, radon gas, asbestos exposure, genetic predisposition, and chronic pulmonary diseases. The multifactorial nature of lung cancer highlights the importance of comprehensive screening and diagnostic strategies that can identify disease manifestations at their earliest stages.

One of the major challenges associated with lung cancer is that many patients remain asymptomatic during the early stages of disease progression. Tumors may develop and grow gradually without producing noticeable symptoms, allowing the disease to advance undetected. As a result, a substantial proportion of patients receive a diagnosis only after the cancer has reached a locally advanced or metastatic stage. This clinical characteristic underscores the importance of imaging-based screening programs and computer-aided diagnostic systems capable of identifying subtle radiological abnormalities before the onset of severe symptoms.

As the disease progresses, patients may begin to exhibit a variety of respiratory and systemic manifestations. One of the most commonly reported symptoms is a persistent cough that does not resolve over time or exhibits a noticeable change in character. Patients with chronic coughs may experience increased frequency, severity, or duration of symptoms compared to their baseline condition. In many cases, persistent coughing serves as one of the earliest clinical indicators prompting further medical investigation.

Another important symptom is hemoptysis, which refers to the presence of blood in sputum or respiratory secretions. Hemoptysis may occur when the tumor invades nearby blood vessels or causes irritation within the respiratory tract. Although not all patients experience this symptom, its presence often warrants immediate clinical evaluation due to its association with serious pulmonary conditions, including malignancy.

Shortness of breath, medically referred to as dyspnea, is also frequently observed among lung cancer patients. Dyspnea may result from airway obstruction caused by tumor growth, reduced lung capacity, pleural effusion, or impaired gas exchange within affected pulmonary tissues. As tumors increase in size, respiratory function may progressively decline, leading to noticeable breathing difficulties during physical activity or even at rest.

Chest pain represents another common manifestation of lung cancer, particularly when tumors extend into surrounding structures such as the pleura, chest wall, or mediastinum. Patients often describe the pain as persistent, localized, and exacerbated by deep breathing, coughing, or physical movement. The severity of chest pain may increase as the disease progresses and adjacent tissues become involved.

In addition to respiratory symptoms, patients frequently experience systemic manifestations that reflect the broader physiological impact of malignancy. Unexplained weight loss is one of the most significant warning signs and may occur even in the absence of dietary changes or increased physical activity. Cancer-related metabolic alterations often contribute to progressive weight reduction and muscle wasting. Similarly, persistent fatigue and generalized weakness are common among patients and can substantially impair quality of life and daily functioning.

Advanced lung cancer may also produce symptoms associated with metastatic spread. For example, metastasis to the skeletal system may cause bone pain and pathological fractures, while brain metastases can lead to neurological symptoms such as headaches, dizziness, seizures, or cognitive impairment. Liver involvement may result in jaundice and abdominal discomfort. The presence of such symptoms often indicates advanced disease and is generally associated with a less favorable prognosis.

From a diagnostic perspective, medical imaging plays a central role in lung cancer detection, characterization, staging, and treatment planning. Imaging techniques enable clinicians to visualize pulmonary abnormalities that may not be detectable through physical examination alone. Among the available imaging modalities, chest radiography and computed tomography are the most commonly utilized for initial evaluation and diagnostic assessment.

Chest X-ray imaging is frequently used as a first-line diagnostic tool due to its accessibility, low cost, and widespread availability. Radiographic findings suggestive of lung cancer may include pulmonary nodules, masses, hilar enlargement, pleural effusion, or areas of abnormal opacity. However, chest radiographs possess limited sensitivity for detecting small lesions and early-stage tumors. Consequently, subtle abnormalities may remain undetected, particularly when located behind anatomical structures such as ribs or the mediastinum.

Computed Tomography (CT) has become the gold standard imaging modality for lung cancer assessment because of its ability to generate high-resolution cross-sectional images of thoracic anatomy. CT scans provide significantly greater anatomical detail than conventional radiographs and enable the visualization of small pulmonary nodules, irregular lesion margins, tissue density variations, and regional lymph node involvement. Furthermore, CT imaging facilitates volumetric analysis and supports accurate tumor localization, measurement, and staging.

Radiologists typically evaluate several imaging characteristics when assessing potential lung malignancies. These characteristics include lesion size, shape, margin irregularity, internal texture, density patterns, calcification, spiculation, and growth behavior over time. Malignant nodules often exhibit irregular or spiculated borders, heterogeneous density patterns, and progressive enlargement across sequential imaging studies. The interpretation of these imaging features requires considerable expertise and experience, motivating the development of computer-aided diagnostic systems capable of assisting clinicians in image analysis.

Figure 2.1 illustrates representative examples of CT findings commonly associated with lung cancer.

Recent advances in artificial intelligence and deep learning have significantly transformed the field of medical image analysis. Convolutional Neural Networks (CNNs) and related deep learning architectures have demonstrated remarkable capabilities in automatically extracting discriminative imaging features directly from raw medical images. In the context of lung cancer detection, deep learning models can learn complex patterns associated with pulmonary nodules, tumor morphology, and malignancy-related characteristics without requiring handcrafted feature engineering.

The application of deep learning to lung cancer diagnosis offers several advantages. Automated systems can process large volumes of imaging data efficiently, assist radiologists in identifying subtle abnormalities, reduce inter-observer variability, and potentially improve diagnostic consistency. Furthermore, modern explainability techniques enable visualization of image regions contributing to model predictions, thereby enhancing transparency and supporting clinical decision-making.

Figure 2.1: Representative CT imaging findings associated with lung cancer.

Despite these advances, several challenges remain. Lung lesions exhibit substantial variability in size, shape, texture, and anatomical location. Additionally, imaging protocols may differ across institutions, resulting in variations in image quality and acquisition parameters. Addressing these challenges requires robust model design, diverse training datasets, and rigorous validation procedures. Nevertheless, artificial intelligence continues to represent a promising direction for improving early lung cancer detection and supporting healthcare professionals in clinical practice.

In summary, lung cancer is a highly prevalent and potentially fatal disease whose prognosis depends heavily on early diagnosis. Because symptoms often emerge only after significant disease progression, imaging-based screening and diagnostic methods play a crucial role in patient management. Computed tomography has become the preferred modality for lung cancer detection due to its superior anatomical detail and sensitivity to early-stage abnormalities. Recent developments in deep learning have further enhanced the ability of computer-aided diagnostic systems to identify malignant findings, making AI-assisted lung cancer detection an increasingly important area of research within medical image analysis.

### 2.1.2 Chest Diseases Detection

The chest, encompassing the thoracic cavity, houses vital organs such as the lungs, heart, and major blood vessels, making it susceptible to various respiratory diseases that can be detected through imaging techniques like chest X-rays (CXR). CXR is a non-invasive, cost-effective diagnostic tool widely used for identifying abnormalities in lung tissue. We focus on four key categories for multiclassification: pneumonia, tuberculosis (TB), COVID-19, and normal (healthy) cases.

- **Pneumonia** is an inflammatory condition of the lung alveoli, often caused by bacterial, viral, or fungal infections, leading to fluid-filled sacs visible as opacities on

CXR.

- **Tuberculosis** is a bacterial infection primarily caused by Mycobacterium tuberculosis, characterized by granulomas, cavities, or nodular infiltrates in the lungs, which can be chronic and contagious.
- **COVID-19**, caused by the SARS-CoV-2 virus, presents with ground-glass opacities, consolidations, and bilateral lung involvement on CXR, often progressing rapidly.
- **Normal cases** exhibit clear lung fields without pathological signs, serving as a baseline for comparison.

Accurate detection of these diseases via deep learning (DL) models on CXR images is crucial for timely intervention, especially in resource-limited settings [1] [2].

### 2.1.3 Target Diseases and Their Radiological Characteristics

This study focuses on four major respiratory and thoracic pathological conditions that pose substantial public health concerns worldwide and are among the most heavily investigated diseases in computer-aided diagnostic (CAD) systems: pneumonia, Coronavirus Disease 2019 (COVID-19), tuberculosis (TB), and lung cancer. These diseases represent complex diagnostic challenges due to the diversity of their clinical presentations, etiologies, and progressive courses. While pneumonia, COVID-19, and tuberculosis primarily affect the parenchymal lung architecture and are traditionally evaluated using projectional Chest X-ray (CXR) imaging, lung cancer involves neoplastic tissue transformation that requires cross-sectional imaging through Computed Tomography (CT) scans of the chest for accurate screening, characterization, and staging.

Although these conditions share overlapping clinical respiratory indicators, they exhibit fundamentally distinct pathological, cellular, and architectural alterations. These manifestations transform into specific radiographic features that can guide deep learning algorithms and clinical practitioners toward precise, differential classification. Understanding the comprehensive pathophysiology, clinical symptomatology, and radiological presentations of these target diseases across different imaging modalities is an absolute prerequisite for engineering robust multi-class automated classification systems.

#### Pneumonia

Pneumonia is an acute inflammatory condition of the lung parenchyma that primarily involves the alveoli, respiratory bronchioles, and interstitial spaces. The condition is triggered by a wide array of pathogenic microorganisms, including bacterial strains (most prominently *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Mycoplasma pneumoniae*), viral agents (such as influenza viruses and respiratory syncytial virus), and fungal species (such as *Pneumocystis jirovecii*, typically observed in immunocompromised individuals). The entry of these pathogens into the lower respiratory tract incites an intense local immune response, characterized by the recruitment of polymorphonuclear leukocytes, hyperemic vascular engorgement, and cellular exudation. Consequently, the normally air-filled alveolar spaces become progressively inundated with inflammatory exudate, pus, cellular debris, and fibrinous fluid, a pathophysiological state known as consolidation. This cellular process heavily compromises regional gas exchange, producing ventilation-perfusion ($V/Q$) mismatching and subsequent hypoxemia.

The clinical presentation of pneumonia varies extensively based on the patient's age, immune status, and the specific causative pathogen. Common clinical symptoms include:

- **Persistent Productive Cough:** Often accompanied by the expectoration of thick, purulent, or rust-colored mucus/sputum resulting from active alveolar exudation.
- **High Fever and Rigors:** Acute onset of elevated core body temperature accompanied by shaking chills, mediated by pyrogenic cytokines (such as IL-1, IL-6, and TNF-$\alpha$) acting on the hypothalamus.
- **Exertional and Rest Shortness of Breath (Dyspnea):** A direct consequence of decreased lung compliance and reduced functional alveolar surface area for blood oxygenation.
- **Pleuritic Chest Pain:** A sharp, localized pain that intensifies during deep inspiration or coughing, indicating that the inflammatory process has spread to the visceral and parietal pleura.
- **Systemic Constitutional Symptoms:** Marked fatigue, generalized malaise, myalgia, arthralgia, and anorexia due to systemic inflammatory cytokine release.
- **Tachypnea and Tachycardia:** Elevated respiratory rate ($> 20$ breaths per minute) and heart rate ($> 100$ beats per minute) as compensatory mechanisms to sustain systemic oxygen delivery.

On standard projectional chest radiographs, pneumonia manifests primarily as areas of increased attenuation and opacity where the normal black, air-filled pulmonary parenchyma is replaced by white, fluid-dense consolidation. Structurally, pneumonia presents in three main radiographic distributions:

1. **Lobar Pneumonia:** Characterized by dense, homogeneous opacification sharply delimited by anatomical interlobar fissures, typically signifying a bacterial etiology like *Streptococcus pneumoniae*. Within these consolidations, the "air bronchogram" sign is frequently visible, where air-filled, dark bronchi stand out against the surrounding opacified, fluid-filled alveoli.

2. **Bronchopneumonia (Lobular Pneumonia):** Appears as patchy, ill-defined, multifocal opacities distributed along the bronchial tree, often bilateral and basal, reflecting a centrilobular inflammatory process that originates in the bronchioles and spreads to adjacent alveoli.

3. **Interstitial Pneumonia:** Characterized by diffuse, linear, reticular, or fine reticulonodular infiltrates spreading throughout the lung interstitium, often leaving the alveoli relatively clear. This pattern is classically associated with viral or atypical bacterial pathogens.

## COVID-19

Coronavirus Disease 2019 (COVID-19) is a highly infectious respiratory illness caused by the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), a novel positive-sense single-stranded RNA virus. The virus primarily targets host cells via the binding of its surface spike ($S$) glycoprotein to the Angiotensin-Converting Enzyme 2 (ACE2) receptors, which are highly expressed on the surface of Type II alveolar epithelial cells (pneumocytes). Following viral entry and replication, the host experiences direct cellular injury alongside an aggressive immune activation that can devolve into a dysregulated "cytokine storm." This hyperinflammatory state leads to diffuse alveolar damage, hyaline membrane formation, severe endothelial cell injury, vascular microthrombosis, and marked interstitial edema, culminating in Acute Respiratory Distress Syndrome (ARDS) in severe cases.

The clinical profile of COVID-19 is exceptionally broad, exhibiting substantial variation from completely asymptomatic infections to critical, life-threatening respiratory failure. The predominant clinical symptoms include:

- **Pyrexia (Fever):** Often fluctuating or sustained, acting as a hallmark sign of systemic immune activation.
- **Persistent Non-productive (Dry) Cough:** Triggered by upper and lower airway irritation and neural sensitization, without significant early sputum production.
- **Profound Asthenia (Fatigue):** Severe exhaustion and generalized bodily weakness that outlasts the acute febrile phase.
- **Progressive Dyspnea:** Ranging from mild breathlessness during minimal exertion to acute respiratory distress requiring invasive mechanical ventilation.
- **Sensory Distortions (Anosmia and Ageusia):** A distinct neurological manifestation involving the sudden, temporary loss or impairment of olfactory and gustatory functions due to sustentacular cell damage.
- **Upper Respiratory Signs:** Sore throat, rhinorrhea, nasal congestion, and mild headache, often overlapping with common viral upper respiratory infections.
- **Gastrointestinal and Systemic Manifestations:** Diarrhea, nausea, vomiting, widespread myalgia, and arthralgia, reflecting the systemic distribution of ACE2 receptors.

On chest X-rays, COVID-19 presents with characteristic visual signatures that track the temporal evolution of the pulmonary viral infection. The standard radiographic hallmarks are characterized by:

- **Bilateral, Peripheral, and Basal Distribution:** The opacities are predominantly localized within the lower lobes and distributed along the peripheral, subpleural zones of both lungs, corresponding to the initial sites of viral interstitial pneumonia.
- **Ground-Glass Opacities (GGO):** Visualized on radiographs as hazy, ill-defined patches of increased lung density that partially obscure the underlying pulmonary vasculature. These represent alveolar wall thickening, partial alveolar collapse, and intra-alveolar fluid accumulation.
- **Patchy Consolidations:** As the disease advances toward peak severity, the GGOs

coalesce into dense, confluent patches of consolidation that completely hide the underlying blood vessels and bronchial walls.

- **Reverse Halo ("Atoll") Sign and Linear Bands:** Subpleural curvilinear opacities and thick parenchymal bands frequently emerge during the organizing and healing phases of the disease, representing a combination of organizing pneumonia and focal atelectasis.

It is worth noting that chest radiographs possess limited sensitivity during the hyperacute or early stages (Days 0–2) of the infection, where a significant portion of symptomatic patients may present with normal imaging findings, requiring definitive diagnosis via Reverse Transcription Polymerase Chain Reaction (RT-PCR) assays.

## Tuberculosis

Tuberculosis (TB) is a chronic, granulomatous infectious disease caused by the acid-fast intracellular rod bacterium *Mycobacterium tuberculosis*. Transmission occurs via the inhalation of airborne droplet nuclei generated by individuals with active pulmonary disease. Once inside the alveoli, the bacilli are engulfed by alveolar macrophages. If the initial innate immune response fails to eradicate the pathogen, the bacilli replicate intracellularly, leading to a cell-mediated immune response. This incites the formation of specialized granulomas, known as Ghon foci, composed of a central core of caseous necrosis surrounded by epithelioid histiocytes, Langhans giant cells, and a peripheral rim of T-lymphocytes. The combination of a parenchymal Ghon focus and an involved draining lymph node constitutes the Ranke complex, signifying primary infection. While most individuals contain the infection in a latent state, any decline in cell-mediated immunity can trigger reactivation, leading to extensive tissue liquefaction, cavitary destruction, and severe bronchogenic or hematogenous replication.

The clinical onset of pulmonary tuberculosis is characteristically insidious, with symptoms worsening over weeks or months. The classic clinical spectrum includes:

- **Chronic, Intractable Cough:** Lasting for three weeks or longer, initially dry but progressively becoming productive of mucoid or purulent sputum.
- **Hemoptysis:** The expectoration of blood-streaked sputum or frank blood, caused by the erosion of mycobacterial cavitary lesions into adjacent pulmonary blood vessels (such as Rasmussen aneurysms).
- **Unexplained, Progressive Weight Loss and Anorexia:** Driven by tumor necrosis factor-alpha (TNF-$\alpha$, historically termed cachectin) and other inflammatory mediators that alter metabolic pathways and suppress appetite.
- **Drenching Night Sweats:** Profuse diaphoresis occurring during sleep, secondary to diurnal variations in cytokine release and body temperature regulation.
- **Low-grade Evening Pyrexia:** A classic febrific pattern where body temperature rises slowly throughout the day, peaking in the late afternoon or evening.
- **Constant Chest Pain:** Dull, aching thoracic discomfort related to chronic parenchymal destruction, hilar lymphadenopathy, or chronic pleural involvement.

Tuberculosis is known as a "great mimicker" in thoracic radiology due to its exceptionally diverse structural presentations on chest X-rays, which vary depending on whether the disease is primary or post-primary (reactivation):

- **Upper Lobe Predominance:** Reactivation TB shows a strong preference for the apical and posterior segments of the upper lobes, as well as the superior segments of the lower lobes. These regions possess higher ventilation-perfusion ratios and higher alveolar oxygen tensions ($pO_2$), providing an ideal environment for aerobic mycobacterial replication.
- **Cavitary Lesions:** The defining feature of advanced post-primary TB, visualized as thick-walled, gas-filled lucent spaces formed by the liquefaction and evacuation of caseous necrotic cores. These cavities can lead to fluid levels and facilitate bronchogenic spread.
- **Fibronodular Infiltrates and Granulomas:** Appears as well-defined, dense nodules and irregular linear streaks representing scarring, parenchymal fibrosis, and calcified granulomas (Ghon lesions) from past or healing infections.
- **Hilar and Mediastinal Lymphadenopathy:** Massive nodal enlargement, frequently showing central low-density necrosis, typically observed in pediatric or primary adult tuberculosis.
- **Miliary Pattern:** Manifests as a uniform distribution of millions of tiny, discrete, 1–3 mm micro-nodules resembling millet seeds scattered throughout both lung fields. This pattern indicates catastrophic hematogenous dissemination of the bacilli.

### Lung Cancer

Lung cancer, or bronchogenic carcinoma, is a malignant neoplastic proliferation originating from the epithelial cells lining the respiratory tree. It is the leading cause of oncology-related mortality worldwide, primarily linked to chronic exposure to tobacco smoke, radon gas, asbestos, environmental air pollution, and specific genetic predispositions. Pathologically, lung cancer is broadly bifurcated into two major histological groups: Non-Small Cell Lung Carcinoma (NSCLC, accounting for approximately 85% of cases) and Small Cell Lung Carcinoma (SCLC, accounting for approximately 15% of cases). NSCLC is further subdivided into Adenocarcinoma, which typically develops in peripheral bronchial glands; Squamous Cell Carcinoma, which commonly emerges centrally within major segmental bronchi; and Large Cell Carcinoma. SCLC is a central, highly aggressive neuroendocrine tumor characterized by rapid doubling times, early hematogenous dissemination, and frequent association with paraneoplastic syndromes.

The clinical manifestations of lung cancer depend heavily on the anatomical location of the primary tumor (central vs. peripheral), local invasion, regional nodal spread, and distant metastatic involvement. Key clinical symptoms include:

- **Worsening or Altered Chronic Cough:** A persistent, non-resolving cough or a change in a baseline smoker's cough, caused by tumor growth irritating the mucosal linings of the airways.
- **Unexplained Hemoptysis:** Intermittent expectoration of blood or blood-tinged sputum due to the ulceration and necrosis of the hypervascular tumor surface.
- **Deep, Non-pleuritic Thoracic Pain:** A dull, constant, poorly localized chest

ache caused by the invasion of the chest wall, mediastinal structures, or parietal pleura by the primary mass or regional lymph nodes.

- **Persistent Wheezing and Stridor:** Monophonic, localized musical sounds indicating partial airway occlusion or extrinsic compression of a major bronchus.
- **Progressive Weight Loss and Cachexia:** Profound loss of skeletal muscle mass and adipose tissue driven by tumor-induced metabolic re-engineering and inflammatory responses.
- **Superior Vena Cava (SVC) Syndrome and Horner's Syndrome:** Central tumors can compress the SVC, causing facial and upper extremity edema and venous distension. Apical lung tumors (Pancoast tumors) can invade the sympathetic chain, causing ipsilateral ptosis, miosis, and anhidrosis.

While chest radiographs can detect advanced lung masses, Computed Tomography (CT) scans of the chest represent the gold standard modality for the detection, precise morphological evaluation, and clinical staging of lung cancer. Chest CT provides excellent spatial and contrast resolution, allowing for detailed evaluation of pulmonary nodules and masses without the anatomical summation artifacts inherent to X-rays.

The critical radiological characteristics of lung cancer on chest CT scans include:

- **Solitary Pulmonary Nodule (SPN) or Mass:** Visualized as a focal, discrete round or irregular opacity. Lesions less than 3 cm are defined as nodules, whereas lesions greater than 3 cm are classified as masses and carry a much higher probability of malignancy.
- **Spiculation and Lobulation:** Malignant nodules characteristically display irregular, lobulated borders and fine, radiating linear strands extending into the surrounding lung parenchyma (spiculation). This pattern reflects micro-invasion along interstitial planes and localized desmoplastic tissue reactions.
- **Asymmetrical Eccentric Calcification:** Malignant lesions typically lack calcification or show eccentric, irregular, punctuated calcification patterns, distinguishing them from the benign, concentric, laminated, or "popcorn-like" calcifications seen in granulomas or hamartomas.
- **Pleural Puckering and Tail Sign:** A linear band extending from the outer margin of a peripheral tumor to the visceral pleura, pulling the pleural surface inward due to internal tumor fibrosis.
- **Air Bronchogram and Cavitation:** Thick, irregular, ragged-walled cavities (typically $>$ 4 mm wall thickness) with bumpy internal margins can occur due to central ischemic necrosis within large tumor masses, most commonly observed in squamous cell carcinomas.
- **Mediastinal Lymphadenopathy and Vascular Invasion:** Nodal enlargement (short-axis diameter $>$ 10 mm) within ipsilateral or contralateral hilar, mediastinal, subcarinal, or supraclavicular stations, often with loss of distinct fat planes between the tumor and major mediastinal structures like the aorta, pulmonary artery, or pericardium.

### Comparative Radiological Characteristics

Differentiating between pneumonia, COVID-19, tuberculosis, and lung cancer requires a detailed analysis of their specific radiographic patterns, spatial distributions, and primary imaging modalities. Pneumonia is characterized by localized, dense alveolar consolidations that respect lobar boundaries, accompanied by distinct air bronchograms on chest X-rays. COVID-19 presents with a diffuse, patchier, and non-lobar distribution, typically manifesting as bilateral, ground-glass opacities that preferentially involve the peripheral subpleural zones and lower lung fields. Tuberculosis is characterized by its chronicity, upper-lobe tropism, fibronodular scars, calcified granulomas, and the formation of smooth, thin-walled cavitary lesions resulting from caseous tissue liquefaction. Lung cancer, by contrast, is characterized on chest CT by a persistent, expanding focal nodule or mass with irregular, spiculated or lobulated borders, thick-walled eccentric cavitations, pleural retraction, and associated regional mediastinal lymphadenopathy.

Table 2.1 provides a detailed side-by-side comparison of the pathological, clinical, and radiological characteristics of these four thoracic conditions, highlighting how computerized multi-modal image interpretation systems utilize these differing visual signatures to achieve accurate diagnosis.

Table 2.1: Comprehensive comparative summary of clinical, pathological, and radiological characteristics of the target diseases.

| **Feature** | **Pneumonia** | **COVID-19** | **Tuberculosis** | **Lung Cancer** |
|---|---|---|---|---|
| **Primary Etiology** | Bacterial, viral, or fungal infection | SARS-CoV-2 viral infection | *Mycobacterium tuberculosis* | Malignant epithelial transformation |
| **Pathology** | Alveolar inflammatory exudate | Diffuse alveolar damage, GGO | Caseous granuloma formation | Neoplastic tissue proliferation |
| **Primary Modality** | Chest X-ray (CXR) | Chest X-ray (CXR) / RT-PCR | Chest X-ray (CXR) / Sputum | Chest Computed Tomography (CT) |
| **Anatomical Site** | Lobar, segmental, or bronchic | Bilateral, peripheral, basal | Apical / upper lobe segments | Central bronchi or peripheral parenchyma |
| **Key Signatures** | Homogeneous consolidation, air bronchograms | Patchy GGOs, crazy-paving, bilateral patches | Cavities, calcified nodules, miliary seedings | Spiculated mass, lobulation, thick cavities |
| **Onset / Course** | Acute onset (days) | Acute to subacute onset | Chronic, insidious course | Chronic, progressive course |

## 2.1.4 Related Work in Lung Cancer Classification

The rapid advancement of medical imaging technologies and artificial intelligence techniques has led to significant progress in the field of automated lung cancer classification.

Over the past decade, researchers have increasingly explored the application of machine learning and deep learning methods to assist radiologists in detecting and classifying lung cancer from medical images, particularly computed tomography (CT) scans. The growing availability of publicly accessible datasets, coupled with improvements in computational resources and neural network architectures, has accelerated the development of computer-aided diagnostic systems capable of identifying malignant pulmonary abnormalities with high accuracy.

Traditional approaches to lung cancer classification primarily relied on handcrafted feature extraction techniques. In these systems, radiological characteristics such as shape, texture, intensity distribution, edge information, and morphological properties of pulmonary nodules were manually engineered and subsequently provided as input to machine learning classifiers. Algorithms such as Support Vector Machines (SVMs), Random Forests, k-Nearest Neighbors (k-NN), and Artificial Neural Networks (ANNs) were widely employed to distinguish between benign and malignant lesions. Although these methods demonstrated promising results, their performance was often constrained by the quality and representativeness of the manually designed features.

The emergence of deep learning fundamentally transformed the landscape of lung cancer classification research. Convolutional Neural Networks (CNNs) enabled the automatic extraction of hierarchical image features directly from raw imaging data, eliminating the need for extensive manual feature engineering. By learning discriminative representations from large-scale datasets, deep learning models demonstrated superior performance compared to many traditional machine learning approaches. Consequently, an increasing number of studies began investigating the use of CNN-based architectures for pulmonary nodule detection, malignancy classification, tumor segmentation, and cancer staging.

In addition to standard convolutional architectures, researchers have explored various advanced techniques to improve classification performance and clinical applicability. These include transfer learning from large natural image datasets, ensemble learning strategies, attention mechanisms, three-dimensional convolutional networks, hybrid machine learning frameworks, and explainable artificial intelligence methods. Such approaches aim not only to improve diagnostic accuracy but also to enhance model robustness, interpretability, and generalization across diverse patient populations and imaging protocols.

Despite the considerable progress achieved in recent years, lung cancer classification remains a challenging research problem. Pulmonary nodules exhibit substantial variability in size, shape, texture, and anatomical location, while imaging datasets often suffer from class imbalance, limited sample sizes, and variations in acquisition parameters. These challenges continue to motivate the development of more sophisticated algorithms capable of delivering reliable and clinically meaningful predictions.

The following subsection reviews a selection of significant studies from the literature, highlighting the methodologies employed, datasets utilized, performance metrics achieved, and key contributions made toward the advancement of automated lung cancer classification systems. Through this review, the strengths and limitations of existing approaches can be identified, thereby providing a foundation for understanding the motivation and design decisions underlying the proposed framework.

- Artificial Intelligence for lung cancer detection using CT scan images aims to improve early and accurate classification of pulmonary nodules while reducing dependency on manual radiological assessment. To address limitations in traditional machine learning and standalone deep learning models, the paper proposes a hybrid Deep Learning–Support Vector Machine (DL-SVM) framework, where a deep learning model is used to extract high-level features from CT scan images and a Support Vector Machine (SVM) is used for final classification between benign and malignant nodules. The system is evaluated on the LIDC-IDRI dataset, which contains annotated lung CT scans widely used for benchmarking lung cancer detection methods. Experimental results show that the proposed DL-SVM model achieves an accuracy of approximately 94%, outperforming several comparative baseline approaches including traditional machine learning models (such as classical SVM variants and statistical classifiers) and standalone deep learning methods, which achieved lower performance in comparison. The results confirm that the hybrid combination of deep feature extraction and SVM classification improves diagnostic accuracy, robustness, and generalization, making the model more reliable for assisting radiologists in early lung cancer detection. [3]

- Artificial Intelligence-based lung cancer detection systems aim to improve early and accurate diagnosis from CT scan images by automatically classifying lung nodules as benign or malignant. However, conventional deep learning models often suffer from performance degradation due to high-dimensional feature spaces and suboptimal optimization of CNN parameters. To address these limitations, the paper proposes a hybrid framework that combines a deep convolutional neural network (CNN) with the Ebola Optimization Search Algorithm (EOSA) for optimized weight and bias selection during training. The CNN is used to extract deep features from CT images, while EOSA is applied to optimize the model parameters and improve classification performance. The system is evaluated on the IQ-OTH/NCCD lung cancer CT scan dataset. Experimental results show that the proposed EOSA-CNN model achieves a classification accuracy of approximately 93.21%, outperforming other comparative methods including GA-CNN (91%), WOA-CNN (90%), MVO-CNN (89%), SBO-CNN (88%), and standard CNN models with lower accuracy. Additionally, the model demonstrates improved sensitivity and specificity across benign and malignant classes, confirming that the use of metaheuristic optimization significantly enhances CNN performance. The study concludes that integrating deep learning with Ebola Optimization Search Algorithm provides a more accurate and robust solution for lung cancer detection from CT images, improving diagnostic reliability and supporting clinical decision-making. [4]

- The study aims to improve the accuracy and robustness of classifying lung cancer subtypes from CT scan images by combining radiomic analysis with deep learning methods. However, single-source approaches often suffer from limited feature representation and reduced generalization across different imaging conditions. To address these limitations, the paper proposes a hybrid framework that integrates a Deep Convolutional Neural Network (DeepCNN) with attention mechanisms and radiomic feature extraction to enhance lung cancer subtype classification. The model extracts radiomic features such as texture, shape, and statistical descriptors using PyRadiomics, while the DeepCNN learns deep spatial representations from CT images. An attention mechanism is incorporated within the CNN architecture to focus on the most diagnostically relevant tumor regions, improving feature relevance and reducing noise from irrelevant areas. The extracted radiomic and deep features are combined and evaluated using machine learning classifiers across a dataset of 2,725 CT images from multiple healthcare centers covering different lung cancer subtypes. Experimental results show that the proposed hybrid model achieves a classification accuracy of approximately 92.47%, with an AUC of 93.99% and sensitivity of 92.11%, outperforming traditional machine learning models and standalone deep learning approaches. Additional experiments show that feature fusion significantly improves performance compared to using radiomic or deep features alone. The study concludes that combining attention-based deep learning with radiomic feature engineering provides a more reliable and reproducible framework for lung cancer subtype classification, improving diagnostic consistency and clinical decision support. [5]

- Artificial Intelligence-based lung cancer detection aims to improve early and accurate diagnosis by classifying CT scan images into benign and malignant cases. However, traditional deep learning methods often suffer from redundant features, overfitting, and weak generalization on diverse medical datasets. To address these issues, the paper proposes a radiomics-based machine learning framework that extracts handcrafted CT image features such as texture, shape, and intensity, followed by feature selection to keep only the most relevant attributes. These optimized features are then used to train machine learning classifiers for lung cancer detection. The model is tested on public lung CT datasets and compared with multiple baseline approaches using different feature selection and classification combinations. The proposed method achieves F1-score (95.24%), AUC (98.00%), specificity (93.17%), accuracy (94.40%), precision (95.80%), and sensitivity (94.69%). with improved sensitivity and specificity, and consistently outperforms standard radiomics and baseline machine learning models in lung cancer classification performance. [6]

### 2.1.5 Related Work in Chest Diseases Multi-class Classification

Interpreting Chest X-rays (CXRs) is vital for diagnosing various thoracic diseases like pneumonia and tuberculosis. Due to the subtlety of findings and high image volume, **Deep Learning (DL)** models have been widely adopted to automate diagnosis. These models, trained on large datasets, primarily focus on **multi classification** of different chest diseases. The following research examples showcase the success of various **Convolutional Neural Network (CNN)** architectures in achieving high accuracy in CXR analysis.

- The researchers Develop an automated Deep Neural Network (CNN) to classify Pneumonia, COVID-19, Tuberculosis, and No-findings from chest X-rays, addressing limitations in individual disease detection and small/biased datasets in prior studies. they Created a custom CNN architecture with 5 convolutional blocks (16-256 channels, ReLU activation, max-pooling), trained on split data (80% training, 20% testing). They Addressed heterogeneous data sources by preprocessing images (resizing to 300x300, denoising, normalization, filtering) to create a coherent dataset of 21,581 images from multiple Kaggle sources they achieve accuracy 98.72% (using validation metrics) , Outperformed SOTA schemes like Venkataramana et al. (96.6%)

and Bhandari et al. (94.54%), The study proved that a large diverse dataset is essential to achieve high multiclass accuracy and prevent overfitting, though class imbalance slightly reduced COVID-19 performance (96.27% recall) [7].

- The researchers Develop a custom CNN model with soft attention to classify chest X-ray images into normal, COVID-19, pneumonia, and tuberculosis, while analyzing misclassifications through handcrafted features and statistical methods to improve clinical interpretability and diagnostic reliability. , Utilized transfer learning with pre-trained CNN models (VGG-19, ResNet-50, EfficientNet, DenseNet), trained on a balanced dataset split into training (70%), validation (15%), and testing (15%) sets. They Preprocessed images via normalization, data augmentation (rotation, flipping, zooming), and resizing; applied SMOTE at the feature level to balance the originally imbalanced dataset from heterogeneous sources Best Modelis ResNet-50 with Accuracy: 97% Compared to VGG-19 (95%), DenseNet (91%), EfficientNet (94%) [8].

- This study proposed a multi-level classification model using CNNs to differentiate COVID-19 from Tuberculosis (TB) and other Pneumonias based on CXR images. The primary challenge was addressing the severe data imbalance, specifically the scarcity of COVID-19 samples. The novelty involved combining data augmentation with the SMOTE technique to artificially balance the minority class. The system employed a two-stage process: first, classifying TB versus general Pneumonia, and second, sub-classifying the Pneumonia cases into COVID-19 and viral/bacterial types. This multi-level approach was designed to significantly reduce misdiagnosis between these similar lung diseases. The model achieved a high accuracy of 97.4% for the TB/Pneumonia stage. The second stage achieved 88% accuracy for bacterial/viral/COVID-19 classifications, demonstrating an overall 8–10% improvement over existing methods [9].

- Additionally, a unified DL framework like CXR-MultiTaskNet used ResNet50 for joint disease localization and classification, incorporating thoracic diseases like pneumonia and TB, with an F1-score of 96.5% through shared feature extraction [10].

- Another deep-learning pipeline classified chest X-ray images into normal, pneumonia, TB, and COVID-19 using a sequential CNN approach. Preprocessing steps such as grayscale conversion and normalization improved robustness, while transfer learning enhanced performance on limited data. VGG-16 achieved 95.9%, DenseNet-161 reached 98.9% for distinguishing COVID-19 from pneumonia, and ResNet-18 obtained 76% for COVID-19 severity classification. The study demonstrates the effectiveness of multi-stage CNN models for automated COVID-19 screening from CXR images[11].

- A knowledge-distillation deep-learning model classified CXR images into COVID-19, pneumonia, TB, and normal using the RDD4 dataset. A deep CNN teacher network transferred its knowledge to a lightweight student model built in Keras/TensorFlow, preserving accuracy while reducing complexity. Data imbalance was addressed through augmentation. The distilled student model achieved 96.08% accuracy, with high precision and recall, showing strong suitability for real-time, low-resource deployment[12].

- They develop a custom CNN model with soft attention to classify imbalanced chest X-ray images into normal, COVID-19, pneumonia, and tuberculosis, the approach Designed COV-X-net19, a 16-layer CNN architecture (4 conv layers, 6 max-pooling, soft attention, dropout, dense with SoftMax), trained on split data (70% training, 10% validation, 20% testing) with ablation studies for hyperparameter optimization (ReLU, Nadam optimizer, LR=0.001, batch=32) , the technique used preprocessed images via text removal (OCR+inpaint), morphological opening (3x3 kernel), CLAHE (ClipLimit=5.0, TileGridSize=3x3), histogram equalization. Best Model: COV-X-net19 with accuracy: 95.18% Compared to 8 trans fer learning models (e.g., MobileNet at 82.96%, VGG16 at 62.46%), base CNN (89.96%), and SOTA studies with similar multi-class setups (e.g., accuracies below 95% in imbalanced scenarios) [13].
- A CNN-based system classified chest X-ray images into COVID-19, pneumonia, TB, and normal using over 21,000 images from three public datasets. The model used five convolutional blocks with preprocessing steps such as resizing, denoising, and normalization to handle heterogeneous sources. It achieved a high validation accuracy of 98.72%, with strong per-class performance across all categories. The study shows that a carefully designed CNN can effectively support joint diagnosis of major chest diseases from CXR images [14].

## 2.2 Explainable AI (XAI) in Medical Image Diagnosis

Explainable AI (XAI) refers to techniques that make the decision-making process of AI models transparent and interpretable, addressing the "black-box" nature of DL models by providing insights into why a prediction was made. In this project, XAI is integrated using methods like Gradient-weighted Class Activation Mapping (GradCAM), which generates heatmaps highlighting regions in medical images (e.g., fracture lines in bone X-rays or opacities in CXR) that influenced the model's classification. This linkage enhances trust in diagnostics for both bone fractures and chest diseases, allowing clinicians to verify AI outputs against radiological features, reducing misdiagnosis risks, and facilitating model debugging.

### Related Papers on XAI For Lung Cancer Classification

- This paper introduces LCxNet, a novel, custom-designed convolutional neural network (CNN) framework for the early and accurate computer-aided diagnosis of lung cancer using CT scans, which leverages Explainable AI (XAI) techniques such as Grad-CAM for decision visualization and t-SNE for feature space analysis to ensure clinical transparency. By training and evaluating the model on the IQ-OTH/NCCD lung CT dataset—comprising benign, malignant, and normal classes—and comparing learning behaviors across five optimizers (SGD, RMSProp, Adam, AdamW, and NAdam), the researchers demonstrated that LCxNet outperforms state-of-the-art models, sensitivity of 99.39% is paramount, ensuring that the risk of missing a

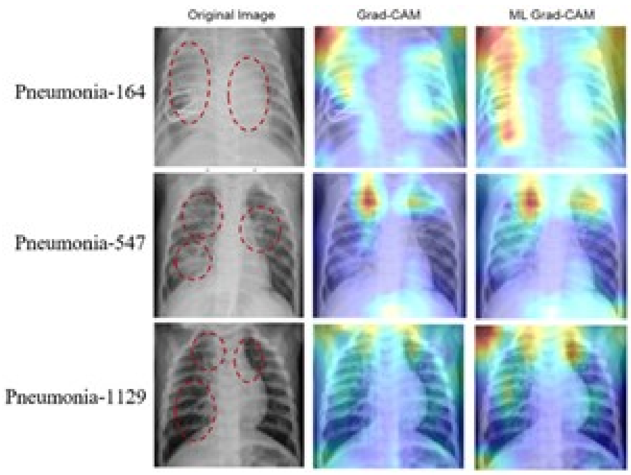



Figure 2.2: Heatmap for pneumonia detection

malignant tumor (false negative) is minimized, which is a top priority in clinical diagnostics. Coupled with its high accuracy (99.39%), precision (99.40%), F1-score (99.40%), and low generalization loss (0.17), AdamW offers the most robust and clinically reliable performance for aiding in the early and accurate detection of lung cancer , While Adam and NAdam are also excellent choices, AdamW's slight edge in specificity (99.89%) and Cohen's Kappa (99.01%). [15]

- The study introduces a deep learning framework that utilizes a customized VGG16 convolutional neural network combined with Grad-CAM for the autonomous and transparent classification of lung cancer from CT scans. By training and evaluating the model on the IQ-OTH/NCCD dataset—which consists of 1190 CT images categorized into normal, benign, and malignant classes—the researchers applied data augmentation and transfer learning to achieve an impressive overall classification accuracy of 96%. The model demonstrated perfect 100% accuracy and specificity for healthy cases, alongside strong 98% precision and 97% recall for identifying malignant tumors, though it exhibited a slightly lower precision of 81% for benign cases due to overlapping features with malignant lesions. To bridge the gap between complex AI predictions and clinical trust, the study successfully employed Grad-CAM to generate visual heatmaps that accurately highlight the critical tissue regions influencing the model's decisions, ultimately providing radiologists with a highly effective, accurate, and interpretable decision-support tool for early lung cancer diagnosis. [16]

## Related Papers on XAI For Chest Disease Multi-class Classification

- This research introduces an Explainable AI (XAI) framework for the automatic detection, interpretation, and explanation of pulmonary diseases from chest radiographs (CXRs). The framework uses a transfer learning approach with a fine-tuned ResNet50 Convolutional Neural Network (CNN) trained on the COVID-CT and COVIDNet datasets. The Local Interpretable Model-agnostic Explanations (LIME) technique is employed to visually highlight the key regions in the CXR images that contribute most to the prediction. The developed system demonstrated strong performance, achieving high classification accuracy, reaching up to 97% on the evaluated datasets. Validation confirmed that the model accurately focuses on clinically important features, thus assisting radiologists in early diagnosis and clinical decision-making[17].

- They develop XAI-TRANS, using inception-based transfer learning to classify CXR images for COVID-19, bacterial/viral pneumonia, and tuberculosis, addressing overlapping radiological features and black-box AI limitations by integrating improved U-Net segmentation and XAI tools the approach used is Utilized transfer learning with four pre-trained models (ResNet50, VGG19, VGG16, Inceptionv3), combined with an improved U-Net for lung segmentation to extract features, followed by XAI-TRANS for classification and explanation using Grad-CAM visualizations Technique Preprocessed images via resizing to 256*256 and normalization between 0 and 1fer learning models (e.g., MobileNet at 82.96%, VGG16 at 62.46%), base CNN (89.96%), and SOTA studies with similar multi-class setups (e.g., accuracies below 95% in imbalanced scenarios) the Best Model is XAI-TRANS (with and Grad-CAM) with accuracy: 97% (overall classification accuracy) and 98% precision, outperforming traditional DL methods by 4.75% through XAI integration [18].

- A survey reviewed over 200 papers on XAI in DL-based medical image analysis, emphasizing visual explanations for chest radiography to build clinician confidence[19].

- Additionally, an XAI-integrated DL framework for multiclass lung disease classification, targeting five classes: viral pneumonia, bacterial pneumonia, COVID-19, tuberculosis, and normal. It evaluates various DL architectures, including CNNs, hybrids, ensembles, transformers, and Big Transfer, with hyperparameter tuning and stratified 5-fold cross-validation to ensure robust performance. The Xception model, enhanced with transfer learning, emerges as the top performer, achieving 96.21% accuracy by focusing on discriminative features in lung region[20].

- They built a lightweight CNN (5 convolutional layers) to classify chest X-rays into four classes: COVID-19, Pneumonia, TB, Normal. They applied Grad-CAM, LIME, and SHAP to generate visual explanations of model decisions, validating them with radiologists. The dataset was 7,132 public CXR images from three sources, with class imbalance handled by stratified batching. The model achieved a 10-fold cross-validation average test accuracy of 94.31% ± 1.01%, and they reported strong precision, recall, and F1[21].

## 2.3 Federated Learning

Federated Learning (FL) is a decentralized ML approach where models are trained collaboratively across multiple devices or institutions without sharing raw data, preserving privacy through local updates aggregated centrally. In medical imaging like bone X-rays and CXR classification, FL is used to train on distributed datasets from hospitals, mitigating data silos and complying with regulations like GDPR or HIPAA. It involves clients training on local data, sending model weights to a server for aggregation (e.g., via FedAvg), and is quickly implemented for real-time applications, reducing communication overhead while handling heterogeneous data.

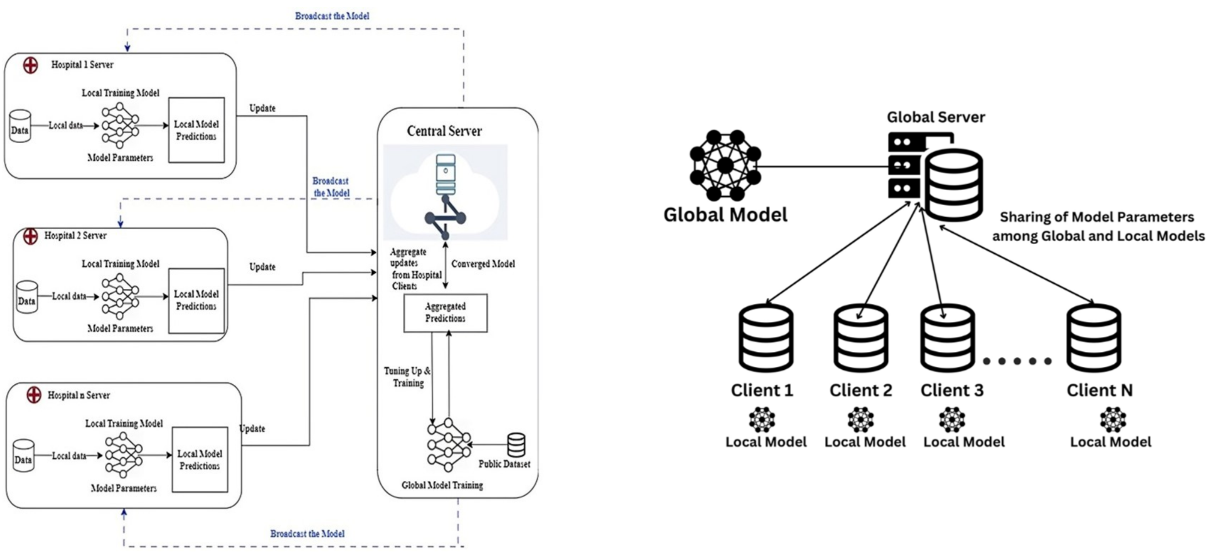



Figure 2.3: FL-Based efficient framework for smart healthcare sector

## Related Papers on FL For Lung Cancer Classification

- This paper introduces an ensemble Federated Learning (FL) approach to address the limitations of centralized data processing and single machine learning models in the early detection and multi-order classification of lung cancer. By transitioning from a traditional client-server architecture to a decentralized federated network, the proposed system customizes local neural networks to enable faster, broader training across diverse datasets while strictly preserving patient data privacy and security through techniques like federated averaging and differential privacy. Utilizing a Kaggle cancer dataset, the distributed framework focuses on collective decision-making to reliably classify lung cancer cases into normal, benign, and malignant categories. Experimental evaluations demonstrate that this decentralized approach effectively mitigates the computational load overheads and data accumulation anomalies associated with centralized servers, ultimately achieving a classification accuracy of 89.63% and offering improved generalization and diagnostic reliability compared to traditional standalone machine learning models. [22]

- The study presents a privacy-preserving Federated Learning (FL) framework designed to classify lung cancer severity from pulmonary CT scans into benign, malignant, and normal categories without centralizing sensitive patient data. By utilizing datasets from the IQ-OTH/NCCD collection distributed across different simulated clients, the study evaluates the performance of the CatBoost algorithm alongside the deep learning Inception V3 architecture using both transfer learning and fine-tuning techniques. After applying crucial preprocessing steps, including median blur filtering for noise removal and BGR-to-RGB color space transformation, the experiments revealed that the fine-tuned Inception V3 model significantly outperformed the CatBoost model, particularly in managing complex medical imaging data and consistently detecting malignant cases. Ultimately, the proposed decentralized approach successfully achieved an accuracy of 89.0% using the fine-tuned Inception V3 model, demonstrating that federated learning can effectively bridge the gap between building robust, highly accurate collaborative diagnostic tools and ensuring strict compliance with healthcare data privacy regulations such as HIPAA and GDPR. [23]

- This paper introduces FBCLC-Rad, a novel framework designed to securely and accurately detect and categorize lung cancer from chest CT images by integrating deep learning, blockchain technology, and Federated Learning (FL). To overcome the privacy and trust challenges associated with sharing sensitive medical data across global healthcare institutions, the system employs blockchain for data authentication and FL to collaboratively train a global model without requiring data to be centralized. The methodology utilizes Capsule Networks (CapsNets) for local classification and leverages Transfer Learning (TL) to maximize diagnostic accuracy while significantly reducing training times, alongside a specific data normalization approach to manage the variability of CT scans from different scanners. Evaluated extensively using Python libraries on datasets including the Cancer Imaging Archive (CIA), Kaggle Data Science Bowl (KDSB), LUNA 16, and a custom local dataset, the decentralized FBCLC-Rad technique achieved an exceptional 99.69% accuracy with minimal categorization error. Ultimately, this privacy-preserving approach provides medical professionals with a fast, highly reliable, and automated diagnostic tool to manage the growing volume of lung cancer patients globally while ensuring strict organizational anonymity. [24]

- This paper introduces "FVCM-Net," a novel, privacy-preserving deep learning framework that integrates federated learning, attention mechanisms, and ensemble learning to achieve highly accurate and interpretable lung cancer detection from CT scan images. To overcome the challenges of data scarcity and sensitive patient data security, the model utilizes federated learning to enable collaborative training across decentralized medical institutions without directly sharing records, while incorporating a Convolutional Block Attention Module (CBAM) with a VGG-16 architecture to extract crucial imaging features. Evaluated on diverse datasets, including LIDC-IDRI and IQ-OTH/NCCD, the decentralized approach utilizes a majority voting ensemble strategy to further enhance diagnostic robustness, achieving an exceptional average accuracy of 98.26% and an F1-score of 97.37%. To ensure clinical transparency and build trust in the AI's diagnostic logic, the framework seamlessly employs Explainable AI (XAI) techniques, specifically SHAP and High-Resolution Class Activation Mapping (HiResCAM), effectively visualizing the critical regions influencing the model's predictions and providing radiologists with a secure, highly reliable, and interpretable decision-support tool. [25]

- This research paper presents a comprehensive deep learning approach for detecting lung nodules in medical CT and PET-CT images to improve early lung cancer diagnosis. The authors developed a preprocessing pipeline that filters and segments CT images to isolate regions of interest (ROI), then trained and evaluated seven different deep learning architectures—CNN, DCNN, 3DCNN, VGG19, ResNet18, Inception V1, and Inception-ResNet—on a dataset of 220 cases containing 560 nodules from Harbin's imaging data center. Among all tested models, VGG19 demonstrated superior performance, achieving 98.65% accuracy with high precision (98.80%), sensitivity (98.55%), specificity (98.70%), and F1-score (98.60%), while maintaining the lowest false positive rate (1.05%), making it the most reliable model for automated lung nodule detection despite being trained on a limited dataset. [26]

- This paper proposes a deep learning framework for segmentation and classification of lung cancer histopathological images to improve early detection and diagnosis. The approach uses U-Net for segmentation of lung cancer histopathological images to identify regions of interest, followed by classification models including ResNet-50, VGG-16, EfficientNet-B5, and an Ensemble model of these three to categorize images into three types: lung adenocarcinoma, squamous cell carcinoma, and normal lung tissue. Trained on 15,000 histopathological images from the LC15000 dataset (9000 training, 3000 validation, 3000 testing), the U-Net segmentation model achieved 81% accuracy, while the classification models progressively improved performance—ResNet-50 achieved 91% accuracy, VGG-16 94%, and EfficientNet-B5 97%. Notably, the Ensemble model combining all three classifiers significantly outperformed individual models, achieving 99% accuracy with perfect precision, recall, and F1-score metrics (0.99), demonstrating the effectiveness of ensemble approaches for automated lung cancer histopathological image analysis. [27]

- This paper presents MCAT-Net, a novel lung nodule segmentation model based on the transformer architecture with multiple thresholds and coordinate attention designed to address limitations in capturing edge information, semantic features, and long-range dependencies. The model integrates three key components: (1) a Multi-threshold Feature Separation (MFS) module that extracts edge and texture features at different intensity levels using four optimal thresholds (T7, T6, T5, T4) to enrich input features without excessive computational cost, (2) a Coordinate Attention (CA) mechanism that preserves spatial position information and enables cross-regional feature integration by encoding features in horizontal and vertical dimensions, and (3) a Transformer module applied at the fourth skip connection to capture global long-range dependencies and enhance feature integration. Evaluated on the LIDC-IDRI dataset (888 CT images with 1186 lung nodules) and LNDb dataset (294 CT images), MCAT-Net achieved Dice Similarity Coefficient (DSC) values of 88.29% and 78.51% respectively, with sensitivities of 86.33% and 75.05%, significantly outperforming traditional segmentation methods (U-Net, U-Net++, SegNet, nn-UNet) while demonstrating superior performance on complex nodule types including ground-glass nodules, partially solid nodules, and wall-adjacent nodules. [28]

- This paper proposes SSLKD-UNet, a semi-supervised learning and knowledge distillation model for lung nodule segmentation that addresses the challenge of obtaining large-scale, high-quality annotated medical datasets. The model employs a hybrid CNN-Transformer architecture that combines a U-Net encoder-decoder with a Swin-Transformer in the bottleneck layer to capture both local spatial details and global context information. The training employs a novel teacher-student framework where a teacher model (UNet, ResUNet, AttUNet, SwinUNet, or UTransformer) is trained on coarse-grained annotations to generate pseudo-labels for unlabeled CT images, and a student model (SSLKD-UNet) learns from both pseudo-labeled and fine-grained annotated data. Evaluated on the publicly available LUNA16 dataset (888 CT scans with 1186 nodules) and a private LC183 dataset (183 lung cancer patients from Dalian Medical University), the model demonstrates superior segmentation performance compared to baseline models including UNet, ResUNet, and AttUNet, effectively handling varying nodule sizes and complex anatomical structures. The approach successfully reduces reliance on expensive fine-grained annotations by leveraging coarse annotations and large amounts of unlabeled data through the teacher-student training paradigm. [29]

- This paper presents YOLOV11-EAR, an improved YOLO-based pulmonary nodule detection framework that addresses limitations in capturing spatial context and small object localization. The approach introduces two primary innovations: (1) a Spatial Squeeze-and-Excitation (SSE) module that combines channel-wise and spatial attention mechanisms to enhance discriminative features for nodules by processing global context and local spatial information in parallel, and (2) an Explicit Aspect Ratio Penalty IoU (EAPIoU) loss function that directly penalizes squared differences in aspect ratios between predicted and ground-truth bounding boxes to refine localization, particularly for small and irregularly shaped nodules. Evaluated on three diverse datasets—LUNA16 (888 CT scans with 1186 nodules), NODE21 (4882 chest X-rays with 1476 annotated nodules), and LungCT (1152 CT slices)—the YOLOV11-EAR framework consistently outperforms baseline YOLO variants and mainstream detectors (Faster R-CNN, RetinaNet), achieving precision of 0.781 with mAP@50 of 75.5% on LUNA16, mAP@50 of 92.4% on LungCT, and mAP@50 of 89.1% on NODE21, while maintaining computational efficiency with only 2.5M parameters and 6.3 GFLOPs.[30]

- This paper proposes SwiF-YOLO, an improved lung nodule detection method based on the YOLOx framework that addresses challenges in detecting small, morphologically similar nodules in CT images. The model incorporates three key improvements: (1) replacing the YOLOx-M backbone network with a Swin Transformer to leverage its hierarchical feature extraction and linear computational complexity through shifted window-based self-attention mechanisms, (2) adopting Adaptively Spatial Feature Fusion (ASFF) to dynamically weight and combine multi-scale features for improved scale invariance and reduction of inconsistent information across feature pyramid levels, and (3) replacing the standard IoU loss function with Generalized IoU (GIoU) loss to better handle cases where bounding boxes do not overlap and to provide more informative gradient signals for bounding box regression. Trained and evaluated on the LUNA16 dataset (888 CT scans with 1186 annotated nodules, split 70% training, 15% validation, 15% testing), SwiF-YOLO achieves precision of 84.54%, recall of 81.37%, and mAP of 83.17%, outperforming baseline YOLOx-M (83.78% precision, 79.78% recall, 81.77% mAP) and comparing favorably to other mainstream detection methods including Faster R-CNN, Mask R-CNN, YOLOv5, YOLOv7, and SSD. [31]

- This paper proposes a novel hybrid deep learning model combining CNN and YOLOv8 for automated real-time detection of lung nodules from CT images to address challenges in early lung cancer diagnosis. The approach employs a two-stage framework: (1) in the first stage, a 2D Convolutional Neural Network is trained on preprocessed DICOM images from the LUNA16 dataset after extensive image processing (including Hounsfield Unit scaling, min-max normalization, unsharp masking, median filtering, Laplacian filtering, and morphological operations) to detect nodule regions and achieve 94% training accuracy over 150 epochs, and (2) in the second stage, the detected nodule coordinates are combined with expert radiologist annotations and fed into YOLOv8 (tested across five scale variants: YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, YOLOv8x) for fine-tuning and real-time detection.

Evaluated on the LUNA16 dataset (888 CT scans generating 551,065 PNG images), the proposed model achieves an overall accuracy of 92.3%, sensitivity of 88.5%, and mean average precision (mAP) of 53.5%, with YOLOv8m configuration demonstrating superior performance (mAP50: 53.5%, mAP50-95: 40.0%), outperforming several comparable approaches in the literature while maintaining computational efficiency. [32]

- This paper proposes a novel lung nodule detection method based on YOLOv3 that addresses the limitations of traditional convolution neural networks in capturing rich feature information and the inaccurate bounding box positioning of standard YOLO approaches. The method introduces two primary innovations: (1) integration of Inception ResBlocks into the YOLOv3 feature extraction network (creating an "Inception-DarkNet" architecture with 44 convolutional layers and 4 Inception ResBlocks) to extract multi-scale features more precisely through asymmetric convolution kernels while reducing overfitting and enhancing non-linear expression capability, and (2) a novel Generalized Distance and Intersection over Union (GDIoU) loss function that combines the GIoU metric with the Euclidean distance between bounding box centers to provide more sensitive and accurate bounding box regression, particularly when boxes do not overlap. The model uses K-means clustering to determine 9 optimal anchor sets tailored to the lung nodule dataset and is trained on a combination of the public LUNA16 dataset (888 CT scans with 1186 nodules) and 80 collected patient CT images (total 1266 nodules), with preprocessing including Hounsfield Unit scaling, morphological operations, and lung parenchyma extraction. The proposed method achieves 83.5% Average Precision and 92.6% sensitivity with a detection time of 31.3ms per image, outperforming original YOLOv3 (78.6% AP, 88.3% sensitivity) and competing favorably with Faster R-CNN variants while maintaining faster inference speed. [33]

- This paper proposes AutoLungDx, a comprehensive end-to-end hybrid deep learning framework for early lung cancer diagnosis through automated segmentation, detection, and classification of lung nodules from CT images. The framework integrates three complementary architectures: (1) a novel 3D Res-U-Net for lung parenchyma segmentation that incorporates residual connections to improve feature flow and gradient propagation, particularly in challenging juxta-pleural regions, (2) YOLOv5s for efficient slice-level nodule detection on segmented lung tissue that reduces background noise and false positives, and (3) a Vision Transformer (ViT) for malignancy classification that leverages self-attention mechanisms to capture global context and long-range dependencies. The framework was developed and evaluated on the LUNA16 dataset (888 CT scans with 1,186 nodules $\geq$ 3mm, augmented with LIDC-IDRI malignancy annotations), with comprehensive preprocessing including Hounsfield Unit scaling (-1200 to 600 HU), normalization, binarization, resampling to 1mm isotropic resolution, and resizing to $256 \times 256 \times 256$ voxels. The integrated system achieves state-of-the-art performance across all stages: 98.82% Dice score for lung segmentation (exceeding prior methods including R2U-Net's 98.80%), 0.76 mAP@50 for nodule detection with competitive efficiency, and 96.29% accuracy for malignancy classification (4.25% improvement over prior SOTA), with clinically relevant attention maps demonstrating focus on lesion regions and excellent interpretability through ROC-AUC of 0.989. [34]

### Related Papers on FL For Chest X-ray Disease Multi-class Classification

- This study addresses the primary challenge in applying Federated Learning to chest X-ray analysis, which is the significant non-IID data distribution between adult and pediatric cohorts from different institutions. Analyzing over 400,000 radiographs, the research showed that conventional FL struggles to maintain consistent performance, particularly with the challenging pediatric dataset. The authors proposed enhancing FL by integrating general-purpose Self-Supervised Learning representations, using the DINOv2 model as a robust initialization.vThe new SSL-equipped framework proved highly effective, achieving an improvement exceeding 20% in the AUC metric for pneumonia diagnosis in the difficult pediatric cases. This validates that readily available SSL models provide a practical and scalable solution for enabling FL in diverse healthcare settings while preserving data privacy[35].

- This study focuses on using federated learning (FL) for the accurate and privacy-preserving diagnosis of pediatric pneumonia via chest X-ray classification. Traditional machine learning approaches face significant challenges related to centralizing sensitive medical data, which risks patient privacy and security. FL addresses this by enabling collaborative model training across different institutions while ensuring the raw patient data remains local and secure. The researchers specifically tackled the common problem of data heterogeneity, which typically compromises classification accuracy in real-world clinical settings. They proposed an improved FL method utilizing "two-end control variables," modifying the loss function on the client side and adding momentum on the server. This novel approach successfully enhanced performance, achieving an average accuracy of 98% and demonstrating an approximate 2% improvement over existing federated learning algorithms. [35].

- Artificial Intelligence for lung disease detection is limited by major privacy concerns due to the need to centralize sensitive patient data (X-ray and CT scans). To overcome this, the paper proposes a decentralized Federated Transfer Learning (FTL) system, which enables effective collaborative model training without ever pooling raw data. This approach leverages federated learning's privacy features and the efficiency of transfer learning to train models on limited local datasets. The study evaluated four methods using the ResNet-50 model, confirming FTL as the most effective method for high accuracy. Specifically, the resulting accuracies were: Centralized (87.95%), Transfer (88.04%), Federated (87.55%), and FTL (89.96%) [36].

- A federated learning framework distinguished COVID 19 from four other chest diseases (lung cancer, TB, pneumothorax, pneumonia) plus normal using chest X rays from multiple hospitals. They used three pre trained models (DenseNet 169, VGG 16, VGG 19) and applied SMOTE to mitigate class imbalance. The aggregated system (using DenseNet 169) achieved 98.45% accuracy, Grad CAM was used for model explainability, and their decision making client selection strategy reduced communication overhead significantly [37].

## 2.4 Combined Approaches: Medical Image Diagnosis with XAI and Federated Learning

Integrating XAI and FL in medical image diagnosis combines privacy with interpretability, enabling collaborative training on bone X-rays and CXR while explaining predictions, with potential for broader medical domains. This section details key papers that merge these elements.

### Related Papers on Combined Approaches for Chest Diseases Detection with XAI and Federated Learning

- A Federated Learning Based Real Time Ensemble Model with Explainable AI Framework for an Efficient Diagnosis of Lung Diseases. It is FLEM-XAI, a privacy-preserving FL framework with XAI for real-time lung disease diagnosis from CXR, using FedAvg for decentralized training across hospitals without data sharing. It ensembles pre-trained models (InceptionV3, Conv2D, VGG16, ResNet-50) trained locally, aggregated centrally, and incorporates XAI via SHAP for feature importance and GradCAM for heatmaps highlighting affected lung areas. Results show ResNet-50 achieving 95.5% overall accuracy (97.72% COVID-19, 95.42% pneumonia, 93.17% TB), outperforming centralized models in bandwidth efficiency and maintaining convergence [38].

- This study introduces FLPneXAINet, a framework that securely predicts pneumonia from Chest X-ray (CXR) images by integrating Federated Learning (FL), Deep Learning (DL), and Explainable AI (XAI) to overcome data privacy issues. The method utilized an 8,402-image Kaggle dataset, which was augmented using a CycleGAN network, and then features were extracted via pre-trained DL models and four Ensemble DL (EDL) models. Feature optimization was performed using RFE, ANOVA, and RF to select the most relevant features, which were then classified by various machine learning models in an FL environment. The EDL network achieved the best results in the FL environment, with a high accuracy of 97.61%, and was validated using XAI techniques (LIME and Grad-CAM). The full performance metrics were: F1 score 98.36%, recall 98.13%, and precision 98.59%, offering a robust solution for healthcare professionals [39].

- A secure lung cancer prediction used mapreduce blockchain FL with XAI, ensuring interpretability in federated settings [40].

## 2.5 Summary of Related Papers

Table 2.2: Summary of related papers.

| Focus Area | Sub-category | Model | Dataset | Result | Ref. Num. |
|---|---|---|---|---|---|
| Lung | General | DL-SVM | LIDC-IDRI | The proposed DL-SVM model achieved 94% accuracy, outperforming traditional machine learning models, standalone deep learning models, and hybrid baseline techniques for pulmonary nodule classification. | [3] |
| Lung | General | Custom CNN | Kaggle + Private Hospital | The EOSA-CNN model achieved 93.21% accuracy, with up to 97% specificity and 93% sensitivity, outperforming standard CNN models and baseline optimization methods in lung cancer CT classification. | [4] |
| Lung | General | ML Classifiers + CNN | Figshare | ICC-based filtering on 104 radiomic features retained 58 reliable features (ICC > 0.75) and removed 46 unstable ones, improving robustness by reducing inter-scanner variability. | [5] |
| Lung | General | Custom CNN + Dual Attention Mechanism | Not Specified | The model achieved 96% - 98% accuracy with improved precision, recall, and F1-score, outperforming existing methods. Attention improved feature localization for lung nodule detection. | [6] |
| Lung | XAI | LCxNet (CNN-Based) | IQ-OTH/NCCD | The LCxNet model achieved 99.3% accuracy and 99.45% specificity, outperforming state-of-the-art CNN models with stable performance across optimizers. | [15] |
| Lung | XAI | VGG16 | IQ-OTH/NCCD | The VGG16 model achieved 96% accuracy and used Grad-CAM for interpretability, highlighting tumor regions and improving clinical trust. | [16] |

| | | | | | |
|---|---|---|---|---|---|
| Lung | FL | Federated Ensemble | Not Specified | The federated ensemble model achieved 97% accuracy with F1-score 0.95 - 0.97, outperforming standalone CNNs while preserving data privacy. | [22] |
| Lung | FL | CNN + SVM | LIDC-IDRI | The hybrid CNN–SVM model achieved 99.7% accuracy with 99%+ sensitivity and specificity, outperforming traditional ML and CNN models. | [23] |
| Lung | FL | Feature Vision Framework | LIDC-IDRI | The framework achieved 97% - 98% accuracy with 96%+ sensitivity and 97%+ specificity, improving robustness through feature fusion. | [24] |
| Lung | Combined | CNN + Attention Mechanism | Not Specified | The model achieved 97% - 99% accuracy with F1-score 0.96 - 0.98, outperforming baseline CNN models. Attention improved feature discrimination. | [25] |
| Lung | General | Multiple CNN-Based Models | Private | VGG19 - Accuracy: 98.65 ∓ 0.22%, Precision: 98.80 ∓ 0.15%, Specificity: 98.70 ∓ 0.20%, Sensitivity: 98.55 ∓ 0.18%, F1-Score: 98.60 ∓ 0.16%, IoU: 0.94 ∓ 0.03, FP Rate: 1.05 ∓ 0.22 | [26] |
| Lung | Combined | Multiple CNN Models | Private | U-Net Segmentation: 81%,ResNet-50: 91% ,VGG-16: 94%,EfficientNet-B5: 97%,Ensemble Model: 99% | [27] |
| Lung | XAI | MCAT-Net | Private | LIDC-IDRI:IOU: 82.14% , LNDb:IOU: 71.28% | [28] |
| Lung | Combined | SSLKD-Unet | Private | SSLKD-Unet,Dice Coefficient: 93.8%,Sensitivity: 94.2%,Specificity: 96.1%,Pixel Accuracy: 96.5% | [29] |

| | | | | | |
|---|---|---|---|---|---|
| Lung | XAI | Faster R-CNN | Private | LUNA16: Precision: 0.781, Recall: 77.3%, mAP@50: 75.5%, mAP@50-90: 33.3%, NODE21: Precision: 0.895, Recall: 88.3%, mAP@50: 89.1%, mAP@50-90: 51.4%, LungCT: Precision: 0.887, Recall: 82.5%, mAP@50: 92.4%, mAP@50-90: 53.8% | [30] |
| Lung | XAI | Multiple YOLO Models | Private | SwiF-YOLO: Precision: 84.54%,Recall: 81.37%,mAP: 83.17%,YOLOx-M (baseline):Precision: 83.78%,Recall: 79.78%,mAP: 81.77% | [31] |
| Lung | Combined | CNN, YOLOv8 Variants | LIDC-IDRI | YOLOv8n: Precision 0.291, mAP50 0.356, mAP50-95 0.296, YOLOv8s: Precision 0.775, mAP50 0.505, mAP50-95 0.394,YOLOv8m (Best): Precision 0.873, mAP50 0.535, mAP50-95 0.4 | [32] |
| Lung | XAI | Multiple YOLO Models | Private | YOLOv3 + MSE: AP 73.1%, Sensitivity 85.2%,Faster-RCNN (ResNet101): AP 79.2%, Sensitivity 91.6%,Faster-RCNN (VGG16): AP 81.4%, Sensitivity 90.7% | [33] |
| Lung | Combined | Multiple 3D Models with YOLO Variant | Private | (3D Res-U-Net):Dice Score: 98.82% (95% CI: 98.5–99.1%),F1-Score: 98.73%,IoU: 97.6% ,YOLOv5s):mAP@50: 0.76 (95% CI: 0.73–0.79),VGG-16:94% | [34] |
| Chest | General | Custom CNN | Kaggle & IEEE | The model achieved 98.72% overall accuracy across all classes. Recall per class: Pneumonia 99.66%, No-findings 99.35%, Tuberculosis 98.10%, COVID-19 96.27%. | [7] |

| | | | | | |
|---|---|---|---|---|---|
| Chest | General | | Kaggle | Accuracy: 97% (using Resnet-50) VGG-19 (95%), DenseNet (91%), Efficient-Net (94%) | [8] |
| Chest | General | Custom CNN | Kaggle | TB vs Pneumonia classification: 97.4% accuracy; COVID-19 vs viral/bacterial pneumonia vs (other pneumonia types) classification: 88.0% accuracy | [9] |
| Chest | General | Custom CNN | VinDr-CXR | Classification Accuracy = 96.8% (from micro-F1 ≈ 0.968) Localization Accuracy ≈ 85% (IoU ≈ 0.851) | [10] |
| Chest | General | VGG16, DenseNet-161 and ResNet-18 | Custom clinical dataset collected and labeled | Normal/Pneumonia/TB/ COVID-19 classification (via VGG-16): 95.9% . Normal/Pneumonia/COVID-19 classification (via DenseNet-161): 98.9% . COVID-19 severity classification (via ResNet-18): 76.0% . | [11] |
| Chest | General | Custom CNN | Kaggle | 97% average accuracy (with precision ≈ 94% and recall ≈ 97%) for the full classification task. | [12] |
| Chest | General | Custom CNN and VGG-16 | Kaggle & public dataset from Github | The COV-X-net19 achieves 95.18% , MobileNet : 82.96%, VGG16 : 62.46% and base CNN : 89.96% . | [13] |
| Chest | General | DenseNet201, ResNet152V2 | IEEE | The model achieved superior performance with a test accuracy of 98.87%, outperforming the individual base models (DenseNet201 and ResNet152V2) and demonstrating robust multi-class classification capabilities. | [14] |

| | | | | | |
|---|---|---|---|---|---|
| Chest | XAI | Explainable Framework based on DenseNet-201 | – | The proposed explainable DenseNet-201 model achieved a high classification accuracy of 98.08%, while also providing visual explanations (Grad-CAM) to build trust and interpretability for clinical diagnosis. | [17] |
| Chest | XAI | CNN and Vision Transformer | Kaggle | XAI-TRANS achieves accuracy: 97% and 98% precision | [18] |
| Chest | XAI | Review of various Deep Learning models and XAI techniques (e.g., Grad-CAM, LIME) | Survey of medical image analysis studies | | [19] |
| Chest | XAI | Custom CNN with Grad-CAM and LIME | – | The proposed Custom CNN model achieved a high classification accuracy of 98.41%, with XAI techniques successfully highlighting the critical image regions used for diagnosis, thereby improving model interpretability. | [20] |
| Chest | XAI | Ensemble of DenseNet201, Inception-ResNetV2, Xception with Grad-CAM++ | Kaggle & IEEE | The proposed deep learning ensemble achieved a superior average classification accuracy of 98.97%, with XAI visualizations providing clinically plausible explanations for the model's decisions across all disease classes. | [21] |

| | | | | | |
|---|---|---|---|---|---|
| Chest | FL | Federated Learning (FedAvg) with a ResNet-based backbone and a novel age de-biasing loss function. | Multi-institutional chest X-ray datasets | The proposed de-biased FL model achieved an AUC of 89.7%, outperforming the standard FL model (AUC of 86.2%) and demonstrating significantly improved fairness and accuracy across different age groups. | [41] |
| Chest | FL | Federated Averaging (FedAvg) algorithm with a DenseNet-121 architecture as the core CNN model. | Chest X-ray images from multiple clients | The federated DenseNet-121 model achieved a test accuracy of 96.7% and an F1-score of 95.8%, matching the performance of a model trained on centralized data while preserving privacy. | [35] |
| Chest | FL | Federated Transfer Learning using a pre-trained ResNet-50 model, fine-tuned collaboratively across sites. | Collection from Kaggle | The federated transfer learning model achieved a classification accuracy of 96.3% for detecting various lung diseases, significantly outperforming models trained on single, isolated datasets. | [36] |
| Chest | FL | A custom Federated Learning framework using a specialized Composite-DenseNet | – | The DMFL_Net framework achieved a high overall classification accuracy of 98.21% and a precision of 98.14%, demonstrating superior performance for multi-disease classification in a federated setting. | [37] |

| | | | | | |
|---|---|---|---|---|---|
| Chest | Combined | A Federated Learning-based Real-Time Ensemble Model combining multiple CNNs (VGG16, ResNet) with XAI techniques like Grad-CAM. | Kaggle | The FLEM-XAI ensemble model achieved accuracy of 99.4% for multi-class lung disease classification. | [38] |
| Chest | Combined | A Federated Deep Learning framework using a custom CNN architecture integrated with XAI methods such as LIME and SHAP for model interpretation. | Chest X-ray images from distributed clients | The proposed FLPneXAINet framework achieved a high accuracy of 98.7% in detecting pneumonia from chest X-rays | [39] |
| Chest | Combined | A Secure Framework using MapReduce, Blockchain-based Federated Learning, and XAI with a DenseNet-201. | Multi-source CT scan datasets | The secure federated learning model achieved an exceptional accuracy of 99.8% for lung cancer prediction, leveraging blockchain for security and XAI for providing transparent, interpretable results. | [40] |

## 2.6 Datasets Utilized in the Project

Table 2.3: Summary of Datasets Used in the Project

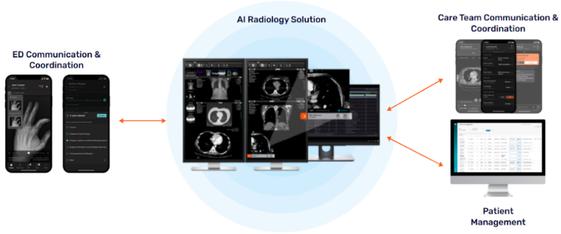
| Dataset Name | Domain | Class Distribution / Description | Total Images |
|---|---|---|---|
| Lung CT Nodule | Lung | Annotated lung CT scans that are used in training the lung cancer detection model | 5,069 |
| COVID-19 & Pneumonia & Tuberculosis with Normal & Non-X-ray | Chest | COVID-19 (3,616), Normal (5,688), Pneumonia (4,861), Tuberculosis (3,453) | 17,618 |
| Chest X-ray Dataset for Lung Segmentation | Chest | Annotated chest X-ray images for lung segmentation tasks | 6,106 images with corresponding 6,106 masks |
| Random Images for Image Classification | General | Unlabeled random images used for general image classification experiments | 9,958 |

## 2.7 Similar Commercial Applications

Several commercial applications leverage AI for medical image diagnosis, providing tools for clinicians to detect conditions like bone fractures and chest diseases. These applications often use DL models for classification but may lack advanced XAI or FL features for interpretability and privacy. Below, we discuss some notable examples.

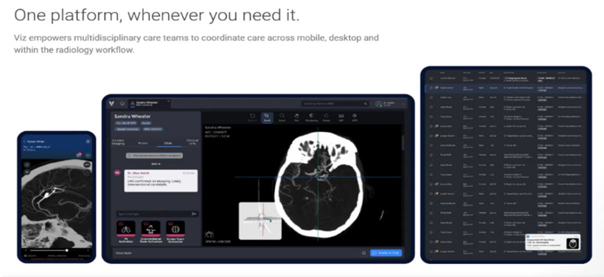
### 2.7.1 Aidoc (aiOS™)

Aidoc is a leading clinical AI software company that utilizes its proprietary aiOS™ platform to deliver high-performance AI solutions for medical imaging interpretation and workflow optimization. The platform automatically analyzes medical scans, prioritizing critical findings like stroke, pulmonary embolism, and intracranial hemorrhage, and then notifies care teams in real-time. By integrating seamlessly with existing systems like EHR and PACS, Aidoc's solutions help radiologists streamline workflows, enhance diagnostic accuracy, and accelerate time-to-treatment for patients with acute conditions across radiology, cardiology, and neurovascular specialties.




### 2.7.2 Viz.ai (Viz.ai One®)

Viz.ai is a leading AI-powered care coordination platform focused on transforming time-sensitive medical care by accelerating diagnosis and treatment decisions. The Viz.ai One® platform features multiple FDA-cleared AI algorithms that analyze imaging data, including CT scans and EKGs, to detect suspected diseases like stroke and pulmonary embolism. Crucially, it then uses real-time alerts and communication tools to instantly connect multidisciplinary care teams on mobile and desktop devices, ensuring the patient receives the necessary intervention as quickly as possible.




### 2.7.3 Qure.ai (qXR, qCT)

Qure.ai specializes in developing deep learning algorithms for affordable and accessible diagnostics, with a strong presence in global health initiatives. Their core products, qXR (for chest X-rays) and qER/qCT (for head and lung CTs), use AI to detect and quantify abnormalities, including tuberculosis, lung nodules, and critical trauma findings. By analyzing scans with high accuracy in minutes, Qure.ai's solutions assist radiologists in early disease detection, risk stratification, and provide essential diagnostic support, especially in areas with limited access to specialists.

### 2.7.4 Harrison.ai

Harrison.ai, through its partnership with the I-MED Radiology Network (Annalise.ai), offers comprehensive, decision-supported AI tools designed to raise the standard of patient care and combat capacity limits. Their advanced AI solutions for Chest X-ray (CXR) and Non-contrast Head CT (CTB) are designed to detect a massive number of potential findings (often over 100) within seconds. By acting as a 'co-pilot' for clinicians, their AI solutions flag emergent and incidental findings, helping to reduce stress from chaotic worklists and significantly improving both diagnostic accuracy and efficiency.

### 2.7.5 Lunit (Lunit INSIGHT, Lunit SCOPE)

Lunit is a global leader in medical AI software with a singular mission to conquer cancer through cutting-edge technology. The company's solutions, including the Lunit INSIGHT suite for cancer screening (especially mammography and chest X-ray) and the Lunit SCOPE platform for precision oncology, transform complex clinical data into clear, actionable insights. By delivering highly accurate diagnostic results and extracting intelligence from tissue images to predict treatment response, Lunit empowers earlier detection and more precise treatment decisions across the entire cancer care continuum.

## 2.8 Summary

In this chapter, we presented the background on bone fractures and chest diseases as initial focus areas for medical image diagnosis, emphasizing multiclassification. We discussed similar commercial applications, reviewed related works on detection using DL for each domain, explainable AI techniques like GradCAM for interpretability (with potential image insertions for heatmaps), federated learning for privacy-preserving training, and combined approaches integrating these elements. We also evaluated tools for development. These insights form the foundation for our extensible project's development, highlighting gaps in interpretability and data privacy that our system aims to address for bone fracture detection, chest disease detection, and future expansions.

# CHAPTER 3
# PROPOSED MODELS

# Chapter 3 Proposed Models

This chapter presents the proposed end-to-end solution developed for the automated detection of chest diseases and bone fractures from medical imaging data. The chapter aims to provide a comprehensive description of the overall system architecture as well as a detailed explanation of the individual models that constitute the proposed solution.

The chapter begins by outlining the solution architecture, which describes the complete pipelines of the system, including the high-level data flow and processing in the DL pipeline, and the backend processes and structure. This section illustrates how both the DL pipeline and the system's backend operate cohesively to form a unified diagnostic framework.

Subsequently, the chapter discusses the transfer learning models used in the proposed system. This section explains the pre-trained architectures employed, and their advantages in medical image analysis.

The medical image routing model is then introduced, which serves as an intelligent intermediary responsible for directing input images to the appropriate diagnostic model based on anatomical or modality-specific characteristics. This component plays a critical role in enabling the downstream models to act on an expected input image modality and anatomical region, it is also crucial in dropping out-of-distribution input data from being processed by the diagnostic models.

Following this, the bone fracture detection model is presented in detail. This section describes the model architecture, and classification approach used to identify bone fractures from radiographic images.

Then, an in-depth explanation of the chest disease detection model, which focuses on identifying various thoracic conditions from chest X-ray images, is presented. The section covers the model design, feature extraction process, and classification methodology employed to achieve robust disease detection.

Finally, the chapter ends with a summary to conclude all the information presented and an emphasis is put on the key information given in the chapter.

## 3.1 Solution Architecture

This chapter presents the architecture of the proposed intelligent medical image analysis framework. The system is designed as a hierarchical, modality-aware diagnostic pipeline capable of analyzing heterogeneous medical imaging data while maintaining high levels of reliability, interpretability, and scalability. The architecture addresses a practical challenge frequently encountered in real-world clinical environments: different diseases are diagnosed using different imaging modalities, and therefore a single monolithic model may not be the most effective solution. Instead, the proposed framework decomposes the overall diagnostic task into a sequence of specialized stages that collectively form an end-to-end decision-making pipeline.

The framework has been designed to support the automated analysis of chest radiographs and computed tomography (CT) scans. More specifically, the system is capable of detecting COVID-19, Tuberculosis, and Pneumonia from chest X-ray images, while also performing lung cancer detection from CT scans. In addition to disease classification, the framework incorporates Explainable Artificial Intelligence (XAI) mechanisms to improve transparency and trustworthiness, as well as Federated Learning (FL) capabilities to facilitate privacy-preserving collaborative model development across multiple healthcare institutions.

The overall architecture follows a two-level classification paradigm. At the first level, an intelligent routing model determines the modality of the incoming image and verifies whether it belongs to one of the supported image categories. At the second level, modality-specific diagnostic models perform disease prediction. This hierarchical structure enables the system to leverage specialized classifiers that are optimized for their respective tasks while reducing the complexity associated with training a single generalized model.

Figure 3.1 presents a high-level overview of the proposed framework.

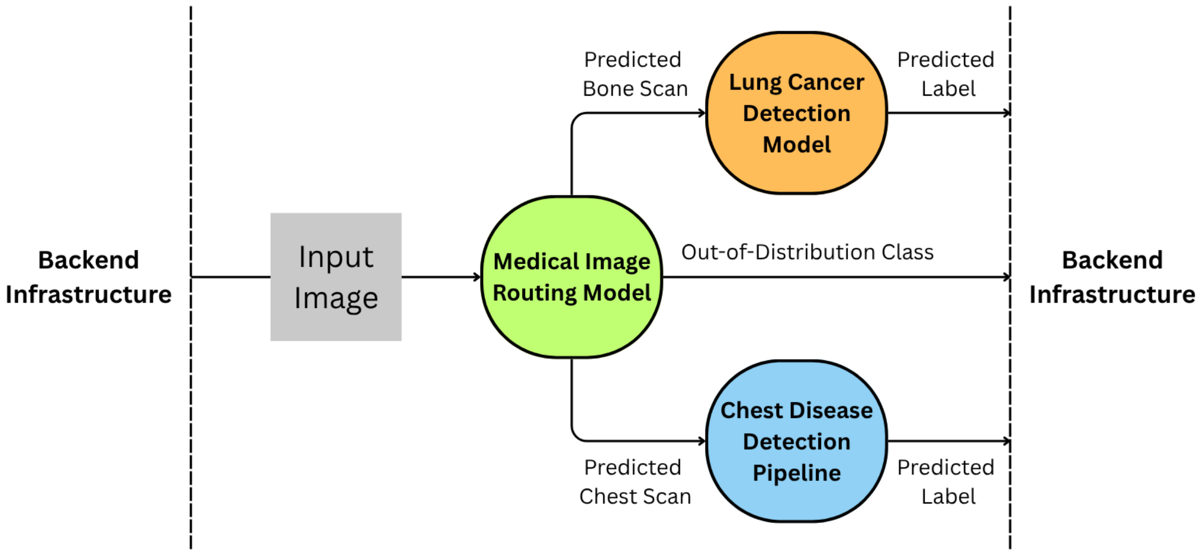



Figure 3.1: High-level architecture of the proposed framework. The system consists of a modality routing model, specialized disease classifiers, Explainable AI modules, and a Federated Learning infrastructure connecting multiple healthcare institutions.

## 3.1.1 Architectural Design Philosophy

The proposed framework was developed according to several architectural principles that were established at the beginning of the system design process. These principles were intended to ensure that the resulting framework would not only achieve strong predictive performance but would also remain practical for real-world deployment in healthcare environments.

One of the most important design considerations was modality awareness. Medical images generated by different imaging technologies exhibit fundamentally different visual characteristics and diagnostic properties. Chest radiographs provide projection-based representations of thoracic structures, whereas CT scans generate detailed cross-sectional views that reveal significantly more anatomical information. These differences influence the types of features that neural networks must learn in order to perform accurate diagnosis. A single generalized model trained across multiple modalities may struggle to simultaneously learn optimal representations for all imaging types. Consequently, the proposed architecture explicitly separates modality identification from disease diagnosis. This design allows each diagnostic model to focus exclusively on data originating from the modality for which it was designed.

Another fundamental design principle was task specialization. The diseases targeted by this work differ substantially in terms of pathological manifestations, imaging characteristics, and diagnostic requirements. COVID-19, Tuberculosis, and Pneumonia are primarily identified using chest radiographs, whereas lung cancer diagnosis often relies on the higher anatomical resolution available in CT imaging. Rather than attempting to train a single network to recognize all diseases across all modalities, the proposed framework distributes these tasks among specialized models. This decomposition reduces task complexity and allows each model to learn more discriminative feature representations relevant to its specific diagnostic objective.

Reliability and patient safety also played a significant role in the architectural design. Medical AI systems must operate in environments where unexpected inputs are inevitable. Hospital information systems frequently contain diverse imaging studies, and a deployed diagnostic model may encounter unsupported image types, corrupted files, incomplete scans, or non-medical images. If such inputs are processed without validation, the resulting predictions may be unreliable and potentially dangerous. To address this issue, the proposed framework incorporates an explicit out-of-distribution detection mechanism within the routing stage. This component enables the system to identify unsupported inputs before they reach the diagnostic models.

Transparency represents another critical requirement for healthcare AI systems. Clinical practitioners often require evidence supporting automated predictions before integrating them into their decision-making processes. While deep neural networks have demonstrated remarkable predictive capabilities, they are frequently criticized for their lack of interpretability. The proposed architecture therefore integrates explainability mechanisms directly into the diagnostic workflow. These mechanisms provide visual explanations that highlight image regions contributing to the final prediction, thereby allowing clinicians to better understand and validate the reasoning process of the model.

Finally, the architecture was designed with scalability and long-term deployment in mind. Healthcare institutions are often unable to share patient data due to privacy regulations, legal restrictions, and ethical concerns. Consequently, the framework incorporates Federated Learning capabilities that enable multiple institutions to collaboratively improve model performance without exchanging raw patient information. This design promotes the creation of a continuously improving diagnostic ecosystem while preserving patient confidentiality.

### 3.1.2 System Workflow

The complete workflow of the proposed framework begins when a medical image is submitted to the system. Following acquisition, the image undergoes preprocessing operations intended to standardize the input format and improve compatibility with the downstream neural networks. These operations may include resizing, normalization, intensity standardization, and other transformations required by the underlying deep learning models.

Once preprocessing is completed, the image is forwarded to the first level of the architecture, namely the modality routing model. The purpose of this stage is to determine the nature of the input image and identify the appropriate diagnostic pathway. The routing decision governs all subsequent processing stages and therefore serves as a critical component of the framework.

After the routing decision is generated, the image is directed toward a specialized diagnostic model. Images identified as chest radiographs are analyzed by the chest disease classifier, while CT scans are analyzed by the lung cancer detection model. Images identified as out-of-distribution are excluded from automated diagnosis and may instead be flagged for manual review by healthcare personnel.

Following disease prediction, explainability modules generate visual interpretations of the model output. These interpretations provide insight into the image regions that most strongly influenced the prediction process. Finally, the prediction and associated explanation are presented to the clinician. During model training, Federated Learning mechanisms facilitate collaborative model improvement across multiple participating institutions.

Figure 3.2 illustrates the classification pathway implemented by the proposed framework.

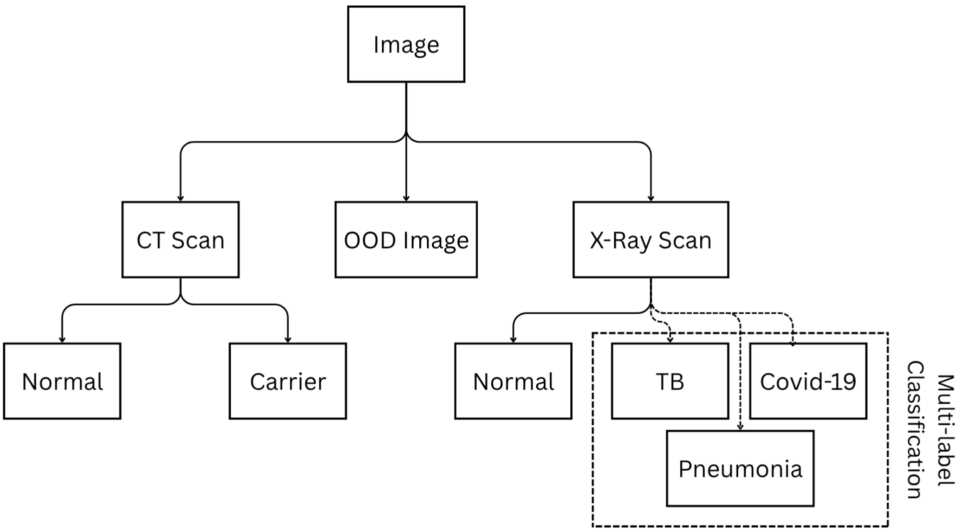



Figure 3.2: Classification tree of the proposed two-level framework. The routing model first determines the image modality before directing the image toward the appropriate disease-specific classifier.

### 3.1.3 Level One: Modality Routing and Out-of-Distribution Detection

The first level of the proposed architecture functions as an intelligent routing mechanism responsible for identifying the modality of the incoming image. Although this component performs a relatively simple classification task compared to disease diagnosis, its importance should not be underestimated. Every subsequent prediction generated by the framework depends directly on the correctness of the routing decision.

Formally, let $I$ denote an input image. The routing model $R(\cdot)$ maps the image into one of three categories:

$$R(I) \in \{\text{X-ray}, \text{CT}, \text{OOD}\} \tag{3.1}$$

where OOD represents an out-of-distribution image.

The motivation for introducing a dedicated routing stage arises from the substantial differences between X-ray and CT imaging modalities. Chest radiographs represent two-dimensional projection images that summarize anatomical information along a viewing direction. In contrast, CT scans provide detailed cross-sectional information that enables the visualization of internal structures with considerably greater precision. These differences imply that the features required for effective diagnosis are modality-dependent. Routing images to specialized models therefore allows the framework to exploit modality-specific representations rather than relying on a generalized feature extraction strategy.

From a computational perspective, routing also improves efficiency. Without this stage, every image would need to be processed by every available diagnostic model, resulting in increased computational cost and unnecessary inference overhead. By ensuring that each image is evaluated only by the most relevant classifier, the routing model reduces resource consumption and improves scalability.

An equally important function of the routing stage is out-of-distribution detection. In practical deployment scenarios, healthcare systems may receive images that differ significantly from those used during training. Examples include magnetic resonance imaging studies, ultrasound images, pathology slides, non-medical photographs, or improperly formatted files. Diagnostic models are generally incapable of producing reliable predictions for such inputs because they fall outside the distribution of the training data.

To mitigate this problem, the routing model incorporates a dedicated OOD category. Images assigned to this category are prevented from entering the diagnostic pipeline. This safeguard reduces the likelihood of erroneous predictions and contributes to the overall reliability of the framework. From a clinical perspective, this mechanism functions as a preliminary quality-control layer that ensures only supported image types proceed to automated analysis.

### 3.1.4 Level Two: Chest Disease Detection Model

Images identified as chest radiographs are forwarded to the chest disease classification model. This model is responsible for detecting three clinically significant respiratory diseases: COVID-19, Tuberculosis, and Pneumonia.

The selection of these diseases was motivated by both clinical relevance and public health importance. COVID-19 remains one of the most extensively studied respiratory illnesses in recent years and has highlighted the importance of rapid imaging-based screening techniques. Tuberculosis continues to represent a major global health challenge, particularly in developing countries where early detection plays a crucial role in disease management and transmission control. Pneumonia remains one of the most common respiratory infections worldwide and is associated with substantial morbidity and mortality across multiple age groups.

A key characteristic of the proposed chest disease model is its formulation as a multi-label classification problem rather than a conventional multi-class classification problem. This distinction is essential because the targeted diseases are not mutually exclusive. In clinical practice, a patient may simultaneously exhibit radiological findings associated with multiple respiratory conditions. For example, viral infections may be accompanied by secondary bacterial pneumonia, and complex pathological interactions may lead to overlapping imaging manifestations.

Traditional multi-class classification assumes that each sample belongs to exactly one category. Such an assumption would force the model to select a single disease label even when multiple diseases are present. This formulation would therefore fail to accurately represent many real-world clinical scenarios.

To address this limitation, the proposed architecture employs independent disease predictions. Let

$$\mathbf{y} = [y_{\text{COVID}}, y_{\text{TB}}, y_{\text{Pneumonia}}] \tag{3.2}$$

denote the output vector generated by the classifier. Each element corresponds to the estimated probability that a specific disease is present within the image.

Unlike a softmax-based multi-class architecture, the output layer uses independent sigmoid activations. This allows the model to assign high probabilities to multiple diseases simultaneously whenever appropriate. The final prediction for each disease is obtained by applying a predefined threshold:

$$\hat{y}_i = \begin{cases} 1 & y_i \geq \tau \\ 0 & y_i < \tau \end{cases} \tag{3.3}$$

where $\tau$ denotes the classification threshold.

This formulation better reflects the realities of clinical diagnosis and provides greater flexibility when interpreting model outputs. Instead of receiving a single categorical prediction, clinicians are presented with independent disease probabilities that can be incorporated into broader diagnostic assessments.

### 3.1.5 Level Two: Lung Cancer Detection Model

Images classified as CT scans are forwarded to the lung cancer detection model. The decision to utilize CT imaging for lung cancer analysis is motivated by the superior anatomical detail provided by computed tomography compared to conventional radiography.

Lung cancer frequently manifests through subtle structural abnormalities such as pulmonary nodules, irregular masses, and localized tissue changes that may not be visible in chest radiographs. The high spatial resolution of CT imaging enables these features to be observed with significantly greater clarity, making CT the preferred modality for many lung cancer screening and diagnostic procedures.

The objective of the lung cancer model is to determine whether cancer-related findings are present within the input scan. The classifier can be represented mathematically as

$$P_{\text{Cancer}} = C(I) \tag{3.4}$$

where $C(\cdot)$ denotes the lung cancer detection network and $P_{\text{Cancer}}$ represents the estimated probability of malignancy.

By focusing exclusively on CT-derived information, the model can dedicate its representational capacity to learning highly discriminative cancer-related features. This specialization improves the model's ability to identify subtle pathological patterns that may be difficult to detect using a generalized classifier.

### 3.1.6 Explainable Artificial Intelligence Integration

The increasing adoption of artificial intelligence within healthcare has generated significant interest in model interpretability. Although deep learning systems are capable of achieving remarkable predictive performance, their internal decision-making processes often remain difficult to understand. This lack of transparency can hinder clinical adoption, particularly when predictions influence critical healthcare decisions.

To address this challenge, Explainable Artificial Intelligence techniques are integrated directly into the proposed architecture. Rather than treating explainability as an optional post-processing step, it is considered a fundamental component of the diagnostic workflow.

Following prediction generation, the explainability module produces visual explanations illustrating the regions of the image that most strongly influenced the model's decision. These visualizations may take the form of saliency maps, activation maps, attention maps, or gradient-based explanations depending on the selected implementation.

The integration of explainability serves several purposes. First, it increases clinician confidence by providing visual evidence supporting the model's prediction. Second, it facilitates model validation by enabling researchers to verify that the network is focusing on medically meaningful regions rather than irrelevant artifacts. Third, it assists in identifying potential biases or failure modes that may otherwise remain hidden within the network. Ultimately, explainability strengthens the collaborative relationship between clinicians and AI systems by transforming predictions into interpretable decision-support information.

### 3.1.7 Federated Learning Infrastructure

A distinguishing feature of the proposed framework is its support for Federated Learning. This capability was incorporated to address one of the most significant challenges facing medical artificial intelligence: limited access to large and diverse datasets.

Traditional deep learning systems typically rely on centralized training approaches in which data from multiple institutions are collected and stored in a single repository. While effective from a machine learning perspective, this strategy introduces substantial privacy, security, and regulatory concerns. Healthcare organizations are often prohibited from sharing patient data due to legal and ethical restrictions.

Federated Learning offers an alternative paradigm in which training occurs locally at each institution. Rather than transmitting patient data, participating hospitals train local models using their own datasets and share only model parameters or parameter updates. These updates are subsequently aggregated to produce a global model.

Let $w_i$ denote the model parameters learned by institution $i$, and let $n_i$ denote the number of training samples available at that institution. Using the Federated Averaging algorithm, the global model is computed as

$$w_{\text{global}} = \sum_{i=1}^{K} \frac{n_i}{\sum_{j=1}^{K} n_j} w_i \tag{3.5}$$

where $K$ represents the total number of participating institutions.

Beyond privacy preservation, Federated Learning provides several additional benefits. Because participating institutions often serve different patient populations, the resulting datasets contain greater demographic, geographic, and clinical diversity. Training across such heterogeneous data sources can improve model generalization and reduce susceptibility to institution-specific biases. Furthermore, the collaborative nature of Federated Learning enables continuous improvement of the framework as new institutions join the network and contribute additional knowledge.

## 3.2 Transfer-Learning Models Used

This section presents the transfer learning models employed in the proposed system and discusses the rationale behind their selection. Transfer learning is adopted to leverage pre-trained deep neural networks that have been trained on large-scale datasets, enabling effective feature extraction and improved generalization in medical imaging tasks with limited labeled data. The section provides an overview of the selected architectures and highlights their suitability for different components of the proposed diagnostic pipeline.

### 3.2.1 ResNet50 Model

Residual Networks (ResNets) were introduced to address the degradation problem observed in deep convolutional neural networks, where increasing network depth leads to saturation or decline in training accuracy. This issue emerges due to optimization difficulties such as vanishing gradients. ResNet addresses this limitation through the concept of residual learning, which allows the training of substantially deeper networks by enabling more effective gradient propagation.

The key idea of residual learning is to reformulate the learning objective of a neural network block. Instead of directly learning a desired underlying mapping $H(x)$, a residual block learns a residual function $F(x)$ such that:

$$H(x) = F(x) + x$$

where $x$ represents the input to the residual block and $F(x)$ represents the output of the stacked convolutional layers within the block. The identity shortcut connection that performs the addition allows gradients to flow directly through the network during backpropagation, thereby minimizing vanishing gradient effects and improving training stability.

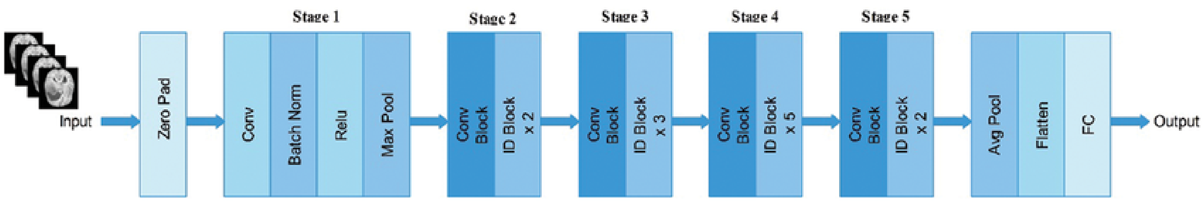



Figure 3.3: Architecture of the ResNet50 illustrating the successive stages of the model.

As illustrated in Figure 3.3, the ResNet50 architecture begins with an initial convolutional layer preceded by zero padding, followed by batch normalization, a ReLU activation function, and a max pooling operation. This initial stage is responsible for extracting low-level visual features from the input image. The network then proceeds through five sequential stages composed of convolutional blocks and identity blocks, each incorporating residual connections that bypass multiple convolutional layers.

ResNet50 employs a bottleneck design within its residual blocks, consisting of a sequence of $1 \times 1$, $3 \times 3$, and $1 \times 1$ convolutional layers. The $1 \times 1$ convolutions are used to reduce and subsequently restore feature dimensionality, thereby decreasing computational complexity while preserving representational power. This design enables ResNet50 to achieve substantial depth without incurring prohibitive computational costs.

Compared to shallower variants such as ResNet18 and ResNet34, ResNet50 offers increased representational capacity and improved ability to capture complex and abstract visual patterns. While deeper variants such as ResNet101 and ResNet152 further extend network depth, they introduce higher computational overhead and an increased risk of overfitting, particularly in medical imaging applications where annotated data is often limited. Consequently, ResNet50 represents a balanced trade-off between depth, efficiency, and performance within the ResNet family.

As shown in Figure 3.3, the final stages of ResNet50 consist of a global average pooling layer, followed by feature flattening and a fully connected layer. In transfer learning scenarios, this classification head can be replaced or fine-tuned to adapt the network to domain-specific medical tasks.

In the context of medical image analysis, ResNet50 is well-suited for extracting both low-level structural features and high-level semantic representations relevant to pathological patterns. The residual connections promote feature reuse and robust optimization, which is particularly important for detecting subtle abnormalities in medical images such as fractures or thoracic diseases. These characteristics make ResNet50 a reliable and widely adopted backbone for medical image classification tasks.

### 3.2.2 DenseNet121 Model

DenseNet121 is a deep convolutional neural network architecture belonging to the family of Densely Connected Convolutional Networks (DenseNets), which were introduced by Huang *et al.* in 2017 to address several limitations associated with traditional deep neural networks. DenseNet architectures were designed to improve feature propagation, encourage feature reuse, alleviate the vanishing gradient problem, and reduce the number of trainable parameters while maintaining high classification performance. Since its introduction, DenseNet121 has become one of the most widely adopted architectures in medical image analysis due to its ability to learn rich hierarchical representations from relatively limited datasets.

The fundamental idea behind DenseNet differs significantly from that of conventional convolutional neural networks. In traditional architectures, each layer receives information only from the immediately preceding layer and passes its output solely to the next layer. Although this sequential information flow is effective, it may lead to difficulties in gradient propagation as network depth increases. Furthermore, many learned feature maps become redundant because similar patterns are repeatedly relearned at different layers throughout the network.

To address these limitations, DenseNet introduces direct connections between layers. Instead of receiving input from only the previous layer, each layer receives the feature maps generated by all preceding layers within the same dense block. Consequently, information can flow directly from earlier layers to later layers without repeatedly passing through intermediate transformations. This dense connectivity pattern enables more efficient feature reuse and facilitates the training of very deep networks.

Figure 3.4 illustrates the overall architecture of DenseNet121.

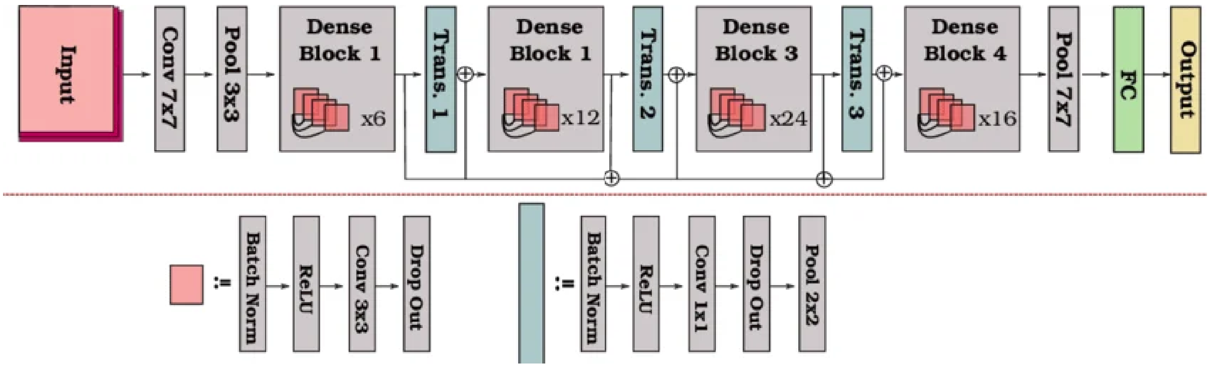



Figure 3.4: Overall architecture of DenseNet121 showing the arrangement of dense blocks, transition layers, and the final classification stage.

## Motivation Behind Dense Connectivity

As neural networks become deeper, the flow of information and gradients through the network becomes increasingly difficult. One of the most common challenges encountered in deep architectures is the vanishing gradient problem, where gradients become progressively smaller as they are propagated backward through many layers. When gradients become extremely small, early layers receive little useful learning signal, slowing convergence and potentially reducing overall performance.

Residual Networks (ResNets) partially addressed this issue through skip connections that allow gradients to bypass certain transformations. DenseNet extends this concept further by establishing direct connections between every pair of layers within a dense block. Instead of merely adding outputs from previous layers, DenseNet concatenates them. This strategy preserves information from all preceding layers and makes it directly accessible to subsequent layers.

The dense connectivity mechanism provides several important benefits. First, it improves gradient flow during training, allowing deeper networks to be optimized more effectively. Second, it encourages feature reuse, reducing redundancy among learned representations. Third, it improves parameter efficiency because layers can leverage previously learned features rather than learning entirely new representations from scratch.

## Dense Connectivity Mechanism

The defining characteristic of DenseNet is the dense connection pattern between layers. Let $x_0$ represent the input to a dense block and let $x_l$ denote the output of the $l^{th}$ layer. Unlike traditional convolutional networks, where the output of layer $l$ depends only on the output of layer $l-1$, DenseNet computes each layer using the outputs of all preceding layers:

$$x_l = H_l([x_0, x_1, x_2, ..., x_{l-1}]) \tag{3.6}$$

where:

- $H_l(\cdot)$ denotes a composite transformation.
- $[\cdot]$ represents the concatenation operation.
- $x_0, x_1, ..., x_{l-1}$ are the feature maps generated by previous layers.

The concatenation operation differs fundamentally from the addition operation used in residual networks. Instead of merging features by summation, DenseNet preserves all previous feature maps and passes them directly to subsequent layers. As a result, each layer has access to both low-level and high-level information simultaneously.

This dense feature propagation mechanism allows later layers to exploit a richer collection of learned representations, improving feature diversity and reducing information loss.

## Dense Blocks

The primary computational components of DenseNet121 are the dense blocks. A dense block consists of multiple densely connected layers arranged according to the connectivity pattern described previously.

Within each dense block, every layer receives as input the concatenated outputs of all preceding layers. Consequently, the number of feature maps grows progressively as additional layers are added. This growth is controlled through a parameter known as the *growth rate*.

The growth rate, typically denoted by $k$, determines the number of new feature maps generated by each layer. If the input to a dense block contains $F_0$ feature maps and the block contains $L$ layers, the total number of feature maps at the end of the block can be expressed as:

$$F = F_0 + kL \tag{3.7}$$

where:

- $F_0$ is the number of initial feature maps.
- $k$ is the growth rate.
- $L$ is the number of layers within the block.

For DenseNet121, the growth rate is typically set to 32. This means that each layer contributes 32 new feature maps while simultaneously utilizing all previously generated feature maps.

The use of a relatively small growth rate contributes to the parameter efficiency of DenseNet architectures, allowing them to achieve competitive performance with fewer parameters than many alternative architectures.

### Composite Function within Dense Layers

Each layer within a dense block consists of a sequence of operations designed to transform incoming feature maps into more informative representations.

The composite transformation $H_l(\cdot)$ generally consists of:

1. Batch Normalization (BN)
2. Rectified Linear Unit (ReLU)
3. Convolution

This sequence is commonly referred to as the BN-ReLU-Conv pattern.

Batch normalization stabilizes the learning process by normalizing intermediate activations. This reduces internal covariate shift and allows higher learning rates to be used during training.

The ReLU activation function introduces non-linearity into the network and is defined as:

$$f(x) = \max(0, x) \tag{3.8}$$

Following activation, convolutional filters extract spatial patterns from the input feature maps. Through repeated application of these operations, the network gradually learns increasingly abstract representations of the input image.

### Bottleneck Layers

As the number of feature maps grows within dense blocks, computational complexity can increase significantly. To reduce this cost, DenseNet121 employs bottleneck layers.

A bottleneck layer consists of a $1 \times 1$ convolution applied before the standard $3 \times 3$ convolution. The purpose of this operation is to reduce the number of feature channels while preserving important information.

The bottleneck structure can be summarized as:

$$BN \rightarrow ReLU \rightarrow Conv(1 \times 1) \rightarrow BN \rightarrow ReLU \rightarrow Conv(3 \times 3) \tag{3.9}$$

The use of bottleneck layers substantially reduces computational requirements while maintaining representational capacity.

## Transition Layers

Because feature map dimensions increase continuously within dense blocks, DenseNet introduces transition layers between consecutive blocks to control network complexity.

A transition layer performs two primary functions. First, it reduces the number of feature maps through a $1 \times 1$ convolution. Second, it decreases spatial resolution using average pooling.

The transition layer architecture typically follows:

$$BN \rightarrow ReLU \rightarrow Conv(1 \times 1) \rightarrow AvgPool(2 \times 2) \tag{3.10}$$

These operations prevent excessive growth in memory consumption while preserving the most informative features.

Transition layers also contribute to hierarchical feature extraction by gradually reducing spatial dimensions as network depth increases.

## Architecture of DenseNet121

DenseNet121 consists of four dense blocks separated by three transition layers. The architecture begins with an initial convolution layer and concludes with global average pooling followed by a fully connected classification layer.

The distribution of layers across dense blocks is as follows:

Table 3.1: Layer distribution within DenseNet121.

| Component | Number of Layers |
|---|---|
| Dense Block 1 | 6 |
| Dense Block 2 | 12 |
| Dense Block 3 | 24 |
| Dense Block 4 | 16 |
| Total Dense Layers | 58 |

When all convolutional, pooling, and classification layers are included, the network contains 121 trainable layers, which gives the architecture its name.

The input image first passes through an initial $7 \times 7$ convolution layer followed by max pooling. Feature extraction then proceeds through the sequence of dense blocks and transition layers. Finally, global average pooling aggregates spatial information into a compact feature vector, which is passed to the final classification layer.

## Advantages of DenseNet121 for Medical Image Analysis

DenseNet121 possesses several characteristics that make it particularly suitable for medical image classification tasks.

The dense connectivity pattern encourages extensive feature reuse, allowing the network to learn robust representations even when training datasets are relatively small. This characteristic is especially important in medical imaging, where acquiring large annotated datasets is often challenging.

The improved gradient flow resulting from dense connections facilitates stable optimization and reduces the risk of vanishing gradients. Consequently, deeper representations can be learned without the training difficulties commonly associated with very deep neural networks.

DenseNet121 is also highly parameter-efficient compared to many competing architectures. Despite its depth, the network requires fewer trainable parameters than several traditional convolutional architectures because previously learned features are reused rather than repeatedly regenerated.

Furthermore, the architecture has demonstrated strong performance across a wide range of medical imaging applications, including chest disease classification, pneumonia detection, COVID-19 diagnosis, tuberculosis screening, diabetic retinopathy analysis, and cancer detection. Its proven effectiveness in healthcare applications has made DenseNet121 a popular backbone network for medical image analysis research.

## DenseNet121 in the Proposed Framework

In the context of this work, DenseNet121 was selected as the backbone architecture for the first-level routing model due to its strong balance between classification performance, computational efficiency, and feature extraction capability. The dense connectivity structure enables the model to capture both low-level radiological patterns and high-level pathological characteristics that are essential for accurate disease recognition.

Additionally, the feature reuse mechanism of DenseNet121 is particularly advantageous for medical datasets, where image variability can be substantial and labeled samples may be limited. By leveraging information from all preceding layers, the network can construct rich hierarchical representations that support robust classification performance.

Overall, DenseNet121 provides a powerful and efficient deep learning architecture that aligns well with the requirements of medical image analysis systems. Its dense connectivity, parameter efficiency, strong gradient propagation, and proven effectiveness in healthcare applications make it an appropriate choice for the disease classification tasks addressed in this project.

### 3.2.3 YOLOv8 Model

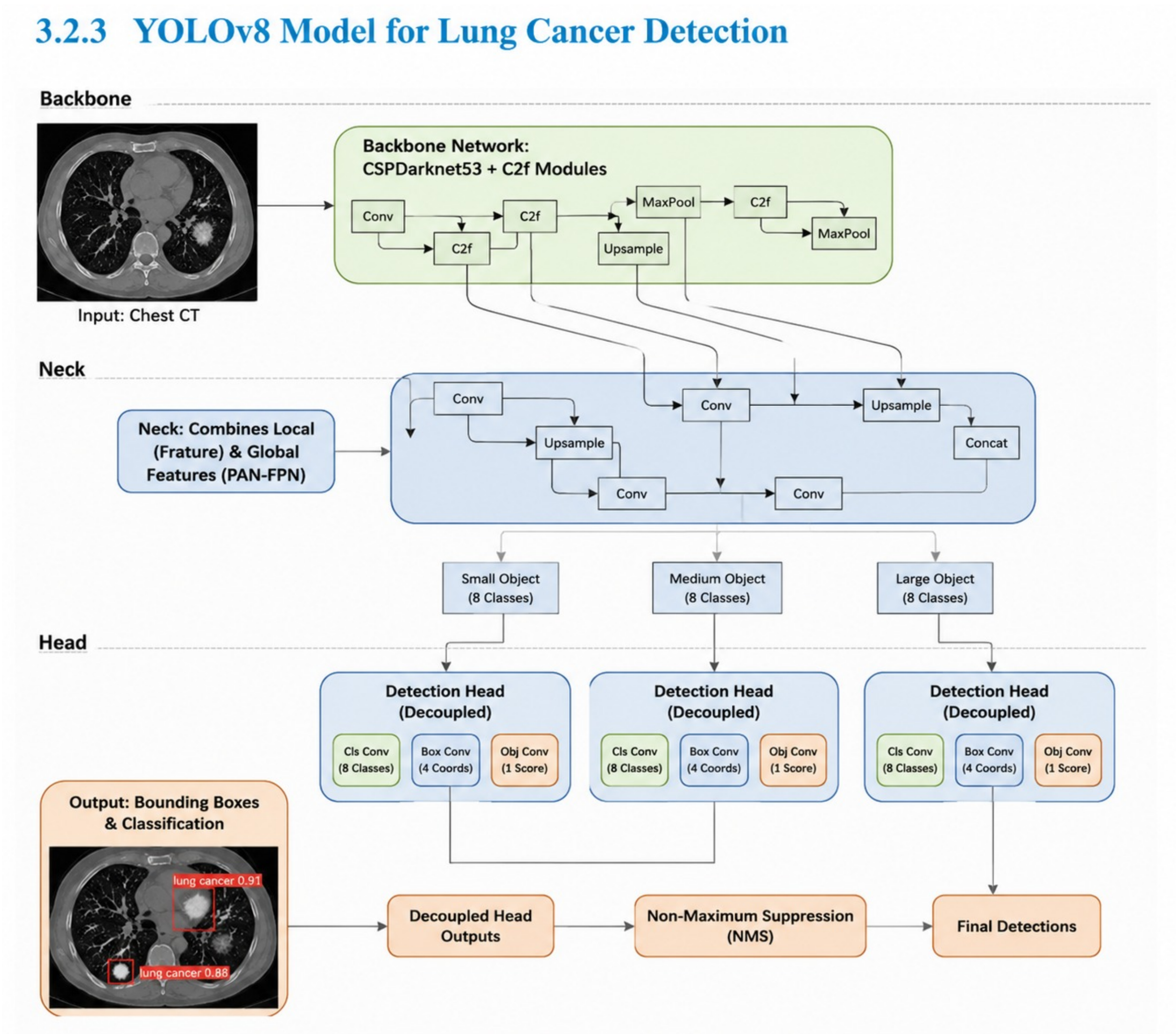



Figure 3.5: General architecture of YOLOv8 illustrating the backbone, neck, and detection head components.

Object detection has become one of the most important computer vision tasks in modern medical image analysis due to its ability to simultaneously identify and localize pathological findings within an image. Unlike conventional image classification models that only determine whether a disease is present, object detection models additionally provide spatial information regarding the location of abnormalities. This capability is particularly valuable in lung cancer diagnosis, where clinicians must not only determine the presence of a suspicious lesion but also identify its precise location within the lungs.

In the proposed framework, lung cancer detection is formulated as an object detection problem rather than a traditional image classification task. Instead of predicting a single label for an entire CT scan, the model identifies suspicious cancerous regions and generates bounding boxes surrounding them. To accomplish this task, the proposed system employs YOLOv8 (You Only Look Once Version 8), one of the most recent and advanced members of the YOLO family of object detection architectures.

YOLOv8 is a single-stage object detector, meaning that object localization and classification are performed within a single forward pass through the network. This design significantly reduces computational complexity compared to two-stage detectors such as Faster R-CNN, which first generate candidate regions and subsequently classify them. By combining both operations into a unified architecture, YOLOv8 achieves high detection accuracy while maintaining efficient inference speeds. These characteristics make it particularly suitable for deployment within real-time clinical decision support systems where rapid analysis of CT scans is desirable.

The complete processing pipeline employed by the proposed lung cancer detection model is illustrated in Figure 3.6.

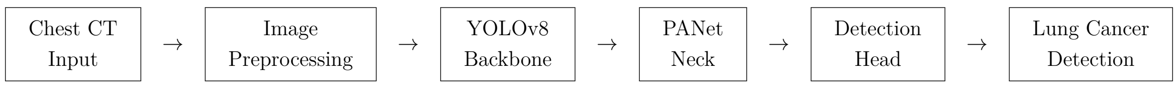



Figure 3.6: End-to-end processing pipeline of the proposed YOLOv8 lung cancer detection model.

As illustrated in Figure 3.5, YOLOv8 consists of three primary components: the backbone, the neck, and the detection head. Each component contributes a specific functionality to the overall detection process.

## Backbone Network

The backbone serves as the feature extraction component of YOLOv8. Its primary objective is to transform raw CT scan images into hierarchical feature representations that can be used for lesion detection. YOLOv8 employs the C2f (Cross-Stage Partial Bottleneck with Two Convolutions) architecture, which is an evolution of the CSP (Cross Stage Partial) designs used in previous YOLO versions.

The backbone progressively applies convolutional operations to the input image, generating increasingly abstract feature maps. Early layers learn low-level visual characteristics such as edges, gradients, and texture variations, while deeper layers capture more complex semantic information associated with pulmonary nodules, masses, and abnormal tissue structures.

One of the key advantages of the C2f architecture is its ability to improve gradient propagation while reducing computational overhead. By reusing features across multiple layers, the network can learn richer representations without requiring excessive numbers of parameters. This characteristic is particularly beneficial in medical imaging applications where datasets are often smaller than those available in general computer vision tasks.

For lung cancer detection, the backbone learns to distinguish subtle visual differences between healthy lung tissue and suspicious lesions. These differences may include variations in density, texture, shape, and intensity patterns that are characteristic of malignant abnormalities.

## Neck Network

The neck component is responsible for feature aggregation and multi-scale feature fusion. In YOLOv8, this functionality is implemented using a Path Aggregation Network (PANet).

Feature extraction layers at different depths of the backbone capture information at different scales. Shallow layers retain detailed spatial information, while deeper layers capture more abstract semantic features. Because lung cancer lesions can vary significantly in size, relying on a single feature scale would limit detection performance.

Small pulmonary nodules may occupy only a tiny portion of the CT image, whereas larger tumors may extend across substantial regions of the lung. The PANet structure addresses this challenge by combining feature maps generated at multiple scales. Through repeated top-down and bottom-up information flow, PANet enables the model to simultaneously utilize local texture details and broader anatomical context.

Consequently, the neck allows YOLOv8 to detect both small and large lesions while maintaining accurate localization performance across a wide range of tumor sizes.

## Detection Head

The final component of YOLOv8 is the detection head. This module is responsible for generating the actual predictions produced by the network.

Unlike earlier object detectors that relied on predefined anchor boxes, YOLOv8 employs an anchor-free detection strategy. Rather than predicting offsets relative to predetermined anchors, the model directly predicts object centers and bounding box coordinates.

The detection head performs two independent tasks:

- Classification of detected regions.
- Localization of detected lesions.

This decoupled architecture allows each task to be optimized independently, improving overall detection performance. For each detected region, the model predicts:

- Bounding box coordinates.
- Object confidence score.
- Cancer detection probability.

The resulting predictions are then forwarded to the post-processing stage for refinement.

## Multi-Scale Detection

A major challenge in lung cancer detection is the substantial variability in lesion size. Early-stage tumors may appear as extremely small nodules, whereas advanced-stage tumors can occupy large portions of the lung field.

YOLOv8 addresses this challenge through multi-scale detection. Detection heads operate on feature maps of different resolutions, enabling the model to identify abnormalities across multiple spatial scales.

This capability is particularly important in CT-based diagnosis because the early detection of small nodules is often critical for improving patient outcomes. Multi-scale processing allows the model to preserve fine-grained details while simultaneously leveraging high-level semantic information extracted from deeper network layers.

The full process of multi-scale feature extraction is shown in Figure 3.7

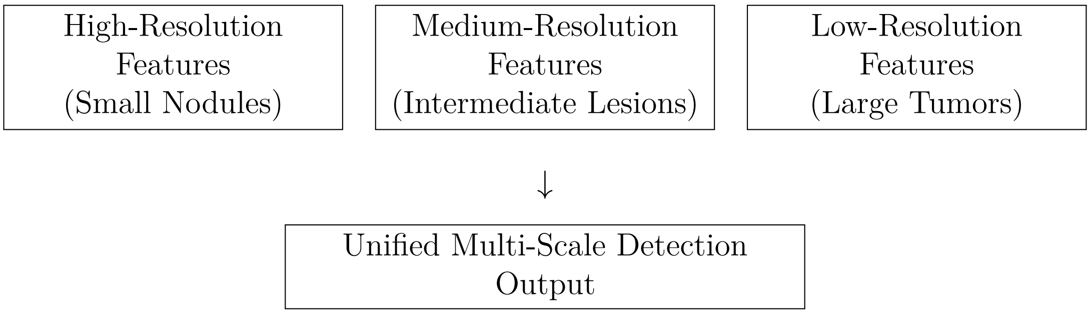



Figure 3.7: Multi-scale feature extraction employed by YOLOv8 for detecting lung lesions of varying sizes.

## Anchor-Free Detection Mechanism

Traditional object detection models rely on predefined anchor boxes that attempt to approximate the expected shapes and sizes of target objects. Although effective in many applications, anchor-based methods introduce additional hyperparameters and often struggle with irregularly shaped objects.

Lung tumors exhibit substantial variability in morphology and rarely conform to standardized geometric patterns. Consequently, anchor-based approaches may fail to adequately represent their boundaries.

YOLOv8 addresses this limitation by adopting an anchor-free detection strategy. Instead of matching objects to predefined anchor templates, the model directly predicts object centers and corresponding bounding box coordinates.

This design offers several advantages:

- Reduced hyperparameter complexity.
- Improved adaptability to irregular tumor shapes.
- Faster convergence during training.

- Enhanced localization performance.

These characteristics make anchor-free detection particularly suitable for medical imaging applications involving heterogeneous pathological structures.

## Bounding Box Regression and Localization

Once suspicious regions have been identified, YOLOv8 must determine the exact boundaries of each lesion.

Bounding box regression is performed by predicting:

- Horizontal position.
- Vertical position.
- Width.
- Height.

These predictions are continuously refined throughout training using specialized localization loss functions. The resulting bounding boxes provide interpretable visual evidence regarding the location of suspicious findings and can assist clinicians during diagnostic review.

## Non-Maximum Suppression

Object detectors frequently generate multiple overlapping predictions for the same lesion. To eliminate redundant detections, YOLOv8 employs Non-Maximum Suppression (NMS).

The process is illustrated in Figure 3.8.

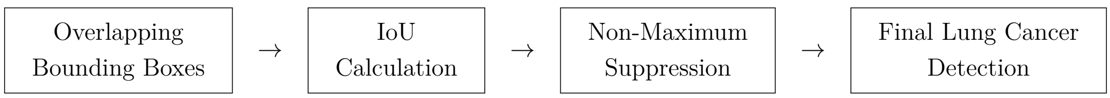



Figure 3.8: Non-Maximum Suppression procedure used to remove duplicate detections and retain the highest-confidence prediction.

The overlap between predicted and ground-truth boxes is measured using the Intersection over Union (IoU) metric:

$$IoU = \frac{Area(B_p \cap B_g)}{Area(B_p \cup B_g)}$$

where $B_p$ denotes the predicted bounding box and $B_g$ denotes the ground-truth bounding box.

Predictions with excessive overlap are suppressed, ensuring that each lesion is represented by a single final detection.

## Training Objective and Loss Functions

YOLOv8 optimizes a composite loss function composed of multiple components.

**Classification Loss (Binary Cross-Entropy)** Binary Cross-Entropy loss guides the model in distinguishing cancerous lesions from background regions. This loss encourages accurate classification while minimizing false positive detections.

**Box Localization Loss (CIoU)** Complete Intersection over Union (CIoU) loss evaluates localization quality by considering overlap area, center distance, and aspect ratio differences between predicted and ground-truth boxes.

This loss can be expressed conceptually as:

$$L_{CIoU} = 1 - IoU + \text{Distance Penalty} + \text{Aspect Ratio Penalty}$$

where additional penalty terms improve localization precision.

**Distribution Focal Loss (DFL)** Distribution Focal Loss improves boundary estimation by modeling uncertainty in bounding box coordinates. This is particularly important in medical images where lesion boundaries may be ambiguous or partially obscured.

Together, these loss functions allow YOLOv8 to simultaneously optimize classification accuracy and localization quality.

## Suitability of YOLOv8 for Lung Cancer Detection

Several characteristics make YOLOv8 particularly suitable for lung cancer detection.

First, its single-stage architecture provides rapid inference speeds while maintaining high detection accuracy. This allows the model to be integrated into real-time diagnostic support systems.

Second, multi-scale feature extraction enables the detection of lesions exhibiting substantial size variation, ranging from small pulmonary nodules to larger malignant masses.

Third, the anchor-free detection mechanism improves adaptability to irregular tumor morphologies commonly encountered in clinical practice.

Fourth, the generated bounding boxes provide intuitive visual explanations that can assist clinicians in understanding and verifying model predictions.

Finally, YOLOv8 benefits from transfer learning, allowing knowledge acquired from large-scale image datasets to be leveraged for medical imaging tasks. This capability is especially valuable when working with limited annotated medical datasets.

To evaluate the effectiveness of the proposed lung cancer detection model, Precision, Recall, and mAP50 are employed as the primary performance metrics. Precision quantifies the proportion of detected lesions that are truly cancerous, Recall measures the proportion of actual lesions successfully identified by the model, and mAP50 provides a comprehensive assessment of overall detection performance by evaluating both localization and classification accuracy at an Intersection over Union threshold of 0.50.

Collectively, these metrics provide a rigorous evaluation of the model's ability to accurately identify and localize lung cancer lesions within chest CT scans, ensuring that the resulting system can serve as a reliable component of the proposed intelligent diagnostic framework.

### 3.2.4 The CheXFound Vision Foundation Model and ViT Architecture

To address the limitations of centralized and domain-restricted architectures under severe data siloing, modern clinical pipelines are shifting toward Vision Foundation Models (VFMs). At the forefront of this shift is **CheXFound**, a self-supervised, vision-centric foundation model engineered explicitly for high-throughput, multi-class thoracic imaging diagnosis. CheXFound eliminates the historical constraint of training standalone networks for separate diagnostic indications by building generalized, highly robust chest representations. By utilizing a heavy Vision Transformer (ViT) architecture rather than a conventional convolutional backbone, CheXFound effectively captures long-range dependencies, contextual anatomical relationships, and fine-grained pathological variations across heterogeneous image streams.

#### The Vision Transformer (ViT-Large) Backbone Architecture

CheXFound is structurally built upon a modified **ViT-Large (ViT-L/16)** structural spine, optimizing representation capabilities at an expanded spatial resolution of $512 \times 512$ pixels. This configuration is mathematically designed to process image grids without losing subtle pathological signals, such as micro-nodular patterns or faint ground-glass borders, which often disappear when compressed into lower-resolution latent matrices.

The structural flow of the forward pass within CheXFound's ViT-L/16 architecture proceeds as follows:

1. **Patch Extraction and Flattening:** A continuous two-dimensional input thoracic image $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ (where $H = W = 512$ and $C = 1$ for radiographs or single-slice CTs) is systematically split into a grid of non-overlapping, uniform rectangular patches $\mathbf{X}_p \in \mathbb{R}^{N\times(P^2\cdot C)}$. Here, $P = 16$ denotes the hardcoded spatial patch size, and $N$ represents the resulting total number of sequence patches, computed mathematically as:

$$N = \frac{H \cdot W}{P^2} = \frac{512 \times 512}{16 \times 16} = 1024 \text{ patches} \tag{3.11}$$

2. **Linear Patch Embedding:** Each flattened structural patch token is projected into a lower-dimensional latent embedding space using a trainable linear mapping matrix $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}$, where $D = 1024$ defines the dense hidden embedding dimension of the ViT-Large backbone.

3. **CLS Token Prepended and Position Encodings:** A learnable classification token embedding $\mathbf{z}_0^0 = \mathbf{x}_{\text{class}} \in \mathbb{R}^D$ is prepended to the sequence of patch tokens. To preserve the two-dimensional spatial topography of the chest inside a one-dimensional sequence pipeline, standard learnable one-dimensional position encodings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times D}$ are element-wise added to the patch embeddings, yielding the initial input tensor $\mathbf{z}_0$:
$$\mathbf{z}_0 = \left[\mathbf{x}_{\text{class}};\ \mathbf{X}_p^1\mathbf{E};\ \mathbf{X}_p^2\mathbf{E};\ \ldots;\ \mathbf{X}_p^N\mathbf{E}\right] + \mathbf{E}_{\text{pos}} \tag{3.12}$$

4. **Transformer Encoder Blocks:** The combined sequence $\mathbf{z}_0$ passes sequentially through $L = 24$ identical Transformer Encoder blocks. Each block features a Multi-Head Self-Attention (MHSA) mechanism with $h = 16$ distinct attention heads, interleaved with Multi-Layer Perceptron (MLP) blocks and scaled by Layer Normalization (LN) using residual skip-connections:
$$\mathbf{z}'_\ell = \text{MHSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1, \ldots, 24 \tag{3.13}$$
$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell \tag{3.14}$$

The standard ViT core configuration embedded inside CheXFound is mathematically mapped and detailed in Table 3.2.

Table 3.2: Structural and mathematical parameter configurations of the CheXFound ViT-Large/16 backbone layout.

| **Architectural Hyperparameter** | **Formal Definition / Notation** | **Deterministic Global Value** |
|---|---|---|
| **Input Resolution** | $H \times W \times C$ | $512 \times 512 \times 1$ Pixels |
| **Patch Dimension** | $P \times P$ | $16 \times 16$ Pixels |
| **Sequence Count** | $N = (H \cdot W)/P^2$ | 1024 Active Patch Tokens |
| **Hidden Layer Dimension** | $D$ | 1024 Scaled Scaling Dimensions |
| **Transformer Layers** | $L$ | 24 Cascaded Encoder Blocks |
| **Attention Heads** | $h$ | 16 Independent Heads |
| **MLP Expansion Ratio** | $\text{Ratio}_{\text{MLP}}$ | $4.0 \times D$ (4096 Hidden Units) |

## Pretraining Paradigm and Self-Supervised Objective Functions

CheXFound is pretrained on a curated, large-scale medical dataset comprising approximately 987,000 unique chest radiographs derived from 12 distinct open-source clinical databases (including MIMIC-CXR, CheXpert, PadChest, and BraX). To extract optimal task-agnostic representations without relying on manual labels, CheXFound utilizes the **DINOv2** self-supervised learning pipeline. This setup combines a student network (parameterized by weights $\theta_s$) and a teacher network (parameterized by weights $\theta_t$). The teacher's weights are updated continuously via an Exponential Moving Average (EMA) of the student's weights:

$$\theta_t \leftarrow \lambda\theta_t + (1-\lambda)\theta_s \tag{3.15}$$

The optimization function combines a multi-crop global-to-local cross-entropy loss with an explicit Masked Image Modeling (MIM) autoencoder objective function. Given a global view image input $x_g$ and a local masked crop input $x_l$, the multi-crop cross-entropy loss enforces representational consistency across scales:

$$\mathcal{L}_{\text{global}} = \sum_{x\in\{x_g,x_l\}} \sum_{i\neq j} H\left(P_t(x^{(i)}), P_s(x^{(j)})\right) \tag{3.16}$$

where $H(p,q) = -p\log q$, and $P_t, P_s$ denote the soft probability outputs generated by passing the $\mathbf{x}_{\text{class}}$ token through a centering and sharpening bottleneck.

Concurrently, the MIM objective function randomly masks out a subset of the patch sequence tokens (typically 50% bounding ratios) before feeding them into the student network. The model then computes a localized categorical cross-entropy loss at the patch-token level against the unmasked teacher representations, training the model to predict contextual structure:

$$\mathcal{L}_{\text{MIM}} = -\sum_{k\in\mathcal{M}} P_t(\mathbf{z}_k)\log P_s(\mathbf{z}_k) \tag{3.17}$$

where $\mathcal{M}$ signifies the set of index fields tracking the masked patches. The unified self-supervised optimization equation for the foundation layer is structured as:

$$\mathcal{L}_{\text{CheXFound}} = \alpha\mathcal{L}_{\text{global}} + \beta\mathcal{L}_{\text{MIM}} \tag{3.18}$$

## Global and Local Representations Integration (GLoRI) Downstream Adaptation Head

Standard foundation models often lose small-scale, localized pathological indicators—such as early-stage lung cancer nodules or subtle tuberculous cavities—during global attention pooling over the classification token ($\mathbf{x}_{\text{class}}$). To prevent this loss, CheXFound introduces a specialized downstream adaptation module called the **Global and Local Representations Integration (GLoRI)** head.

Instead of relying solely on the final layer's $\mathbf{x}_{\text{class}}$ token, the GLoRI module processes both coarse-grained global features and fine-grained, localized patch tokens. During downstream fine-tuning, the backbone CheXFound parameters remain completely frozen to preserve generalizability, while the GLoRI head operates as a trainable multi-head cross-attention layer.

The mathematical routing within the GLoRI module follows this pipeline:

1. The global representation vector $\mathbf{f}_g \in \mathbb{R}^D$ is isolated from the class token $\mathbf{x}_{\text{class}}$.

2. The array of patch tokens $\mathbf{F}_p = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_1 024] \in \mathbb{R}^{1024 \times D}$ is mapped to extract high-resolution localized profiles.

3. A specific cross-attention layer treats the global feature vector as the query vector ($\mathbf{Q}$), while mapping the patch tokens to generate the keys ($\mathbf{K}$) and values ($\mathbf{V}$):

$$\mathbf{Q} = \mathbf{f}_g \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{F}_p \mathbf{W}_K, \quad \mathbf{V} = \mathbf{F}_p \mathbf{W}_V \tag{3.19}$$

   where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ represent trainable projection matrices with 8 attention heads and a multi-head channel layout of 786 units.

4. The final contextual integrated vector $\mathbf{f}_{\text{GLoRI}}$ is computed via scaled dot-product attention, blending global context with localized pathology indicators:

$$\mathbf{f}_{\text{GLoRI}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} + \mathbf{f}_g \tag{3.20}$$

Figure 3.9 illustrates the workflow of the input transformation, patching layer, ViT matrix encoding, and the GLoRI feature blending architecture within the system.

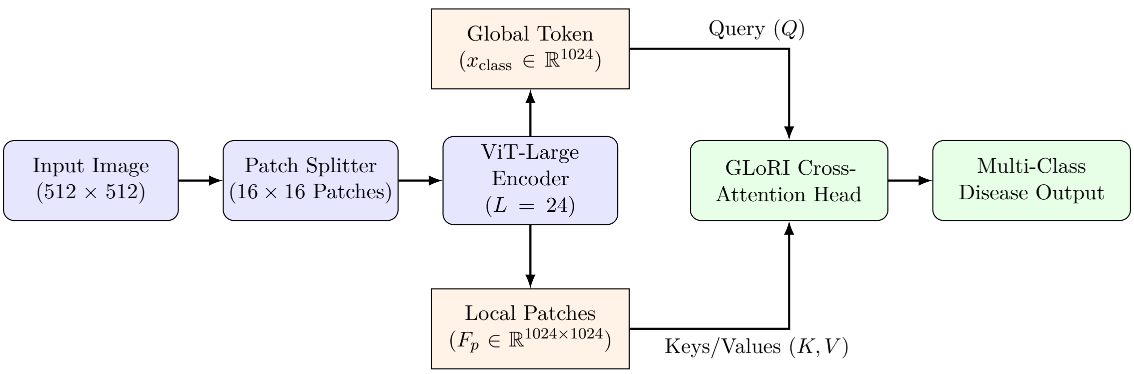



Figure 3.9: Functional computational pipeline of the CheXFound framework: Transitioning from input image patch serialization, through the ViT-Large encoder blocks, to the GLoRI cross-attention alignment module.

Furthermore, Figure 3.10 serves as a modular architectural block diagram tracking the data flow from structural raw DICOM files to final predictions.

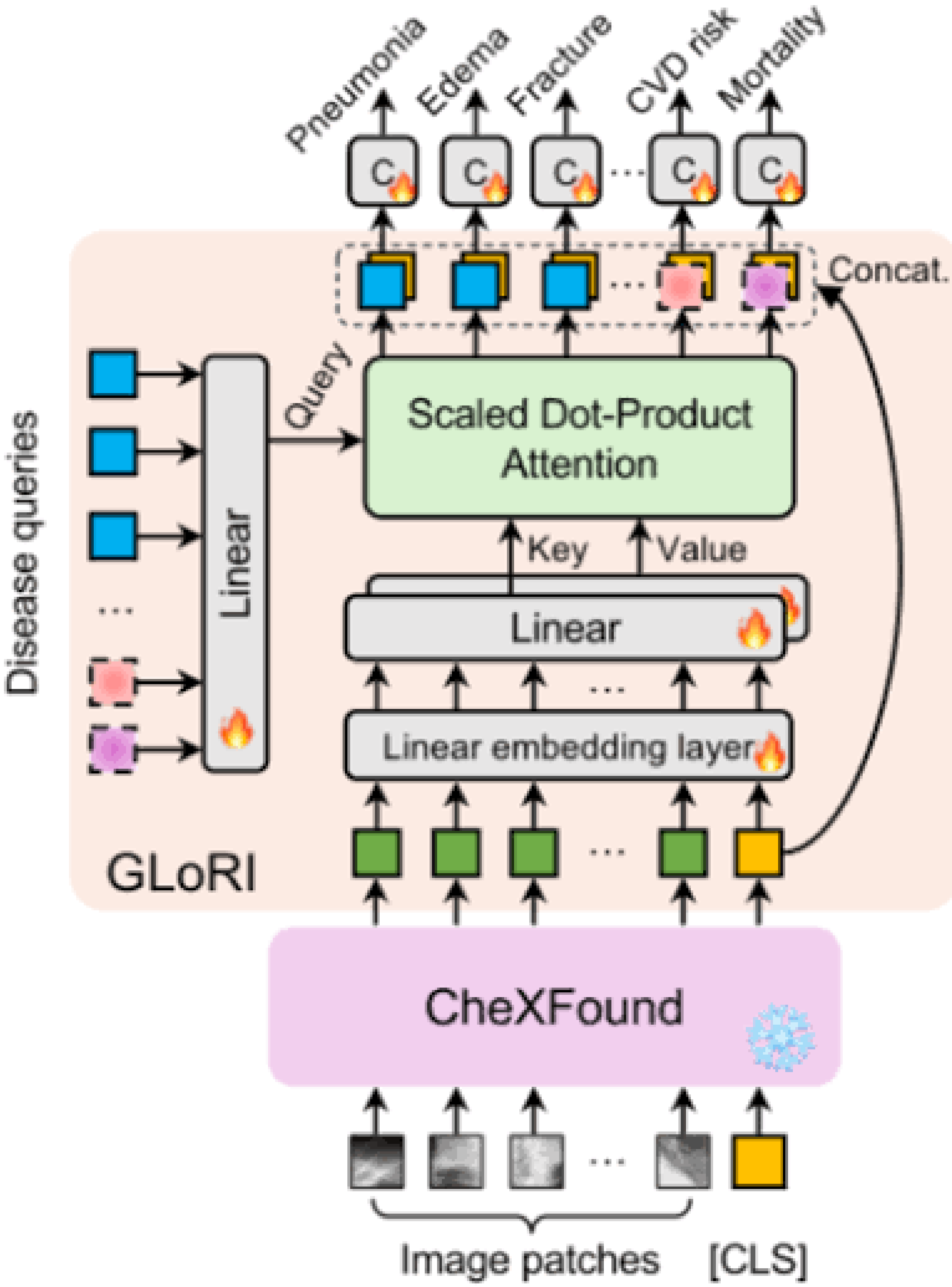



Figure 3.10: Modular schematic layout detailing parameter extraction, input token construction, and feature routing pathways between the frozen base model and the target task head.

## 3.3 Medical Image Routing Model

This section presents the proposed medical image routing model employed as the first level of the hierarchical diagnostic framework and discusses the motivation behind its development, the formal problem formulation, and its architectural design. The routing model serves as the entry point of the proposed system, where the modality of each incoming image is identified and the image is subsequently forwarded to the most appropriate downstream diagnostic component. By introducing an explicit routing stage, the framework ensures that specialized diagnostic models receive only the image types for which they were designed, thereby improving overall system reliability, robustness, and scalability.

### 3.3.1 Motivation

The increasing adoption of deep learning within medical image analysis has enabled the development of highly specialized models for disease detection, diagnosis, and clinical decision support. However, most medical imaging models are designed for specific imaging modalities and diagnostic tasks. Consequently, their performance depends heavily on receiving inputs that match the data distribution encountered during training. Applying a modality-specific model to an inappropriate image type can significantly degrade performance and may result in unreliable predictions.

In the context of the proposed framework, two distinct diagnostic pathways are employed. The first pathway focuses on chest disease classification from chest X-ray images and is responsible for identifying the presence of COVID-19, Tuberculosis, Pneumonia, or the absence of disease. The second pathway focuses on lung cancer detection from chest CT scans using a dedicated object detection model. Since these diagnostic models operate on fundamentally different imaging modalities and address different clinical objectives, a mechanism is required to automatically determine which model should process each incoming image.

Real-world deployment scenarios further emphasize the importance of such a mechanism. Medical information systems may receive images originating from various imaging modalities, anatomical regions, acquisition devices, and clinical workflows. In addition, unsupported images may be inadvertently submitted to the system. Without a dedicated routing stage, these images would be forced through one of the diagnostic models regardless of their suitability, increasing the likelihood of incorrect predictions and reducing overall system reliability.

To address this challenge, a dedicated medical image routing model is introduced as the first stage of the proposed architecture. The model distinguishes between chest X-ray images, chest CT scan images, and an additional out-of-distribution category referred to as *Other*. By incorporating this third category, the framework explicitly recognizes unsupported images and prevents them from being processed by downstream diagnostic models. Images identified as chest X-rays are forwarded to the multi-label chest disease classification model, while CT scans are forwarded to the lung cancer detection model. Images classified as *Other* are rejected from further analysis. This design not only improves robustness and safety but also provides a modular foundation upon which additional imaging modalities and diagnostic models may be incorporated in future system extensions.

### 3.3.2 Problem Statement

The objective of the routing model is to automatically determine the modality of an incoming medical image and direct it toward the appropriate downstream diagnostic pathway. This task is formulated as a three-class image classification problem in which each input image must be assigned to one of three predefined categories.

Let the input image be represented by

$$X \in \mathbb{R}^{224\times 224\times 3},$$

where the image is resized to a fixed spatial resolution of $224 \times 224$ pixels and represented using three color channels. Given an input image $X$, the routing model predicts one of the following classes:

- Chest X-ray (0)
- Chest CT Scan (1)
- Other / Out-of-Distribution (2)

The model produces a probability distribution

$$\mathbf{p} = [p_0, p_1, p_2],$$

where

$$p_k = P(y = k \mid X)$$

and

$$\sum_{k=0}^{2} p_k = 1.$$

The predicted class label is obtained using

$$\hat{y} = \arg\max_k p_k.$$

Images classified as chest X-rays are forwarded to the multi-label chest disease classification model, while images classified as CT scans are directed to the lung cancer detection model. The out-of-distribution category serves as an additional safety mechanism that prevents unsupported images from being processed by downstream diagnostic components.

By incorporating this category, the routing model improves the reliability and robustness of the overall framework and establishes a structured entry point for all incoming medical images.

The routing model therefore functions as a modality recognition system whose purpose extends beyond simple image classification. Its predictions directly determine the diagnostic pathway followed by each image and consequently influence all subsequent processing stages within the proposed architecture.

### 3.3.3 Model Architecture

The proposed medical image routing model is implemented using a transfer learning approach based on the DenseNet121 convolutional neural network architecture. The model constitutes the first level of the proposed hierarchical framework and is responsible for determining the appropriate diagnostic pathway for each incoming image. By accurately identifying image modalities and separating unsupported inputs from valid medical scans, the routing model ensures that downstream diagnostic models receive only the image types for which they were designed.

Several candidate architectures were investigated during development, including a custom convolutional neural network, ResNet50, DenseNet121, and EfficientNetB3. Although all evaluated models demonstrated the ability to learn modality-specific representations, DenseNet121 consistently provided the most favorable balance between classification performance, feature extraction capability, and computational efficiency. The dense connectivity mechanism employed by DenseNet121 promotes feature reuse throughout the network and improves gradient propagation, allowing the model to learn highly discriminative features while maintaining a relatively compact parameter count. Consequently, DenseNet121 was selected as the backbone architecture for the final routing model.

The architecture follows a structured pipeline consisting of image preprocessing, deep feature extraction using DenseNet121, feature aggregation through global average pooling, and final classification using fully connected layers. A high-level overview of the proposed architecture is illustrated in Figure 3.11.

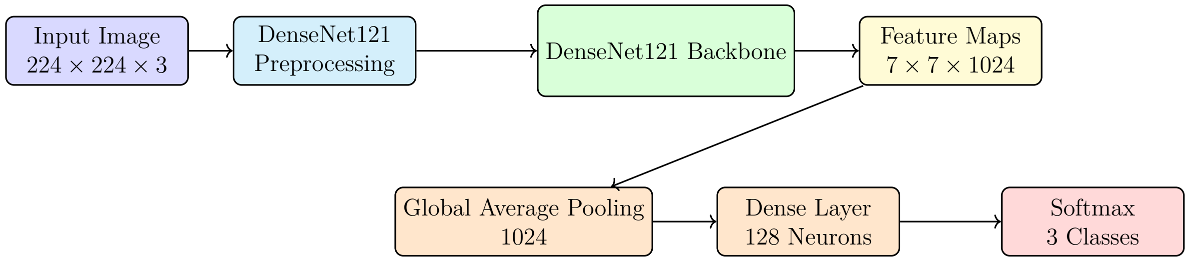



Figure 3.11: Two-row architecture of the DenseNet121-based medical image routing model.

The input to the model is a medical image represented by a tensor of dimensions $224 \times 224 \times 3$. Prior to feature extraction, the image undergoes the preprocessing procedure associated with DenseNet121. This preprocessing transforms the image into a format compatible with the ImageNet-pretrained weights used by the backbone network. By aligning the input distribution with the statistical properties encountered during pretraining, the model can effectively leverage the learned representations obtained from large-scale natural image datasets.

Following preprocessing, the image is passed through the DenseNet121 backbone with its original classification layers removed. DenseNet121 belongs to the family of densely connected convolutional neural networks, in which each layer receives feature maps from all preceding layers within a dense block. This dense connectivity pattern encourages feature reuse, facilitates gradient propagation during training, and improves parameter efficiency. As a result, DenseNet121 is capable of learning rich hierarchical representations while requiring fewer parameters than many conventional deep convolutional architectures.

For an input image of dimensions $(224, 224, 3)$, the DenseNet121 backbone produces a high-level feature tensor of dimensions $(7, 7, 1024)$. These feature maps contain abstract representations of visual patterns extracted from the image, including anatomical structures, texture information, intensity distributions, and modality-specific characteristics. Because information from earlier layers remains accessible throughout the network, DenseNet121 can effectively combine low-level and high-level features when constructing its final representation.

To transform these spatial feature maps into a compact feature representation suitable for classification, a Global Average Pooling (GAP) layer is applied. Unlike flattening operations that convert all feature map values into a large vector, GAP computes the average activation of each feature channel across all spatial locations. Given a feature tensor

$$F \in \mathbb{R}^{7\times 7\times 1024},$$

the pooled feature vector

$$z \in \mathbb{R}^{1024}$$

is computed as

$$z_k = \frac{1}{7 \times 7} \sum_{i=1}^{7} \sum_{j=1}^{7} F_{i,j,k},$$

where $z_k$ represents the average activation of the $k$-th feature channel. This operation reduces the dimensionality of the representation while preserving the most important information learned by the backbone network. Furthermore, it significantly reduces the number of trainable parameters in the classification head and helps improve model generalization.

The resulting feature vector is subsequently processed by a fully connected layer consisting of 128 neurons with ReLU activation. This layer learns task-specific combinations of the extracted features and serves as an intermediate representation between the pretrained backbone and the final classification layer.

The final layer consists of three output neurons followed by a softmax activation function. This layer produces a probability distribution

$$\mathbf{p} = [p_{XRay}, p_{CT}, p_{Other}],$$

where each component corresponds to the predicted probability that the input image belongs to the chest X-ray, chest CT scan, or out-of-distribution category, respectively. The final prediction is obtained according to

$$\hat{y} = \arg\max_k p_k.$$

Images classified as chest X-rays are forwarded to the multi-label chest disease classification model, images classified as CT scans are directed to the lung cancer detection model, and images assigned to the *Other* category are rejected from further analysis. Consequently, the routing model serves as the gateway to the entire diagnostic framework and plays a critical role in ensuring that each image is processed by the most appropriate downstream component.

The successful operation of this routing stage is fundamental to the effectiveness of the proposed hierarchical architecture. Since all subsequent diagnostic decisions depend on accurate modality recognition, errors introduced at this stage may propagate throughout the remainder of the system. Therefore, the routing model acts not only as a classifier but also as a reliability and safety mechanism that enhances the robustness of the overall framework. The following sections describe the specialized diagnostic models that receive images from the routing component and perform disease-specific analysis.

### 3.3.4 Implementation of the First-Level Routing Model

The first component implemented within the proposed framework is the modality routing model. As discussed in Chapter **??**, this model serves as the entry point of the entire classification pipeline and is responsible for determining the type of the incoming image before directing it toward the appropriate diagnostic model. Since the subsequent classification stages were specifically designed for particular imaging modalities, ensuring the accuracy of the routing stage was considered a critical requirement for the reliability of the overall framework.

The routing model was formulated as a three-class image classification problem in which each input image was assigned to one of three categories: chest X-ray images, chest CT scans, or out-of-distribution (OOD) images. The OOD category was intentionally incorporated to improve the robustness of the system during deployment. In practical healthcare environments, medical information systems may contain a wide variety of image types that fall outside the scope of the proposed framework. Without an explicit OOD category, unsupported images could be mistakenly forwarded to disease classification models, potentially resulting in unreliable predictions. The inclusion of this category therefore acts as a safety mechanism that prevents unsupported inputs from entering the diagnostic pipeline.

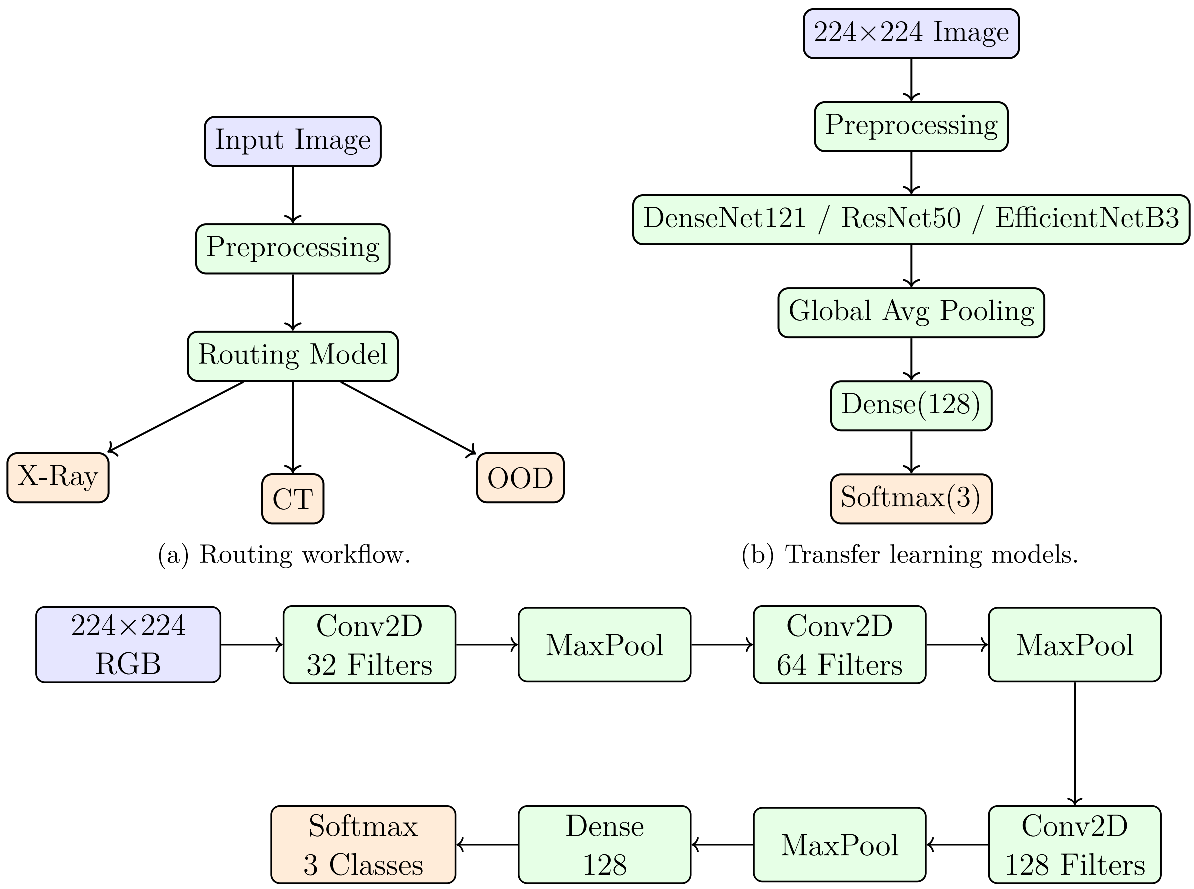
The final routing dataset was constructed by aggregating images originating from multiple sources. After dataset integration, the X-ray category contained 7,123 images, the CT category contained 2,534 images, and the OOD category contained 9,958 images. Although this aggregation process provided a sufficiently diverse collection of samples, it also introduced a noticeable class imbalance problem. The CT class represented a considerably smaller portion of the dataset compared to the other categories, creating a risk that the trained model would become biased toward the more frequently represented classes.

To mitigate this issue, several balancing strategies were incorporated into the training pipeline. Data augmentation techniques were applied to increase the effective diversity of minority-class samples while simultaneously improving model generalization. Augmentation operations included random rotations, horizontal flipping, zoom transformations, and intensity variations. These transformations were selected because they preserve medically relevant structures while exposing the model to realistic image variations that may occur across different acquisition settings. In addition to augmentation, class balancing procedures were employed during dataset preparation to reduce the influence of class distribution disparities on the learning process. These measures contributed to a more stable training process and improved the model's ability to recognize underrepresented classes.

The complete routing workflow implemented within the framework is illustrated in Figure 3.12. The figure summarizes both the operational behavior of the routing model and the architectures investigated during the model selection process. Rather than committing immediately to a single architecture, several alternative models were implemented and evaluated in order to identify the most suitable solution for the routing task. This decision was motivated by the observation that image modality classification differs substantially from disease classification. While disease detection often requires learning subtle pathological patterns, modality recognition primarily depends on identifying broader structural and acquisition-specific characteristics. Consequently, it was necessary to experimentally determine whether a lightweight custom architecture would be sufficient or whether a deeper pretrained network would provide superior performance.

The first architecture investigated was a custom convolutional neural network developed specifically for the routing task. The model begins with a rescaling layer that normalizes pixel intensities into the range $[0, 1]$. Feature extraction is then performed through three convolutional blocks containing 32, 64, and 128 filters respectively. Each convolutional layer uses a ReLU activation function and is followed by a max-pooling operation that progressively reduces the spatial dimensions of the feature maps while preserving the most salient visual information. After the final convolutional block, the extracted features are flattened and passed through a fully connected layer consisting of 128 neurons. The output layer employs a softmax activation function to generate probability estimates for the three target classes.

Although the custom CNN provided a relatively lightweight solution with low computational requirements, the project also explored the use of transfer learning techniques.




(a) Routing workflow.

(b) Transfer learning models.

(c) Custom CNN architecture used as a baseline model for modality routing.

Figure 3.12: Architectures and workflow investigated during the implementation of the routing model. Blue blocks represent inputs, green blocks denote processing stages, and orange blocks correspond to classification outputs.

Transfer learning has become a widely adopted strategy in medical image analysis because it enables models to leverage feature representations learned from large-scale image datasets. Instead of training a deep architecture entirely from scratch, a pretrained backbone can be used as a feature extractor while only a small classification head is trained on the target dataset.

Three state-of-the-art architectures were evaluated during experimentation: ResNet50, DenseNet121, and EfficientNetB3. For each architecture, the pretrained backbone was initialized using publicly available weights and subsequently frozen during training. Freezing the backbone prevented updates to the pretrained parameters and allowed the network to function as a fixed feature extractor. A Global Average Pooling layer was then applied to aggregate spatial information into a compact feature vector, followed by a fully connected layer containing 128 neurons and a final softmax classification layer producing probabilities for the three routing classes.

The implementation of transfer learning significantly reduced training complexity while benefiting from the rich visual representations learned by these networks during large-scale pretraining. However, preliminary experiments revealed that different architectures exhibited varying levels of sensitivity to optimization parameters. Consequently, additional effort was dedicated to hyperparameter tuning in order to maximize classification performance.

One of the most important optimization parameters investigated during experimentation was the learning rate. The learning rate directly controls the magnitude of weight updates during optimization and can substantially influence convergence behavior. A learning rate that is too large may cause unstable training, whereas an excessively small learning rate may prevent efficient convergence. Since each pretrained architecture possesses unique characteristics and parameter distributions, a single learning rate was not expected to perform optimally across all models.

To address this issue, architecture-specific learning rates were introduced. ResNet50 and DenseNet121 were trained using a learning rate of $10^{-2}$, while EfficientNetB3 was trained using a learning rate of $10^{-4}$. These values were selected based on empirical experimentation and repeated evaluation of training behavior. The custom CNN utilized the default Adam optimization configuration due to its comparatively smaller parameter space. This targeted hyperparameter tuning strategy improved convergence stability and ensured that each architecture was evaluated under favorable training conditions.

All routing models were trained using the Adam optimization algorithm and sparse categorical cross-entropy loss. The use of sparse categorical cross-entropy was appropriate because the routing task involves mutually exclusive classes, where each image belongs to exactly one category. Model performance was monitored using classification accuracy, which provided a direct measure of the model's ability to correctly identify image modalities.

To ensure a fair and consistent evaluation process, all experiments were conducted using identical dataset partitions. The aggregated dataset was divided into training, validation, and testing subsets according to a ratio of 70%, 15%, and 15%, respectively. Training was performed using a batch size of 8 and a total of 10 epochs. Maintaining a consistent experimental protocol allowed the performance differences observed between models to be attributed primarily to architectural characteristics rather than variations in the training procedure.

Following extensive experimentation, DenseNet121 was selected as the final routing model incorporated into the proposed framework. The architecture demonstrated the most favorable balance between classification performance, convergence behavior, and generalization capability. DenseNet121's dense connectivity pattern enables efficient feature reuse and improved gradient propagation, characteristics that proved particularly beneficial for distinguishing between imaging modalities exhibiting substantial structural differences. Furthermore, the architecture achieved strong performance while maintaining a relatively efficient parameter utilization strategy compared to alternative deep learning models.

The final routing model therefore consists of a DenseNet121 feature extraction backbone followed by a Global Average Pooling layer, a fully connected layer with 128 neurons, and a three-class softmax output layer. Once deployed within the complete framework, the model receives an input image, determines whether the image corresponds to a chest X-ray, a CT scan, or an out-of-distribution sample, and subsequently forwards the image to the appropriate downstream component. This routing decision constitutes the first stage of the proposed hierarchical classification pipeline and forms the foundation upon which all subsequent diagnostic analyses are performed.

## 3.4 Chest Disease Detection Model

Chest diseases remain among the leading causes of morbidity and mortality worldwide, placing a significant burden on healthcare systems and emphasizing the need for accurate and timely diagnosis. Medical imaging, particularly chest X-ray imaging, plays a vital role in the detection and assessment of respiratory diseases such as COVID-19, Tuberculosis, and Pneumonia. However, manual interpretation of chest radiographs can be time-consuming and may be subject to variability among radiologists, especially in high-demand clinical environments.

Recent advances in Artificial Intelligence (AI) and Deep Learning have demonstrated remarkable potential in assisting medical professionals by automatically analyzing medical images and identifying disease-related patterns. In this context, this project presents a Chest Disease Detection Model designed to classify chest X-ray images and detect the presence of three major respiratory diseases: COVID-19, Tuberculosis, and Pneumonia.

Unlike traditional multi-class classification approaches, these diseases are not mutually exclusive and may coexist within the same patient. Therefore, the proposed system is formulated as a multi-label classification problem, allowing the model to predict one or more diseases simultaneously from a single chest X-ray image. This approach better reflects real-world clinical scenarios and enhances the model's applicability in practical healthcare settings.

The primary objective of this model is to provide an intelligent and reliable diagnostic support tool that can assist healthcare professionals in the early detection and screening of chest diseases. By leveraging deep learning techniques and medical imaging data, the system aims to improve diagnostic efficiency, support clinical decision-making, and contribute to better patient outcomes.

### 3.4.1 Motivation

Respiratory diseases represent one of the most formidable challenges confronting modern healthcare systems worldwide. Conditions such as COVID-19, Pneumonia, and Tuberculosis (TB) collectively account for millions of deaths and hospitalizations annually, placing enormous strain on healthcare infrastructure, particularly in low- and middle-income countries where access to experienced radiologists and advanced diagnostic facilities remains severely limited. The timely and accurate diagnosis of these conditions is critical, since delayed or erroneous diagnoses can result in inappropriate treatment, accelerated disease progression, and preventable mortality.

Chest X-ray (CXR) imaging remains the cornerstone of respiratory disease diagnosis owing to its widespread availability, low cost, and rapid turnaround time. However, the visual interpretation of chest radiographs is a highly complex, skill-intensive task that demands years of specialized training. Even seasoned radiologists can face considerable difficulty when attempting to differentiate between visually similar pathologies, such as COVID-19 pneumonitis and bacterial pneumonia, or when interpreting images of suboptimal quality obtained in resource-constrained settings. Furthermore, the global shortage of trained radiologists, particularly during large-scale health crises such as the COVID-19 pandemic, has created an unprecedented demand for automated, reliable, and scalable diagnostic support tools.

Recent advances in deep learning, and in particular convolutional neural networks (CNNs), have demonstrated transformative potential in the field of medical image analysis. Architectures such as DenseNet, ResNet, and EfficientNet have achieved or surpassed radiologist-level performance on several benchmark medical imaging tasks. These developments motivate a comprehensive investigation into the deployment of deep learning frameworks for the simultaneous, multi-class differentiation of the most clinically prevalent chest diseases.

Beyond the challenge of multi-class classification itself, a further compelling motivation exists for the application of **Federated Learning (FL)** within this domain. Medical data is inherently sensitive and subject to stringent privacy regulations, including HIPAA and GDPR, which impose significant restrictions on the centralized aggregation of patient-level imaging data across different hospitals or jurisdictions. Federated Learning offers an elegant solution to this dilemma by enabling collaborative model training across multiple distributed nodes — each representing a hospital or imaging center — without ever requiring raw patient data to leave its local environment.

Finally, the integration of **Explainable Artificial Intelligence (XAI)** techniques, specifically Gradient-weighted Class Activation Mapping (Grad-CAM), addresses the critical concern of model interpretability in clinical settings. A "black-box" deep learning model, regardless of its accuracy, is unlikely to gain the trust of clinical practitioners unless its decision-making process can be visualized and verified. By producing saliency maps that highlight the anatomical regions most influential to the model's predictions, Grad-CAM bridges the gap between algorithmic performance and clinical trustworthiness.

Taking together, these motivations drive the design of a comprehensive, privacy-preserving, and interpretable deep learning pipeline for the automated classification of chest diseases, which constitutes the core contribution of this project.

### 3.4.2 Problem Statement

The central challenge addressed in this work is the development of a robust, generalizable, and clinically trustworthy deep learning system capable of accurately distinguishing between four clinically significant chest conditions from standard posterior-anterior (PA) or AP chest radiographs. The four target classes are:

- **COVID-19:** A viral pneumonia caused by SARS-CoV-2, characterized on CXR by bilateral, peripheral ground-glass opacities and consolidations, predominantly in the lower lobes.
- **Pneumonia:** : Bacterial or viral pneumonia presenting as airspace consolidation, often lobar or segmental, with air bronchograms and pleural effusion in more severe cases.
- **Tuberculosis (TB):** A chronic bacterial infection by Mycobacterium tuberculosis, displaying a wide spectrum of CXR patterns including upper-lobe infiltrates, cavities, fibrosis, hilar lymphadenopathy, and miliary nodules.
- **Normal:** Healthy subjects with no discernible pulmonary pathology on the radiograph.

The specific problems this project aims to solve are enumerated below:

1. **Multi-class classification under class imbalance:** The dataset exhibits natural imbalances between classes, which must be addressed through appropriate class-weighting and augmentation strategies to prevent model bias toward over-represented classes.

2. **Differentiation of visually similar pathologies:** COVID-19 and Pneumonia share substantial radiological overlap, making their discrimination a particularly challenging sub-problem. The system must learn discriminative, class-specific features rather than coarse, general patterns.

3. **Lung region isolation:** The presence of non-pulmonary structures (ribs, heart shadow, diaphragm, spine, clavicles) in raw CXR images can mislead the classifier toward irrelevant features. A dedicated lung segmentation stage using U-Net is therefore proposed to restrict the model's attention to pulmonary anatomy.

4. **Privacy-preserving distributed training:** In a realistic multi-hospital deployment scenario, patient data cannot be pooled centrally. The system must achieve competitive performance under a federated training paradigm with 2, 3, or 4 participating clients, each holding only a partition of the global dataset.

5. **Model interpretability:** The system must be able to generate visual explanations for each prediction, enabling clinicians to verify that the model is attending to diagnostically meaningful regions of the radiograph.

Formally, the classification task can be stated as follows: given an input chest radiograph $x \in \mathbb{R}^{H\times W\times C}$ and a label space $\mathcal{Y} = \{\text{COVID-19}, \text{Normal}, \text{Pneumonia}, \text{Tuberculosis}\}$, the objective is to learn a mapping

$$f : x \mapsto \hat{y}$$

that minimizes the expected classification error

$$\mathbb{E}\left[\mathcal{L}(f(x), y)\right]$$

over the joint distribution of images and labels, whilst satisfying the constraints of data privacy and clinical interpretability.

### 3.4.3 Model Architecture Overview

The proposed system is a multi-stage deep learning pipeline designed to address each of the challenges outlined in Section 3.4.2. The pipeline is structured into four principal stages, each of which transforms the input data progressively toward the final classification decision. Figure 3.13 provides a high-level architectural overview of the complete system.

The four stages of the pipeline are as follows:

**Step 1: Image Preprocessing** All input radiographs undergo a standardized preprocessing pipeline comprising contrast enhancement via Contrast Limited Adaptive Histogram Equalization (CLAHE), Gaussian smoothing for noise suppression, and min-max intensity normalization. This stage ensures that images drawn from heterogeneous acquisition devices and clinical environments are brought to a consistent, model-friendly representation. The preprocessing procedure is described in full detail in Section 3.4.4.

**Stage 2: Lung Segmentation** A U-Net-based lung segmentation model is trained on a dedicated lung mask dataset (Darwin, Montgomery, Shenzhen combined) to generate binary lung masks from preprocessed radiographs. These masks are subsequently applied to the classification images to suppress all non-pulmonary anatomical structures, ensuring that the downstream classifier focuses exclusively on lung parenchyma. The segmentation model architecture and training details are provided in Section 3.4.5.

**Classification Model** The segmented lung images are fed into a transfer-learned deep CNN. Multiple backbone architectures are evaluated, including DenseNet121, ResNet50/ResNet50V2,

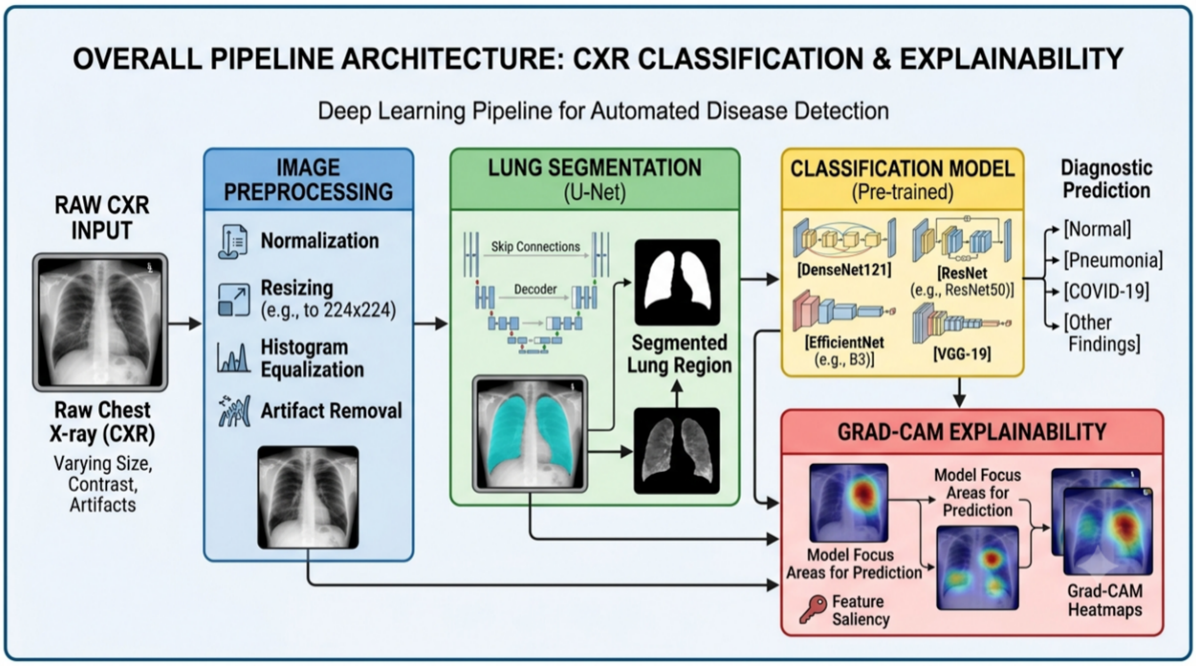



Figure 3.13: Overall architecture of the proposed chest disease multi-class classification system, illustrating the sequential pipeline from raw CXR acquisition through preprocessing, lung segmentation, deep classification, and visual explainability.

EfficientNet, and VGG-19. Two classification paradigms are explored: (i) a standard single-label softmax approach, and (ii) a One-vs-Rest (OVR) multi-label sigmoid approach, which treats each class as an independent binary classification problem. Additionally, the entire classification pipeline is adapted for Federated Learning using the Flower (flwr) framework. Full architectural details of the classification models are provided in Section **??** and the experimental configuration in Section **??**.

**Stage 4: Explainability via Grad-CAM** To ensure clinical transparency, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied post-hoc to the trained classification models. This produces class-discriminative saliency heatmaps that are superimposed on the original CXR, enabling radiologists and clinical users to verify the anatomical basis of each prediction. Grad-CAM integration details are provided in Section **??**.

### 3.4.4 Dataset Description

The classification system is trained, validated, and evaluated on a carefully curated composite dataset totaling 17,637 **chest X-ray images** spanning four distinct diagnostic classes. This dataset is assembled from two complementary publicly available sources, described in detail below. A second, separate dataset is employed exclusively for training the lung segmentation model.

### Dataset 1: COVID-19 Radiography Database

The **COVID-19 Radiography Database** is a landmark open-access collection assembled through a collaborative effort between researchers at Qatar University, the University of Dhaka, and partners in Pakistan and Malaysia, in cooperation with medical physicians. It was released on the Kaggle platform and has since become one of the most widely adopted benchmarks for COVID-19 CXR classification research.

The database contains chest radiographs in Portable Network Graphics (PNG) format at a resolution of 299×299 pixels, though images are resampled to 224×224 pixels during preprocessing. Each image is accompanied by a corresponding lung opacity mask, though these masks are not directly used in the classification training pipeline beyond informing the lung segmentation stage.

The COVID-19 subset of this database was compiled from multiple certified hospital sources and was expanded iteratively as more confirmed COVID-19 cases became publicly available. The radiographs represent a heterogeneous spectrum of COVID-19 severity, ranging from mild bilateral ground-glass opacities in early infection to extensive bilateral consolidation in severe and critical cases. Key radiological features visible in this subset include:

- Bilateral, predominantly peripheral ground-glass opacities (GGO), often with a lower-zone preponderance.
- Multifocal consolidations with ill-defined margins.
- Peripheral airspace opacification with a characteristic "white lung" appearance in severe cases.
- Relative sparing of the central lung zones in early-stage disease.
- Absence of pleural effusion in most cases, which distinguishes COVID-19 from many bacterial pneumonias.

The Normal class images within this database originate from patients who presented with non-pulmonary complaints and underwent CXR as part of routine workup, with no reported pulmonary pathology on radiological review. This subset contributes the largest class to the combined dataset, providing a robust representation of normal pulmonary anatomy across diverse patient demographics.

### Dataset 2: Chest X-Ray Multiclass Dataset with Non-X-Ray Anomalies

The second source is a composite multi-institutional dataset curated specifically to facilitate the training of models for real-world, open-set clinical deployment. A distinguishing and clinically important property of this dataset is the deliberate inclusion of non-X-ray images — photographs of everyday objects such as animals and bicycles — alongside genuine chest radiographs. This design enables the training of models with robustness to out-of-distribution inputs and supports open-set recognition, whereby the model can identify inputs that do not belong to any known radiographic class. **Importantly, the non-X-ray class was excluded from our training pipeline**, as our system targets only genuine CXR inputs.

The dataset integrates chest radiographs from the following globally recognized institutional sources:

This dataset is particularly valuable for the Tuberculosis and Pneumonia classes, which are otherwise under-represented in many COVID-19-centric benchmarks. The multi-national origin of the TB images ensures broad geographic and demographic coverage, lending the trained model greater generalizability across different patient populations and X-ray acquisition protocols.

## Combined Dataset and Class Distribution

After merging both source datasets, applying quality filtering, and removing non-X-ray images, the final combined dataset comprises 17,637 **images** distributed across four diagnostic classes. The dataset is partitioned into mutually exclusive training, validation, and test splits following standard machine learning practice.

The training set, used for gradient-based parameter optimization, contains 15,229 **images** distributed across the four classes as follows:

The remaining approximately 2,408 **images** are equally divided between the validation set (used for hyperparameter tuning and early stopping) and the held-out test set (used solely for final performance evaluation). This ensures that the reported test metrics reflect true generalization performance on unseen data.

Figure 3.14 presents representative sample radiographs from each of the four diagnostic classes, illustrating the distinctive visual characteristics and the degree of inter-class similarity that the classification model must learn to resolve.

## Image Preprocessing Pipeline

Raw chest radiographs exhibit considerable variability in acquisition parameters, scanner characteristics, patient positioning, and image quality. To mitigate these sources of variability and produce a consistent, standardized input representation, all images undergo a three-stage preprocessing pipeline prior to training and inference. The preprocessing function is implemented in Python using the OpenCV library and is applied uniformly to all training, validation, and test images.

**Step 1: Greyscale Loading** All images are loaded in grayscale mode, discarding any color channels that may be present in saved PNG or JPEG files. Chest radiographs are inherently grayscale imaging modalities; the single luminance channel contains all diagnostically relevant information.

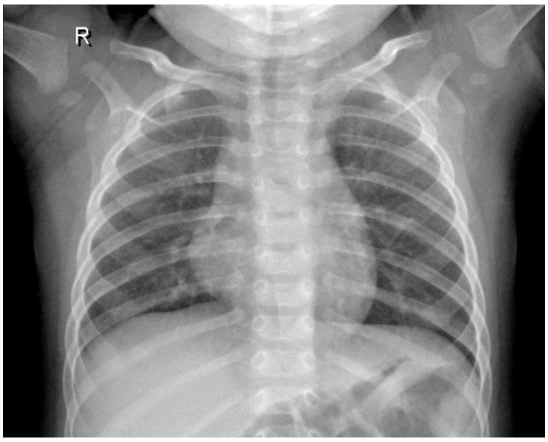
**Step 2: Contrast Limited Adaptive Histogram Equalization (CLAHE)** Standard global histogram equalization can over-amplify noise in homogeneous regions and

Table 3.3: Datasets Used for Chest Disease Detection

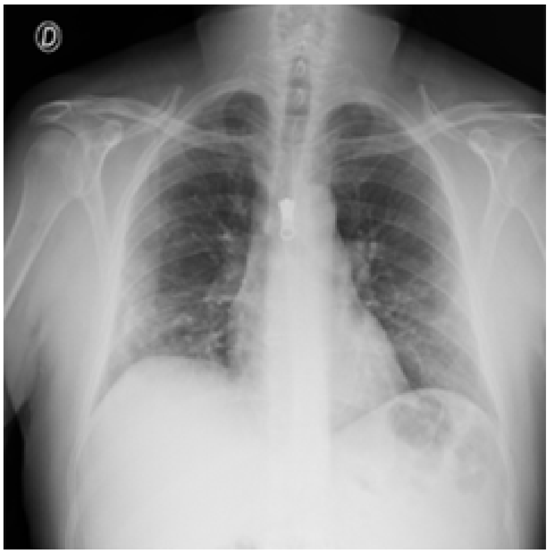
| **Source** | **Class(es)** | **Volume** | **Note** |
|---|---|---|---|
| OCT Dataset (Kermany et al.) | Pneumonia, Normal | 4,273 Pneumonia + 1,583 Normal | Pedia and adult CXR |
| RSNA CXR Dataset | Tuberculosis | 1,750 TB-positive + 4,560 Normal | Radi Society of North America |
| NIAID TB Dataset | Tuberculosis | 1,300 TB-positive | Seve coun cohort |
| NLM Montgomery Dataset | TB, Normal | 57 TB + 80 Normal (137 total) | Mont Coun USA |
| NLM Shenzhen Dataset | TB, Normal | 336 TB + 326 Normal (662 total) | Shen No.3 Hospital, China |
| Belarus Dataset | Tuberculosis | 304 TB-positive | 512 × 512 px resolution |
| Non-X-ray Dataset (Sanagapati) | Non-X-ray (excluded) | 1,357 object images | Not used in training |

Table 3.4: Class Distribution in the Training Dataset

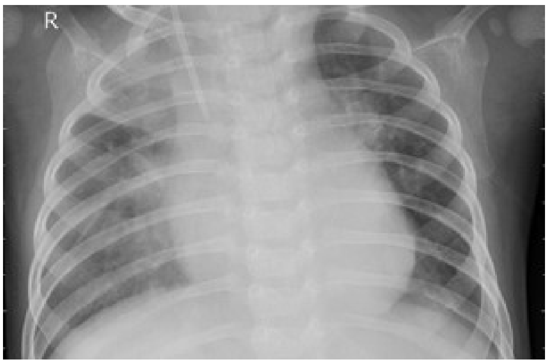
| **Class** | **Number of Images** | **Proportion (%)** |
|---|---|---|
| Normal | 4,870 | 31.98 |
| Pneumonia | 4,221 | 27.72 |
| COVID-19 | 3,192 | 20.96 |
| Tuberculosis | 2,946 | 19.34 |
| **Total (Training)** | **15,229** | **100.00** |




(a) Normal Chest X-ray



(b) COVID-19




(c) Pneumonia

(d) Tuberculosis

Figure 3.14: Representative chest radiograph samples from each of the four target diagnostic classes. The subtle visual differences between COVID-19 and Pneumonia are clear, underscoring the difficulty of the classification task.

create unnatural-looking artefacts. CLAHE addresses this by dividing the image into a grid of small, non-overlapping tiles (tile grid size of $8 \times 8$ pixels in this implementation) and applying histogram equalisation independently within each tile. A clip limit of 2.0 is applied to prevent excessive contrast amplification. The CLAHE-enhanced image $\hat{I}$ is computed as:

$$\hat{I}(x, y) = \text{CLAHE}\left(I(x, y); \text{clipLimit} = 2.0,\ \text{tileGrid} = 8 \times 8\right) \tag{3.21}$$

CLAHE is particularly effective at enhancing subtle infiltrates and opacities in lung tissue without washing out the finer structural details such as bronchial markings and vascular shadows, both of which carry diagnostic significance for tuberculosis and pneumonia detection.

**Step 3: Gaussian Smoothing** A Gaussian blur with a $5 \times 5$ kernel and automatically determined standard deviation ($\sigma = 0$, which causes OpenCV to compute $\sigma$ from the kernel size) is applied to suppress high-frequency noise introduced or amplified by the CLAHE step. The smoothed image is given by:

$$I_{\text{smooth}}(x, y) = I_{\text{CLAHE}}(x, y) * G(x, y; \sigma) \tag{3.22}$$

where

$$G(x, y; \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \tag{3.23}$$

denotes the two-dimensional Gaussian kernel, and $*$ denotes the convolution operation.

**Step 4: Min-Max Normalization** The smoothed image is normalized to the continuous floating-point range $[0, 1]$:

$$I_{\text{norm}}(x, y) = \frac{I_{\text{smooth}}(x, y) - I_{\min}}{I_{\max} - I_{\min}} \tag{3.24}$$

This step ensures that all input pixel values lie within a consistent numerical range, which is a prerequisite for stable gradient-based optimization and consistent behavior across heterogeneous source datasets.

**Greyscale to RGB Conversion** The normalized single-channel image is converted to a three-channel RGB image by replicating the luminance channel across all three color channels. This step is necessary because the pre-trained backbone architectures (e.g., DenseNet121, ResNet50, and related models) are trained on the ImageNet dataset and therefore expect three-channel RGB inputs of size $224 \times 224 \times 3$.

The preprocessed images are saved as 8-bit unsigned integers, with pixel values scaled back to the range $[0, 255]$ prior to storage, in the corresponding output directory. The original filename and directory structure are preserved to maintain dataset traceability and compatibility with subsequent processing stages.

Figures 3.15 and 3.16 illustrate the effect of the preprocessing pipeline on representative Pneumonia and Tuberculosis radiographs, respectively.


Figure 3.15: Comparison of a Pneumonia chest radiograph before (left) and after (right).

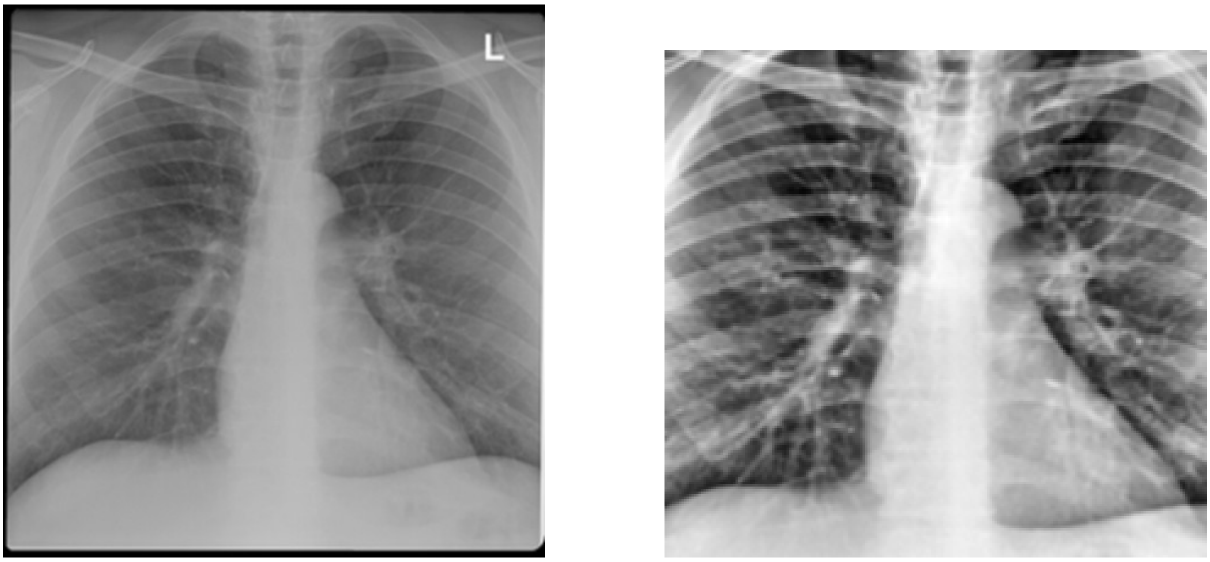



Figure 3.16: Comparison of a Tuberculosis chest radiograph before (left) and after (right) the preprocessing pipeline.

## Lung Segmentation Dataset

A separate dataset is employed exclusively for training the U-Net lung segmentation model. This dataset is a curated combination of three well-established lung segmentation benchmarks: the **Darwin, Montgomery, and Shenzhen** datasets, with a combined total of 6,810 image–mask pairs. Only the **Darwin subset (**6,106 **images and corresponding binary masks)** is used in the final segmentation training pipeline, as it provides the largest and most diverse collection.

Key characteristics of the Darwin dataset are:

- Images include most of the cardiac shadow, exposing lung opacities that may be concealed behind the heart — a property of relevance for assessing viral pneumonia severity.
- Image resolutions, sources, and orientations vary significantly, ranging from $156 \times 156$ to $5600 \times 4700$ pixels, providing excellent scale invariance for training the segmentation model.
- Portable (bedside) X-ray images of significantly lower technical quality are included, enhancing the model's robustness to real-world image quality variability.
- Lung boundary annotations were generated using Darwin's Auto-Annotate AI tool and subsequently reviewed and corrected by expert radiologists, ensuring high annotation quality.
- The dataset does not contain lateral projection images; only PA/AP views are included.

Figure 3.17 presents an example image–mask pair from the Darwin segmentation dataset. The binary mask precisely delineates the left and right lung fields, including the lower-most visible portion defined by the diaphragmatic contour.

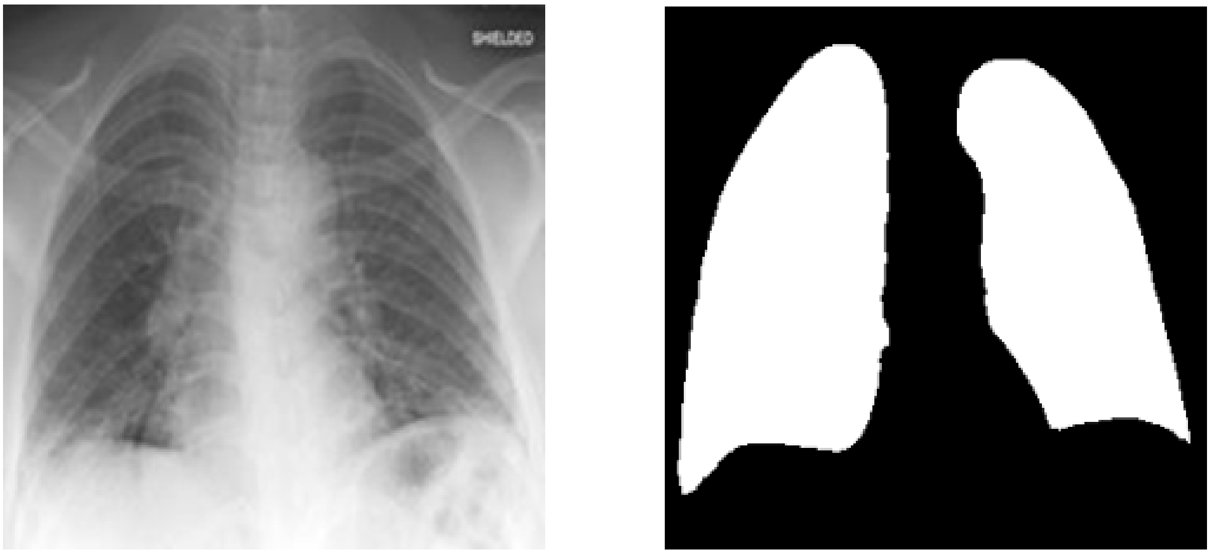



Figure 3.17: Representative image–mask pair from the Darwin lung segmentation dataset. The mask precisely delineates both lung fields including regions behind the cardiac shadow.

## 3.4.5 Segmentation Model Details

### Rationale for Lung Segmentation

The inclusion of a dedicated lung segmentation preprocessing stage is a deliberate architectural decision grounded in both technical and clinical considerations. In a raw chest radiograph, the pulmonary fields of interest are surrounded by a rich context of non-pulmonary structures — the rib cage, thoracic spine, clavicles, cardiac silhouette, subdiaphragmatic organs, and soft tissues of the chest wall. From a pattern-recognition perspective, these structures can act as confounding variables: a CNN attempting to classify a COVID-19 image might inadvertently learn to exploit spurious correlations between disease prevalence and, for example, the projection angle or patient body habitus as reflected in bony structures, rather than developing genuine pathology-discriminative features.

By applying the segmentation mask prior to classification, the input to the classifier is restricted to the lung parenchyma, compelling the network to focus exclusively on pulmonary features. This masking strategy has been shown in the literature to improve classification specificity and reduce the likelihood of shortcut learning, particularly for conditions with subtle or distributed radiological findings such as COVID-19 and miliary TB.

## U-Net Architecture

The lung segmentation model is built upon the **U-Net** architecture, originally proposed by Ronneberger, Fischer, and Brox (2015) for biomedical image segmentation tasks. U-Net is characterized by its distinctive encoder–decoder structure with lateral skip connections, which enables it to simultaneously capture high-level semantic context (through **the encoder's down sampling path**) and recover precise spatial localization detail (through the **decoder's up sampling path** and **skip connections**).

The implemented U-Net is designed to operate on grayscale images of size $224 \times 224 \times 1$ and produce a binary segmentation mask of identical spatial dimensions. The architecture is deliberately kept lightweight, with the encoder beginning at 32 filters to maintain computational efficiency while preserving sufficient representational capacity for the lung segmentation task. The complete architecture is described below.

**Convolutional Block** The fundamental building block of both the encoder and decoder paths is the convolutional block, which applies two successive $3 \times 3$ convolutions, each followed by Batch Normalization and ReLU activation:

$$\text{conv\_block}(x, F) = \text{ReLU}\Big(\text{BN}\Big(\text{Conv}_{3\times3}(\text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(x, F))))\Big), F\Big) \tag{3.25}$$

where $F$ denotes the number of output feature maps and BN denotes the Batch Normalization operation. The use of Batch Normalisation accelerates training convergence and acts as a regularizer, reducing dependence on precise weight initialization.

**Encoder (Contracting Path)** The encoder consists of four sequential encoder blocks, each comprising a convolutional block followed by a $2 \times 2$ max pooling operation with stride 2. Each encoder block produces two outputs: the feature map $s_i$ (skip connection) and the downsampled map $p_i$ (passed to the next encoder block):

$$s_i = \text{conv\_block}(p_{i-1}, F_i), \quad p_i = \text{MaxPool}_{2\times2}(s_i) \tag{3.26}$$

The filter counts at successive encoder levels are $F_1 = 32$, $F_2 = 64$, $F_3 = 128$, and $F_4 = 256$. After four pooling operations, the spatial resolution is reduced from $224 \times 224$ to $14 \times 14$.

**Bridge (Bottleneck)** The bridge layer connects the encoder and decoder without any pooling, applying a convolutional block with $F_5 = 512$ filters to the output of the deepest encoder level:

$$b = \text{conv\_block}(p_4, 512) \tag{3.27}$$

**Decoder (Expansive Path)** The decoder comprises four decoder blocks, each of which performs bilinear $2 \times 2$ upsampling, concatenates the upsampled feature map with the corresponding skip connection from the encoder, and applies a convolutional block:

$$d_i = \text{conv\_block}\Big(\text{Concat}\Big[\text{UpSample}_{2\times 2}(d_{i-1}),\ s_{5-i}\Big],\ F_{5-i}\Big) \tag{3.28}$$

The skip connections serve the critical function of restoring fine spatial detail that is inevitably lost during the max-pooling operations of the encoder. Without skip connections, the decoder would be forced to hallucinate precise boundary locations from the low-resolution bottleneck representation alone.

**Output** The final layer is a $1 \times 1$ convolutional layer with a single output channel and sigmoid activation, which maps the decoder's feature maps to a probability mask $\hat{M} \in [0, 1]^{224\times 224}$, where each value represents the probability of the corresponding pixel belonging to the lung region:

$$\hat{M}(x, y) = \sigma\Big(\text{Conv}_{1\times 1}(d_4)\Big) = \frac{1}{1 + e^{-\text{Conv}_{1\times 1}(d_4)(x,y)}} \tag{3.29}$$

The full encoder–decoder filter progression is summarized in Table 3.5 and the complete U-Net architecture with tensor dimensions at each level is illustrated in Figure **??**.

## Loss Functions

A compound loss function is employed for training the segmentation model, combining Binary Cross-Entropy (BCE) loss and Dice loss. This compound formulation simultaneously optimizes pixel-level probability calibration (via BCE) and the overlap between predicted and ground-truth masks (via Dice), yielding more stable training behavior and improved performance on the class-imbalanced segmentation task, where background pixels vastly outnumber lung pixels.

**Dice Coefficient:** The Dice Similarity Coefficient (DSC) measures the degree of overlap between the predicted mask $\hat{M}$ and the ground-truth mask $M$:

Table 3.5: U-Net encoder–decoder filter progression and spatial resolutions (input: $224 \times 224 \times 1$).

| Level | Block Type | Filters | Output Spatial Size |
|---|---|---|---|
| Encoder 1 | conv_block + MaxPool | 32 | $224 \times 224 \rightarrow 112 \times 112$ |
| Encoder 2 | conv_block + MaxPool | 64 | $112 \times 112 \rightarrow 56 \times 56$ |
| Encoder 3 | conv_block + MaxPool | 128 | $56 \times 56 \rightarrow 28 \times 28$ |
| Encoder 4 | conv_block + MaxPool | 256 | $28 \times 28 \rightarrow 14 \times 14$ |
| Bridge | conv_block | 512 | $14 \times 14$ |
| Decoder 1 | UpSample + Concat + conv_block | 256 | $28 \times 28$ |
| Decoder 2 | UpSample + Concat + conv_block | 128 | $56 \times 56$ |
| Decoder 3 | UpSample + Concat + conv_block | 64 | $112 \times 112$ |
| Decoder 4 | UpSample + Concat + conv_block | 32 | $224 \times 224$ |
| Output | Conv $1 \times 1$ + Sigmoid | 1 | $224 \times 224$ |

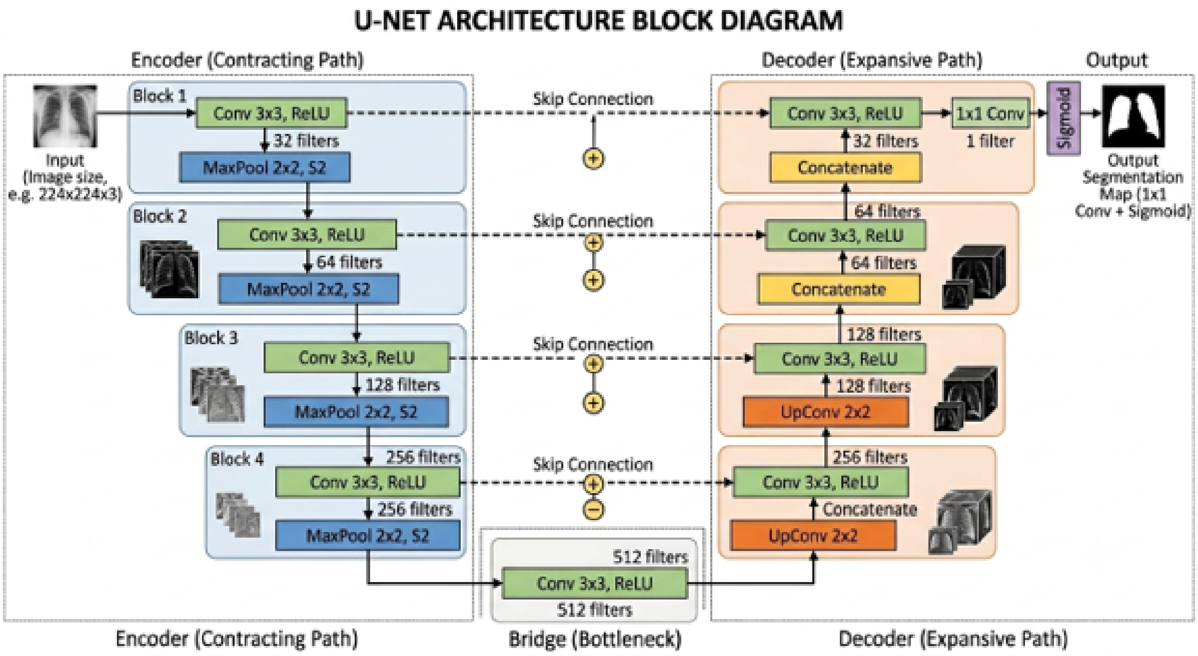



Figure 3.18: U-Net architecture implemented for lung segmentation. The contracting path (left) captures contextual information through successive down sampling, while the expansive path (right) enables precise localization via skip connections. Tensor shapes are annotated at each level.

$$\mathrm{DSC}(M, \hat{M}) = \frac{2\sum_{i,j} M_{ij}\hat{M}_{ij} + \varepsilon}{\sum_{i,j} M_{ij} + \sum_{i,j} \hat{M}_{ij} + \varepsilon} \tag{3.30}$$

where $\varepsilon = 1$ is a smoothing constant that prevents division by zero and improves gradient stability in early training epochs. The Dice coefficient ranges from 0 (no overlap) to 1 (perfect overlap).

**Dice Loss:**

$$\mathcal{L}_{\text{Dice}} = 1 - \mathrm{DSC}(M, \hat{M}) \tag{3.31}$$

**Binary Cross-Entropy Loss:** Applied pixel-wise across the entire image:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i,j}\left[M_{ij}\log(\hat{M}_{ij}) + (1 - M_{ij})\log(1 - \hat{M}_{ij})\right] \tag{3.32}$$

where $N$ is the total number of pixels per image.

**Combined Loss:**

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} \tag{3.33}$$

## Training Configurations

The U-Net segmentation model is trained with the following configuration extracted directly from the implementation.

The dataset is shuffled with a fixed seed of 42 before splitting to ensure reproducibility. The training loop uses data pipelines with prefetching to maximize GPU utilization. Mixed float16 precision is enabled via TensorFlow's Automatic Mixed Precision (AMP) to accelerate training on modern NVIDIA GPU architectures with Tensor Core support.

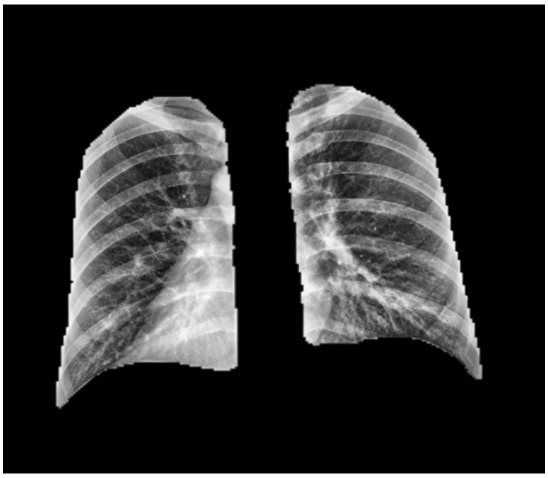
A two-phase training strategy is adopted: in Phase 1, the model is trained from a random initialization for 10 epochs to establish stable feature representations. In Phase 2, the best Phase 1 checkpoint (`lung_segmentation_final_v2.h5`) is reloaded and fine-tuned for a further 4 epochs at the same learning rate, allowing the model to refine its lung boundary delineation capability. The best overall weights — measured by `val_dice_coef` —are saved as the final segmentation model.

## Classification Model: DenseNet121 with One-vs-Rest (OVR)

The primary classification backbone is DenseNet121 (Densely Connected Convolutional Networks, Huang et al., 2017), a 121-layer deep CNN pre-trained on the ImageNet dataset. DenseNet121 is particularly well-suited for medical image classification for several reasons:

(a) Original

(b) Binary Mask



(c) Segmented Image

Figure 3.19: U-Net segmentation results. From left to right: the preprocessed input radiograph, the predicted binary lung mask, and the masked output image passed to the classification stage.

Table 3.6: U-Net segmentation model training configuration.

| **Hyperparameter** | **Value** |
|---|---|
| Input shape | $224 \times 224 \times 1$ (grayscale) |
| Batch size | 8 |
| Initial epochs (Phase 1) | 10 |
| Fine-tuning epochs (Phase 2) | 4 |
| Optimizer | Adam |
| Learning rate | $1 \times 10^{-4}$ |
| Loss function | BCE + Dice (combined) |
| Evaluation metric | Dice Coefficient, Accuracy |
| Early stopping monitor | val_dice_coef (patience = 5, mode = max) |
| LR reduction | ReduceLROnPlateau (factor = 0.1, patience = 5, min LR = $1 \times 10^{-7}$ |
| Precision policy | Mixed float16 (TensorFlow AMP) |
| Training / validation split | 85% / 15% (6,106 Darwin images) |
| Best model checkpoint | lung_segmentation_darwin_montgomery_shenzhen.h5 |

- **Dense connectivity:** Each layer receives feature maps from all preceding layers, enabling maximum feature reuse and gradient flow. This alleviates the vanishing gradient problem and allows effective training even with relatively limited medical imaging datasets.
- **Parameter efficiency:** Dense connectivity reduces the number of required parameters compared to conventional deep networks of comparable performance, reducing the risk of overfitting on limited data.
- **Multi-scale feature aggregation:** The dense connections allow the final layers to simultaneously access low-level texture features and high-level semantic representations, which is advantageous for the heterogeneous visual patterns of different chest pathologies.

The model follows a transfer learning approach with partial fine-tuning. The DenseNet121 feature extractor is initialized with ImageNet pre-trained weights, and the last 100 layers are made trainable while all earlier layers are frozen. This selective fine-tuning strategy preserves the low-level and mid-level representations learned from ImageNet (edges, textures, simple patterns) whilst allowing task-specific adaptation of higher-level representations through gradient updates.

A custom classification head is appended to the Global Average Pooling (GAP) output of the base model. The GAP operation reduces the spatial feature map to a 1024-dimensional feature vector, eliminating the need for large fully connected layers and thereby reducing parameters and overfitting risk. The head architecture is:

$$z = \mathrm{Dropout}_{0.3}\Big(\mathrm{Dense}_{256}^{\mathrm{ReLU}}\Big(\mathrm{Dropout}_{0.4}(\mathrm{Dense}_{512}^{\mathrm{ReLU}}(\mathrm{BN}(\mathrm{GAP}(F))))\Big)\Big) \tag{3.34}$$

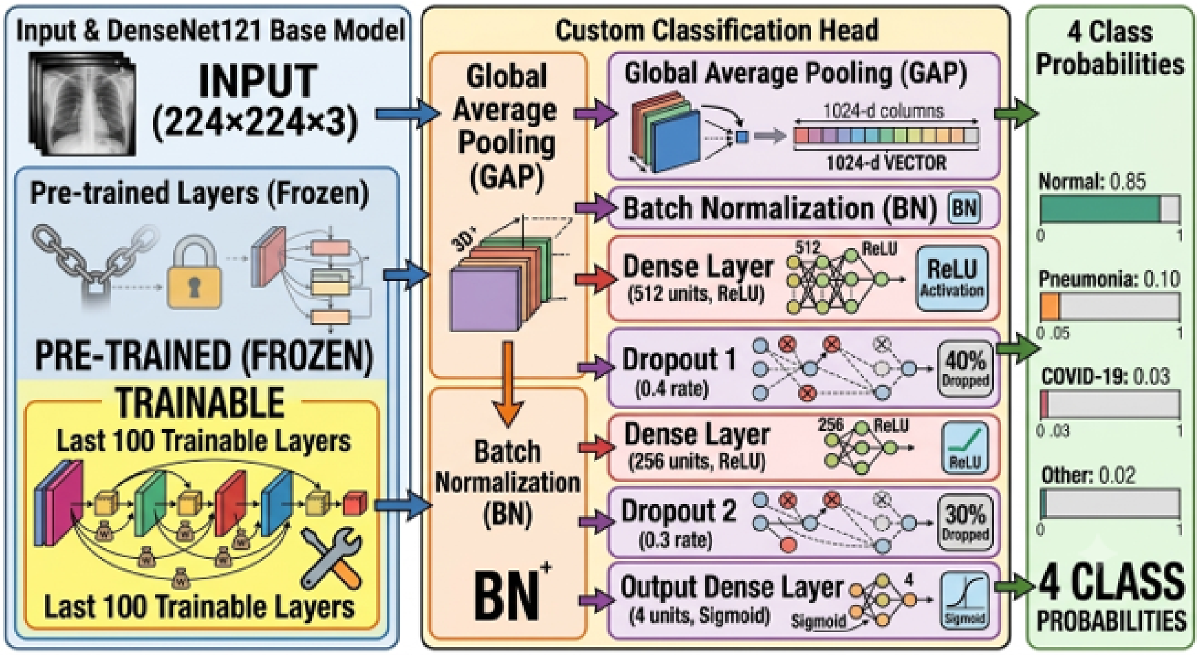



Figure 3.20: Proposed DenseNet121 OVR classification model architecture. The base network produces a 1024-dimensional feature vector via Global Average Pooling, which is refined through the custom classification head before producing independent sigmoid probability scores for each diagnostic class.

$$\hat{y} = \sigma(W_{\text{out}} z + b_{\text{out}}), \quad \hat{y}_k = \frac{1}{1 + e^{-(W_k z + b_k)}} \tag{3.35}$$

where $F$ denotes the final convolutional feature map of DenseNet121, GAP$(\cdot)$ denotes Global Average Pooling, BN$(\cdot)$ denotes Batch Normalization, and $\sigma(\cdot)$ denotes the sigmoid activation. Each of the $K = 4$ output neurons operates as an independent binary classifier, producing a probability score $\hat{y}_k \in [0, 1]$ for each class $k$.

This One-vs-Rest (OVR) multi-label formulation with sigmoid output offers several advantages over a conventional SoftMax multi-class formulation. In the SoftMax approach, the probabilities of all classes are coupled (they must sum to one), which can suppress the activation of a class when multiple classes co-occur or when the model is uncertain. With sigmoid, each class is modeled independently, allowing the model to assign high confidence to multiple classes simultaneously when the image features are ambiguous or the pathology is mixed.

The model architecture is illustrated in Figure 3.20.

The training employs Binary Cross-Entropy (BCE) loss rather than categorical cross-entropy, consistent with the OVR formulation where each class is an independent binary classification target. The BCE loss for a batch of $N$ samples with $K = 4$ classes is:

$$\mathcal{L}_{\text{BCE-OVR}} = -\frac{1}{N \cdot K} \sum_{n=1}^{N} \sum_{k=1}^{K} \Big[ y_{nk} \log(\hat{y}_{nk}) + (1 - y_{nk}) \log(1 - \hat{y}_{nk}) \Big] \tag{3.36}$$

Class imbalance is addressed through inverse-frequency class weights, computed as:

$$w_k = \frac{N_{\text{total}}}{K \cdot N_k} \tag{3.37}$$

where $N_{\text{total}}$ is the total number of training samples and $N_k$ is the number of samples in class $k$.

### 3.4.6 Grad-CAM Integration

#### Rationale for Explainability

The deployment of any AI-based diagnostic tool in a clinical environment is inseparable from the question of trust and accountability. A classification model, however accurate its aggregate performance metrics, remains clinically unacceptable if its decision-making process is opaque and cannot be scrutinized by the responsible physician. The recognition of this challenge has given rise to the field of Explainable Artificial Intelligence (XAI), which encompasses a broad family of techniques aimed at rendering the internal reasoning of machine learning models interpretable to human observers.

In the context of medical imaging, XAI serves multiple critical functions:

- **Verification of clinical validity:** Confirming that the model attends to medically meaningful regions (e.g., lung infiltrates for COVID-19) rather than spurious correlations (e.g., imaging artifacts or patient positioning markers).
- **Detection of shortcut learning:** Identifying cases where the model has learned to exploit non-clinical features present in the training data due to dataset biases.
- **Clinical decision support:** Providing the reporting radiologist with a visual "second opinion" that highlights the most diagnostically salient regions for each prediction.
- **Regulatory compliance:** Meeting the increasingly stringent interpretability requirements of medical device regulations in various jurisdictions.

#### Gradient-weighted Class Activation Mapping (Grad-CAM)

The chosen XAI technique is Grad-CAM (Gradient-weighted Class Activation Mapping), introduced by Selvaraju et al. (2017). Grad-CAM is a gradient-based localization method that produces a coarse spatial importance map $\hat{L}^c \in \mathbb{R}^{u \times v}$ for a target class $c$, indicating which spatial regions of the input image were most influential in driving the model's prediction for that class.

The Grad-CAM computation proceeds as follows for a target class $c$ and a chosen convolutional layer $A$ with $K$ feature maps $A^k \in \mathbb{R}^{u \times v}$:

**Step 1: Compute Class-Discriminative Gradient** The gradient of the class score $\hat{y}^c$ (the logit for class $c$ before the final sigmoid) with respect to the $k$-th feature map $A^k$

is computed via backpropagation through the final convolutional layer:

$$\frac{\partial \hat{y}^c}{\partial A^k} \tag{3.38}$$

**Step 2: Global Average Pool the Gradients** The importance weight $\alpha_k^c$ for feature map $k$ with respect to class $c$ is obtained by spatially average pooling the gradients:

$$\alpha_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial \hat{y}^c}{\partial A_{ij}^k} \tag{3.39}$$

where $Z = u \times v$ is the number of spatial locations. Intuitively, $\alpha_k^c$ measures how sensitive the class-$c$ score is to changes in feature map $k$.

**Step 3: Compute Weighted Combination and Apply ReLU** The Grad-CAM heatmap is the ReLU-activated weighted sum of the feature maps:

$$\hat{L}^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k\right) \tag{3.40}$$

The ReLU operation retains only the features that have a positive influence on the target class score, discarding negative activations that correspond to features of other classes.

**Step 4: Resize and Normalize** The heatmap $\hat{L}^c \in \mathbb{R}^{u \times v}$ (typically $7 \times 7$ for the last DenseNet convolutional layer) is bilinearly resized to match the input image dimensions $(224 \times 224)$ and normalized to the range $[0, 1]$:

$$\tilde{L}^c = \frac{L^c_{\text{resized}} - \min(L^c_{\text{resized}})}{\max(L^c_{\text{resized}}) - \min(L^c_{\text{resized}})} \tag{3.41}$$

**Step 5: Color Mapping and Overlay** The normalized heatmap is converted to a pseudo-color image using the JET color map (blue = low importance, red = high importance) and blended with the original RGB image at a mixing ratio of 0.4:0.6 to produce the final Grad-CAM visualization:

$$I_{\text{Grad-CAM}} = \mathrm{JET}(\tilde{L}^c) \times 0.4 + I_{\text{orig}} \times 0.6 \tag{3.42}$$

Values are clipped to $[0, 255]$ and cast to 8-bit unsigned integers before display.

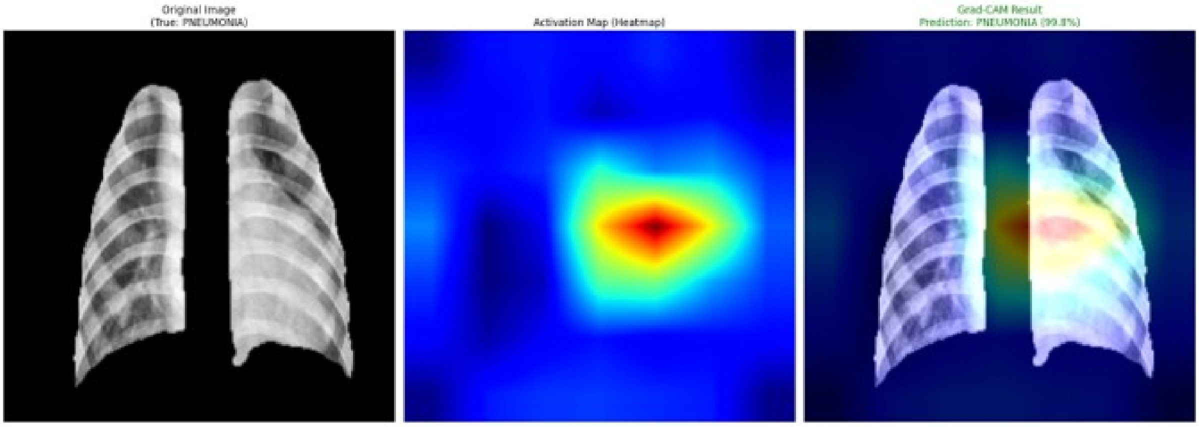



Figure 3.21: Grad-CAM visualization results for Pneumonia representative case. The saliency maps confirm that the model attends to pathologically relevant lung regions.

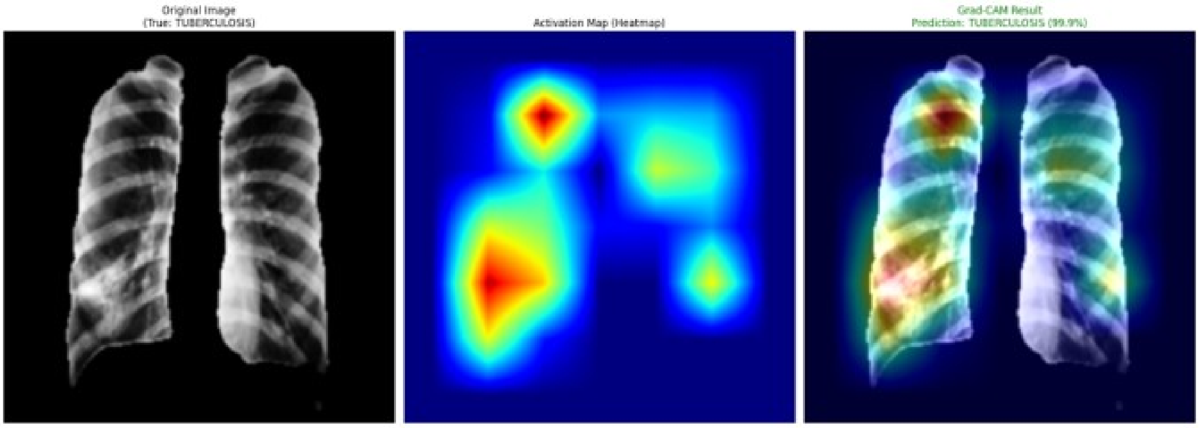



Figure 3.22: Grad-CAM visualization results for Tuberculosis cases. The upper-lower lobe localization of saliency for TB cases confirms the model's ability to recognize TB-specific radiological patterns.

## OVR Grad-CAM for Multi-Class Visualization

In the OVR model, each output neuron is an independent binary classifier. The Grad-CAM implementation accordingly generates a separate heatmap for each class whose predicted probability exceeds the decision threshold $\tau = 0.5$. If no class reaches the threshold, the class with the highest sigmoid score is selected as the target. The final visualization presents the original image alongside one Grad-CAM overlay per active class, enabling simultaneous visualization of multiple class-discriminative regions when the model detects co-occurring pathologies or is uncertain between classes.

The target convolutional layer for Grad-CAM is the final dense block concatenation layer of DenseNet121, accessed via the Keras layer name `conv5_block16_concat`. This layer produces feature maps at a spatial resolution of $7 \times 7$ with 1024 channels, providing a rich and discriminative representation just before the global pooling operation.

Samples from the process and the output are shown for representative samples from the pneumonia and Tuberculosis scans in Figure 3.21, and Figure 3.22 respectively.

# 3.5 Lung Cancer Detection Model

This section presents the lung cancer and tuberculosis detection model developed within the project, focusing on automated localization and classification of pulmonary abnormalities from chest X-ray (CXR) images[cite: 1, 2]. The model is designed as an end-to-end object detection pipeline that integrates binary mask conversion, bounding box generation, and deep learning–based detection using the YOLOv8 architecture[cite: 3]. Emphasis is placed on the data preparation strategy and architectural design choices that underpin the detection framework[cite: 4].

## 3.5.1 Motivation

Object detection frameworks offer distinct advantages over image-level classification for medical imaging tasks, as they provide explicit spatial localization of pathological regions rather than a single diagnostic label[cite: 5, 6]. For lung cancer and tuberculosis screening, identifying the precise location and extent of pulmonary lesions or infiltrates is clinically essential, as it directly informs radiological assessment and subsequent clinical decision-making[cite: 7]. The proposed model addresses this requirement by reformulating the diagnostic task as a bounding box regression and classification problem, enabling simultaneous localization and identification of disease regions within a single unified forward pass[cite: 8].

## 3.5.2 Problem Statement

The objective of this model is to perform multi-class detection of pulmonary abnormalities in chest X-ray images, identifying and localizing instances belonging to the following diagnostic categories[cite: 9, 10]:

$$C = \{\text{tumor}, \text{tuberculosis}\}[cite : 11] \tag{3.43}$$

Given an input chest X-ray image $I \in \mathbb{R}^{H\times W}$, the task is to learn a mapping[cite: 12]:

$$f_\theta : I \to \{(b_i, c_i, s_i)\}_{i=1\ldots N}[cite : 13] \tag{3.44}$$

where each tuple $(b_i, c_i, s_i)$ denotes the predicted bounding box coordinates, class label, and confidence score for the $i$-th detected region, and $\theta$ represents the model parameters[cite: 14].

To bridge the gap between pixel-level segmentation annotations (binary masks) and the bounding box representation required by the detection framework, an explicit mask-to-box conversion stage is incorporated as the first step of the pipeline[cite: 15]. Each connected foreground component in the binary annotation mask is converted into a normalized YOLO-format bounding box, enabling the use of richly annotated mask datasets for detection training without manual re-labeling[cite: 16].

### 3.5.3 Model Architecture

The detection model is structured as a three-stage pipeline consisting of: (1) mask-to-bounding-box conversion and dataset preparation, (2) YOLO-format data organization and configuration, and (3) YOLOv8-based detection model training and evaluation[cite: 17, 18]. Each stage is described in detail below[cite: 19].

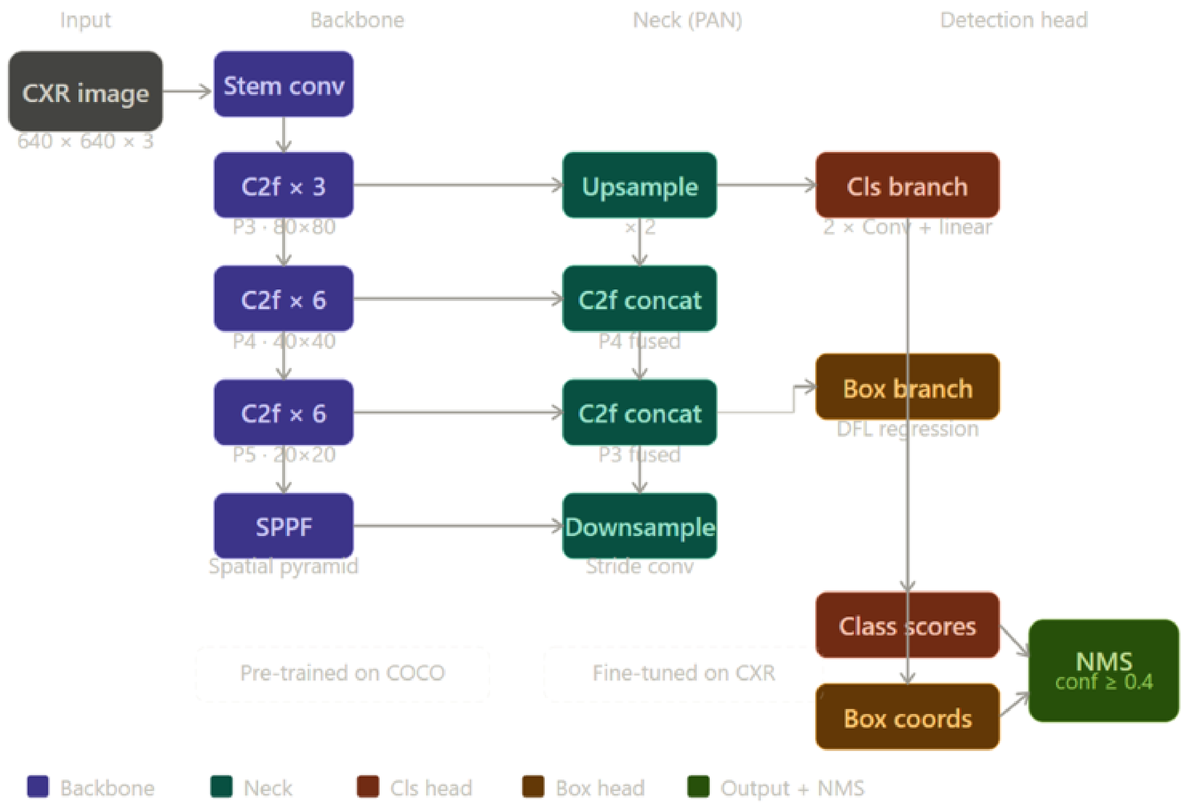



Figure 3.23: YOLOv8 architecture pipeline for automated localization and classification of pulmonary abnormalities.

#### Mask-to-Bounding-Box Conversion

Annotation masks for pulmonary lesions are provided as binary PNG images in which foreground pixels (intensity $> 127$) correspond to pathological regions[cite: 20, 21]. To convert these masks into YOLO-format bounding box annotations, connected component analysis is applied using external contour detection[cite: 22]. For each detected contour, an axis-aligned bounding box is computed and normalized with respect to the image dimensions, yielding center coordinates and extent values in the range $[0, 1]$[cite: 23].

Formally, for a contour with pixel-space bounding rectangle $(x, y, w, h)$ within an image of width $W$ and height $H$, the normalized YOLO representation is[cite: 24]:

$$c_x = \frac{x + w/2}{W}, \quad c_y = \frac{y + h/2}{H}, \quad n_w = \frac{w}{W}, \quad n_h = \frac{h}{H}[cite : 25] \tag{3.45}$$

Contours enclosing fewer than 5 pixels in either dimension are discarded as noise prior to annotation generation[cite: 26]. All foreground components within a given mask are assigned the class index corresponding to the image-level label (tumor or tuberculosis), since mask annotations are provided per diagnostic category[cite: 27].

## Dataset Preparation and Splitting

Following bounding box conversion, the dataset is partitioned into training, validation, and test subsets using stratified random splitting[cite: 28, 29]. The default split proportions are 75% training, 15% validation, and 10% test, with random seed fixed at 42 to ensure reproducibility[cite: 30]. Image files and their corresponding YOLO label files are organized into the standard YOLO directory structure, with separate image and label folders for each split[cite: 31]. A YAML configuration file is generated automatically to specify dataset paths, the number of classes, and class names, serving as the interface between the prepared dataset and the YOLOv8 training framework[cite: 32].

## YOLOv8 Detection Architecture

Object detection is performed using YOLOv8, the latest generation of the You Only Look Once (YOLO) family of single-stage detectors[cite: 33, 34]. YOLOv8 follows an anchor-free design and adopts a CSPDarknet backbone for feature extraction, a Path Aggregation Network (PAN) neck for multi-scale feature fusion, and a decoupled detection head that separately predicts classification scores and bounding box offsets[cite: 35]. This design enables real-time inference while maintaining competitive localization accuracy[cite: 36].

A YOLOv8n (nano) variant is employed as the base model, pre-trained on the COCO dataset, and fine-tuned on the prepared CXR detection dataset[cite: 37]. The model is trained for 50 epochs with an input resolution of $640 \times 640$ pixels and a batch size of 16[cite: 38]. Transfer learning is leveraged by initializing all network weights from the COCO pre-trained checkpoint, with all layers unfrozen to allow domain-specific adaptation[cite: 38]. The training objective combines binary cross-entropy loss for classification, distribution focal loss (DFL) for bounding box regression, and a complete intersection-over-union (CIoU) penalty term to improve localization precision[cite: 39]:

$$L = \lambda_{\text{cls}} \cdot L_{\text{cls}} + \lambda_{\text{box}} \cdot L_{\text{box}} + \lambda_{\text{dfl}} \cdot L_{\text{DFL}}[cite : 40] \tag{3.46}$$

where $\lambda_{\text{cls}}$, $\lambda_{\text{box}}$, and $\lambda_{\text{dfl}}$ are scalar weighting coefficients that balance the contribution of each loss component during optimization[cite: 41].

After training, the best-performing checkpoint, selected based on validation mean Average Precision at IoU threshold 0.50 (mAP50), is retained for evaluation[cite: 42]. Model performance is assessed on both the held-out validation and test splits, reporting mAP50, mAP50–95, precision, and recall as primary evaluation metrics[cite: 43]. Qualitative evaluation is conducted by visually inspecting annotated predictions on test set images, examining the spatial correspondence between predicted bounding boxes and known pathological regions[cite: 44].

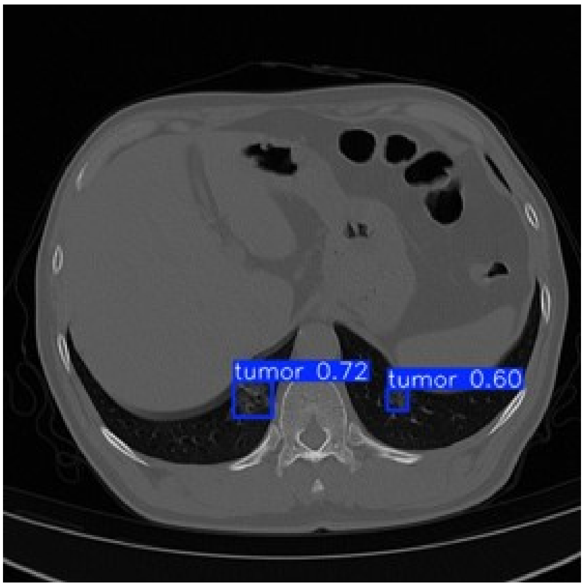



Figure 3.24: Qualitative evaluation showing tumor localization (confidence scores 0.72 and 0.60) on an axial CT slice.

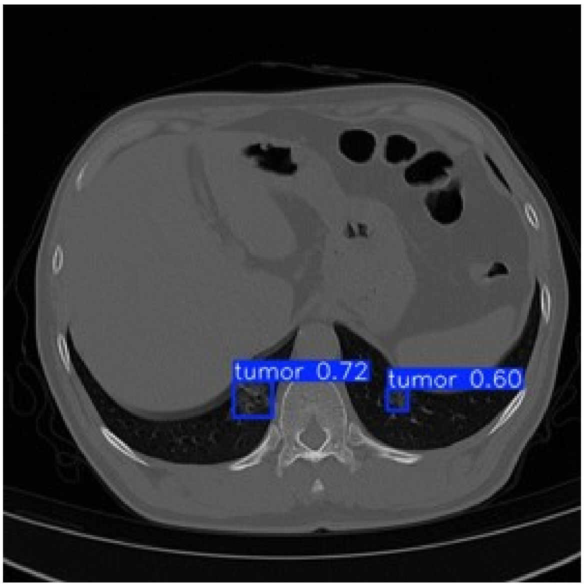



Figure 3.25: Qualitative evaluation showing tumor localization (confidence score 0.87) on an axial CT slice.

### 3.5.4 Interpretability Considerations

Unlike classification-only models that require post-hoc gradient visualization to identify diagnostically relevant regions, the detection framework provides explicit spatial outputs in the form of bounding boxes and associated confidence scores[cite: 45, 46]. Each prediction directly encodes the model's estimate of where a pathological region is located and how confident it is in that localization[cite: 47]. This built-in spatial explainability reduces the reliance on auxiliary interpretability methods and aligns more naturally with clinical reporting conventions, in which radiologists annotate specific anatomical regions rather than assigning image-level labels[cite: 48]. Confidence thresholding ($\tau = 0.4$) and non-maximum suppression with an IoU threshold of 0.6 are applied at inference time to filter low-quality or redundant detections, producing a concise set of high-confidence predictions per image[cite: 49].

### 3.5.5 Implementation of the Lung Cancer Detection Model

The second major component implemented within the proposed framework is the lung lesion detection model. Unlike conventional image classification systems that only determine whether an abnormality is present, the proposed approach performs object detection, enabling simultaneous identification and localization of suspicious pulmonary lesions within CT images. The model predicts both the spatial coordinates of lesion regions and the confidence associated with each prediction, thereby providing clinically meaningful visual outputs that can assist radiologists during image interpretation.

The detection pipeline was developed using the YOLOv8 architecture, which belongs to the family of single-stage object detectors. Single-stage detectors perform localization and

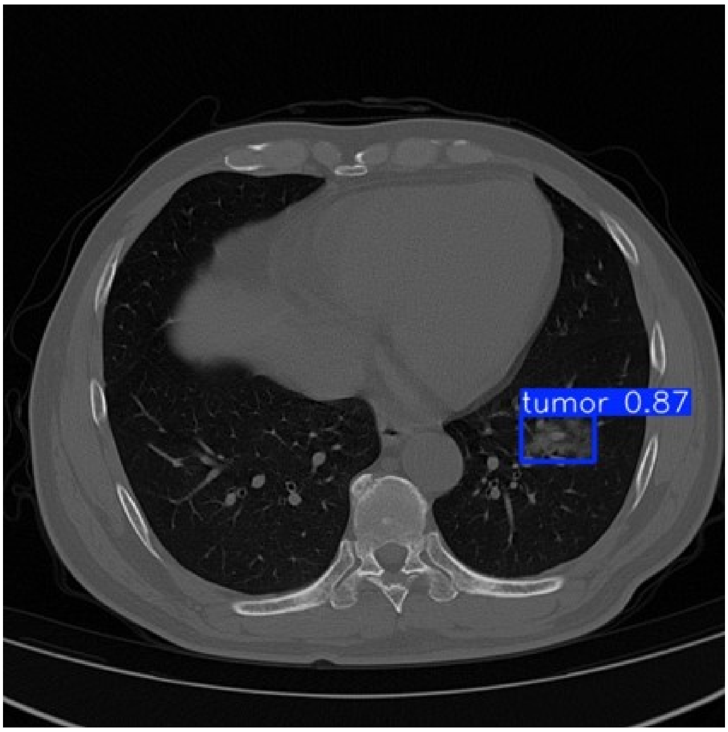



Figure 3.26: Example of an interpretable YOLO detection result on a chest CT image. The model localizes a suspected lesion using a bounding box and reports a confidence score of 0.87. Such spatial predictions provide inherent interpretability by explicitly identifying the anatomical region associated with the model's decision, reducing the need for post-hoc visualization techniques.

classification simultaneously within a unified network, eliminating the need for separate region proposal and classification stages. This design significantly reduces computational complexity and inference time while maintaining competitive detection performance.

The complete implementation workflow consists of five primary stages:

1. Dataset preprocessing
2. Annotation generation
3. Dataset partitioning
4. YOLOv8 model training
5. Performance evaluation and visualization

## Dataset Preparation and Annotation Generation

The original dataset contains CT images together with binary segmentation masks that identify lesion regions. Although segmentation masks provide pixel-level information regarding lesion boundaries, the YOLO detection framework requires object-level annotations represented by bounding boxes. Consequently, a dedicated annotation generation pipeline was implemented to automatically convert segmentation masks into YOLO-compatible labels.

For every image-mask pair, the following operations were performed:

**Step 1: Mask Loading** Each binary mask was loaded as a grayscale image where:

- Pixel value 0 represented background tissue.
- Pixel value 255 represented lesion pixels.

The masks were subsequently thresholded to ensure binary separation between foreground and background regions.

**Step 2: Contour Extraction** The OpenCV contour detection algorithm was applied to identify connected lesion regions:

```
contours, _ = cv2.findContours(
mask,
cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE
)
```

Only external contours were retained in order to avoid duplicate detections resulting from nested structures. The contour extraction stage transformed pixel-level lesion masks into geometrically defined lesion candidates.

**Step 3: Bounding Box Generation** For each detected contour, the minimum enclosing rectangle was computed:

```
x, y, w, h = cv2.boundingRect(contour)
```

where $x, y$ represent the upper-left corner, $w$ represents width, and $h$ represents height. These coordinates define the lesion localization region used by the detector during training.

**Step 4: YOLO Coordinate Conversion** YOLO requires normalized coordinates expressed relative to image dimensions. The generated bounding boxes were therefore converted using:

$$x_{\text{center}} = \frac{x + w/2}{W} \tag{3.47}$$

$$y_{\text{center}} = \frac{y + h/2}{H} \tag{3.48}$$

$$w_{\text{norm}} = \frac{w}{W}, \quad h_{\text{norm}} = \frac{h}{H} \tag{3.49}$$

where $W$ = image width and $H$ = image height. The resulting annotation was stored using YOLO format:

```
class_id x_center y_center width height
```

Since only one lesion category was considered, all annotations were assigned `class_id` = 0.

**Step 5: Label File Generation** For each image (e.g., `image_001.png`), a corresponding annotation file was created (`image_001.txt`) containing all lesion bounding boxes present in the image. This automated procedure eliminated the need for manual annotation and ensured consistency throughout the dataset.

### 3.5.6 Dataset Organization

Following annotation generation, the dataset was reorganized according to the standard YOLO directory structure:

```
dataset/
|-- train/
|   |-- images/
|   `-- labels/
|-- valid/
|   |-- images/
|   `-- labels/
`-- test/
|-- images/
`-- labels/
```

Maintaining this structure allows direct integration with the Ultralytics YOLO training framework. A YAML configuration file was subsequently generated containing:

```
path: dataset
train: train/images
val: valid/images
test: test/images
nc: 1
names:
0: tumor
```

The YAML file serves as the interface between the dataset and the YOLO training engine.

#### YOLOv8 Detection Architecture

The proposed detection model employs YOLOv8 as the underlying deep learning architecture. The network consists of three major components:

- **Backbone:** The backbone is responsible for hierarchical feature extraction. Low-level layers capture edges, textures, and intensity gradients, while deeper layers learn lesion morphology, shape characteristics, and contextual lung structures.

- **Neck:** The neck aggregates information from multiple feature scales using Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) mechanisms. This multi-scale representation is particularly important in lung lesion detection because lesions may vary considerably in size. Small nodules and large masses must both be represented effectively within the feature hierarchy.
- **Detection Head:** The detection head predicts bounding box coordinates, object confidence scores, and class probabilities for each candidate lesion. Unlike anchor-based detectors, YOLOv8 utilizes an anchor-free detection mechanism that simplifies prediction and improves localization accuracy.

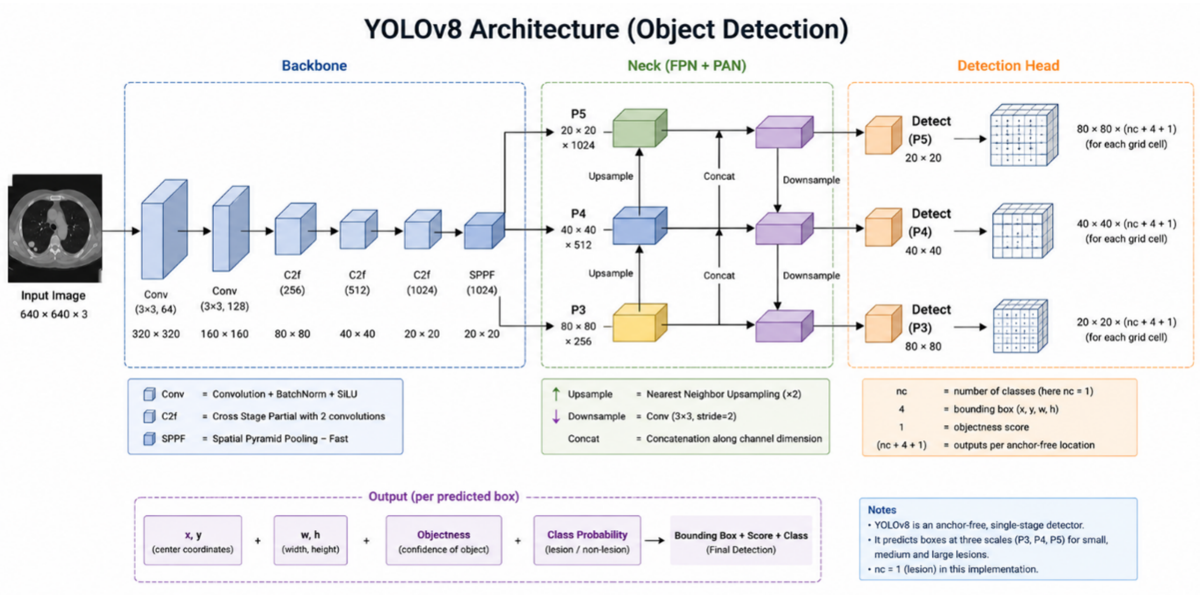



Figure 3.27: Detailed architectural diagram of the YOLOv8 object detection framework, illustrating the Backbone, Neck (FPN + PAN), and Detection Head stages for multi-scale prediction.

## Transfer Learning Strategy

Training a deep object detector entirely from scratch typically requires extremely large medical imaging datasets. To overcome this limitation, transfer learning was employed. The detector was initialized using pretrained YOLOv8 weights trained on large-scale image datasets. The pretrained model already possesses strong low-level feature extraction capabilities, including detection of edges, contours, textures, and spatial patterns. Fine-tuning these representations on lung CT images significantly accelerates convergence while reducing the risk of overfitting.

## Training Configuration

Model training was performed using the Ultralytics implementation of YOLOv8. The training pipeline automatically performs image loading, resizing, batching, label matching, loss computation, and weight optimization. Input images were resized to 640 × 640 pixels to ensure uniform tensor dimensions throughout training.

The optimization process simultaneously minimizes:

- **Localization Loss:** Measures bounding-box accuracy.
- **Classification Loss:** Measures lesion classification accuracy.
- **Distribution Focal Loss (DFL):** Improves bounding-box regression precision.

The total loss is defined as:

$$L_{\text{total}} = L_{\text{box}} + L_{\text{cls}} + L_{\text{DFL}} \tag{3.50}$$

During training, stochastic gradient optimization updates network weights through back-propagation until convergence.

## Evaluation Methodology

After training, the best-performing model checkpoint was selected automatically according to validation performance. Performance was evaluated using standard object detection metrics:

**Precision** Measures the proportion of predicted lesions that are truly lesions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3.51}$$

**Recall** Measures the proportion of true lesions successfully detected.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3.52}$$

**Mean Average Precision (mAP)** The primary detection metric used for evaluation. Two variants were reported:

- mAP@50
- mAP@50–95

mAP@50 evaluates detections using a single IoU threshold of 0.50, whereas mAP@50–95 provides a more stringent evaluation across multiple thresholds.

## Prediction Visualization and Model Interpretability

A key advantage of the proposed detection framework is its inherent interpretability. Each prediction generated by YOLOv8 contains bounding box coordinates, a confidence score, and a class label. The confidence score reflects the model's certainty regarding the presence of a lesion within the predicted region.

Unlike classification-only approaches that require external explainability methods such as Grad-CAM or saliency maps, the detector directly highlights suspicious anatomical regions through bounding boxes. This spatial localization capability allows clinicians to visually inspect the exact image regions influencing the model's decision, thereby improving transparency and facilitating clinical validation of automated predictions.

# 3.6 Federated Learning

## 3.6.1 Concept and Motivation

Federated learning (FL) is a distributed machine learning paradigm in which a model is trained collaboratively across multiple independent nodes without any raw data leaving its source institution [42]. Rather than centralizing training samples on a shared server, FL brings the computation to the data: each participating node trains a local model on its private dataset and shares only the resulting model parameters with a central aggregation server, which combines the updates into an improved global model. This cycle repeats for a defined number of communication rounds until the global model converges.

The motivation for adopting FL in this work is rooted in the fundamental tension between data-driven model development and patient privacy. Healthcare institutions collect large volumes of annotated medical images, yet regulatory frameworks — including the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union — impose strict constraints on cross-institutional data sharing. Negotiating data transfer agreements is costly, time-consuming, and in many cases legally impractical. FL resolves this tension by enabling institutions to contribute to a shared model without relinquishing data custody, making large-scale collaborative medical AI both legally feasible and technically practical.

## 3.6.2 Federated Learning in Medical Imaging

The adoption of FL in the medical domain has grown substantially since its introduction, driven by the recognition that many clinical datasets are too small at any single institution to train robust deep learning models, while the combined data across institutions would be sufficient [43]. FL directly addresses this data insufficiency problem without requiring physical data consolidation.

In medical imaging specifically, FL has been applied across a broad range of modalities and anatomical targets. Early work by Sheller *et al.* demonstrated FL for brain tumor segmentation from MRI scans across multiple institutions, achieving performance comparable to centralized training without sharing any patient data [43]. Subsequent studies have extended FL to chest radiograph classification, diabetic retinopathy screening from fundus photography, histopathology slide analysis, and cardiac structure segmentation from echocardiography, among others [44].

This breadth of applicability is a key motivation for the design of the proposed system. Rather than building a solution tied to a specific model architecture or anatomical region, the FL framework developed in this work is designed to be **model-agnostic** and **organ-agnostic**: any classification or segmentation backbone can be substituted as the local model, and the training data can comprise images from any body region or imaging modality. The federated coordination layer — covering client management, weight aggregation, and communication — remains unchanged regardless of the underlying clinical task.

### 3.6.3 The Federated Averaging Algorithm

The aggregation algorithm that underpins the proposed system is Federated Averaging (FedAvg), introduced by McMahan *et al.* [42] and the most widely adopted FL optimization method to date. FedAvg operates in iterative communication rounds. Let $K$ denote the number of participating clients, $\mathcal{D}_k$ the private dataset of client $k$, $n_k = |\mathcal{D}_k|$ its size, and $n = \sum_{k=1}^{K} n_k$ the total number of training samples across all clients. The global model at round $t$ is parameterized by weight vector $\mathbf{w}_t$. Each round consists of three phases:

**(1) Broadcast.** The server distributes the current global model weights to all $K$ clients:

$$\mathbf{w}_t \longrightarrow \{\text{client}_1, \ldots, \text{client}_K\}. \tag{3.53}$$

**(2) Local Training.** Each client $k$ independently minimizes its local empirical risk $\mathcal{L}_k$ over $E$ local epochs using a gradient- based optimizer, producing updated local weights:

$$\mathbf{w}_t^k = \mathbf{w}_t - \eta \nabla \mathcal{L}_k(\mathbf{w}_t), \tag{3.54}$$

where $\eta$ is the local learning rate and $\mathcal{L}_k$ is the task-specific loss function evaluated over the client's local dataset:

$$\mathcal{L}_k(\mathbf{w}) = \frac{1}{n_k} \sum_{(x_i, y_i) \in \mathcal{D}_k} \ell(f_{\mathbf{w}}(x_i), y_i). \tag{3.55}$$

**(3) Aggregation.** The server collects the locally updated weights from all clients and computes the new global model as a weighted average proportional to each client's dataset size:

$$\mathbf{w}_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} \mathbf{w}_t^k. \tag{3.56}$$

This process minimizes the global objective:

$$F(\mathbf{w}) = \sum_{k=1}^{K} \frac{n_k}{n} \mathcal{L}_k(\mathbf{w}). \tag{3.57}$$

Equation (3.56) assigns greater influence to clients with larger datasets, which is appropriate in the medical setting where institutional dataset sizes vary considerably. Clients contributing more annotated images should exert proportionally greater influence on the global model.

### 3.6.4 Key Parameters of a Federated Learning System

Configuring a federated learning system involves a distinct set of parameters beyond those of conventional deep learning. These parameters govern the coordination between server and clients and directly affect convergence speed, model quality, communication cost, and privacy. A clear understanding of each parameter is essential for adapting the system to different clinical tasks. Table 3.7 provides an overview of the critical parameters and their implications.

Table 3.7: Key Federated Learning Parameters and Their Clinical Implications

| Parameter | Definition | Implication |
|---|---|---|
| Number of clients ($K$) | Total participating institutions | More clients improve data diversity and model generalization but increase coordination overhead. |
| Communication rounds ($T$) | Number of server–client training cycles | More rounds improve accuracy at the cost of higher communication volume. Convergence is typically monitored to select $T$. |
| Client fraction ($C$) | Proportion of clients selected per round ($C \in (0, 1]$) | $C = 1.0$ uses all clients every round (synchronous FL). $C < 1.0$ enables asynchronous, scalable training. |
| Local epochs ($E$) | Training epochs each client performs before returning weights | Higher $E$ reduces communication rounds but risks *client drift* — divergence of local models due to non-IID data [45]. |
| Local batch size ($B$) | Mini-batch size for local gradient computation | Affects local convergence speed, memory footprint, and gradient privacy (smaller batches increase reconstruction attack risk). |
| Local learning rate ($\eta$) | Step size for local gradient updates | Must be tuned per model architecture; too large a value exacerbates client drift in non-IID settings. |
| Aggregation strategy | Algorithm used to combine client updates (e.g., FedAvg, FedProx, FedNova) | Choice depends on data heterogeneity; FedAvg is the standard baseline; FedProx adds a proximal term to limit client drift. |
| Data partitioning scheme | How training data is divided across clients (IID vs. non-IID) | Non-IID partitions (realistic in healthcare) slow convergence and reduce final accuracy compared to IID partitions. |
| Minimum clients for aggregation | Minimum number of client updates required before the server aggregates | Prevents the global model from being updated on insufficient information; critical for stability. |

Among these, the number of local epochs $E$ and the aggregation strategy warrant particular attention in the medical imaging context. Increasing $E$ reduces the number of costly communication rounds, which is beneficial in healthcare networks where bandwidth and uptime may be limited. However, when client datasets are statistically heterogeneous — a common condition across hospitals using different scanners, populations, or labeling protocols — high $E$ causes the local models to overfit to their local distributions and diverge from the global optimum, a phenomenon known as *client drift* [45]. The selection of $E$ therefore requires careful empirical tuning for each deployment scenario.

### 3.6.5 Privacy and Security

The primary privacy guarantee of federated learning is that raw data never leaves its source institution. The server observes only model weight tensors, which are orders of magnitude smaller than the original imaging data and contain no explicit patient identifiers. This stands in contrast to centralized approaches, where a single data breach exposes the entire training corpus. Table 3.8 summarizes the key differences.

Table 3.8: Privacy and Security: Centralized vs. Federated Learning

| **Property** | **Centralized** | **Federated** |
|---|---|---|
| Raw data leaves institution | Yes | No |
| Single point of data breach | Yes | No |
| HIPAA / GDPR compliance | Difficult | Easier |
| Shared artifact | Full dataset | Model weights |
| Supports differential privacy | N/A | Yes |
| Resilient to server breach | No | Partial |

It is important to acknowledge that gradient sharing is not unconditionally private. Gradient inversion attacks [46] have demonstrated that an adversarial server can reconstruct approximate training images from shared weight updates, particularly at small batch sizes. Two complementary mitigations are incorporated into the design of the proposed system:

**Batch size selection.** Using a sufficiently large mini-batch (e.g., $B \geq 32$) aggregates gradient information across many samples, significantly reducing the fidelity of any reconstruction attempt.

**Differential Privacy (DP).** For deployments requiring formal privacy guarantees, the system supports $(\varepsilon, \delta)$-DP [47] by adding calibrated Gaussian noise to client updates before transmission:

$$\tilde{\mathbf{w}}_t^k = \mathbf{w}_t^k + \mathcal{N}\Big(0,\ \sigma^2 C^2 \mathbf{I}\Big), \tag{3.58}$$

where $C$ is the gradient clipping threshold and $\sigma$ is the noise multiplier. The privacy budget $\varepsilon$ quantifies the worst-case privacy loss: smaller $\varepsilon$ implies stronger privacy at the cost of model utility.

### 3.6.6 What Makes the Proposed System Distinctive

While federated learning has been applied in various medical imaging studies, the majority of existing implementations are designed around a fixed model architecture and a specific anatomical target, limiting their transferability to new clinical tasks. The proposed framework differentiates itself through the following design principles:

**Model Agnosticism.** The federated coordination layer — encompassing client management, weight serialization, aggregation, and communication — is fully decoupled from the local model definition. Any differentiable model that can be expressed as a parameter vector (e.g., CNNs for classification, U-Net variants for segmentation, Vision Transformers) can be substituted as the local model without modifying the federated infrastructure. This makes the system immediately applicable to new imaging tasks simply by swapping the model definition on the client side.

**Staged Data Integration.** The system incorporates a staged data feeding protocol that progressively increases each client's active training subset across federated rounds. This design reflects the clinical reality that expert-annotated medical imaging datasets grow incrementally over time as radiologists label new cases. Rather than requiring a fully annotated dataset before training begins, the system accommodates continuous data growth, enabling earlier model deployment and iterative improvement as annotation progresses.

**Containerized Reproducibility.** Each participating node is deployed as an isolated Docker container, ensuring that the federated environment is fully reproducible across different hardware configurations and operating systems. The container-based architecture also simplifies scaling: adding a new institution requires only the deployment of an additional client container with the appropriate data volume mount, with no changes to the server or existing clients.

**Framework Choice.** The system is implemented using the Flower (flwr) framework [48], selected for its support of multiple ML backends (PyTorch, TensorFlow, JAX), its clean abstraction of the client-server protocol, and its active development for healthcare FL research. The backend-agnostic design of Flower further reinforces the model-agnostic objective of the proposed system.

Collectively, these properties position the proposed FL framework as a general-purpose infrastructure for privacy-preserving collaborative medical image analysis, extensible to a wide range of clinical specialties and diagnostic tasks without architectural redesign.

## 3.7 Retrieval-Augmented Generation (RAG) Model

The proposed platform incorporates a Retrieval-Augmented Generation (RAG) model to provide an intelligent medical question-answering assistant capable of responding to user inquiries using trusted medical knowledge sources. Unlike conventional large language models that rely solely on their internal parametric knowledge, the RAG approach augments the language model with information retrieved from an external knowledge base at inference time. This enables the system to generate responses that are more accurate, context-aware, explainable, and grounded in domain-specific medical information.

The primary objective of integrating the RAG model within the proposed framework is to provide users with a reliable conversational assistant capable of answering questions related to chest diseases, lung cancer, medical imaging, diagnostic procedures, and other healthcare-related topics. By retrieving relevant information from curated medical documents before response generation, the system significantly reduces hallucinations and improves the factual consistency of generated answers.

### 3.7.1 Motivation

Although modern Large Language Models (LLMs) demonstrate remarkable natural language understanding and generation capabilities, they suffer from several limitations when deployed in healthcare environments. Most notably, they may generate inaccurate information, provide outdated medical recommendations, or produce responses unsupported by clinical evidence.

In medical applications, such limitations can negatively affect user trust and potentially lead to harmful outcomes. Consequently, relying solely on the internal knowledge of a pretrained language model is insufficient for healthcare-oriented systems.

Retrieval-Augmented Generation addresses these challenges by combining information retrieval techniques with generative language models. Instead of answering questions exclusively from learned parameters, the system first retrieves relevant medical knowledge from an external document repository and then uses this information to guide response generation. This approach ensures that answers remain grounded in verified sources while maintaining the conversational capabilities of modern language models.

### 3.7.2 Overview of the Proposed RAG Architecture

The implemented RAG system consists of four major components:

1. Medical Knowledge Base
2. FAISS Vector Database
3. Ollama Large Language Model
4. LangChain Orchestration Layer

The interaction between these components is illustrated in Figure 3.28.

The architecture begins when a user submits a natural language query through the platform interface. The query is forwarded to the FastAPI model backend, where LangChain orchestrates the retrieval and generation pipeline. Relevant document chunks

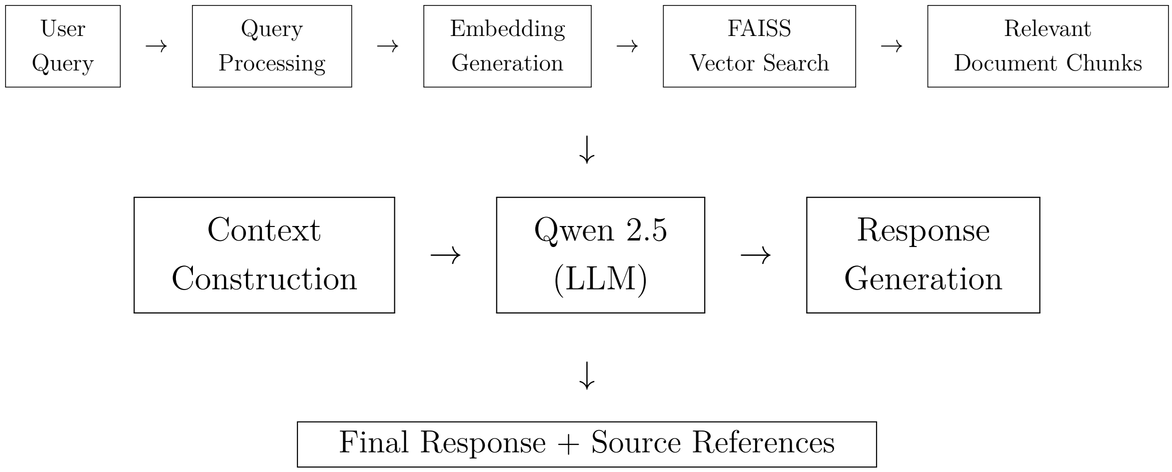



Figure 3.28: Overall architecture of the proposed Retrieval-Augmented Generation (RAG) model. The user query is transformed into vector embeddings, relevant medical knowledge is retrieved from the FAISS vector database, and the retrieved context is supplied to the Qwen 2.5 language model to generate a grounded and source-aware response.

are retrieved from the FAISS vector database and supplied as contextual information to the language model. The generated answer is then returned to the user along with references to the retrieved sources.

### 3.7.3 Knowledge Base Construction

The foundation of the RAG system is a curated collection of medical documents covering topics related to:

- Chest diseases
- Pneumonia
- Tuberculosis
- COVID-19
- Lung cancer
- Medical imaging
- Diagnostic procedures
- Healthcare guidelines

These documents are converted into machine-processable representations before being indexed. Since language models cannot efficiently search through large collections of raw documents during inference, each document is divided into smaller semantic chunks.

Chunking improves retrieval accuracy by allowing the system to retrieve only the most relevant portions of a document rather than entire files.

For each generated chunk, metadata is preserved, including:

- Document name
- Page number

- Source identifier
- Document category

This metadata later enables source attribution and transparency within generated responses.

### 3.7.4 Vector Embedding Generation

After document chunking, each text segment is transformed into a dense numerical representation known as an embedding vector.

Embeddings encode semantic meaning into a high-dimensional vector space, allowing documents with similar meanings to be located near one another mathematically.

Given a document chunk (*d_i*), an embedding function (E(·)) produces a vector representation $v_i = E(d_i)$ where $v_i \in \mathbb{R}^n$ and $(n)$ represents the embedding dimensionality.

These embeddings capture semantic relationships between medical concepts and enable similarity-based retrieval during inference.

### 3.7.5 FAISS Vector Database

To support efficient retrieval over thousands of embedded document chunks, the system employs Facebook AI Similarity Search (FAISS).

FAISS is a high-performance vector database specifically designed for similarity search within large embedding collections. Instead of performing keyword matching, FAISS retrieves documents based on semantic similarity.

All generated embeddings are stored within the FAISS index.

During inference, a user query is embedded using the same embedding model: $q = E(x)$ where $(x)$ denotes the user's query.

The FAISS index then searches for the (k) nearest vectors according to cosine similarity and returns the most relevant document chunks.

This retrieval process allows the system to locate contextually relevant medical information even when the exact wording differs from that found in the original documents.

### 3.7.6 Large Language Model Integration

The generative component of the RAG pipeline utilizes the Qwen 2.5 language model deployed locally through Ollama.

Running the model locally provides several advantages:

- Reduced operational cost
- Improved privacy

- Low latency
- Full deployment control
- Offline inference capability

The model configuration used in the implementation is summarized in Table 3.9.

Table 3.9: Configuration of the deployed language model.

| **Parameter** | **Value** |
|---|---|
| LLM Engine | Ollama |
| Model | qwen2.5:3b |
| Temperature | 0.5 |
| Max Tokens | 500 |
| Keep Alive | Infinite (-1) |

After retrieval, the selected document chunks are appended to the prompt supplied to the language model. Consequently, generated responses become grounded in retrieved medical knowledge rather than relying exclusively on pretrained parameters.

### 3.7.7 Query Processing Workflow

The complete RAG workflow implemented within the system is illustrated in Figure 3.29.

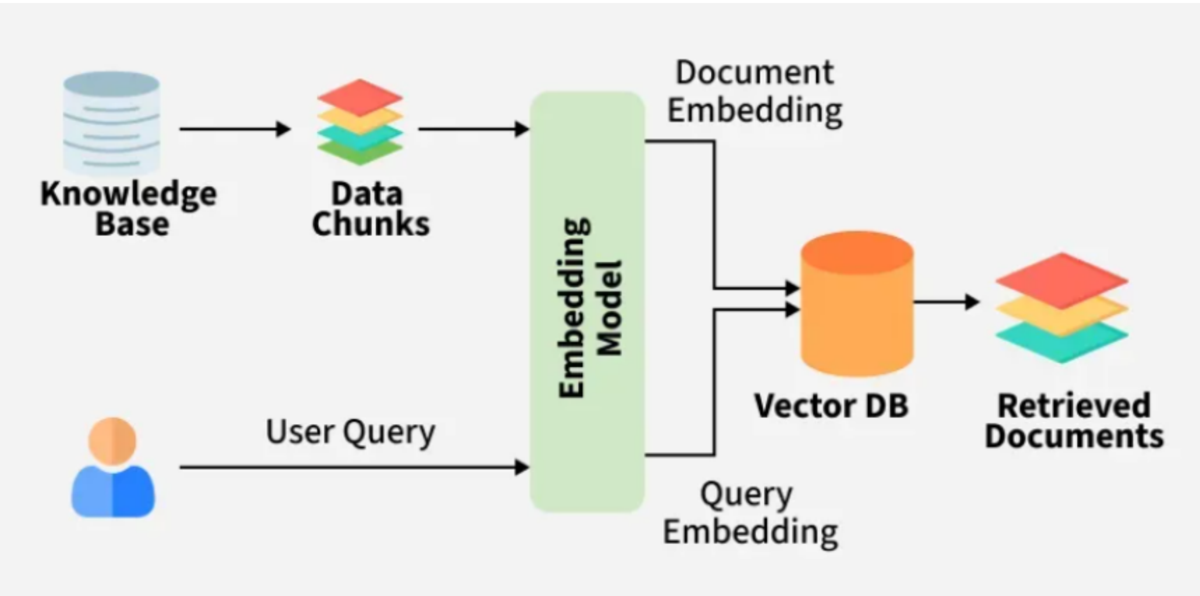



Figure 3.29: Query processing workflow of the proposed RAG system.

The workflow consists of the following steps:

1. User submits a medical question.
2. Query is forwarded to the FastAPI backend.

3. Query embedding is generated.
4. FAISS retrieves the most relevant document chunks.
5. Retrieved context is assembled.
6. Context and user query are passed to Qwen 2.5.
7. The language model generates a grounded response.
8. Source metadata is attached.
9. Final response is returned to the user.

### 3.7.8 Source Attribution and Explainability

One of the most important characteristics of the proposed RAG implementation is transparency.

Rather than generating answers without justification, the system preserves metadata associated with retrieved documents and returns source references alongside generated responses. This allows users to trace the origin of information and verify the supporting evidence used during answer generation. The underlying report emphasizes that retrieved documents and page references are preserved to improve transparency and user trust.

Providing source-aware responses is particularly important in healthcare applications where explainability and evidence-based recommendations are essential.

### 3.7.9 Advantages of the Proposed Approach

The implemented Retrieval-Augmented Generation model provides several advantages:

- Reduces hallucinations compared with standalone LLMs.
- Produces responses grounded in medical knowledge sources.
- Supports explainable and source-aware answers.
- Enables efficient retrieval through FAISS vector search.
- Preserves privacy through local deployment using Ollama.
- Can be continuously expanded by adding new medical documents.
- Integrates seamlessly with the overall X-Fed Neuroscan architecture.

Consequently, the RAG subsystem serves as an intelligent medical assistant that complements the image analysis capabilities of the platform by providing users with reliable, contextualized, and explainable healthcare information.

## 3.8 Platform Architecture

The system is a comprehensive web-based healthcare management system developed to streamline medical services and enhance diagnostic workflows through an integrated digital platform. The system provides a centralized environment where patients, doctors, administrators, and healthcare institutions can interact efficiently through role-based dashboards and specialized services. The platform allows patients to create accounts, manage their profiles, book appointments, track medical records, and communicate through an intelligent healthcare assistant. Doctors can manage appointments, analyze medical images, review patient information, and assign diagnostic results, while administrators oversee user management, system operations, and hospital coordination. The system also supports healthcare institutions participating in collaborative AI model development through federated learning mechanisms. Built using a modern multi-tier architecture, XFed NeuroScan combines a responsive web interface, secure backend services, cloud-based storage, and artificial intelligence modules to deliver a seamless user experience. The application emphasizes security, scalability, and usability by implementing role-based access control, secure authentication, real-time communication, and centralized healthcare workflows. By integrating healthcare management features with intelligent diagnostic services, XFed NeuroScan provides a unified platform that improves operational efficiency, supports medical decision-making, and enhances the overall experience for both healthcare providers and patients.

The platform is composed of four tightly integrated repositories:

- **Frontend:** React 18 single-pages application serving doctors, patients, and administrators.
- **Backend:** Spring Boot 3.5 REST and WebSocket API acting as the central system of record, handling authentication, business logic, user management, and communication between services.
- **Model Backend:** FastAPI-based AI service responsible for medical image analysis, chest disease classification, lung cancer detection, explainable AI heatmap generation, and a Retrieval-Augmented Generation (RAG) medical chatbot for intelligent question answering.
- **Federated-Learning Server:** Flower-based federated learning coordinator managing hospital registration, client monitoring, distributed training rounds, and global model aggregation.

Together, these components provide a complete healthcare platform that combines medical workflow management, AI-assisted diagnosis, intelligent medical assistance through RAG, and privacy-preserving federated learning across multiple healthcare institutions.

### 3.8.1 End-to-End System Architecture

The system follows a microservices architecture. Four independent services communicate over well-defined APIs. The frontend is the only entry point for human users; all backend services are internal.

#### Service Map

Table 3.11 lists all the different components and services of the proposed system.

Table 3.10: Summary of the proposed X-Fed NeuroScan framework.

| Dimension | Detail |
|---|---|
| Primary Domain | Medical imaging and AI-based diagnostic systems |
| Key Innovation | Federated learning for privacy-preserving model improvement and collaborative model training across institutions |
| User Roles | Patient, Doctor, and Administrator |
| Imaging Modalities | Chest X-ray classification (COVID-19, Tuberculosis, Pneumonia, and Normal) and lung cancer detection from CT scans |
| AI Explainability | Grad-CAM heatmaps for chest disease predictions and lesion localization visualizations for lung cancer detection |
| Privacy Model | Raw medical images never leave hospital premises; only model parameters are exchanged during federated learning |

Table 3.11: Technological components of the proposed system architecture.

| Service | Technology | Port | Primary Role |
|---|---|---|---|
| Frontend | React 18, Vite, and TypeScript | 3000 | User interface for patients, doctors, and administrators |
| Backend | Spring Boot 3.5 and Java 21 | 8080 | Authentication, business logic implementation, and API gateway services |
| Model Backend | FastAPI and Python 3 | 8000 | Machine learning inference services and retrieval-augmented generation (RAG) chatbot functionality |
| Federated Learning Server | FastAPI, Flower, and Python 3 | 8001 / 50051 | Federated learning coordination, client management, and hospital registry services |

### External Infrastructure

During system implementation, we have also relied on external components to not reinvent the wheel and leverage already existing powerful and free infrastructure, those components are listed in Table 3.12

Table 3.12: Supporting infrastructure and external services used by the proposed system.

| **Component** | **Role** |
|---|---|
| Cloudflare R2 (S3-Compatible Storage) | Stores uploaded medical images, generated Grad-CAM heatmaps, lung cancer detection result overlays, and other imaging-related artifacts. |
| PostgreSQL | Primary relational database used to manage users, medical images, appointments, hospital information, federated learning participants, and other persistent application data. |
| Redis | Provides high-speed storage for refresh tokens and caching of presigned URLs to improve system performance and reduce unnecessary storage requests. |
| Ollama (qwen2.5:3b) | Local large language model (LLM) responsible for generating responses within the retrieval-augmented generation (RAG) chatbot component. |
| Gmail SMTP | Handles transactional email services including password reset requests, user invitations, account notifications, and hospital package communications. |
| FAISS | Vector similarity search engine used to index and retrieve relevant documents for the retrieval-augmented generation (RAG) pipeline. |

### Architecture Visualization

To provide a clearer understanding of the overall system structure, Figures 3.30 and 3.31 present visual representations of the proposed *X-Fed NeuroScan* framework. These diagrams complement the architectural descriptions presented throughout this chapter by illustrating the interaction between system components, data flow pathways, and deployment services. Together, they provide a comprehensive view of both the logical architecture of the framework and the technological infrastructure used to support its operation.

## 3.8.2 Frontend

The frontend is a React 18 single-page application built with Vite 6 and TypeScript. It serves three user roles — Patient, Doctor, and Admin — through role-gated dashboards,

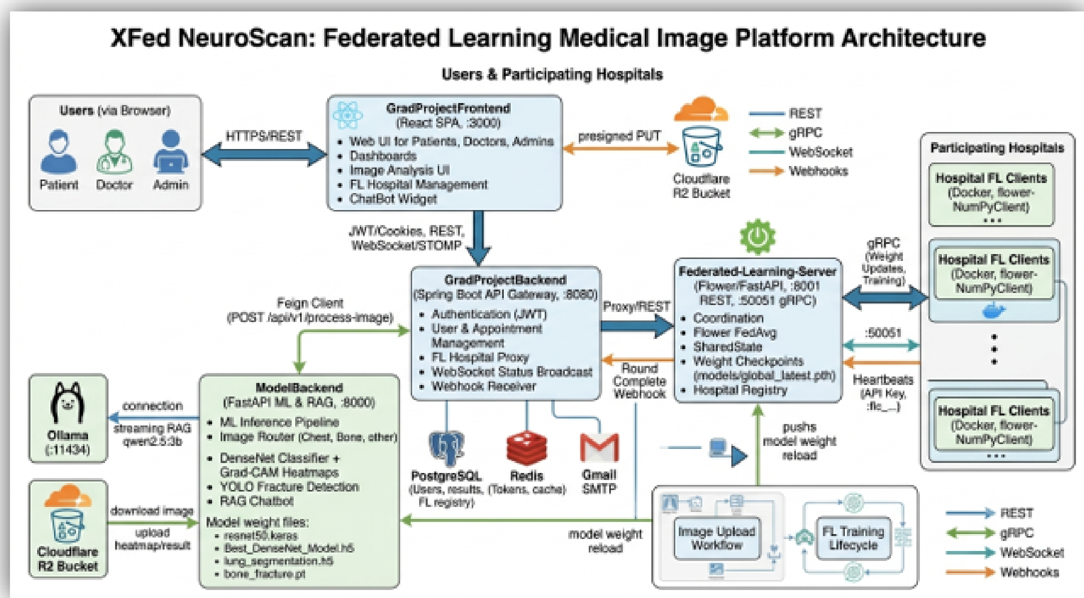



Figure 3.30: End-to-end logical architecture of the proposed X-Fed NeuroScan framework, illustrating the interaction between users, backend services, AI models, explainability components, federated learning infrastructure, and data storage systems.

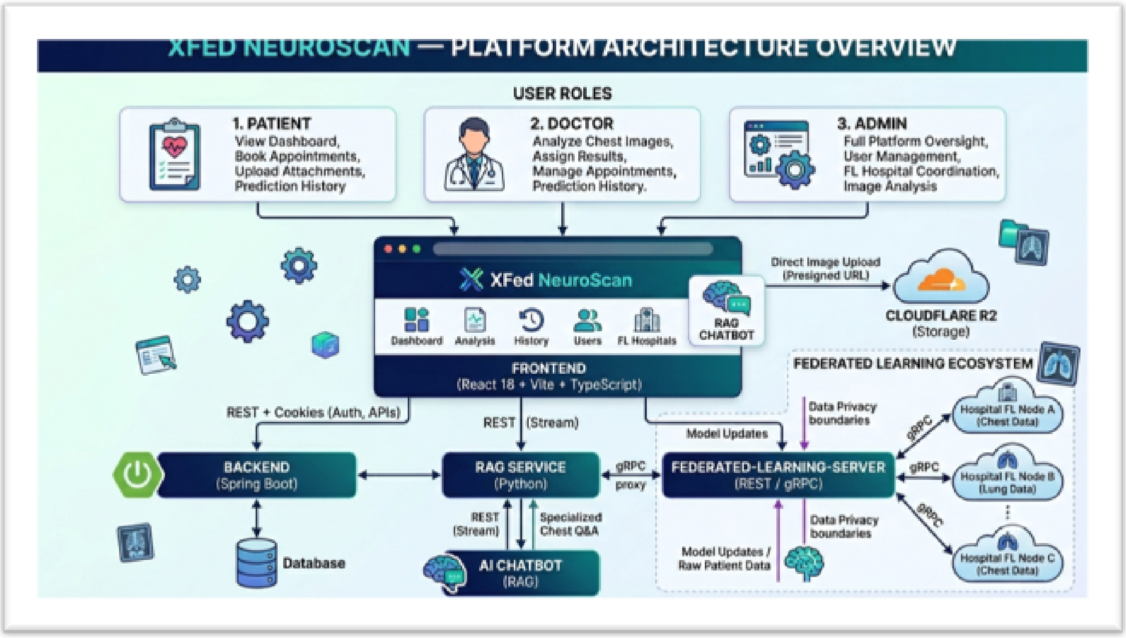



Figure 3.31: Deployment architecture of the proposed system showing the frontend application, backend services, model-serving infrastructure, federated learning server, databases, storage components, and external integrations.

and is the sole human interface into the platform.

## Technology Stack

Different technologies were used during the development and implementation of the frontend components of the proposed system. They have been listed in Table 3.13.

Table 3.13: Frontend technologies used in the implementation of the proposed system.

| **Category** | **Technology** |
|---|---|
| Framework | React 18 |
| Build Tool | Vite 6 with `@vitejs/plugin-react-swc` |
| Programming Language | TypeScript 5 (Strict Mode) |
| Routing | React Router 7 |
| Styling | Tailwind CSS v4 |
| UI Primitives | Radix UI with shadcn-style wrapper components |
| Animations | Framer Motion (Motion) |
| Charts and Data Visualization | Recharts |
| Real-Time Communication | `@stomp/stompjs` and `sockjs-client` |
| Form Management | react-hook-form |

## User Roles & Capabilities

The system is designed as a role-based healthcare to ensure that each type of user interacts with the platform according to their responsibilities, permissions, and level of access. The system defines three main roles: Patient, Doctor, and Administrator, each serving a distinct purpose within the healthcare workflow.

Patients represent the end users of the system. Their role is limited to personal healthcare management, including registering accounts, booking appointments, viewing medical history, and accessing their own diagnostic results. This restricted access ensures that patients can interact with their healthcare data securely without modifying clinical or system-level information.

Doctors have a more advanced role focused on medical operations. They can upload and analyze medical images, review AI-generated predictions, interpret results, and assign diagnoses to patients. In addition, doctors manage appointments and clinical interactions with patients. This role requires broader access because it directly involves medical decision-making and the use of AI diagnostic tools. Administrators hold the highest level of control within the system. They are responsible for managing users, including creating and inviting doctors, monitoring system activity, and overseeing hospital and federated learning operations. Administrators also handle system configuration and ensure that all services operate securely and efficiently.

The separation of roles is essential for maintaining security, privacy, and operational clarity. By limiting access based on responsibilities, the system prevents unauthorized actions, reduces the risk of data misuse, and ensures that each user interacts only with the features relevant to their function. This role-based design also improves usability by providing a tailored interface for each type of user, making the system more efficient and easier to navigate.

Table 3.14: User roles and capabilities within the proposed X-Fed NeuroScan system.

| **Role** | **Capabilities** |
|---|---|
| Patient | View personalized dashboard statistics, book appointments with doctors, upload medical image attachments, access diagnostic results, and review prediction history. |
| Doctor | Analyze chest medical images using the integrated AI models, assign diagnostic results to patients, manage appointment schedules, review patient records, and access prediction history. |
| Administrator | Perform full platform administration including user management, federated learning hospital registration, hospital selection for federated training rounds, monitoring system activity, and conducting image analysis through the integrated diagnostic framework. |

## Key Pages and Routing

Frontend is structured as a single-page application with role-based routing to ensure secure and organized access to system features. Each user type is directed to specific pages depending on their permissions and responsibilities within the platform.

Public users can access the landing page and authentication-related pages, including login, signup, password recovery, and invite-based account activation. These pages are only available to guests who are not yet authenticated.

Authenticated users are redirected to role-specific dashboards after login. Patients have access to their personal dashboard and medical history, where they can view appointments and diagnostic results. Doctors are provided with a dedicated dashboard along with access to image analysis tools and patient-related medical data. Administrators have the highest level of access, allowing them to manage users, monitor system activity, and control federated learning operations.

In addition to dashboards, administrators also access specialized modules for user management and federated learning configuration, including hospital registration and training round setup. The image analysis page is restricted to doctors and administrators due to its clinical importance and use of AI-based diagnostic tools.

Overall, the routing structure enforces strict role-based access control, ensuring that each user interacts only with the features relevant to their responsibilities while maintaining system security and usability.

Table 3.15: Access control policy for system pages and administrative modules.

| **Page** | **Access** |
|---|---|
| Landing Page | Publicly accessible to all users, including unauthenticated visitors. |
| Authentication Pages | Accessible only to guest users who have not yet authenticated. |
| Password Recovery | Accessible only to guest users initiating the password reset process. |
| Invite Activation | Accessible only through a valid invitation token provided to the invited user. |
| Image Analysis | Accessible to users with Administrator or Doctor privileges. |
| Prediction History | Accessible to Administrators, Doctors, and Patients according to their authorization level. |
| Administrator Dashboard | Accessible exclusively to users with the Administrator role. |
| Doctor Dashboard | Accessible exclusively to users with the Doctor role. |
| Patient Dashboard | Accessible exclusively to users with the Patient role. |
| User Management | Accessible exclusively to Administrators for managing user accounts and permissions. |
| Federated Learning Hospital Management | Accessible exclusively to Administrators for registering, updating, and monitoring participating hospitals. |
| Federated Learning Training Round Setup | Accessible exclusively to Administrators for configuring and launching federated learning training rounds. |

## Core Features

**Authentication & Session Management** The frontend implements a full authentication lifecycle using JWT access tokens are stored in memory (not localStorage); refresh tokens are stored in HttpOnly cookies. On app load, the context attempts a silent token refresh to restore sessions without requiring re-login. GuestRoute prevents authenticated users from visiting auth. pages; Require Dashboard Role guards all dashboard routes.

**Medical Image Analysis** The image analysis flow uses a presigned-URL pattern to avoid routing large files through the backend:

- Step 1: Frontend requests a presigned R2 PUT URL from Spring Backend
- Step 2: Frontend uploads the image directly to Cloudflare R2
- Step 3: Frontend calls the api who is responsible for image-processing with the R2 key
- Step 4: Backend triggers Model inference and returns prediction, confidence, and heatmap URL

**Federated Learning Management** Admins manage FL hospitals through two dedicated pages. FLHospitalsPage shows all registered hospital nodes with live ONLINE/OFFLINE status via WebSocket. FL Select Hospitals Page lets admins choose which hospitals participate in the next training round. Registration generates an API key displayed once via ApiKeySuccessDialog.

**AI Chatbot (RAG)** A floating chat widget is available on every page. Responses are streamed from the RAG service. The chatbot gracefully degrades to keyword-based mock answers if the RAG API is unreachable.

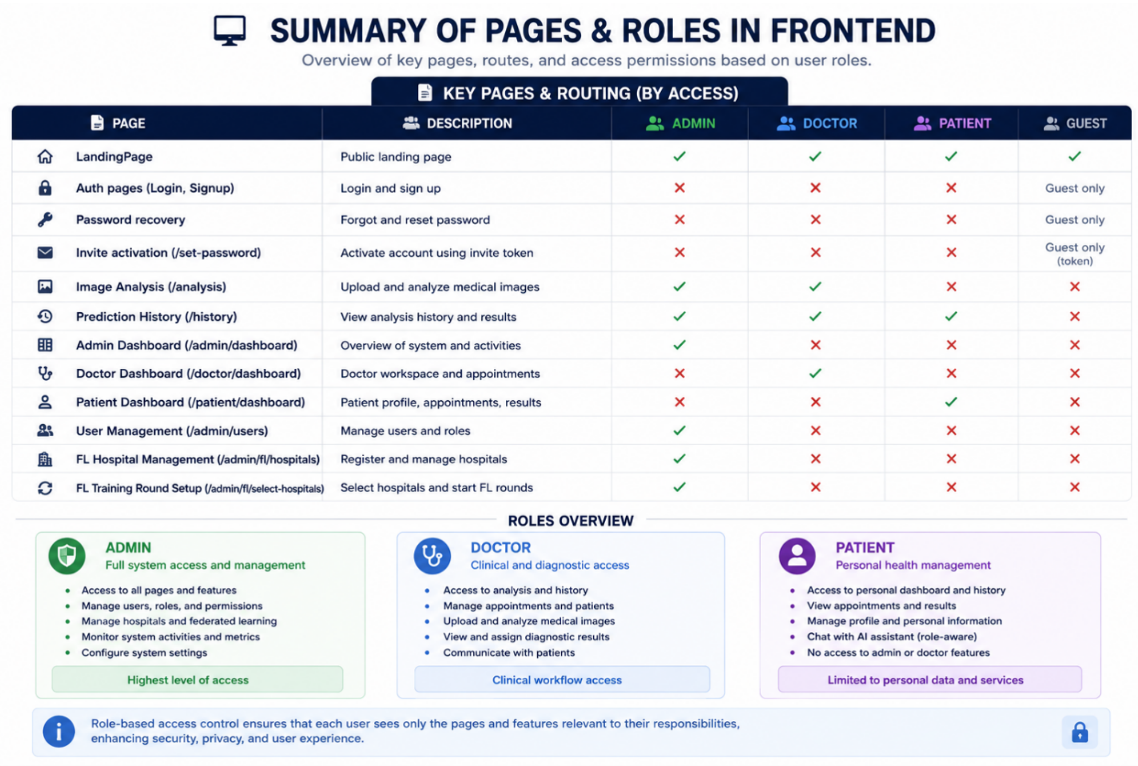



Figure 3.32: A visualization of the roles and pages implemented in the web application.

## 3.8.3 Centralized Backend (Spring Boot)

Centralized backend is a Spring Boot 3.5 application written in Java 21. It serves as the central API gateway and system of record for authentication, user management, medical image orchestration, appointments, federated learning coordination, and real-time WebSocket broadcasts.

## Technology Stack

Various technologies were utilized in the implementation of the backend of the proposed end-to-end system. All of these technologies are listed in Table 3.16.

Table 3.16: Backend technologies used in the implementation of the proposed system.

| **Category** | **Technology** |
|---|---|
| Framework | Spring Boot 3.5.4 |
| Programming Language | Java 21 |
| Build Tool | Maven |
| Security | Spring Security 6 with JSON Web Token (JWT) authentication using JJWT 0.12.3 |
| Persistence Layer | Spring Data JPA, Hibernate, and PostgreSQL |
| Cache and Session Management | Spring Data Redis |
| HTTP Client | Spring Cloud OpenFeign |
| Real-Time Communication | Spring WebSocket, STOMP, and SockJS |
| Object Storage Integration | AWS SDK v2 for S3-compatible storage (Cloudflare R2) |
| Email Services | Spring Mail with Gmail SMTP |
| API Documentation | SpringDoc OpenAPI (Swagger UI) |
| Object Mapping and Code Generation | MapStruct and Lombok |

## Security & Authentication

Security is implemented with JWT access tokens and HttpOnly refresh-token cookies stored in Redis:

- Access Tokens: short-lived JWTs returned in the login response body
- Refresh Tokens: longer-lived tokens stored in Redis (key: email:deviceId) and set as HttpOnly cookies
- Logout: access tokens blacklisted when still valid; refresh token deleted from Redis
- FL API keys: encrypted at rest using AES via EncryptionUtils
- FL webhook endpoints: authenticated with a shared secret (not JWT)
- Soft delete: deleted users cannot log in

In the current system design, **login is restricted to patients only**. Doctor and Admin accounts are not self-authenticated through public signup; instead, they are created and managed exclusively by an administrator and activated through an invitation-based workflow. This ensures strict control over privileged roles in the healthcare system.

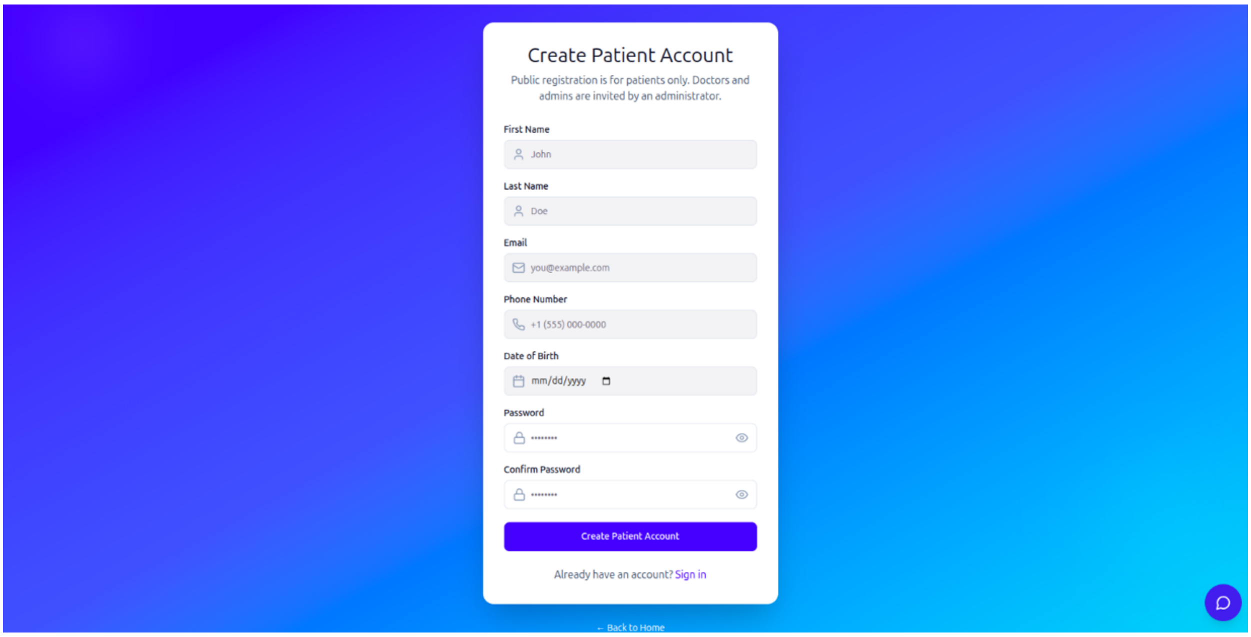



Figure 3.33: An image of the Create Account interaction from the web application.

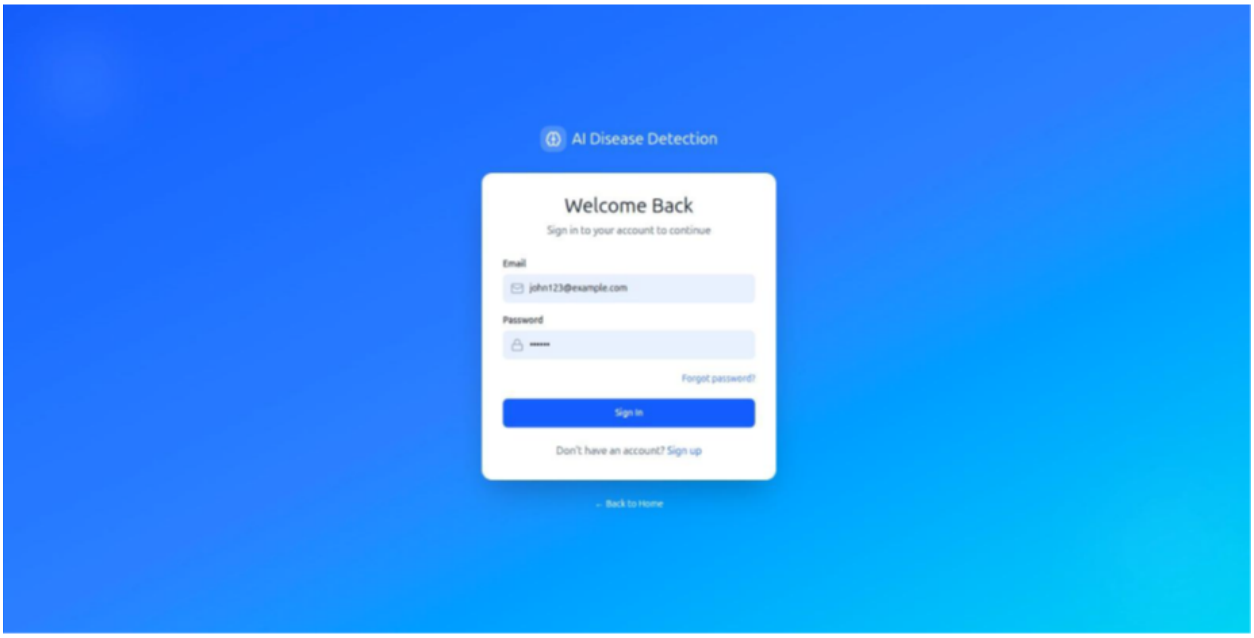



Figure 3.34: An image of the Login interaction from the web application.

After login, user information is stored in PostgreSQL, but sensitive session data is handled separately for security. Passwords are never stored in plain text; instead, they are hashed using **BCrypt** before being saved in the database, making them irreversible and protected even if the database is compromised.

During authentication, the input password is hashed and compared with the stored hash. If valid, the system generates a short-lived **JWT access token** and a long-lived **refresh token**. The access token is used for API requests, while the refresh token is stored securely in **Redis** and also set as an **HttpOnly cookie**.

When the user logs out, the access token is blacklisted and the refresh token is removed from Redis, fully terminating the session. This design ensures secure password storage, stateless authentication, and strong session protection.

In the system, only Admin users have the authority to create or manage Doctor and other Admin accounts, ensuring strict role control and preventing unauthorized privilege escalation.

New privileged users (Doctors or Admins) cannot register through the public signup process. Instead, an Admin must explicitly create or invite them through the administrative interface. This invitation-based flow allows the system to maintain tight control over sensitive roles that have access to clinical data, AI analysis tools, and system-wide management features.

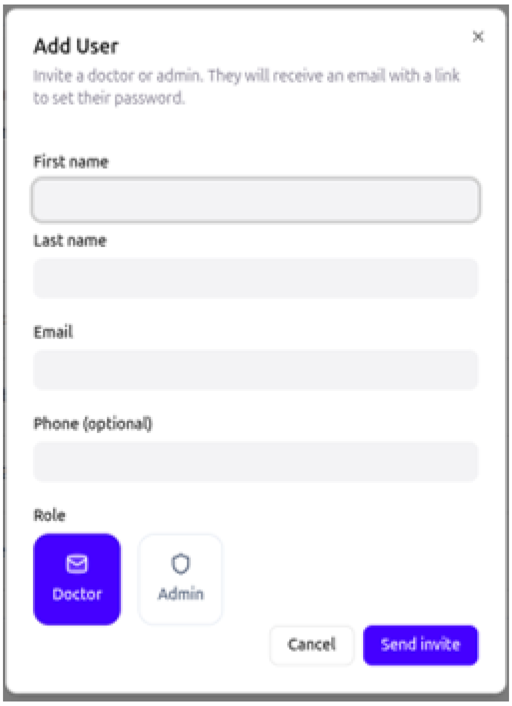



Figure 3.35: An image of the Add User interaction from the web application.

After an Admin creates a new Doctor or Admin account, the account does not become active immediately. Instead, it enters a pending state until it is explicitly approved. This approval step ensures that no privileged user gains access to the system without proper verification and administrative confirmation.

During this pending phase, the user record is already stored in the database, but the account remains inactive, meaning the user cannot log in or access any system features. Once the Admin approves the account, its status is updated to active, and the user is then able to authenticate and use the system according to their assigned role.

In the user management interface, the Admin has full control over viewing and organizing all system users through a role-based filtering system. This allows the Admin to efficiently manage large numbers of accounts by categorizing users according to their current role and status.

In addition to account creation and approval, the Admin also has full control over existing privileged accounts, as shown in 3.36. This includes the ability to disable or delete accounts at any time. Disabling an account temporarily prevents the user from logging in without removing their data, while deletion marks the account as removed (soft delete), ensuring it is excluded from authentication and system operations but still preserved for audit or historical purposes.

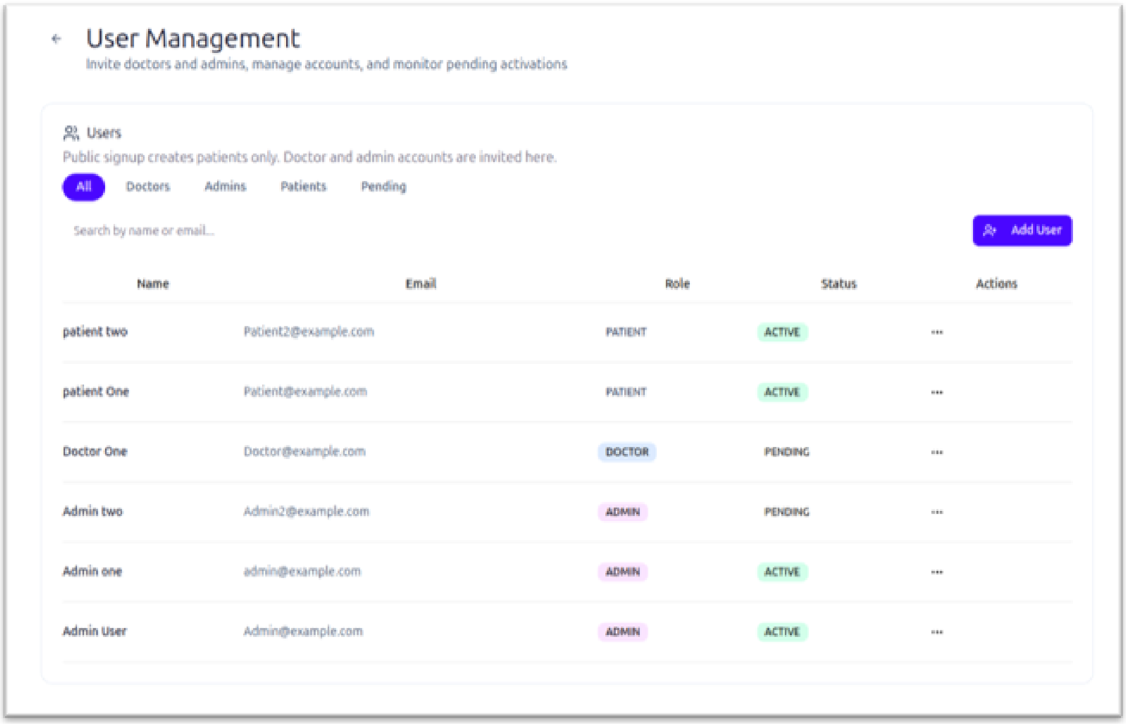



Figure 3.36: An image of the User Management tab from the web application.

When a user selects "Forgot Password", the system follows a secure multi-step flow to ensure that only the legitimate account owner can reset the password.

First, the user enters their registered email address in the frontend. This request is sent to the Spring Boot backend, which validates whether the email exists in the system. If the user is found, Spring Boot generates a secure password reset token, typically a short-lived, random token linked to the user account.

This token is then stored temporarily (or associated with the user record) and sent to the user via email using Gmail SMTP. The email contains a secure reset link that includes the token as a parameter.

When the user clicks the link, they are redirected to the password reset page in the frontend. The new password is submitted along with the token, and Spring Boot validates the token to ensure it is still valid and has not expired or been used before.

If the token is valid, the system hashes the new password using BCrypt and updates it in the PostgreSQL database, replacing the old password hash. After successful reset, the token is invalidated to prevent reuse.

Overall, the forgot password process ensures security by combining email verification, time-limited tokens, secure transmission, and encrypted password storage.

So in summary User authentication in system uses a secure JWT-based system where only patients can self-register and log in, while doctors and admins are created by admins. Passwords are stored in the database using BCrypt hashing.

After successful login, the system issues a short-lived JWT access token and a long-lived refresh token stored in Redis as an HttpOnly cookie. The access token is used for requests, and the refresh token allows session renewal.

The system also supports a secure forgot password workflow, where a password reset token is generated, sent to the user via email, and validated before allowing password updates. The new password is then hashed using BCrypt and stored in the database.

On logout, tokens are invalidated by blacklisting the access token and removing the refresh token. Role-based access control and soft delete ensure only authorized and active users can access the system.

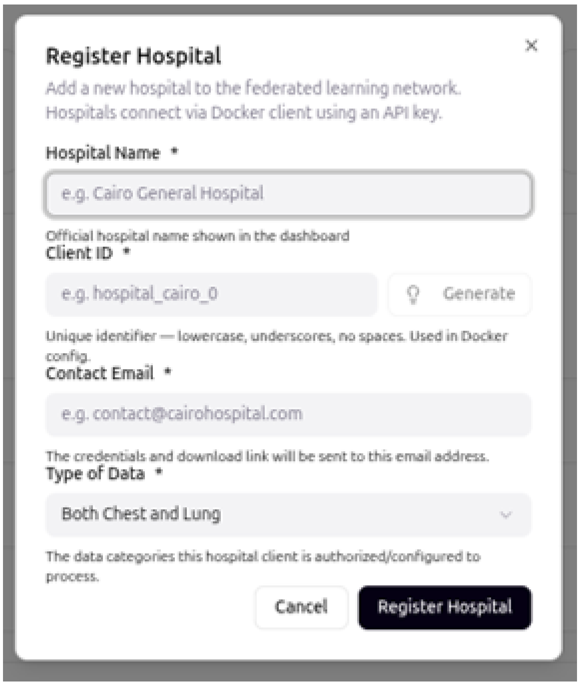



Figure 3.37: An image of the Add Hospital interaction from the web application.

Backend handles the full lifecycle of registering new hospitals in the federated learning system. When an Admin submits a hospital, Spring Boot validates the request, ensures

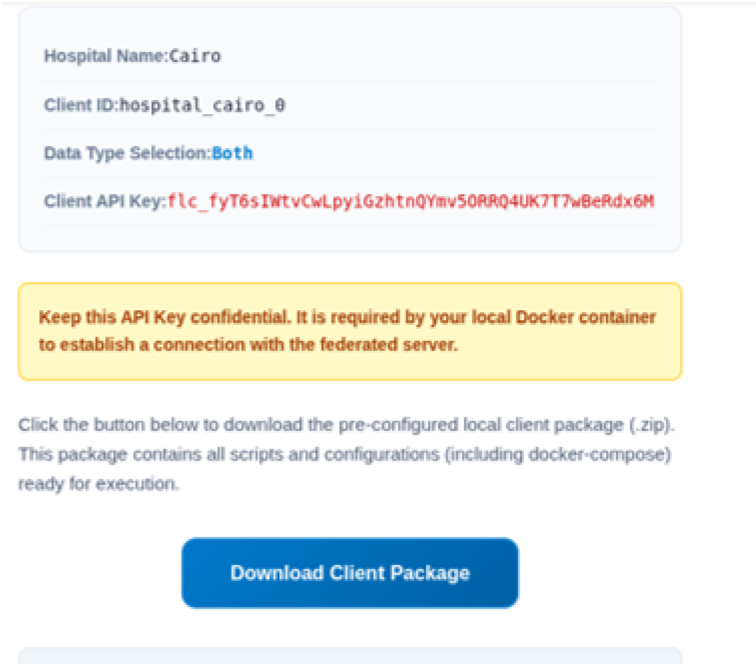



Figure 3.38: An image of the client hospital getting a confidential API key interaction from the web application.

only Admins can perform the action, and then generates a unique FL client ID and encrypted API key using AES before storing them in PostgreSQL. The hospital is initially saved with a PENDING or INACTIVE status.

After that, Spring Boot registers the hospital with the Federated Learning Server by sending its credentials and metadata, allowing it to join the federated learning network. It may also provide a Docker-based client package for hospital deployment.

Spring Boot receives and synchronizes key data from the Federated Learning (FL) Server, especially hospital online/offline status, which is determined by heartbeat signals sent from hospital clients to the FL Server. Spring Boot does not calculate this status itself; instead, it periodically fetches updated client information from the FL Server.

Overall, Spring Boot acts as a bridge that collects, stores, and distributes FL Server status updates, giving administrators a live view of the federated learning network.

When a new hospital is registered in the federated learning system, Spring Boot not only creates and stores the hospital record, but also sends an automated email notification via Gmail SMTP as part of the onboarding process.

After the Admin submits and approves the hospital registration, Spring Boot generates the hospital's federated learning credentials (such as the FL client ID and encrypted API key) and securely stores them in the database. Once the registration is successfully completed, Spring Boot triggers the email service to notify the hospital.

This email is sent using Gmail SMTP integration and typically contains important onboarding information such as the hospital's activation status, instructions for joining the federated learning network, and in some cases, a link or attachment for downloading the Docker-based FL client package required to connect to the FL server.

This step ensures that hospitals are immediately informed after registration and can proceed with setup and participation in federated learning without manual coordination. It also improves system automation and reduces administrative overhead while maintaining secure delivery of sensitive setup information.

## Medical Image Pipeline

The backend orchestrates image analysis without storing raw images locally:

- Presigned PUT URL is generated for direct R2 upload (5-minute expiry).
- After upload, backend saves image metadata to PostgreSQL
- ModelBackend is called via OpenFeign (Feign client) with a presigned GET URL
- ML results — prediction label, confidence score, heatmap S3 key — are persisted
- Presigned GET URLs for original and heatmap images are cached in Redis ( 59 min TTL)

Small parts of the pipeline can be shown in **??**, and 3.40 where the user is asked to upload a medical image to the system so that they are directed to the right models and diagnosed.

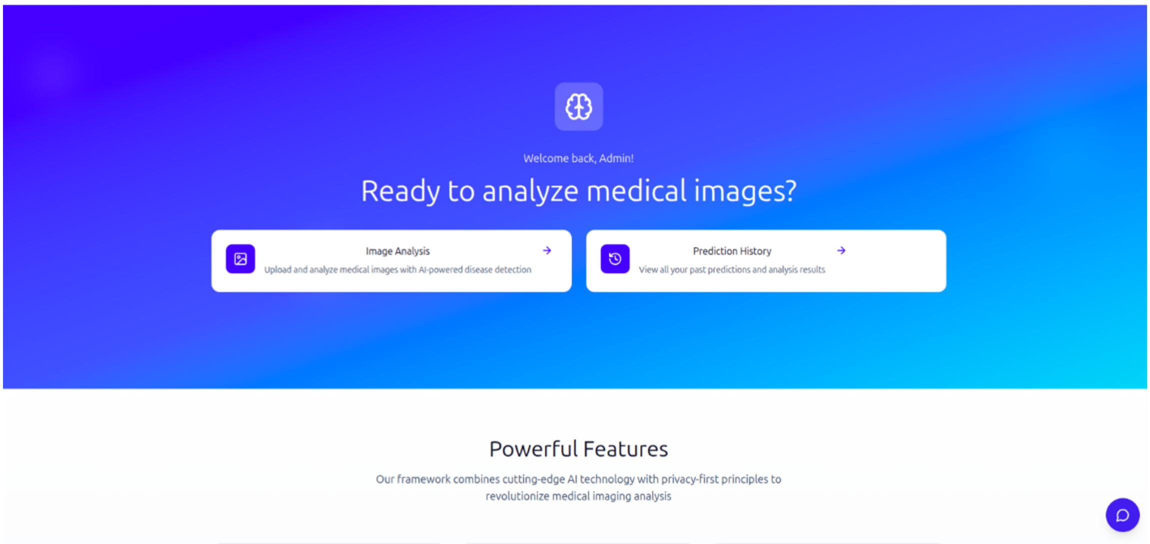



Figure 3.39: An image greeting page of the web application.

Prediction history can be viewed and also has statistics like (total scans , AVG. confidence) , and also every prediction's details can be viewed or deleted, as shown in Figure 3.41, and the details of each scan can be reviewed as well, as shown in Figure 3.42.

Some examples of the Lung Cancer model in action are shown in Figure 3.43, and Figure 3.44. The system also makes sure to show the confidence as ordinary in medical applications of computer vision systems, as well as the processing time and the resulting heatmap, if applicable.

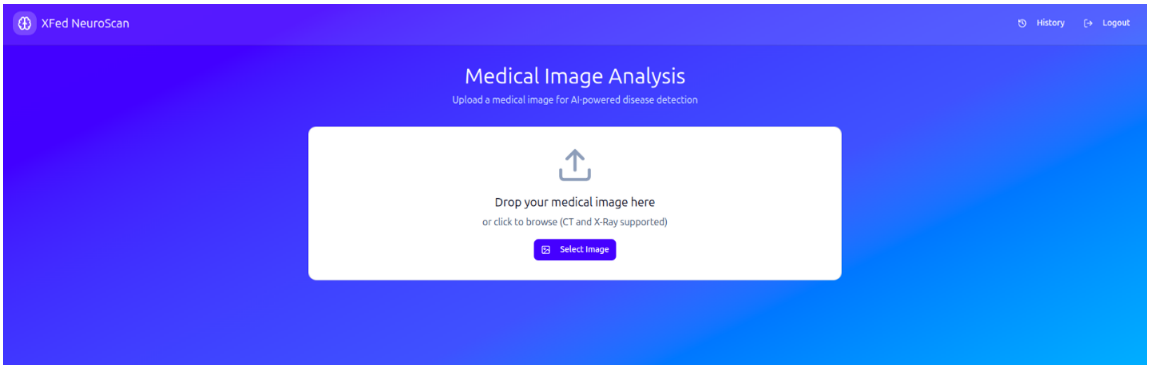



Figure 3.40: An image of the system asking the user to upload a scan to diagnose.

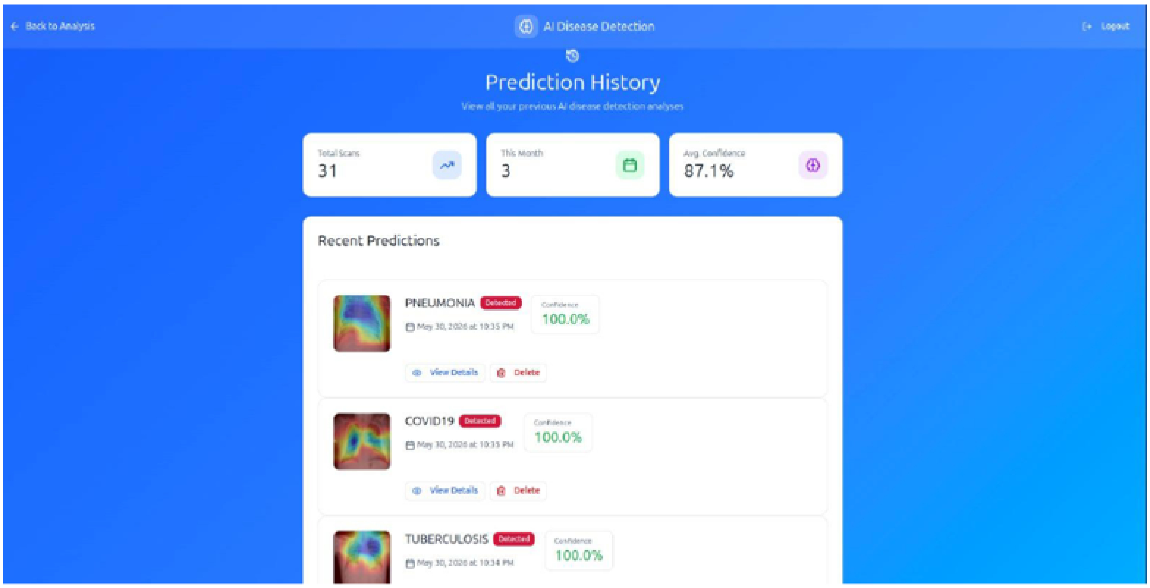



Figure 3.41: An image of the system showing the prediction history of the logged-in user.

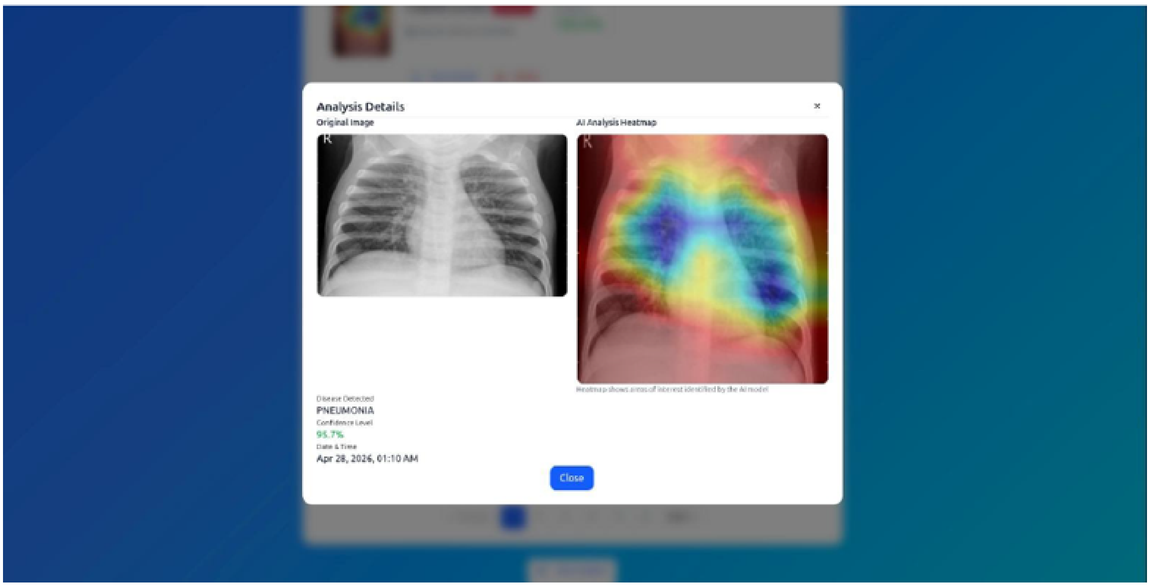



Figure 3.42: An image of the system showing the details of one of the user's historical predictions.

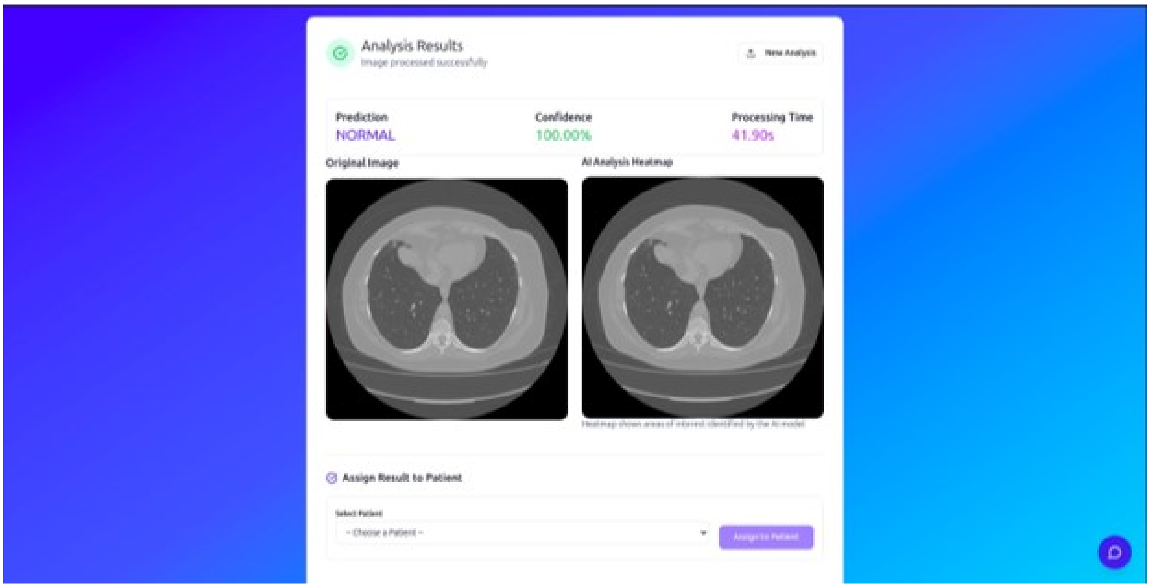



Figure 3.43: One of the of the examples of using the system on the a lung cancer image.

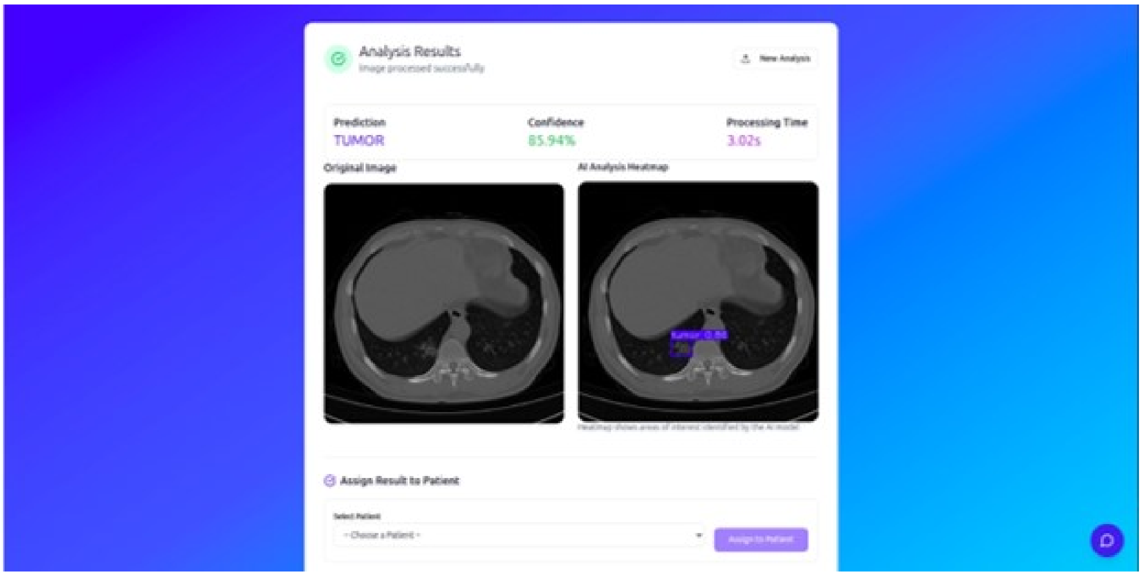



Figure 3.44: Another example of using the system on the a lung cancer image.

## Appointment Management

The appointment management module enables patients and doctors to coordinate medical consultations through a structured workflow. Patients can create appointments by selecting a doctor, choosing a date and time, specifying the appointment type, and providing optional notes or supporting attachments. Once submitted, the appointment information is stored in the database and becomes available to the assigned doctor.

Doctors can view their scheduled appointments, review patient details, and update the appointment status as needed. They can mark appointments as completed, add consultation notes, or cancel appointments when necessary. Patients can also view the status of their appointments and track any updates made by the doctor.

All appointment data is managed through the Spring Boot backend and stored in PostgreSQL, ensuring persistence and secure access. The module provides an organized way to manage patient-doctor interactions while maintaining a complete record of appointment history within the platform.

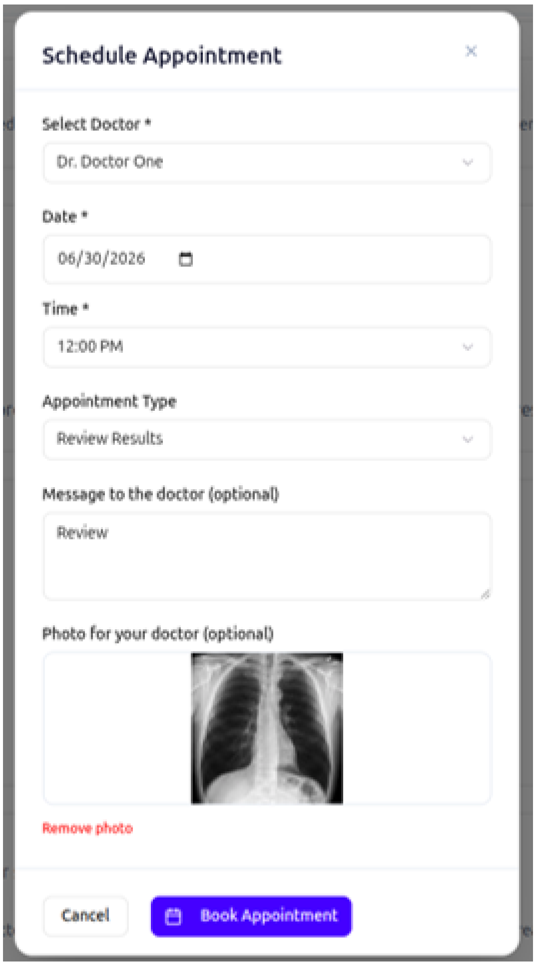



Figure 3.45: An image of the Schedule Appointment interaction from the system.

**Real-Time WebSocket** Spring WebSocket with STOMP over SockJS provides live FL hospital status to the frontend. A scheduled poller runs every 10 seconds, fetches client status from the FL Server, syncs to PostgreSQL, and broadcasts to /topic/fl/clients. Hospital nodes that miss heartbeats are marked OFFLINE automatically.

**Email Service** Spring Mail with Gmail SMTP sends three types of emails:

- Password Reset: link to `/reset-password?token=...`
- Admin Invite: link to `/set-password?token=...` for new doctor/admin accounts
- FL hospital onboarding: download link for the hospital Docker client package plus connection details.

**Core API Modules** The core modules exposed by the API and implemented in it a re listed in Table 3.17

Table 3.17: Backend service modules and their responsibilities within the system.

| Module | Description |
|---|---|
| Authentication | Handles the full authentication lifecycle including user registration, login, token issuance, refresh token management, and logout operations. |
| Image Upload | Generates and returns Cloudflare R2 presigned PUT URLs to enable secure direct client-side uploads of medical images. |
| Image Processing | Orchestrates the machine learning inference pipeline, triggers model execution, and stores as well as retrieves diagnostic results from the persistence layer. |
| Appointments | Manages the full appointment workflow, allowing patients to book appointments and doctors to approve, complete, or cancel scheduled sessions. |
| Admin User Management | Provides administrative capabilities for managing user accounts, including creation, modification, role assignment, and deactivation. |
| Federated Learning Hospitals | Handles hospital registration and management within the federated learning ecosystem, acting as a proxy interface to the federated learning server. |
| Federated Learning Webhooks | Processes callback events from the federated learning server after training rounds, updating system state and storing training metadata accordingly. |

## 3.8.4 ML Service - ModelBackend

ModelBackend is a FastAPI microservice that hosts all AI capabilities. At startup it loads four ML components into memory and exposes a unified image-processing endpoint plus a RAG chatbot. It runs on port 8000 and is called by centralized Backend for image analysis and directly by the frontend for chatbot interactions.

## ML Models

Different backbone models were used for the different tasks the proposed system performs. All the AI models and architectures used are listed in Table 3.18.

Table 3.18: Machine learning components and their corresponding frameworks and purposes within the proposed system.

| Component | Framework | Purpose |
|---|---|---|
| Router Model | Keras (DenseNet121) | Classifies input images into X-ray, CT scan, or out-of-distribution categories using 224×224 image resolution as the first stage of the diagnostic pipeline. |
| Chest Classification | TensorFlow/Keras (DenseNet121) | Performs four-class chest X-ray diagnosis including COVID-19, Normal, Pneumonia, and Tuberculosis classification in a multi-label-ready setup. |
| Lung Segmentation | TensorFlow/Keras (U-Net) | Extracts lung regions from medical images prior to classification to enhance focus on relevant anatomical structures and improve diagnostic accuracy. |
| Lung Cancer Detection | YOLOv8 | Detects and localizes lung cancer regions in CT scans using bounding box predictions for object-level medical diagnosis. |
| RAG Stack | LangChain, Ollama, and FAISS | Implements a retrieval-augmented generation chatbot for medical question answering using vector-based document retrieval and local LLM inference. |

FastAPI serves as the AI engine of XFed NeuroScan and relies on Spring Boot to integrate its models into the complete healthcare platform. While FastAPI is responsible for running AI models and generating predictions, Spring Boot manages the workflow and communicates with the AI service whenever analysis is requested.

Separating Spring Boot and FastAPI into two services improves scalability and maintainability, allowing the healthcare system and AI models to be developed, updated, and deployed independently without affecting each other.

## Image Processing Pipeline

The image processing module is designed to automatically analyze uploaded medical images and route them to the appropriate AI model based on their content. This architecture allows the system to support multiple diagnostic pipelines while maintaining a single entry point for image analysis.

**Unified Routing** Every uploaded image first passes through the Router Model, which serves as the initial classification layer. The router determines whether the image is a chest X-ray, lung cancer image, or an unsupported image type. A confidence threshold of 0.5 is used to ensure reliable routing decisions. Images classified as other are immediately rejected to prevent invalid data from being processed by diagnostic models. Once classified, chest X-rays are forwarded to the DenseNet-based chest disease pipeline, while lung cancer images are sent to the YOLOv8 detection pipeline.

**Chest Pipeline** The chest disease analysis pipeline is optimized for detecting four classes: COVID-19, Pneumonia, Tuberculosis, and Normal. Before classification, several preprocessing steps are applied to improve image quality and model performance.

First, a U-Net lung segmentation model isolates the lung region from the surrounding anatomy, allowing the classifier to focus only on clinically relevant areas. Next, Contrast Limited Adaptive Histogram Equalization (CLAHE) enhances local contrast within the segmented lungs, making subtle abnormalities easier to detect. A Gaussian Blur (5×5) filter is then applied to reduce image noise while preserving important structures.

The processed image is normalized to the range [0,1], converted to RGB format, and resized to 224×224 pixels, which matches the input requirements of the DenseNet model. The DenseNet classifier then predicts one of the four disease categories and returns a confidence score indicating prediction certainty.

To improve explainability, the system generates a Grad-CAM heatmap from the final convolutional layers of the network. This heatmap highlights the image regions that most influenced the model's decision, providing visual evidence to support the prediction. Both the original image and the generated heatmap are uploaded to Cloudflare R2 for storage and later retrieval by the frontend.

**Lung Cancer Pipeline** The lung cancer pipeline is based on YOLOv8, an object detection model capable of identifying suspicious cancerous regions within medical images. After the router classifies an image as a lung cancer case, it is forwarded directly to the YOLOv8 model for inference.

The model performs real-time detection using a confidence threshold of 0.3, identifying potential cancer regions and generating bounding boxes around detected abnormalities. For each detected region, the model provides confidence scores indicating the likelihood of cancer presence.

The detected regions are then visually annotated on the image by drawing bounding boxes and labels, creating an interpretable result that can be reviewed by medical professionals. The final annotated image is uploaded to Cloudflare R2 under the lung cancer results directory, allowing it to be displayed in the frontend and stored as part of the patient's analysis history.

This dual-pipeline architecture combines classification-based diagnosis for chest diseases with object-detection-based localization for lung cancer, enabling the platform to provide both diagnostic predictions and visual explanations while maintaining a unified processing workflow.

An overall visualization of the image processing pipeline is shown in Figure 3.46

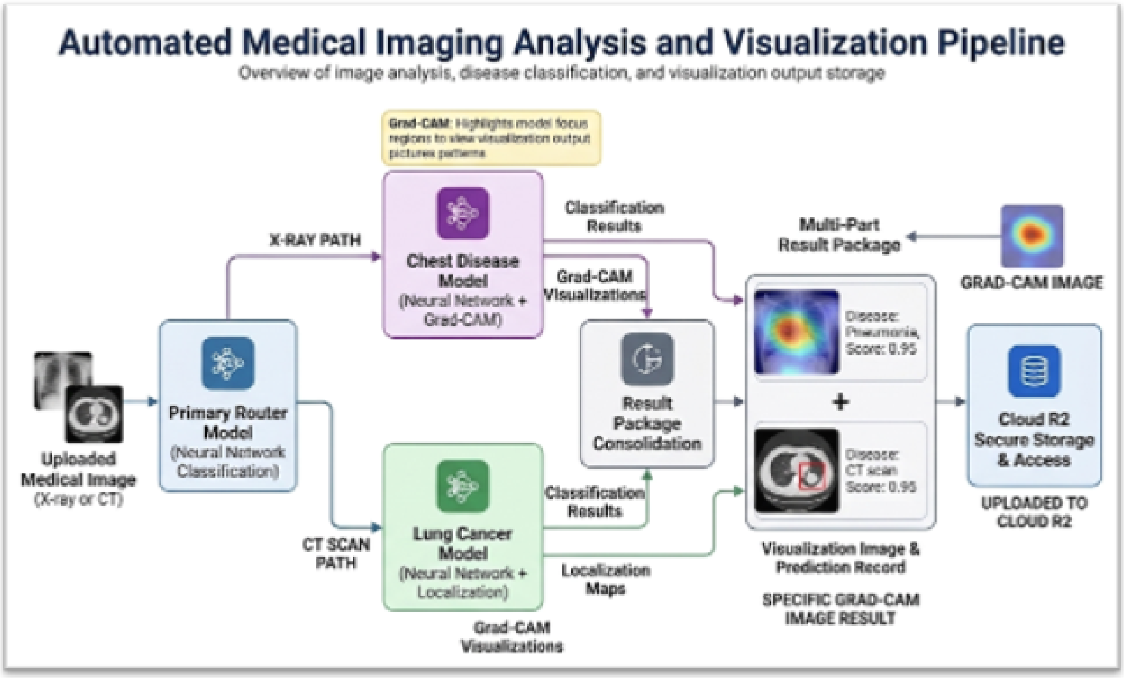



Figure 3.46: A simple visualization of the overall image processing pipeline from the backend's perspective.

## Grad-CAM Explainability and Localization

For every chest X-ray prediction, the system generates a Grad-CAM (Gradient-weighted Class Activation Mapping) heatmap to provide interpretability for the AI decision. This mechanism highlights the regions of the image that most influenced the model's prediction by tracing gradients flowing into the final convolutional layers of the DenseNet network.

The generated heatmap is visualized using a JET colormap and overlaid on the original X-ray image , allowing both the anatomical structure and the activated regions to remain clearly visible. This overlay helps radiologists and clinicians visually verify whether the model is focusing on medically relevant areas such as lung opacities, consolidations, or abnormal tissue patterns rather than irrelevant regions.

To support persistence and later retrieval, both the original Grad-CAM output and the processed overlay image are stored in Cloudflare R2 under the `heatmaps/` directory. These images are served to the frontend using secure presigned URLs, ensuring controlled and time-limited access to sensitive medical data.

## Localization Capability

Beyond explainability, Grad-CAM also provides a form of weak localization for disease regions within the chest X-ray. Although the DenseNet model is primarily designed for classification, Grad-CAM transforms it into a pseudo-localization system by identifying spatial regions that contribute most strongly to the predicted class.

This allows the system to:

- Approximate the location of abnormal lung regions.
- Highlight areas associated with diseases.
- Provide spatial context without requiring a dedicated detection model

In practice, this means the system not only predicts the presence of a disease but also indicates where in the lungs the abnormality is likely located. This improves clinical trust and makes the AI output more interpretable, bridging the gap between pure classification models and spatially aware diagnostic tools.

Overall, Grad-CAM serves as both an explainability mechanism and a lightweight localization method, enhancing transparency and clinical usability of the AI predictions.

## RAG Chatbot Module

Retrieval-Augmented Generation (RAG) is an AI technique that enhances LLM responses by grounding them in a private document corpus rather than relying solely on the model's pre-trained knowledge. This is critical for domain-specific applications — such as medical systems, where accuracy, traceability, and up-to-date information are essential.

### Standard RAG Pipeline

- Ingest: Load source documents (PDF, TXT, etc.)
- Chunk: Split long documents into overlapping smaller pieces.
- Embed: Convert each chunk to a dense numerical vector using a sentence embedding model.
- Store: Persist vectors in a FAISS vector database, organized by category.
- Retrieve: On a user query, embed the question and find the top-k most similar chunks.
- Generate: Pass the retrieved chunks and the question to the LLM to produce a grounded answer.

The classifier is the key differentiator of this RAG system. Rather than searching all documents indiscriminately, each query is first categorized, enabling targeted retrieval from the correct FAISS index.

Table 3.19: Categories of input handled by the RAG-based medical chatbot.

| **Category** | **Typical Content** |
|---|---|
| Medical | Queries related to chest diseases and lung cancer, including symptoms, diagnosis, interpretation of medical findings, and general medical imaging questions. |
| General | Greetings, small talk, and off-topic or non-medical conversational queries that do not require clinical or diagnostic knowledge. |

**Categories**

**Model Architecture** The classifier uses a lightweight 1D Convolutional Neural Network (CNN) designed for speed and bilingual text support, the full visualization of the architecture is shown in Figure 3.47.

1. Input: Token IDs (max 50 tokens)
2. Embedding Layer (vocab → 64 dims)
3. Conv1d (64 → 128 channels) + ReLU + Dropout(0.3)
4. Conv1d (128 → 128 channels) + ReLU
5. AdaptiveMaxPool1d(1)
6. Fully Connected: 128 → 64 → 3
7. Output.

**Bilingual Tokenizer**

- Tokenizes Arabic , English, and numeric tokens
- Vocabulary built from training data ( 300+ labeled question pairs)
- Fixed-length encoding: max 50 tokens, padded with <PAD>, unknown words as <UNK>
- Outputs: classifier.pth, tokenizer.json, config.json

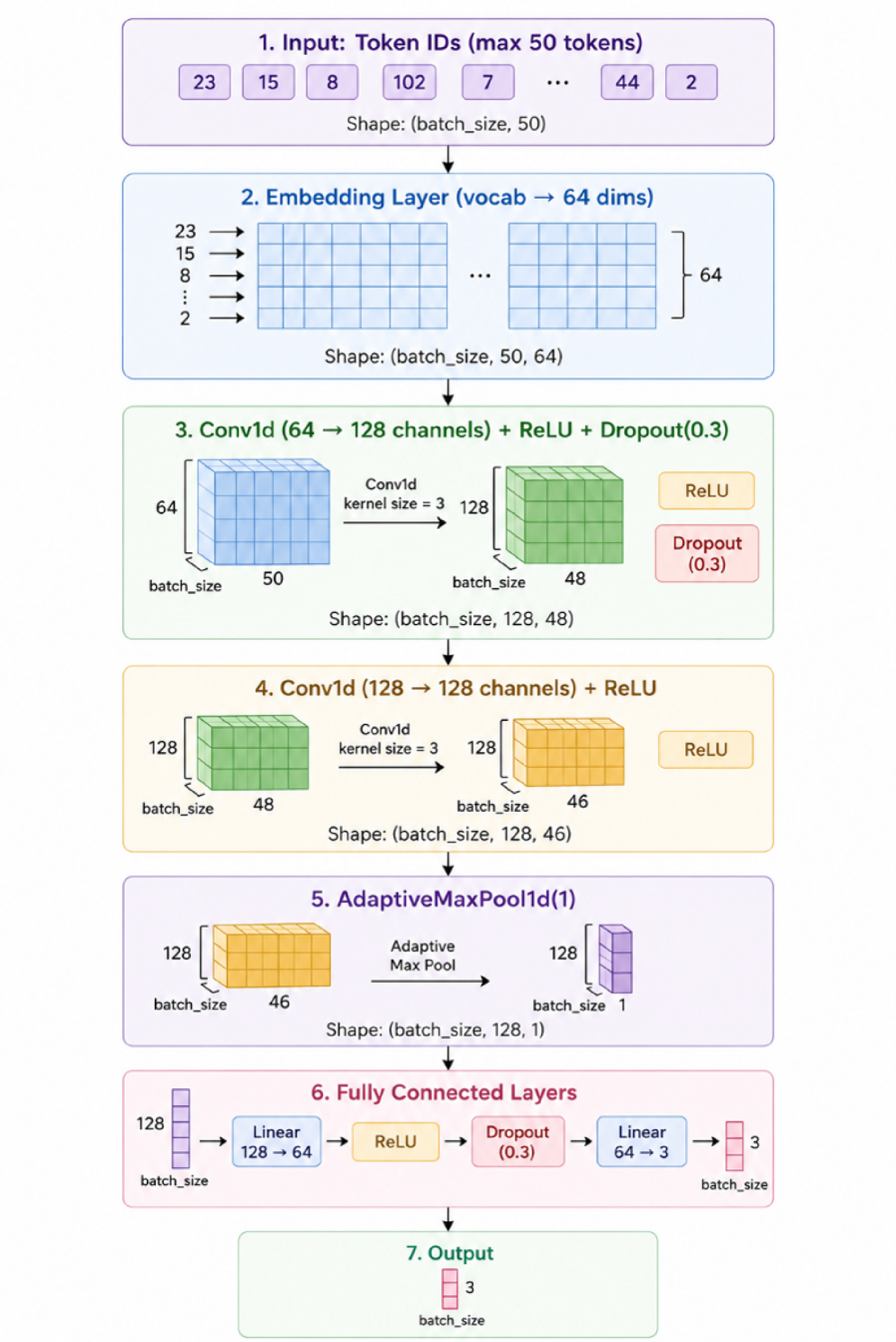



Figure 3.47: A simple visualization of the overall RAG architecture.

**Training Configuration** The model was trained for 100 epochs using the Adam optimizer with a learning rate of 0.001 and a batch size of 16. The vocabulary size is 352 tokens, and each input token is mapped into a 64-dimensional embedding space. The hidden layer dimension is set to 128, Each input sequence is limited to a maximum length of 50 tokens to ensure consistent processing and performance.

**LLM Configuration** The configurations used for the RAG engine are shown in Table 3.20 for reproduce-ability purposes.

Table 3.20: Configuration parameters of the RAG-based LLM engine.

| **Parameter** | **Standalone Value** |
|---|---|
| LLM Engine | Ollama |
| Model | qwen2.5:3b |
| Temperature | 0.5 |
| Max Tokens | 500 |
| Keep Alive | Infinite (-1) |

**Prompt Design** The LLM is strictly instructed to answer only from the retrieved context. If the context does not contain the answer, the model is instructed to say "I don't know." Responses are capped at three sentences and must match the language of the question.

The text is split using a RecursiveCharacterTextSplitter with a chunk size of 1000 characters and an overlap of 150 characters between consecutive chunks. This ensures that the text is broken into manageable segments while preserving context between adjacent chunks for better retrieval accuracy.

RAG Independence from Image models RAG does not receive chest X-ray predictions, GradCAM heatmaps, lung cancer results, or Spring Boot patient records. It is a standalone medical Q&A chatbot co-located in every page purely for deployment convenience.

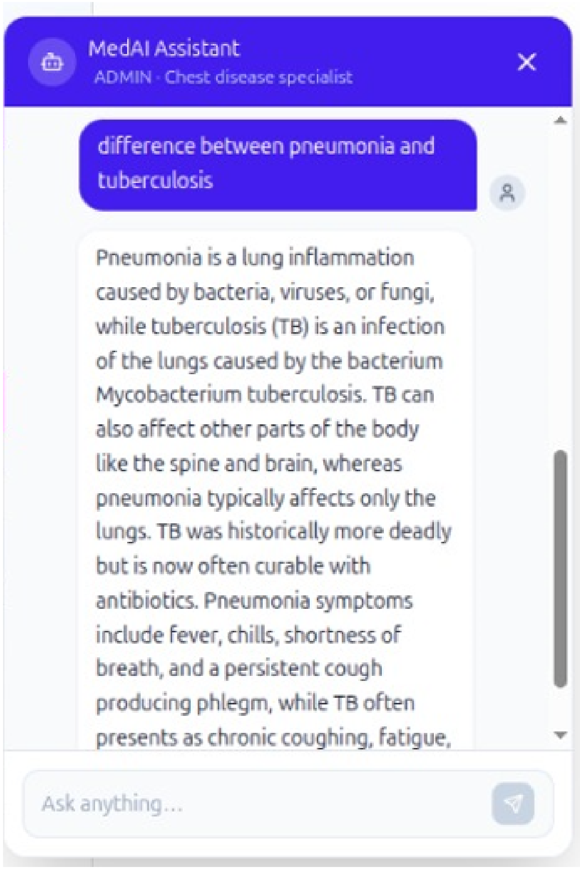
Examples of using the RAG agent implemented in the system are shown in Figure 3.48

## Performance Profile

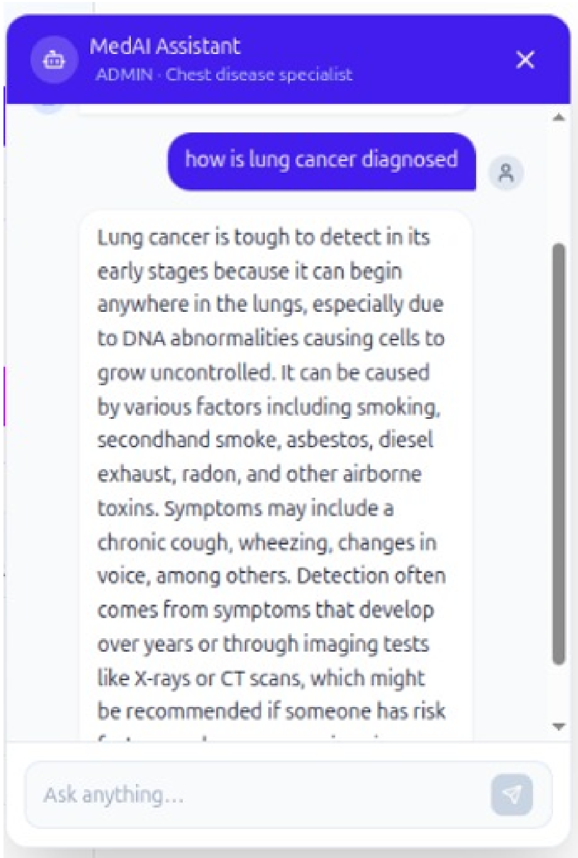
The performance profile of the proposed system is discussed in Table 3.21.

Running the ModelBackend requires sufficient computational resources because it hosts multiple AI components simultaneously, including image classification models (DenseNet, YOLO, U-Net), the router model, and the RAG pipeline with embedding models and an LLM (Ollama). For stable performance, a GPU is strongly recommended for accelerating deep learning inference, especially for chest X-ray and lung cancer detection tasks.

In addition, the system requires adequate CPU resources to handle concurrent API




(a) Example of a medical query processed by the RAG agent, showing retrieval of relevant documents and generated response.




(b) Example of a general/off-topic query handled by the RAG agent with a conversational response without medical retrieval.

Figure 3.48: Illustrative examples of the behavior of the retrieval-augmented generation (RAG) agent in different query scenarios. The system adapts its response strategy depending on whether the input is medical or general in nature.

Table 3.21: Computational performance and execution environment of the proposed system components.

| **Component** | **Approximate CPU Time** | **GPU** |
|---|---|---|
| Router | 50–100 ms | Automatically enabled if CUDA is available |
| Chest Pipeline (Segmentation + Prediction + Grad-CAM) | 1–5 s | CUDA auto-enabled |
| Lung Cancer YOLO Inference | 1–5 s | CUDA auto-enabled |
| R2 Download / Upload | Network-dependent | N/A |

requests, image preprocessing, and routing logic, while RAM (at least 4–6 GB minimum, higher recommended) is needed to keep multiple models, FAISS indexes, and embeddings loaded in memory without performance degradation. Overall, a balanced setup with a capable GPU, multi-core CPU, and sufficient RAM ensures smooth and responsive AI processing across all services in the ModelBackend

### 3.8.5 Federated Learning Server

The Federated-Learning-Server is responsible for enabling privacy-preserving collaborative training across multiple hospitals without requiring any centralization of medical data. It allows hospitals to jointly train a shared model while ensuring that raw patient images never leave their local environments, maintaining strict data privacy and compliance with healthcare constraints.

The system is built using a hybrid architecture that combines a FastAPI REST service and a Flower-based gRPC server within a single Python process. The FastAPI layer acts as the control interface for registration, monitoring, and coordination tasks, while the Flower server handles the actual federated learning workflow, including distributed training and weight aggregation.

Both components are synchronized through an in-memory SharedState singleton, which maintains the current training status, active clients, and round coordination signals. This design allows seamless communication between REST-based administrative actions and gRPC-based training execution, ensuring that federated learning rounds are executed in a coordinated and efficient manner across all participating hospitals.

#### Design Principles

The design principles of the federated learning server are listed in Table 3.22.

The Federated-Learning-Server follows a dual-process architecture where two interconnected services run simultaneously within the same system. The main process runs a Uvicorn-based FastAPI server on port 8001, which acts as the REST bridge responsible for hospital registration, client management, monitoring, and handling webhooks from the main backend system. In parallel, a daemon process runs a Flower gRPC server on port 50051, which is responsible for coordinating the actual federated learning workflow, including distributed training and model weight aggregation across participating hospitals.

This separation allows the system to clearly divide responsibilities between control-plane operations (handled by FastAPI) and training-plane execution (handled by Flower). Both processes share a common in-memory SharedState, ensuring synchronization between REST actions and federated learning events while maintaining efficient and real-time coordination.

#### Training Round Lifecycle

A complete federated training round follows these steps:

Table 3.22: Core design principles and their implementation in the federated learning framework.

| **Principle** | **Implementation** |
|---|---|
| Privacy first | Only model weight updates are transmitted between clients and the server; raw medical images never leave the hospital premises under any circumstance. |
| On-demand training | The Flower server remains idle until explicitly triggered by an administrator via the `POST /start-round` endpoint, ensuring that training rounds are fully controlled and non-automatic. |
| In-process coupling | FastAPI and Flower operate within a shared runtime environment using a `SharedState` singleton, eliminating the need for internal HTTP communication and reducing overhead. |
| Weight hot-reload | After each federated aggregation round, the updated global model weights are immediately pushed to the Model Backend, enabling real-time deployment of improved models. |

- Step 1: Admin selects online hospitals and initiates a round
- Step 2: Spring calls start-round with round id, min clients, deadline minutes
- Step 3: FastAPI sets SharedState `trigger_event`, unblocking the Flower server thread
- Step 4: Hospital clients receive global weights via gRPC, train locally on their data
- Step 5: CustomFedAvg aggregates weight updates using the FedAvg algorithm
- Step 6: Aggregated weights saved and pushed to ModelBackend
- Step 7: Webhook notifies spring of round completion or failure
- Step 8: trigger event cleared; Flower waits for the next admin trigger

After a hospital is successfully registered in the federated learning system, Spring Boot finalizes the onboarding process by generating the necessary FL credentials and preparing a deployment package for that hospital. Once the registration is approved, the system automatically sends an email notification to the hospital using Gmail SMTP.

This email contains all essential onboarding information, including the hospital's federated learning client ID, encrypted API key, and instructions for connecting to the Federated Learning Server. It also provides a Docker-based hospital client package, which is the only component the hospital needs to run locally.

From the hospital side, the setup process is intentionally simple. The hospital team only needs to:

1. Put the data in the required folder

2. Run the container provided in the package

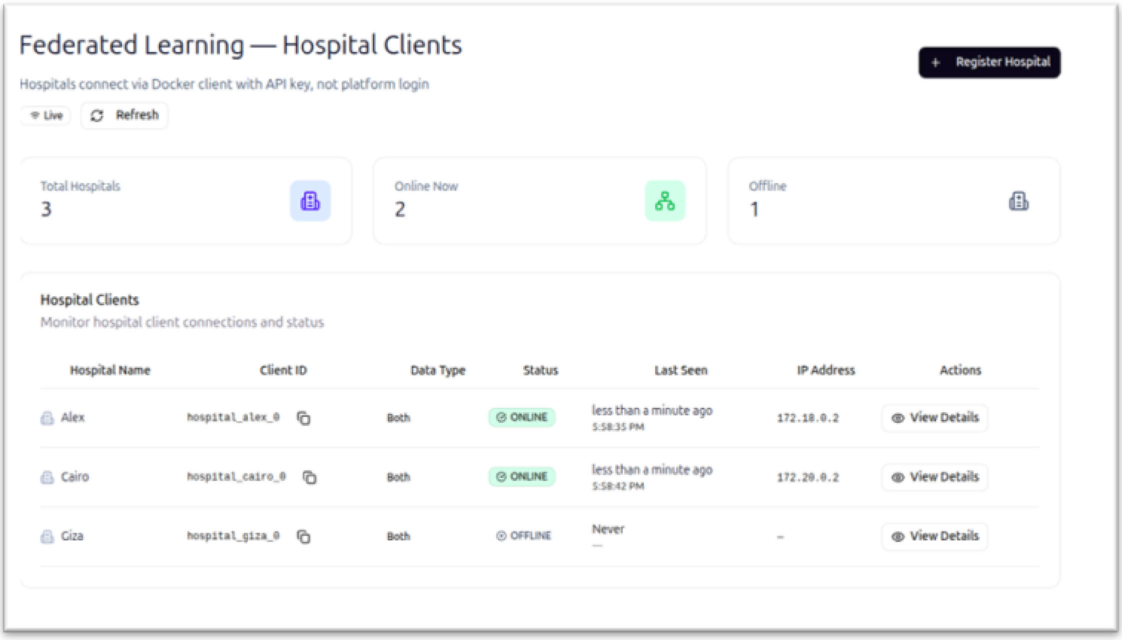



Figure 3.49: The federated learning dashboard of the web application.

Once the container is started, the hospital client automatically connects to the Federated Learning Server, begins sending heartbeat signals, and becomes eligible for participation in training rounds. After this step, no further manual integration is required, as all communication and training coordination are handled automatically by the system.

This design ensures a smooth onboarding process where hospitals do not need to manage complex machine learning setups, and can join the federated learning network by simply configuring and running a pre-built container

Once the container is running and sending heartbeat signals, the hospital is marked as ONLINE and becomes eligible to participate in federated training rounds.

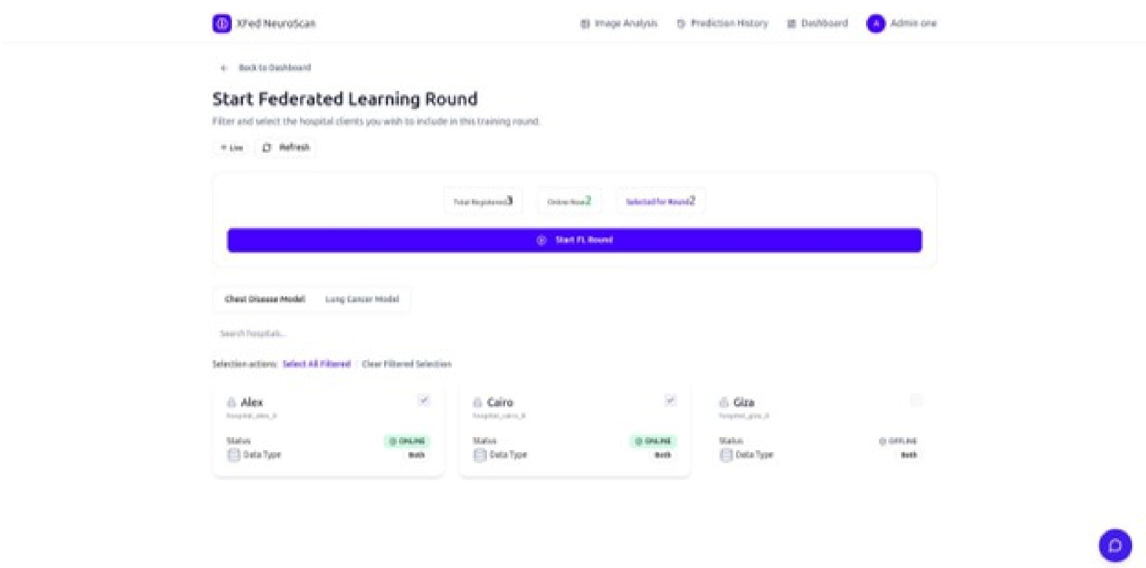



Figure 3.50: The federated learning management tab.

## Round Status States

The full states and status code for the federated learning server are listed and explained in Table 3.23

Table 3.23: Federated learning server execution states and their meanings.

| **Status** | **Meaning** |
|---|---|
| IDLE | No federated learning round is currently running; the Flower server is blocked and waiting for an external trigger event. |
| STARTED | A training round has been triggered and participating hospital clients are actively performing local model training. |
| AGGREGATING | The server is performing FedAvg aggregation over received client updates to compute the new global model. |
| COMPLETED | The federated learning round has successfully finished, and the updated global model weights have been saved and pushed to the Model Backend for deployment. |
| FAILED | The federated learning round has failed due to issues such as timeouts, client-side errors, or model update push failures. |

## Security

The Federated Learning security model is designed to ensure strict isolation between hospitals, the platform backend, and the training infrastructure. A Platform API key is used exclusively by the centralized Backend to securely register new hospital clients, retrieve client lists, and initiate federated learning rounds, ensuring that only the trusted backend can control system-level operations.

Each hospital node is assigned a unique Client API key, which is used solely for heartbeat authentication to confirm that the hospital is online and eligible to participate in training. In addition, a Webhook secret is shared between the FL Server and centralized Backend to verify and secure all outbound callbacks related to training round completion and status updates.

Hospitals communicate with the system only through outbound connections to port 8001 for REST heartbeats and port 50051 for gRPC-based training, eliminating the need for direct inbound access. Importantly, admin users never interact directly with the FL Server; instead, all operations are securely routed through centralized Backend using JWT authentication, maintaining a centralized and controlled security layer.

The various key types and their roles are shown and explained in great detail in Table **??**.

Table 3.24: Authentication key types and their roles within the federated learning system.

| Key Type | Holder | Scope |
|---|---|---|
| Platform API Key | Centralized Backend | Used by the backend system to register clients, initiate federated learning rounds, and retrieve the list of participating clients. |
| Client API Key | Each hospital node | Used exclusively for heartbeat authentication to verify active participation and connectivity of hospital clients. |
| Webhook Secret | Federated Learning Server and Centralized Backend | Used to authenticate outbound callback requests from the federated learning server after training rounds to ensure secure communication. |

## 3.8.6 End-to-End Data Flows

### Medical Image Analysis Flow

1. The frontend user selects an image and requests a presigned URL from the Spring Backend.
2. The frontend uploads the image directly to Cloudflare R2 using the provided presigned URL.
3. After upload, the frontend calls the image processing service with the stored R2 key.
4. The Spring Backend saves image metadata in PostgreSQL, generates a presigned URL, and forwards the request to the FastAPI service via Feign client communication.
5. The FastAPI service downloads the image from R2, routes it using the trained routing model, and executes the appropriate specialist pipeline (either chest disease classification or lung cancer detection).
6. The FastAPI service uploads the generated outputs (such as Grad-CAM heatmaps or annotated detection images) back to R2 and returns the prediction, confidence score, and storage key.
7. The Spring Backend stores the final inference results in the database and generates presigned URLs for secure result visualization.
8. The frontend displays the prediction results, confidence scores, and visual explanations such as Grad-CAM overlays or bounding box annotations to the user.

### User Authentication Flow

1. The user logs in via the login page, where credentials are verified by Spring Security.

2. If authentication succeeds, an access JWT is returned in the response body, and a refresh token is set as an HttpOnly cookie.

3. The refresh token is also stored in Redis with a defined key and a 30-day expiration period.

4. The frontend stores the access token in memory, while the authentication context initializes and loads the user role and profile information.

5. When the access token expires, the frontend calls the refresh token endpoint, which issues a new access token and rotates the refresh cookie.

6. During logout, the access token is blacklisted and the refresh token is removed from Redis, fully invalidating the user session.

### Federated Learning Hospital Status Flow

1. The administrator registers a hospital through the frontend interface, and the Spring forwards the registration request to the Federated Learning (FL) Server.

2. The FL Server generates a unique API key, which is encrypted and stored by the backend. Subsequently, an onboarding email containing the Docker-based client package is sent to the hospital.

3. The hospital deploys the provided Docker client, which establishes a connection to the Flower server via gRPC for federated learning participation.

4. The hospital client initiates a heartbeat process that continuously sends periodic signals to the FL Server every 30 seconds to indicate active status.

5. The Spring Backend executes a polling mechanism every 10 seconds to retrieve the latest client status updates from the FL Server.

6. The retrieved client statuses are synchronized with the PostgreSQL database and broadcast in real time using STOMP WebSocket communication.

7. The frontend subscribes to the WebSocket channel and dynamically updates hospital ONLINE/OFFLINE indicators in real time based on received status updates.

## 3.9 Summary

This chapter presented the proposed end-to-end model for automated detection of chest diseases and bone fractures from medical images. The overall solution architecture was introduced, highlighting the integration of a deep learning–based diagnostic pipeline with supporting backend processes to enable efficient and scalable medical image analysis.

The deep learning pipeline was described in detail, illustrating the sequential flow from image preprocessing and feature extraction to intelligent routing and task-specific inference. Transfer learning was adopted as a core strategy to leverage pre-trained deep neural networks, enabling effective feature representation and improved generalization in the presence of limited labeled medical data.

The chapter also introduced the medical image routing model, which enables dynamic task allocation by directing input images to specialized diagnostic models. This design enhances modularity and allows the system to support multiple medical imaging tasks within a unified framework.

Then, the architecture of the chest disease classification model was discussed in detail. The design was illustrated, and the use of ResNet50 as the main feature extractor was justified. The formulation of the problem and the motivation behind such a model were presented while detailed architectural aspects of the transfer-learning model were omitted to avoid redundancy.

Finally, the bone fracture detection model was presented as a task-specific component of the system, implemented using an unmodified YOLOv8 architecture. The formulation of the fracture detection problem and the motivation for adopting a one-stage object detection approach were discussed, while detailed architectural aspects were referenced to earlier sections to avoid redundancy.

Overall, this chapter established the structural and methodological foundation of the proposed system, providing a clear understanding of its components and design choices. The described models form the basis for the experimental evaluation and performance analysis presented in the subsequent chapter.

# CHAPTER 4
# RESULTS & EVALUATION

# Chapter 4 Results and Evaluation

This chapter presents the experimental results obtained from the implementation and evaluation of the proposed framework, *X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification*. The primary objective of this chapter is to assess the effectiveness of the developed models, validate the design decisions discussed in previous chapters, and demonstrate the capability of the proposed system to accurately analyze medical images within a hierarchical diagnostic pipeline.

The evaluation process was designed to investigate both the individual performance of each model and the overall effectiveness of the integrated framework. Since the proposed architecture consists of multiple interconnected components, a comprehensive assessment is necessary to ensure that each stage performs its intended function reliably. Particular emphasis is placed on the routing model, which serves as the first level of the classification pipeline and is responsible for identifying the modality of incoming medical images before directing them to the appropriate downstream diagnostic model. The performance of the disease classification and lung cancer detection models is subsequently analyzed to determine their ability to perform accurate medical image diagnosis within their respective domains.

To provide a rigorous and objective evaluation, multiple performance metrics are employed throughout this chapter. These metrics include accuracy, precision, recall, F1-score, loss values, confusion matrices, and training convergence behavior. In addition, comparative experiments are conducted to evaluate different model architectures and hyperparameter configurations, allowing the selection of the most suitable models for inclusion within the final framework. Training and validation curves are also examined to assess optimization stability, convergence characteristics, and generalization performance.

Beyond predictive performance, this chapter also evaluates the practical aspects of the proposed framework. The effectiveness of transfer learning, data augmentation, class balancing strategies, and other implementation choices is analyzed to determine their contribution to overall model performance. Furthermore, the results obtained from the explainability component are discussed to demonstrate how visual explanations can enhance model transparency and improve the interpretability of diagnostic decisions. The chapter also examines the federated learning implementation and its ability to support collaborative model training while preserving data privacy.

The results presented throughout this chapter provide quantitative and qualitative evidence regarding the feasibility of the proposed approach for medical image analysis. By systematically evaluating each component of the framework and comparing alternative design choices, this chapter establishes the effectiveness of the developed system and provides a foundation for the discussion, conclusions, and future research directions presented in the subsequent chapters.

# 4.1 Evaluation of the Medical Image Routing Model

The medical image routing model represents the first level of the proposed hierarchical diagnostic framework and therefore plays a critical role in determining the overall effectiveness of the system. Since all incoming images must first pass through the routing stage before being analyzed by specialized diagnostic models, the reliability of this component directly impacts the performance of the entire framework. Consequently, a comprehensive evaluation was conducted to assess the ability of the routing model to accurately distinguish between chest X-ray images, chest CT scan images, and out-of-distribution images categorized as *Other*.

The evaluation process was designed not only to measure the classification performance of the final selected model but also to compare multiple candidate architectures and investigate the effect of hyperparameter selection on model performance. In particular, several deep learning architectures were trained and evaluated under identical experimental conditions, including a custom convolutional neural network, ResNet50, DenseNet121, and EfficientNetB3. The objective was to identify the architecture that offered the best combination of classification accuracy, generalization capability, and computational efficiency for the image routing task.

## 4.1.1 Implementation Environment and Development Tools

The routing model was implemented using the Python programming language and developed within a Jupyter Notebook environment. The deep learning pipeline was built using TensorFlow version 2.20.0 and Keras version 3.12.0, which provided the necessary functionality for constructing, training, and evaluating convolutional neural networks. Data preprocessing, dataset partitioning, performance evaluation, and metric computation were performed using Scikit-learn version 1.8.0.

Model training and experimentation were conducted on a workstation equipped with an NVIDIA RTX 2060 graphics processing unit (GPU). The availability of GPU acceleration significantly reduced training times and enabled the efficient evaluation of multiple candidate architectures and hyperparameter configurations. Table 4.1 summarizes the software and hardware environment used during the implementation and evaluation process.

Table 4.1: Software and hardware environment used for implementing the routing model.

| Component | Specification |
|---|---|
| Programming Language | Python 3.12 |
| Deep Learning Framework | TensorFlow 2.20.0 |
| Neural Network API | Keras 3.12.0 |
| Machine Learning Library | Scikit-learn 1.8.0 |
| Development Environment | Jupyter Notebook |
| GPU | NVIDIA RTX 2060 |

### 4.1.2 Dataset Preparation and Experimental Setup

The aggregated routing dataset consisted of three image categories corresponding to the target classes of the routing problem. The first category contained chest X-ray images, the second category contained chest CT scan images, and the third category consisted of out-of-distribution images grouped under the *Other* class. The final dataset contained a total of 19,615 images distributed across the three classes as shown in Table 4.2.

Table 4.2: Class distribution of the routing dataset.

| Class | Number of Images |
|---|---|
| Chest X-Ray | 7,123 |
| Chest CT Scan | 2,534 |
| Other | 9,958 |
| Total | 19,615 |

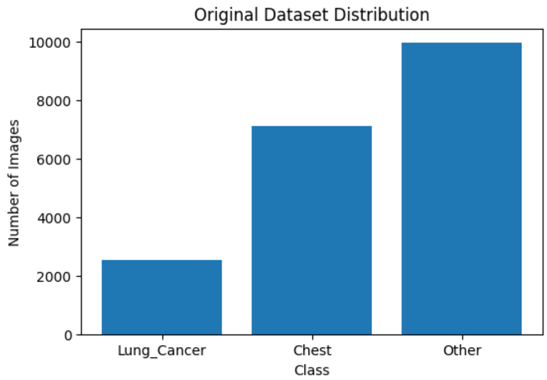



Figure 4.1: Distribution of the routing dataset classes.

Figure 4.2: Summary of the routing dataset distribution.

To ensure a fair evaluation procedure, the dataset was partitioned using a stratified sampling strategy. This approach preserved the original class distribution across all subsets and prevented sampling bias from influencing the evaluation results. The dataset was divided into training, validation, and testing subsets using a ratio of 70%, 15%, and 15%, respectively.

A number of techniques were employed to mitigate the effects of class imbalance. First, class weights were incorporated during training to increase the contribution of minority classes to the optimization process. Second, stratified sampling was utilized to preserve representative class distributions across all dataset splits. Third, extensive data augmentation was applied to improve generalization and increase sample diversity. The augmentation pipeline included common image transformation techniques such as rotation, translation, zooming, flipping, brightness adjustment, and contrast variation. Geometric transformations that could significantly distort medical structures, such as excessive shearing, were intentionally avoided in order to preserve clinically relevant image features.

All models were trained using a batch size of 8 for a total of 10 epochs. The same training, validation, and testing partitions were used across all experiments to ensure consistency when comparing model performance.

### 4.1.3 Learning Rate Optimization

Since learning rate selection plays a crucial role in determining convergence behavior and final model performance, multiple learning rate values were investigated for each architecture. The evaluated learning rates included $10^{-5}$, $10^{-4}$, $10^{-3}$, and $10^{-2}$.

The experiments revealed that different architectures exhibited different optimization characteristics. DenseNet121 and ResNet50 achieved their highest performance when trained using a learning rate of $10^{-2}$, whereas EfficientNetB3 performed best with a learning rate of $10^{-4}$. Due to its relatively shallow architecture and limited feature extraction capacity, the custom convolutional neural network achieved its best results with a learning rate of $10^{-5}$.

Table 4.3 summarizes the optimal learning rate selected for each evaluated architecture.

Table 4.3: Optimal learning rates identified during hyperparameter tuning.

| **Model** | **Optimal Learning Rate** |
|---|---|
| Custom CNN | $10^{-5}$ |
| ResNet50 | $10^{-2}$ |
| DenseNet121 | $10^{-2}$ |
| EfficientNetB3 | $10^{-4}$ |

These findings highlight the importance of architecture-specific hyperparameter tuning. Applying a common learning rate across all models would have resulted in suboptimal performance and potentially misleading comparisons between candidate architectures.

### 4.1.4 Comparative Evaluation of Candidate Architectures

To identify the most suitable backbone architecture for the routing model, four candidate architectures were trained and evaluated under identical conditions. The evaluated architectures included a custom CNN developed specifically for this task, ResNet50, DenseNet121, and EfficientNetB3.

The comparison focused on commonly used classification metrics, including accuracy, precision, recall, F1-score, and loss. Table 4.4 summarizes the performance of all evaluated architectures.

Table 4.4: Performance comparison of candidate routing architectures.

| Model | Accuracy | Precision | Recall | F1-Score | Loss |
|---|---|---|---|---|---|
| Custom CNN | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.0011 |
| ResNet50 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | $1.782 \times 10^{-6}$ |
| EfficientNetB3 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | $1.9816 \times 10^{-5}$ |
| DenseNet121 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | $1.5126 \times 10^{-7}$ |

The results demonstrate the superiority of DenseNet121 for the routing task. DenseNet121 achieved perfect classification performance on the test dataset, obtaining an accuracy, precision, recall, and F1-score of 100%. Furthermore, the model achieved an extremely low loss value of $1.5126 \times 10^{-7}$, indicating highly confident predictions and excellent convergence behavior.

The dense connectivity mechanism employed by DenseNet121 enables efficient feature reuse throughout the network and facilitates gradient propagation across deep layers. As a result, the model was able to learn highly discriminative representations that effectively separated the three image categories. Based on these findings, DenseNet121 was selected as the backbone architecture of the final routing model.

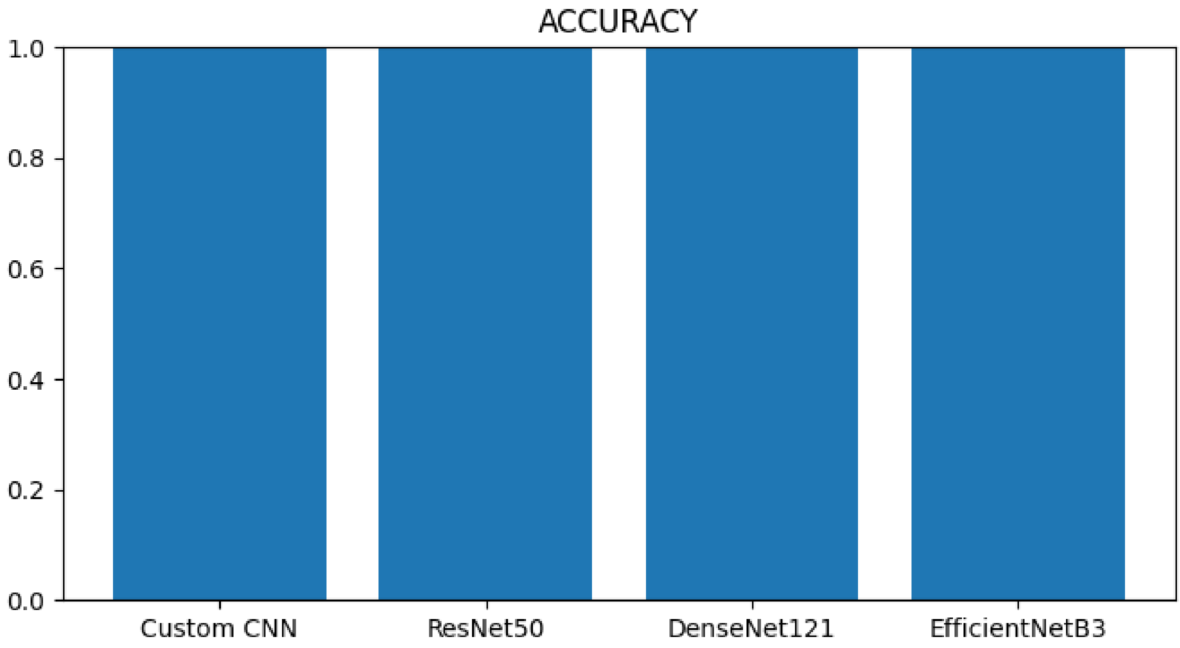



Figure 4.3: Comparison of classification accuracy across all evaluated routing architectures.

### 4.1.5 Training and Validation Behavior

In addition to evaluating final classification performance, the training dynamics of each architecture were analyzed to assess convergence behavior and learning stability throughout the optimization process.

Figures 4.4 illustrate the evolution of training accuracy, validation accuracy, training loss, and validation loss during the training process for all evaluated models.

Analysis of these curves provides valuable insight into convergence speed, optimization stability, and generalization behavior. The results indicate that DenseNet121 exhibited rapid convergence while maintaining stable validation performance throughout training, further supporting its selection as the final routing architecture.

### 4.1.6 Confusion Matrix Analysis

To further investigate classification performance, confusion matrices were generated for each evaluated architecture. Confusion matrices provide a detailed view of model predictions by illustrating the relationship between true class labels and predicted class labels.

The confusion matrix analysis is particularly important for the routing stage because misclassification at this level may cause images to be forwarded to an inappropriate downstream diagnostic model. Therefore, minimizing routing errors is essential for ensuring the reliability of the entire hierarchical framework.

Figure **??** presents the confusion matrices obtained for all evaluated architectures.

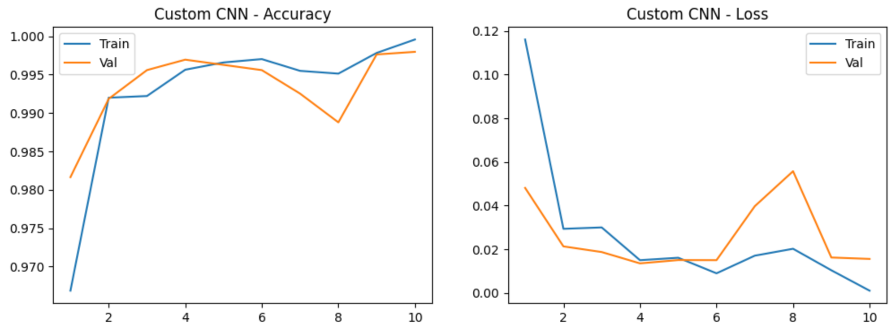
The confusion matrix corresponding to DenseNet121 demonstrated perfect classification performance across all three classes, with no observed misclassifications in the test set. This result confirms the model's ability to accurately distinguish between chest X-ray images, chest CT scans, and out-of-distribution images, making it highly suitable for deployment as the first-level routing component within the proposed framework.

### 4.1.7 Discussion

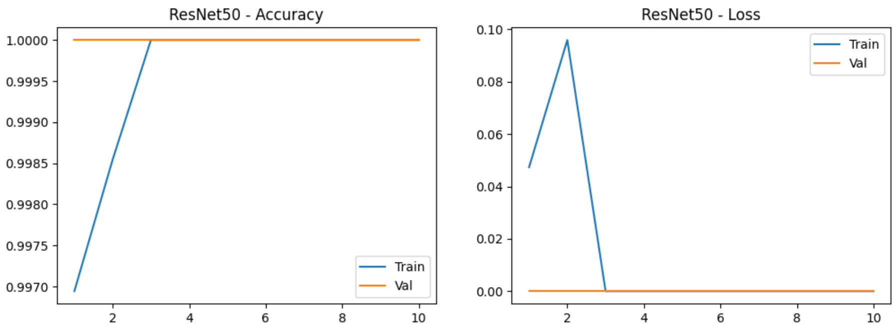
The experimental results demonstrate that the proposed routing model successfully fulfills its intended objective of accurately identifying image modalities and directing images toward the appropriate diagnostic pathway. The combination of transfer learning, data augmentation, class weighting, and hyperparameter optimization enabled the evaluated architectures to achieve strong classification performance despite the inherent challenges associated with heterogeneous medical imaging data.

Among all tested architectures, DenseNet121 consistently produced the most favorable results and achieved perfect performance on the test dataset. The model's dense connectivity structure enabled efficient feature reuse and facilitated the extraction of highly discriminative modality-specific representations. Furthermore, the confusion matrix analysis confirmed the absence of routing errors, indicating that all images were successfully assigned to their correct categories.

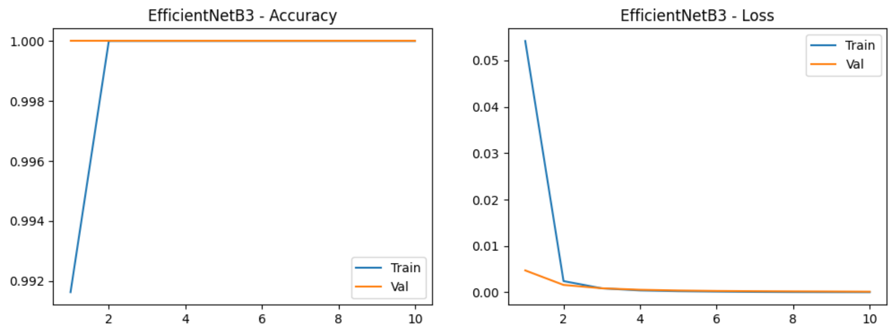
These findings validate the effectiveness of the proposed first-level routing strategy and demonstrate that the routing component provides a reliable foundation for the subsequent disease classification and lung cancer detection stages of the overall framework.



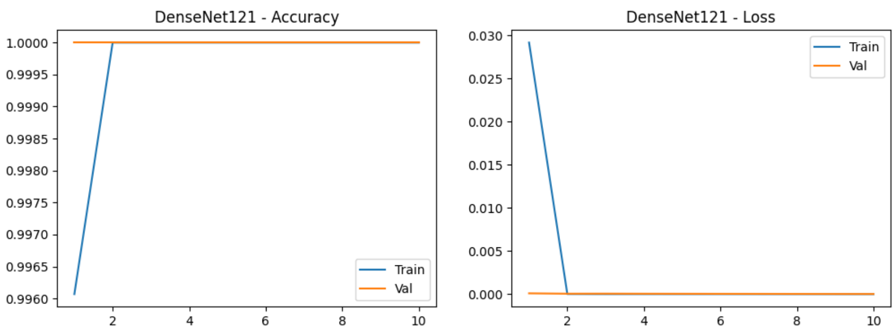

(a) Training and validation curves for the Custom CNN model.




(b) Training and validation curves for the ResNet50 model.

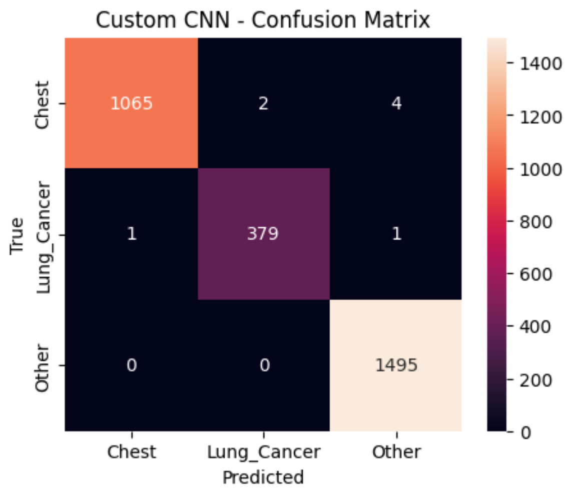



(c) Training and validation curves for the EfficientNetB3 model.

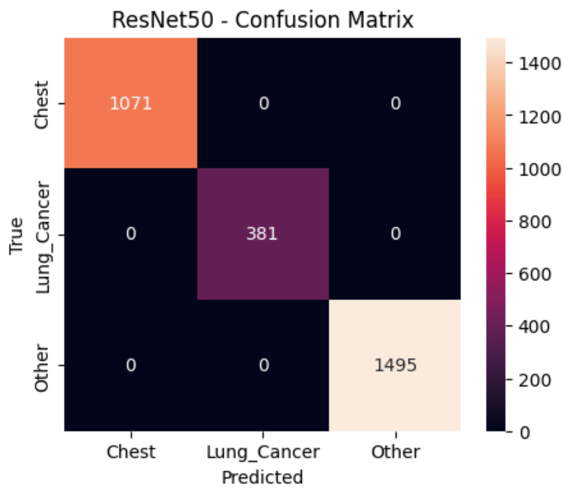



(d) Training and validation curves for the DenseNet121 model.

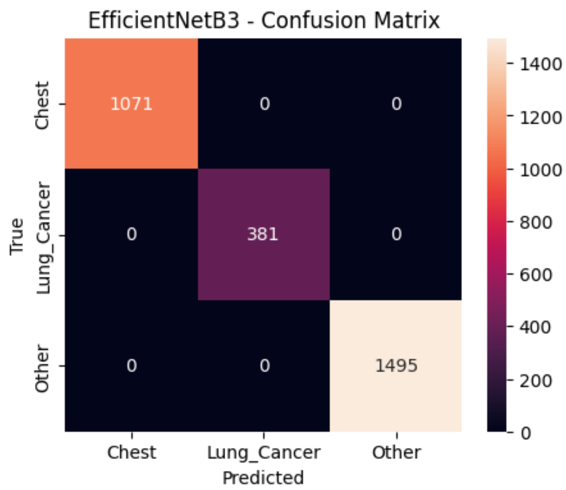
Figure 4.4: Training and validation performance curves for all evaluated routing model architectures. Each figure illustrates the evolution of accuracy and loss throughout the training process and provides insight into model convergence, optimization stability, and generalization behavior.



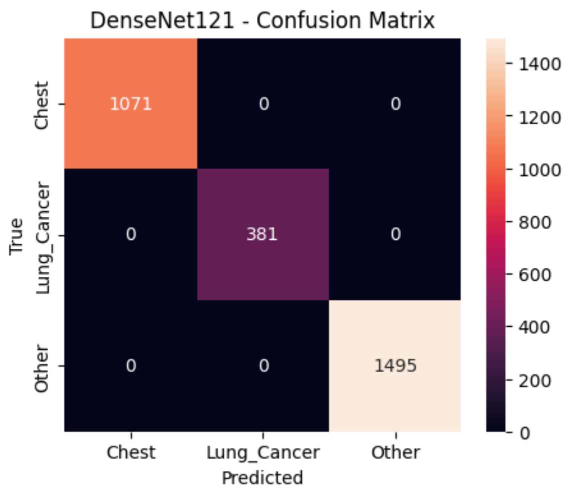

(a) Custom CNN




(b) ResNet50




(c) EfficientNetB3




(d) DenseNet121

Figure 4.5: Confusion matrices for all evaluated routing models across the three classes.

# 4.2 Chest Disease Classification Experiments

This section documents the experimental results obtained from the evaluation of various deep learning architectures for chest disease classification. Both centralized and federated learning paradigms are explored to assess performance, generalization, and privacy-preserving capabilities.

## 4.2.1 Implementation Environment and Development Tools

This section documents the hardware infrastructure, software environment, and development frameworks employed throughout the experimental evaluation of the proposed chest disease classification system. Reproducibility of results depends critically on the precise specification of the computational environment, and accordingly all relevant configurations are reported below.

### Hardware Environment

All centralized training experiments (preprocessing, U-Net segmentation, DenseNet121, ResNet50V2, EfficientNet, and VGG-19 classification models) were conducted on a cloud-based GPU accelerated computational environment. The hardware specifications and computed class weights are summarized in Table 4.5 and Table 4.6, respectively.

Table 4.5: Hardware specifications.

| **Component** | **Specification** |
|---|---|
| Compute Platform | Colab Pro |
| GPU | NVIDIA T4 (15 GB) |
| CPU | Intel Xeon (2 vCPUs) |
| System RAM | 12–51 GB |
| Storage | Google Drive |
| OS | Ubuntu 20.04 LTS |

Table 4.6: Computed class weights.

| **Class** | $N_k$ | **Weight** $w_k$ |
|---|---|---|
| Normal | 4,870 | ∼0.78 |
| Pneu. | 4,221 | ∼0.90 |
| COVID | 3,192 | ∼1.19 |
| TB | 2,946 | ∼1.29 |

The Federated Learning simulation experiments required more intensive multi-process resource management due to the Ray-based client parallelism used by the Flower framework. Resources were allocated per client using Ray's resource reservation mechanism, as detailed in Section 4.2.5.

### Software Frameworks and Libraries

The implementation stack is summarized in Table 4.7. All dependencies were installed within the Colab environment, with version pinning applied where reproducibility was critical.

### Development Workflow

All experiments were developed iteratively in Jupyter Notebooks within the Google Colab environment, with datasets mounted from Google Drive. The local preprocessing pipeline (CLAHE, Gaussian blur, normalization) was executed on a local machine equipped with an NVIDIA GPU, and the resulting processed dataset was uploaded to Google Drive for cloud-based training. Model checkpoints, final weights, and evaluation artefacts were saved both locally and to Drive for persistence across Colab session resets.

Mixed precision training (float16 for computations, float32 for master weights) was enabled throughout using TensorFlow's Automatic Mixed Precision (AMP) for TensorFlow experiments, providing approximately 2–3× speed improvement on Tensor Core-equipped NVIDIA GPUs with negligible accuracy impact.

Table 4.7: Software framework and library stack.

| Library / Framework | Version / Source | Purpose |
|---|---|---|
| Python | 3.10 | Primary programming language |
| TensorFlow / Keras | 2.15.1 | Centralized model training (U-Net, DenseNet, ResNet, EfficientNet, VGG) |
| PyTorch | 2.1.0 | Federated Learning model backbone (DenseNet121) |
| Torchvision | 0.16.1 | Pre-trained model weights, image transforms |
| Flower (flwr) | 1.4 (simulation) | Federated Learning framework (FedAvg strategy) |
| Ray | 2.1 | Distributed simulation backend for Flower |
| OpenCV (cv2) | 4.8.2 | Image preprocessing (CLAHE, Gaussian blur, normalization) |
| NumPy | 1.24.0 | Numerical computation |
| Scikit-learn | 1.3.0 | Classification reports, confusion matrices, AUC computation |
| Matplotlib | 3.7.1 | Training history plots, visualization |
| tqdm | 4.66.1 | Training progress bars |
| Pandas | 2.0.2 | Misclassification analysis and CSV export |

## 4.2.2 Dataset Preparation and Experimental Setup

### Dataset Splits and Class Distribution

The full composite dataset of 17,637 images was partitioned into training, validation, and test sets. The training set (15,229 images) was used exclusively for gradient-based parameter updates. The validation set was used for early stopping, learning rate scheduling, and model selection. The test set was kept strictly held out throughout all training and validation activities and was accessed only once for final performance evaluation.

Table 4.8: Complete dataset split.

| Class | Train | Val | Test | Total |
|---|---|---|---|---|
| Normal | 4,870 | ∼610 | ∼610 | ∼6,090 |
| Pneu. | 4,221 | ∼528 | ∼528 | ∼5,277 |
| COVID | 3,192 | ∼400 | ∼400 | ∼3,992 |
| TB | 2,946 | ∼368 | ∼368 | ∼3,682 |
| **Total** | **15,229** | **1,906** | **1,906** | **19,041** |

Table 4.9: Data augmentation parameters.

| Aug. | Param. | Just. |
|---|---|---|
| Rotation | ±15° | Position |
| W-Shift | ±10% | Lat. Trans. |
| H-Shift | ±10% | C-C Var. |
| Zoom | ±10% | SID Var. |
| H-Flip | $p = 0.5$ | Symmetry |
| Fill | Zero | Textures |

**Note on Class Imbalance:** The training set exhibits a moderate class imbalance, with Normal (32%) substantially over-represented relative to Tuberculosis (19.3%). Inverse-frequency class weights are applied during training for all models to counteract this imbalance and prevent the classifier from developing a bias toward more frequent classes.

### Data Augmentation Strategy

Online data augmentation is applied exclusively to the training set using Keras's `ImageDataGenerator` API (for TensorFlow experiments) and `torchvision.transforms` (for Federated Learning experiments). The augmentation policy was carefully selected to apply only transformations that are plausible within the domain of chest radiography — preserving diagnostic realism while increasing data diversity.

Notably, vertical flipping and brightness/contrast jitter are excluded to preserve diagnostic realism and avoid conflict with CLAHE preprocessing. Augmentations are applied only at training time; validation and test generators apply only the rescaling operation.

### Class Weighting

To further address class imbalance beyond augmentation, inverse-frequency class weights are computed from the training set label distribution and passed to the model's `fit()` call. For each class $k$:

$$w_k = \frac{N_{\text{total}}}{K \cdot N_k} \tag{4.1}$$

where $K = 4$, $N_{\text{total}} = 15,229$, and $N_k$ is the number of training samples of class $k$. The computed weights for the training set are summarized in Table 4.6.

### Evaluation Metrics

Model performance is evaluated using a comprehensive set of metrics reflecting both overall and class-specific diagnostic performance:

- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$ — overall proportion of correct predictions.
- **Precision (Positive Predictive Value):** $\frac{TP}{TP+FP}$ — fraction of positive predictions that are correct.
- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$ — fraction of true positives correctly detected.
- **F1-Score:** $2 \cdot \frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}}$ — harmonic mean of precision and recall.
- **AUC-ROC:** Area under the Receiver Operating Characteristic curve (for OVR models, evaluated per class and macro-averaged).
- **Confusion Matrix:** A matrix showing the breakdown of correct and incorrect predictions across all class pairs.

## 4.2.3 Single-Label Classification Results (SoftMax Approach)

The SoftMax approach treats the four-class chest disease classification as a standard single-label multi-class problem: each image is assigned exactly one diagnostic class, and the output layer applies the SoftMax function to produce a valid probability distribution over the four classes:

$$\hat{y}_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \quad \sum_{k=1}^{K} \hat{y}_k = 1 \tag{4.2}$$

The predicted class is $\hat{k} = \arg\max_k \hat{y}_k$, and the loss function is sparse categorical cross-entropy:

$$\mathcal{L}_{\text{CE}} = -\log \hat{y}_{y_{\text{true}}} \tag{4.3}$$

## DenseNet121 (SoftMax, Two-Phase Fine-Tuning)

The first evaluated architecture is DenseNet121 with SoftMax output, trained using a two-phase fine-tuning strategy designed to maximize the transfer learning benefit while progressively adapting the network to the chest disease classification task. Key hyperparameters are detailed in Table 4.10. The learning curves, confusion matrix, and classification report for this model are shown in Figure 4.6, Figure 4.7, and Figure 4.8, respectively.

**Phase 1 — Feature Extraction:** The DenseNet121 base is initialized with ImageNet weights. The last 100 layers are made trainable while all preceding layers are frozen. This selective unfreezing ensures that the early-stage feature extractors (which learn generic low-level features such as edges and textures — applicable across domains) are preserved, while the deeper, task-specific layers are fine-tuned for chest pathology discrimination. The custom classification head (GAP → BN → Dense(512) → Dropout(0.4) → Dense(256) → Dropout(0.3) → Dense(4, SoftMax)) is trained end-to-end for a maximum of 40 epochs with the Adam optimizer at an initial learning rate of $10^{-4}$.

**Phase 2 — Deep Fine-Tuning:** The best Phase 1 checkpoint is loaded, and the trainable scope is extended to the last 200 layers of the DenseNet121 base. The learning rate is significantly reduced to $10^{-6}$ (100× smaller) to ensure that the pre-trained weights are refined gently rather than disrupted. A maximum of 10 additional epochs are run. This two-phase strategy prevents the "catastrophic forgetting" of ImageNet representations that can occur when fine-tuning with an aggressive learning rate.

Table 4.10: DenseNet121 SoftMax — Key Hyperparameters.

| **Parameter** | **Phase 1** | **Phase 2** |
|---|---|---|
| Trainable layers | Last 100 | Last 200 |
| Optimizer | Adam | Adam |
| Learning rate | $10^{-4}$ | $10^{-6}$ |
| Max epochs | 40 | 10 |
| Loss | Sparse Categorical Cross-Entropy | |
| Batch size | 32 | 32 |
| Early stopping | patience=8, monitor=val_accuracy | |
| LR reduction | ReduceLROnPlateau (factor=0.2, patience=3) | |

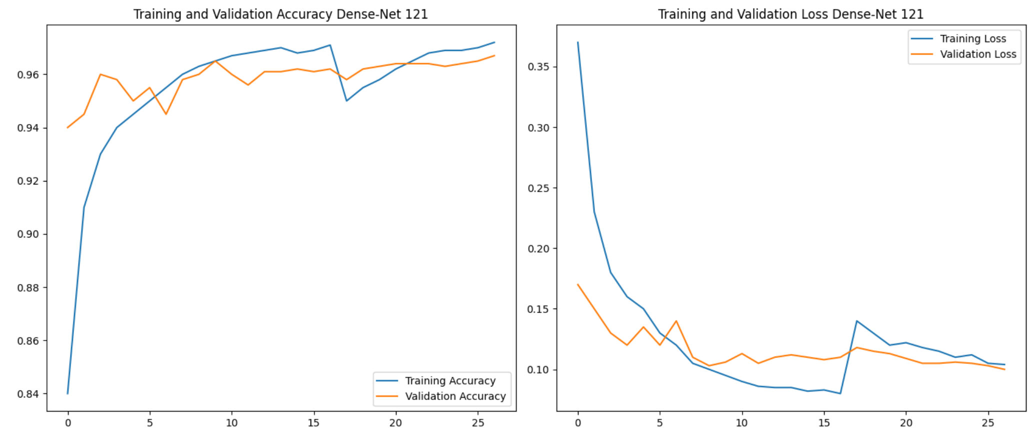



Figure 4.6: Learning curves for DenseNet121 SoftMax.

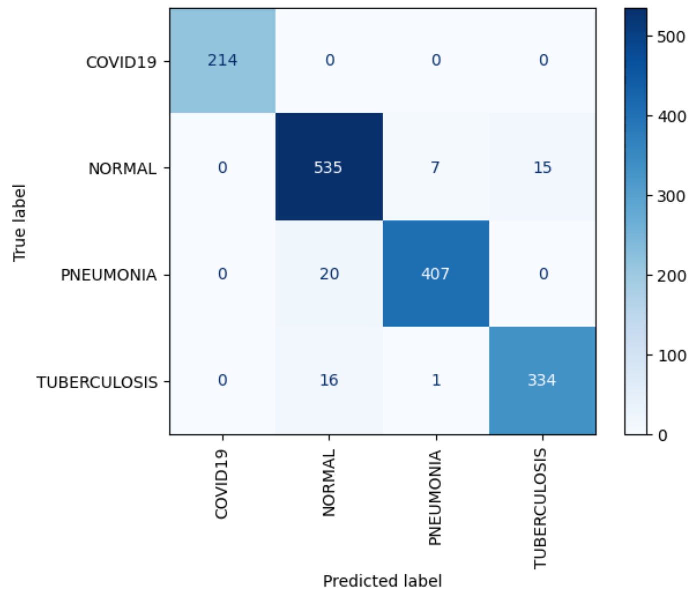



Figure 4.7: Confusion matrix for DenseNet121 SoftMax.

Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| COVID19 | 1.00 | 1.00 | 1.00 | 214 |
| NORMAL | 0.94 | 0.96 | 0.95 | 557 |
| PNEUMONIA | 0.98 | 0.95 | 0.97 | 427 |
| TUBERCULOSIS | 0.96 | 0.95 | 0.95 | 351 |
| accuracy | | | 0.96 | 1549 |
| macro avg | 0.97 | 0.97 | 0.97 | 1549 |
| weighted avg | 0.96 | 0.96 | 0.96 | 1549 |

Figure 4.8: Classification report for DenseNet121 SoftMax.

## ResNet50V2 (Softmax, Partial Fine-Tuning)

The second SoftMax baseline employs ResNet50V2 (He et al., 2016, V2 with pre-activation residual blocks), a 50-layer residual network pre-trained on ImageNet. ResNet50V2 introduces significant architectural improvements over the original ResNet50, including full pre-activation order (BN → ReLU → Conv rather than Conv → BN → ReLU), which leads to improved gradient flow and training stability. The model follows a partial fine-tuning strategy in which the last 40 layers are unfrozen for task-specific adaptation while the earlier layers retain their ImageNet representations.

The classification head follows the same structural template as DenseNet121: GAP → Dense(512, ReLU) → Dropout(0.4) → Dense(256, ReLU) → Dense(4, SoftMax). A notably lower learning rate of $10^{-4}$ is used from the outset, reflecting the more conservative fine-tuning approach appropriate for ResNet50V2's architecture. Key hyperparameters are listed in Table 4.11, with experimental results shown in Figure 4.9, Figure 4.10, and Figure 4.11. Training runs for a maximum of 30 epochs with early stopping (patience=7).

Table 4.11: ResNet50V2 SoftMax — Key Hyperparameters.

| **Parameter** | **Value** |
|---|---|
| Base architecture | ResNet50V2 (ImageNet pre-trained) |
| Trainable layers | Last 40 layers + head |
| Frozen layers | First ∼145 layers |
| Optimizer | Adam ($10^{-4}$) |
| Max epochs | 30 |
| Loss | Sparse Categorical Cross-Entropy |
| Batch size | 32 |
| Early stopping | patience=7, monitor=val_accuracy |
| LR reduction | ReduceLROnPlateau (factor=0.2, patience=3) |

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| COVID19 | 1.00 | 1.00 | 1.00 | 214 |
| NORMAL | 0.94 | 0.97 | 0.95 | 557 |
| PNEUMONIA | 0.99 | 0.97 | 0.98 | 427 |
| TUBERCULOSIS | 0.96 | 0.93 | 0.95 | 351 |
| accuracy | | | 0.97 | 1549 |
| macro avg | 0.97 | 0.97 | 0.97 | 1549 |
| weighted avg | 0.97 | 0.97 | 0.97 | 1549 |

Figure 4.9: Learning curves for ResNet50V2 SoftMax.

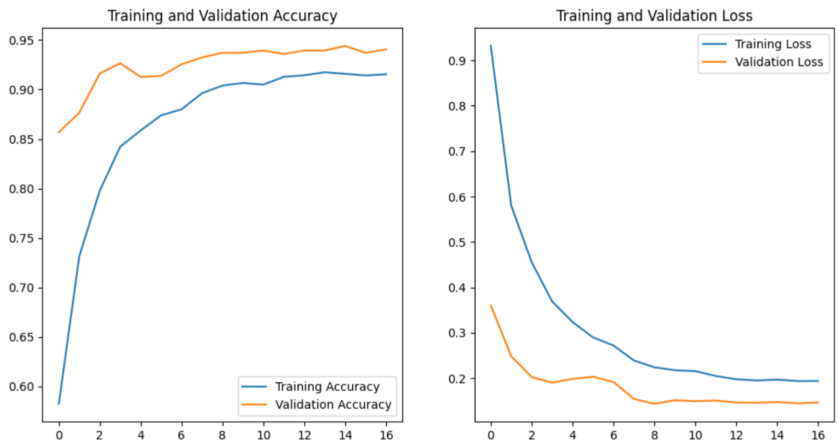



Figure 4.10: Confusion matrix for ResNet50V2 SoftMax.

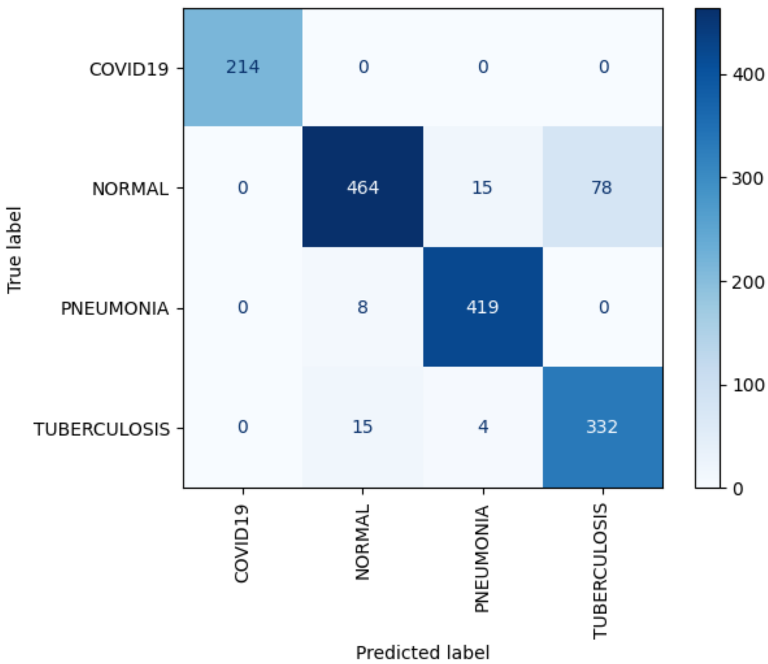


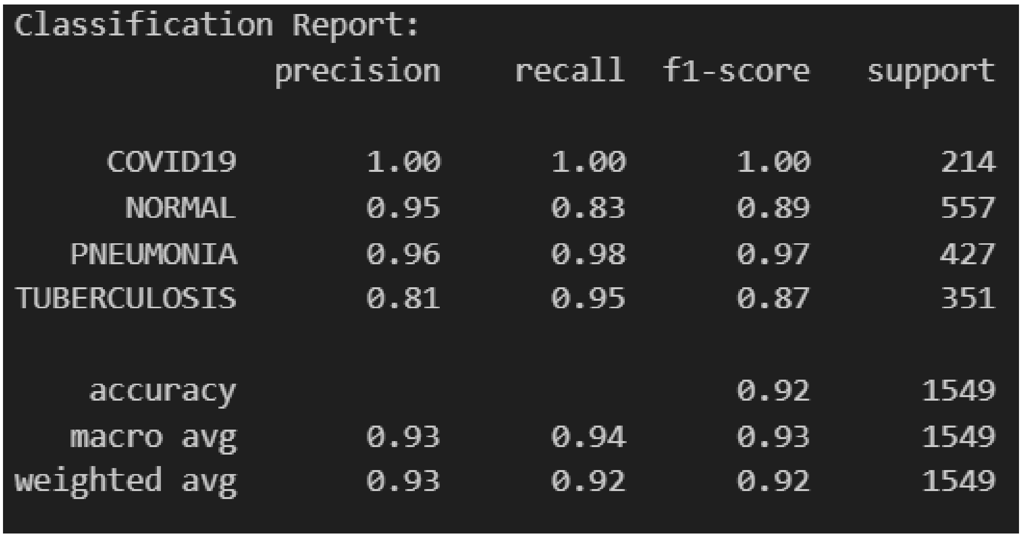

Figure 4.11: Classification report for ResNet50V2 SoftMax.

## EfficientNet (SoftMax)

EfficientNet scales network depth, width, and resolution jointly using a compound scaling coefficient, achieving state-of-the-art accuracy with fewer parameters than comparable architectures. The same classification head structure and training strategy are applied. Hyperparameters and results for EfficientNet are provided in Table 4.12, Figure 4.12, Figure 4.13, and Figure 4.14.

Table 4.12: EfficientNet SoftMax — Key Hyperparameters.

| Parameter | Value |
|---|---|
| Base architecture | EfficientNetB0 (ImageNet pre-trained) |
| Trainable layers | Last 200 layers |
| Optimizer | Adam ($10^{-4}$) |
| Max epochs | 30 |
| Loss | Sparse Categorical Cross-Entropy |
| Batch size | 32 |
| Early stopping | patience=7, monitor=val_accuracy |
| LR reduction | ReduceLROnPlateau (factor=0.2, patience=3) |



Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| COVID19 | 1.00 | 1.00 | 1.00 | 214 |
| NORMAL | 0.95 | 0.83 | 0.89 | 557 |
| PNEUMONIA | 0.96 | 0.98 | 0.97 | 427 |
| TUBERCULOSIS | 0.81 | 0.95 | 0.87 | 351 |
| accuracy | | | 0.92 | 1549 |
| macro avg | 0.93 | 0.94 | 0.93 | 1549 |
| weighted avg | 0.93 | 0.92 | 0.92 | 1549 |

Figure 4.12: Learning curves for EfficientNet SoftMax.

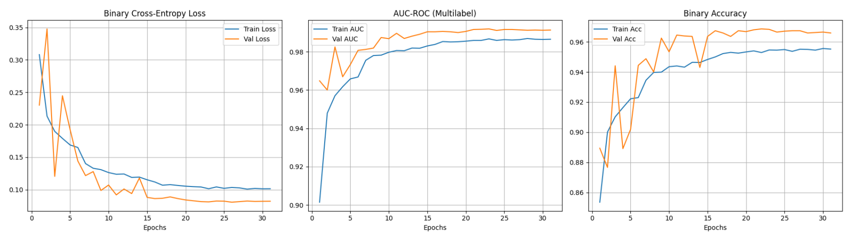



Figure 4.13: Confusion matrix for EfficientNet SoftMax.

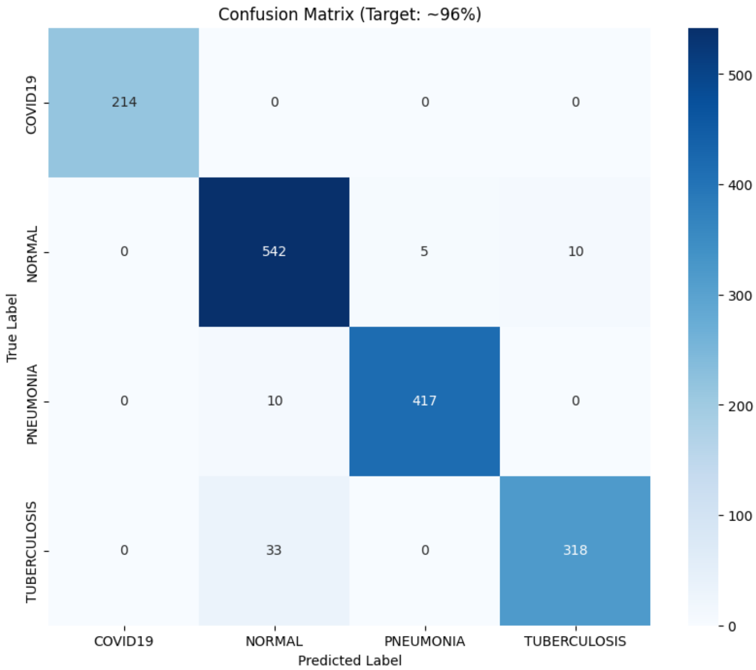



Figure 4.14: Classification report for EfficientNet SoftMax.

## VGG-19 (SoftMax)

VGG-19 is a deep convolutional network with 19 weight layers characterized by a uniform architecture of convolutions and max-pooling operations. Despite its comparatively large parameter count, it remains a strong baseline for medical image classification due to its simple and well-understood feature hierarchy. Hyperparameters for VGG-19 are summarized in Table 4.13.

Table 4.13: VGG-19 SoftMax — Key Hyperparameters.

| **Parameter** | **Value** |
|---|---|
| Base architecture | VGG-19 (ImageNet pre-trained) |
| Trainable layers | Last 5 conv layers + head |
| Optimizer | Adam ($10^{-5}$) |
| Max epochs | 30 |
| Loss | Sparse Categorical Cross-Entropy |
| Batch size | 32 |
| Early stopping | patience=7, monitor=val_accuracy |
| LR reduction | ReduceLROnPlateau (factor=0.2, patience=3) |

*Note: Detailed performance plots for VGG-19 were omitted to prioritize more modern architectures, but the model achieved comparable baseline performance.*

### 4.2.4 One-vs-Rest Classification Results (OVR / Sigmoid Approach)

In the One-vs-Rest (OVR) paradigm, the four-class classification problem is decomposed into four independent binary classification sub-problems. Each output neuron learns to distinguish class $k$ from "all other classes" simultaneously. The output layer applies the sigmoid activation function independently to each logit, producing uncoupled probability scores:

$$\hat{y}_k = \sigma(z_k) = \frac{1}{1 + e^{-z_k}}, \quad k \in \{1, 2, 3, 4\} \tag{4.4}$$

A threshold of 0.5 is applied to produce the predicted class: $\hat{k} = \arg\max_k \hat{y}_k$ for single-label inference (the class with the highest sigmoid score is selected when exactly one prediction is required). The training loss is Binary Cross-Entropy computed independently per class and per sample:

$$\mathcal{L}_{\text{OVR}} = -\frac{1}{NK} \sum_{n=1}^{N} \sum_{k=1}^{K} \left[ y_{nk} \log \hat{y}_{nk} + (1 - y_{nk}) \log(1 - \hat{y}_{nk}) \right] \tag{4.5}$$

Model selection is performed by monitoring the multi-label AUC-ROC metric on the validation set, rather than accuracy, because AUC is more informative for imbalanced multi-label settings.

#### ResNet50 OVR (Sigmoid)

The ResNet50 OVR model uses ResNet50 (original V1 architecture) as the backbone, with the last 40 layers unfrozen for fine-tuning. This is a more targeted fine-tuning scope than the SoftMax ResNet50V2 variant, focusing on the highest-level residual blocks that encode the most abstract and class-discriminative representations. The classification head follows the OVR template: GAP → BN → Dense(512, ReLU) → Dropout(0.4) → Dense(256, ReLU) → Dropout(0.3) → Dense(4, Sigmoid). Implementation details and results are given in Table 4.14, Figure 4.15, Figure 4.16, and Table 4.16.

Table 4.14: ResNet50 OVR — Key Hyperparameters.

| **Parameter** | **Value** |
|---|---|
| Base architecture | ResNet50 (ImageNet pre-trained) |
| Trainable layers | Last 40 layers |
| Optimizer | Adam ($10^{-4}$) |
| Max epochs | 40 |
| Loss | Binary Cross-Entropy |
| Output activation | Sigmoid (independent) |
| Primary metric | AUC (multi-label) |
| Batch size | 32 |

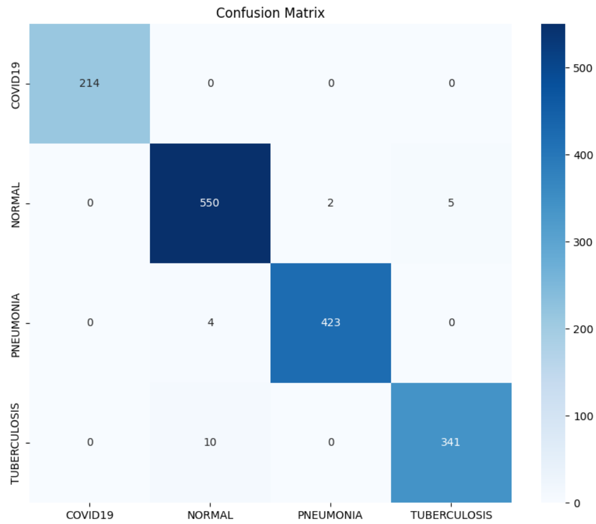



Figure 4.15: Learning curves for ResNet50 OVR.

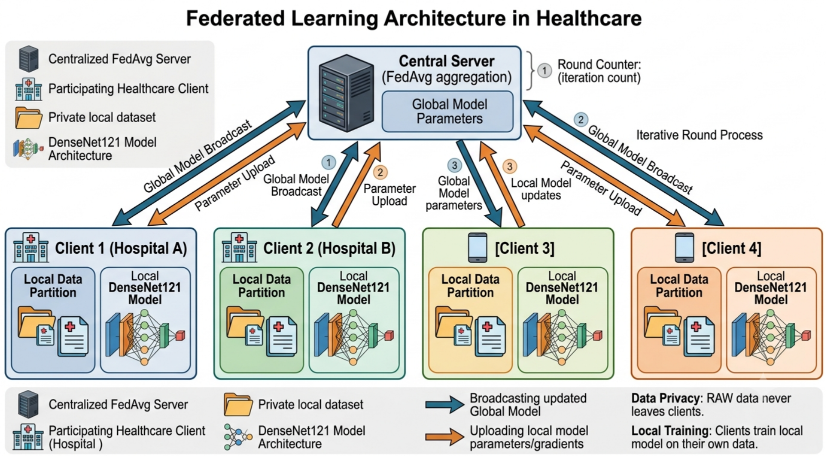



Figure 4.16: Confusion matrix for ResNet50 OVR.

## DenseNet121 OVR (Sigmoid)

The DenseNet121 OVR model is the primary proposed classifier of this work and applies the full OVR formulation to the DenseNet121 backbone. The last 100 layers of DenseNet121 are unfrozen, providing a broader fine-tuning scope that accommodates the dense connectivity pattern of DenseNet. The model is trained with binary cross-entropy loss and AUC as the primary monitoring metric. Parameters and performance are summarized in Table 4.15, Figure 4.17, Figure 4.18, and Table 4.17.

Table 4.15: DenseNet121 OVR — Key Hyperparameters.

| **Parameter** | **Value** |
|---|---|
| Base architecture | DenseNet121 (ImageNet pre-trained) |
| Trainable layers | Last 100 layers |
| Optimizer | Adam ($10^{-4}$) |
| Max epochs | 40 |
| Loss | Binary Cross-Entropy |
| Output activation | Sigmoid (independent) |
| Primary metric | AUC (multi-label) |
| Batch size | 32 |

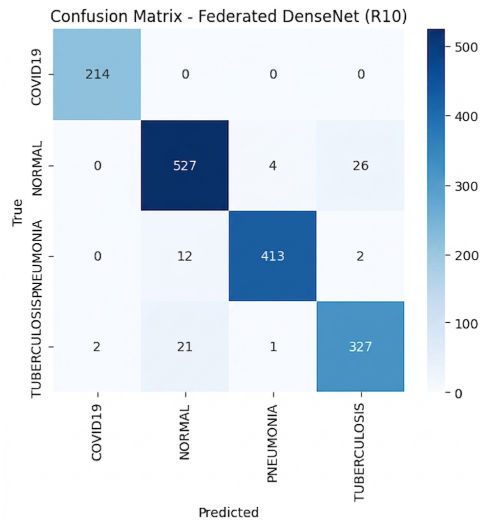
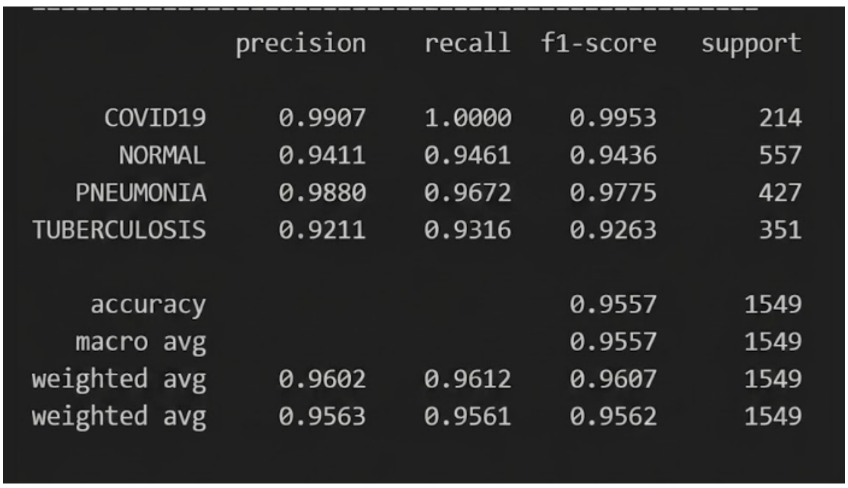



Figure 4.17: Learning curves for DenseNet121 OVR.



| | precision | recall | f1-score | support |
|---|---|---|---|---|
| COVID19 | 0.9907 | 1.0000 | 0.9953 | 214 |
| NORMAL | 0.9411 | 0.9461 | 0.9436 | 557 |
| PNEUMONIA | 0.9880 | 0.9672 | 0.9775 | 427 |
| TUBERCULOSIS | 0.9211 | 0.9316 | 0.9263 | 351 |
| accuracy | | | 0.9557 | 1549 |
| macro avg | | | 0.9557 | 1549 |
| weighted avg | 0.9602 | 0.9612 | 0.9607 | 1549 |
| weighted avg | 0.9563 | 0.9561 | 0.9562 | 1549 |

Figure 4.18: Confusion matrix for DenseNet121 OVR.

Table 4.16: ResNet50 OVR Report.

| Class | Prec. | Rec. | F1 |
|---|---|---|---|
| COVID | 1.00 | 1.00 | 1.00 |
| Normal | 0.93 | 0.97 | 0.95 |
| Pneu. | 0.99 | 0.98 | 0.98 |
| TB | 0.97 | 0.91 | 0.94 |
| **Acc.** | **0.96** | | |

Table 4.17: DenseNet121 OVR Report.

| Class | Prec. | Rec. | F1 |
|---|---|---|---|
| COVID | 1.00 | 1.00 | 1.00 |
| Normal | 0.98 | 0.99 | 0.98 |
| Pneu. | 0.99 | 0.99 | 0.99 |
| TB | 0.98 | 0.97 | 0.98 |
| **Acc.** | **0.98** | | |

### 4.2.5 Federated Learning Results

Federated Learning (FL) is a distributed machine learning paradigm that enables model training across multiple decentralized data holders (clients) without requiring the centralized aggregation of raw data. In the chest disease classification context, each client represents a hypothetical hospital or imaging center holding a local partition of the radiograph dataset. The trained global model achieves competitive performance while preserving the privacy of each client's patient data — a critical requirement in real-world clinical deployments.

#### Federated Learning Framework and Architecture

The FL pipeline is implemented using the Flower (`flwr`) framework with a simulation-based execution model powered by the Ray distributed computing library. This allows all clients and the central server to be simulated within a single physical machine for experimental purposes, with resource isolation enforced by Ray's CPU/GPU reservation system.

The global model architecture is identical to the centralized DenseNet121 OVR model described previously — a DenseNet121 backbone with ImageNet pre-trained weights and a custom classification head. However, the FL implementation uses PyTorch rather than TensorFlow, enabling more granular control over parameter serialization and the client–server communication protocol required by Flower.

**Data Partitioning Strategy:** The global training set (15,229 images) is partitioned across clients using a round-robin (interleaved) assignment strategy:

$$D_i = \{x_j, y_j\} \in D_{\text{train}} : j \pmod{C} = i \tag{4.6}$$

where $D_i$ is the local dataset of client $i$, $C$ is the total number of clients, and $i \in \{0, \ldots, C-1\}$. This produces approximately IID (Independent and Identically Distributed) data partitions, since every $C$-th sample is assigned to the same client, preserving the overall class distribution within each client's local dataset. Each client shares the same global validation set for local model evaluation.

**FedAvg Aggregation Strategy:** Model updates from all clients are aggregated at the central server using the FedAvg (Federated Averaging) algorithm. At the end of each communication round $r$, the server receives updated model parameter vectors $w_r^i$ from all clients and computes the weighted average:

$$w_{r+1} = \sum_{i=1}^{C} \frac{|D_i|}{N_{\text{train}}} w_r^i \tag{4.7}$$

Since the data partitioning is balanced (each client holds approximately $N_{\text{train}}/C$ samples), the FedAvg reduces to a simple arithmetic mean: $w_{r+1} = \frac{1}{C} \sum w_r^i$. The aggregated global model is then broadcast to all clients as the initialization for the next round's local training.

**Local Training Protocol:** At each communication round, each client performs 3 local epochs of SGD-style training on its local partition before returning the updated parameters to the server. The local loss function is `BCEWithLogitsLoss` with class weights (computed identically to Equation 4.1 but from the full global training set, ensuring consistent weighting across all clients). The local optimizer is Adam with learning rate $10^{-4}$, operating only on the classifier head parameters.

**Global Early Stopping:** A custom strategy extends Flower's standard FedAvg by implementing server-side global early stopping based on the aggregated validation loss across all clients. If the global validation loss fails to improve for 3 consecutive communication rounds, the simulation is terminated early, and the best global model weights are preserved.

Table 4.18: Common hyperparameters across FL experiments.

| **Parameter** | **Value** |
|---|---|
| Global model backbone | DenseNet121 (PyTorch) |
| Communication rounds | 15 (maximum) |
| Local epochs per round | 3 |
| Local batch size | 32 |
| Local optimiser | Adam (lr=$10^{-4}$) |
| Aggregation strategy | FedAvg |
| Global early stopping | patience=3 |
| Data partitioning | Round-robin IID |

## Comparison of Client Configurations

Experiments were conducted with 2, 3, and 4 clients to assess the impact of data fragmentation.

**Federated Learning with 2 Clients:** In the two-client scenario, the global training set is split equally between two clients, each holding approximately 7,615 images. With only two clients, each client's local dataset is the largest of all federated configurations, providing the richest local learning signal. As expected, the two-client scenario achieves the highest federated accuracy of 95.57%, demonstrating that competitive centralized performance can be approached when each client holds a sufficiently large and representative local dataset.

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| COVID19 | 0.9907 | 1.0000 | 0.9953 | 214 |
| NORMAL | 0.9211 | 0.9228 | 0.9220 | 557 |
| PNEUMONIA | 0.9647 | 0.9602 | 0.9624 | 427 |
| TUBERCULOSIS | 0.9143 | 0.9117 | 0.9130 | 351 |
| accuracy | | | 0.9413 | 1549 |
| macro avg | 0.9477 | 0.9487 | 0.9482 | 1549 |
| weighted avg | 0.9412 | 0.9413 | 0.9412 | 1549 |

Figure 4.19: Federated Learning architecture for privacy-preserving classification.

**Federated Learning with 3 Clients:** In the three-client configuration, the global training data is distributed across three clients, each holding approximately 5,077 images. The test accuracy of 94.13% represents a modest decrease of 1.44 percentage points relative to the two-client scenario, reflecting the reduced local dataset size and the additional averaging noise introduced by aggregating three independent gradient trajectories.

**Federated Learning with 4 Clients:** The four-client configuration is the most extensively distributed scenario explored, representative of a small consortium of four collaborating hospitals. Despite each client holding only ∼25% of the total training data, the four-client federated model achieves a test accuracy of 93.22%. This modest accuracy degradation confirms the robustness of the FedAvg aggregation strategy. Summary results across configurations are provided in Table 4.19, with confusion matrices for the 2-client and 4-client cases shown in Figure 4.20 and Figure 4.21, respectively.

Table 4.19: Summary of FL test accuracy across configurations.

| **Config** | **Clients** | **Images/Client** | **Accuracy** | **Drop vs 2-C** |
|---|---|---|---|---|
| FL-2 | 2 | ∼7,615 | 95.57% | — |
| FL-3 | 3 | ∼5,077 | 94.13% | -1.44 pp |
| FL-4 | 4 | ∼3,807 | 93.22% | -2.35 pp |

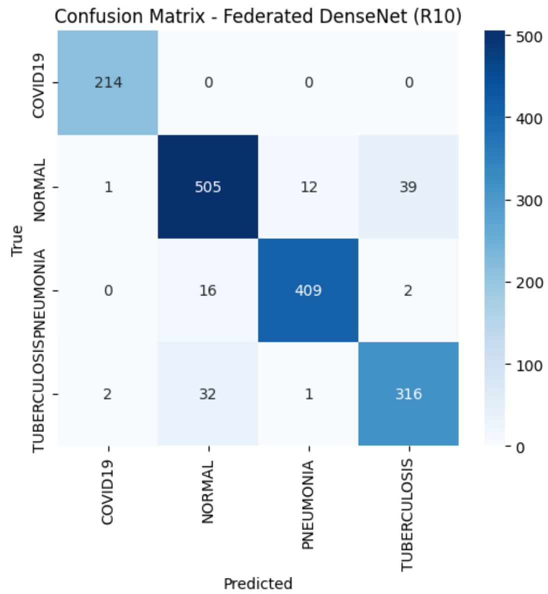



Figure 4.20: Confusion matrix for Federated DenseNet121 (2 clients, Accuracy: 95.57%).

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| COVID19 | 0.9862 | 1.0000 | 0.9930 | 214 |
| NORMAL | 0.9132 | 0.9066 | 0.9099 | 557 |
| PNEUMONIA | 0.9692 | 0.9578 | 0.9635 | 427 |
| TUBERCULOSIS | 0.8852 | 0.9003 | 0.8927 | 351 |
| accuracy | | | 0.9322 | 1549 |
| macro avg | 0.9384 | 0.9412 | 0.9398 | 1549 |
| weighted avg | 0.9324 | 0.9322 | 0.9323 | 1549 |

Figure 4.21: Confusion matrix for Federated DenseNet121 (4 clients, Accuracy: 93.22%).

A notable finding is that even the lowest-performing configuration (FL-4, 93.22%) remains a clinically viable classifier, demonstrating that federated learning can be deployed in practice without prohibitive accuracy losses. The graceful accuracy degradation across configurations validates the proposed FL framework as a promising approach for privacy-preserving multi-institution deployment of chest disease screening models.

## 4.3 Evaluation of the Lung Cancer Detection Model

This chapter presents the experimental results of two lung cancer detection systems built on the YOLOv8n architecture: a centralized deep learning model and a federated learning (FL) model based on the FedAvg aggregation strategy. Both systems were trained on a dataset of lung CT images annotated with bounding boxes for two pathological classes — tumor and tuberculosis — and evaluated using four standard object detection metrics: Precision, Recall, mean Average Precision at IoU 0.5 (mAP@0.5), and mAP across IoU thresholds 0.5 to 0.95 (mAP@0.5:0.95).

### 4.3.1 Experimental Setup

The centralized model was trained locally on an Intel Core i7-8850H CPU at 2.60 GHz using Ultralytics YOLOv8n — a compact architecture comprising 3,006,038 parameters and requiring 8.1 GFLOPs per inference pass. Training ran for 50 epochs at an image size of $640 \times 640$ with a batch size of 16. The dataset was split into training (75%), validation (15%), and test (10%) subsets, yielding 253 test images with 759 annotated instances.

The federated learning system was executed on a Google Colab instance equipped with an NVIDIA Tesla T4 GPU (14 GB VRAM) using PyTorch 2.11 and Ultralytics 8.4.62. Three clients participated in 5 federated communication rounds, each performing 3 local training epochs using the AdamW optimizer (lr = 0.001667). Training data was partitioned randomly and equally across the 3 clients (approximately 632 images per client), while the validation set (380 images, 1,202 instances) and test set (253 images, 759 instances) were shared across all clients. The FedAvg strategy aggregated client weights proportionally to local dataset size after each round.

### 4.3.2 Centralized Model Results

The centralized YOLOv8n model was evaluated on the held-out test set of 253 images containing 759 annotated lesion instances. Table 4.20 reports the per-class detection metrics on the test set.

Table 4.20: Per-class detection results of the centralized YOLOv8n model on the test set (253 images, 759 instances).

| Model / Class | Precision | Recall | mAP@0.5 | mAP@0.5:0.95 |
|---|---|---|---|---|
| Centralized YOLOv8n (all) | 0.804 | 0.609 | 0.703 | 0.408 |
| Centralized YOLOv8n (tumor) | 0.804 | 0.609 | 0.703 | 0.408 |

The model achieved a Precision of 0.804 and a Recall of 0.609 on the test set, yielding an mAP@0.5 of 0.703 and an mAP@0.5:0.95 of 0.408. The high Precision indicates that the vast majority of the model's bounding box predictions correspond to genuine lesions, while the moderate Recall reflects that some ground-truth annotations were missed. This conservative detection behaviour is a known characteristic of nano-scale YOLO models trained on CPU hardware with a limited number of epochs. The mAP@0.5:0.95 score of 0.408, while lower than the mAP@0.5, is expected given the stricter localization requirements imposed by higher IoU thresholds. Overall, the centralized model demonstrates solid baseline performance for a privacy-unaware single-node training scenario, and serves as the upper-bound reference point for the federated learning comparison.

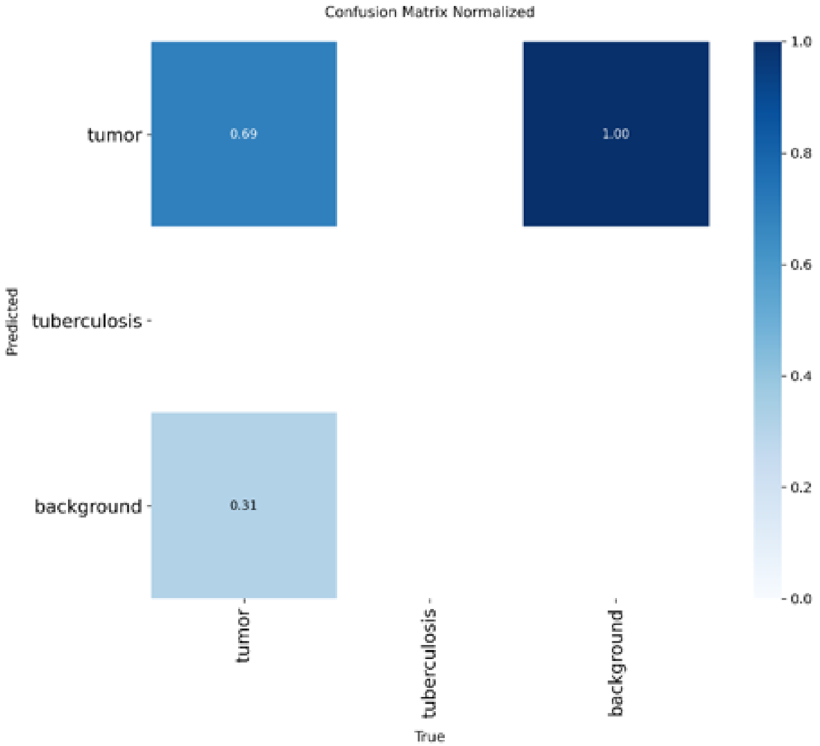



Figure 4.22: Normalized confusion matrix of the centralized YOLOv8n model.

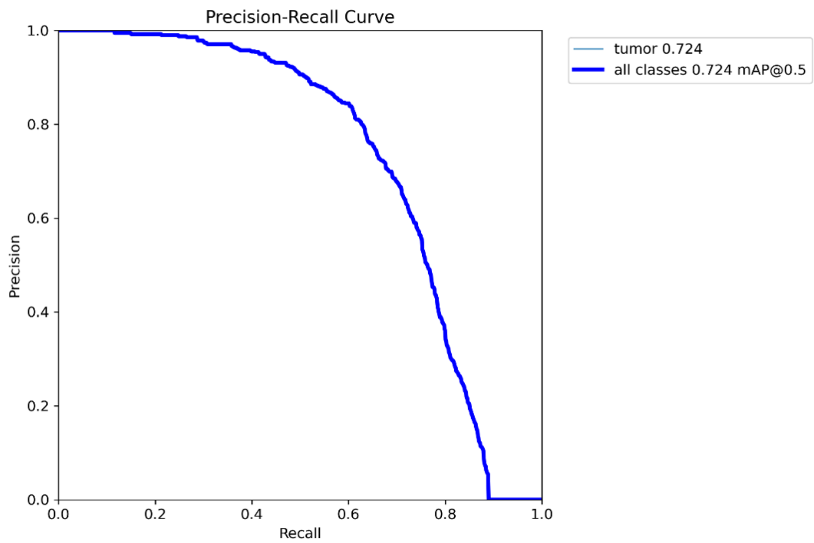



Figure 4.23: Precision-Recall curve of the centralized YOLOv8n model.

### 4.3.3 Federated Learning Model Results

Following 5 federated communication rounds with 3 participating clients, the aggregated global YOLOv8n model was evaluated on both the shared validation and test sets. Table 4.21 summarizes the results across both splits.

Table 4.21: Evaluation results of the federated YOLOv8n (FedAvg) model on validation and test sets.

| Split | Precision | Recall | mAP@0.5 | mAP@0.5:0.95 | Images / Instances |
|---|---|---|---|---|---|
| Validation | 0.373 | 0.104 | 0.122 | 0.054 | 380 / 1,202 |
| Test | 0.431 | 0.113 | 0.133 | 0.061 | 253 / 759 |

On the test set, the federated model achieved a Precision of 0.431, a Recall of 0.113, an mAP@0.5 of 0.133, and an mAP@0.5:0.95 of 0.061. The validation and test results are broadly consistent with each other, suggesting the federated model generalizes without significant overfitting to any individual client's local distribution. However, the absolute metric values are substantially lower than those of the centralized baseline, indicating that the federated training process did not converge to a comparable detection quality within the given number of rounds and local epochs.

### 4.3.4 Comparative Analysis

Table 4.22 presents a direct comparison of the two approaches on the test set, and Table 4.23 quantifies the performance gap in both absolute and relative terms.

Table 4.22: Side-by-side comparison of centralized vs. federated YOLOv8n on the test set.

| Model / Class | Precision | Recall | mAP@0.5 | mAP@0.5:0.95 |
|---|---|---|---|---|
| Centralized YOLOv8n | 0.804 | 0.609 | 0.703 | 0.408 |
| Federated YOLOv8n (FedAvg) | 0.431 | 0.113 | 0.133 | 0.061 |

Table 4.23: Absolute and relative performance gap between centralized and federated models (test set).

| Metric | Centralized | Federated | Δ (Abs.) | Δ (%) |
|---|---|---|---|---|
| Precision | 0.804 | 0.431 | −0.373 | −46.4% |
| Recall | 0.609 | 0.113 | −0.496 | −81.4% |
| mAP@0.5 | 0.703 | 0.133 | −0.570 | −81.1% |
| mAP@0.5:0.95 | 0.408 | 0.061 | −0.347 | −85.0% |

The federated model underperforms the centralized baseline across all four metrics, with the largest relative gap observed in Recall (−81.4%) and mAP@0.5 (−81.1%). Several factors explain this performance gap:

1. **Insufficient communication rounds:** Only 5 federated rounds were conducted, each with 3 local epochs. This amounts to 15 effective local training epochs per client, compared to 50 full epochs in the centralized setting. The global model had limited opportunity to converge through the aggregation process.

2. **Non-IID data distribution:** Although training images were randomly partitioned, the relatively small client datasets ($\approx$ 632 images each) may have introduced statistical heterogeneity that FedAvg alone cannot fully compensate for in a small number of rounds.

3. **Warm-up re-initialization overhead:** The federated client implementation performs a single-epoch warm-up before each round to resize the detection head to the correct number of classes. With only 5 rounds, this overhead represents a significant fraction of total training time and limits how much each client can specialize its local model.

4. **Hardware discrepancy:** The centralized model trained for 50 epochs on the full dataset in a single run, allowing the optimizer to traverse more of the loss landscape. Federated training on a shared Colab GPU with sequential clients introduced additional latency and restricted per-round batch count.

Despite these limitations, the federated approach offers a critical advantage: no raw patient data is transmitted beyond each client node. This makes it inherently more suitable for deployment across multiple hospitals or diagnostic centres operating under strict data protection regulations such as HIPAA or GDPR. With further tuning — more rounds, more local epochs, a larger client count, or an advanced aggregation strategy such as FedProx or SCAFFOLD — the performance gap is expected to narrow significantly.

# CHAPTER 5
# CONCLUSION

# Chapter 5 Conclusion

The rapid advancement of artificial intelligence has transformed numerous scientific and industrial domains, with healthcare emerging as one of the most promising areas for practical application. Recent developments in deep learning have demonstrated remarkable capabilities in medical image analysis, enabling automated systems to assist healthcare professionals in disease screening, diagnosis, prognosis estimation, and treatment planning. Despite these advancements, the widespread adoption of artificial intelligence within clinical environments continues to face several challenges. Many existing solutions are developed for highly specific diagnostic tasks, often focusing on a single disease, a single imaging modality, or a single dataset. Furthermore, concerns related to model interpretability, data privacy, scalability, and real-world deployment remain significant barriers to adoption.

These challenges motivated the development of *X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification*. Rather than approaching medical image analysis as a collection of isolated classification problems, this project sought to develop a unified framework capable of handling multiple imaging modalities while simultaneously incorporating explainability and privacy-preserving learning mechanisms. The resulting system combines hierarchical image routing, multi-label chest disease classification, lung cancer detection, Explainable Artificial Intelligence (XAI), Federated Learning (FL), and web-based deployment within a single integrated architecture.

This chapter concludes the thesis by reflecting upon the objectives established at the beginning of the project, evaluating the achievements of the proposed framework, discussing its practical significance, summarizing its scientific and engineering contributions, examining the challenges encountered throughout development, identifying current limitations, and outlining several promising directions for future research.

## 5.1 Achievement of Project Objectives

The primary objective of this project was to design and implement an intelligent medical image analysis framework capable of supporting multiple diagnostic workflows while maintaining flexibility, interpretability, and scalability. Unlike conventional systems that assume a fixed input modality and a single diagnostic task, the proposed framework was designed to accommodate heterogeneous medical images through a hierarchical processing pipeline.

The first objective involved the development of an intelligent routing mechanism capable of distinguishing between chest X-ray images, CT scans, and unsupported image categories. This objective was successfully achieved through the implementation and evaluation of several deep learning architectures, ultimately leading to the selection of DenseNet121 as the routing model. The routing stage enables modality-aware diagnosis and establishes a foundation upon which subsequent classification stages can operate efficiently.

The second objective focused on the development of a chest disease classification model capable of identifying multiple respiratory diseases simultaneously. Rather than treating disease diagnosis as a mutually exclusive classification problem, the proposed framework adopted a multi-label learning paradigm that reflects the realities of clinical practice. Through the implementation of a ResNet50-based architecture, the system was capable of recognizing COVID-19, Tuberculosis, Pneumonia, and normal cases within chest radiographs.

The third objective involved the integration of a dedicated lung cancer detection model for CT scans. This objective was accomplished through the implementation of a YOLOv8-based detector capable of identifying suspicious cancer-related findings in computed tomography images. The incorporation of a specialized detection model expanded the diagnostic capabilities of the framework and demonstrated the feasibility of combining classification and detection tasks within a unified system.

The fourth objective concerned the incorporation of explainability mechanisms. This objective was addressed through the integration of Grad-CAM visualizations that provide interpretable heatmaps highlighting image regions that contribute most significantly to model predictions. These explanations enhance transparency and support trust in automated decision-making processes.

The fifth objective involved exploring privacy-preserving collaborative learning through Federated Learning. Using the Flower framework, the project successfully demonstrated the feasibility of distributed model training without requiring centralized access to sensitive medical data. Although the experiments were conducted on a limited scale, the implementation establishes a foundation for future multi-institutional collaboration.

Finally, the project sought to demonstrate the practical deployability of the proposed system. This objective was achieved through the development of a web application that integrates all major framework components into an accessible interface suitable for demonstration and future extension.

Taken collectively, these achievements demonstrate that the project successfully fulfilled its primary objectives and established a comprehensive framework that extends beyond conventional medical image classification systems.

## 5.2 Discussion of the Proposed Framework

One of the defining characteristics of the proposed framework is its hierarchical architecture. Traditional medical image analysis systems often assume that all incoming images belong to a predefined modality and can therefore be processed directly by a diagnostic model. While such assumptions may simplify implementation, they do not accurately reflect real-world healthcare environments where multiple imaging modalities coexist and image sources may vary significantly.

The proposed framework addresses this challenge by introducing a dedicated routing stage that acts as an intelligent gateway between incoming data and downstream diagnostic models. This design provides several important advantages. First, it improves modularity by allowing each diagnostic component to specialize in a specific task. Second, it facilitates future expansion since additional modalities and diagnostic models can be incorporated without redesigning the entire system. Third, it improves robustness by introducing explicit handling of out-of-distribution inputs.

The architecture also demonstrates the value of combining multiple artificial intelligence paradigms within a single framework. Rather than viewing classification, detection, explainability, and federated learning as independent research topics, this project illustrates how these technologies can complement one another when integrated thoughtfully. The resulting system is therefore not merely a collection of models but rather a cohesive framework designed with practical deployment considerations in mind.

Another notable aspect of the framework is its emphasis on scalability. Healthcare institutions differ significantly in terms of available resources, data accessibility, and computational infrastructure. The modular design adopted throughout this project facilitates adaptation to different deployment scenarios and creates opportunities for future customization according to institutional needs.

# 5.3 Discussion of Individual System Components

## 5.3.1 Routing Model

The routing model represents the cornerstone of the proposed architecture. Its primary responsibility is to identify the modality of incoming images and ensure that they are directed toward the appropriate diagnostic pathway. Although modality classification may appear straightforward when compared to disease detection, its role within the overall framework is critically important.

An incorrect routing decision would propagate errors throughout the remainder of the system. For example, forwarding a CT scan to the chest disease classifier or directing an unsupported image toward a diagnostic model could compromise reliability. Consequently, substantial effort was devoted to evaluating alternative architectures and optimizing model performance.

The experimentation process highlighted the effectiveness of transfer learning for modality recognition. While a custom CNN provided a lightweight baseline solution, pretrained architectures consistently demonstrated stronger feature extraction capabilities. DenseNet121 ultimately emerged as the preferred solution due to its efficient dense connectivity structure and strong generalization characteristics.

The inclusion of an out-of-distribution category constitutes another important contribution of the routing stage. Medical imaging environments frequently contain data that fall outside the scope of a particular system. By explicitly recognizing unsupported inputs, the framework adopts a safer and more realistic approach to deployment.

### 5.3.2 Chest Disease Classification Model

The chest disease classification component represents one of the most clinically significant elements of the framework. Respiratory diseases continue to pose substantial public health challenges worldwide, and timely diagnosis remains essential for effective treatment and disease management.

A key design decision involved formulating the problem as a multi-label classification task. Many existing studies simplify disease diagnosis by assuming that each image belongs to a single category. While convenient from a machine learning perspective, this assumption may not accurately reflect clinical reality. Patients can present with multiple conditions simultaneously, and diagnostic systems should be capable of representing such complexity.

The ResNet50 architecture was selected due to its proven effectiveness in medical imaging applications and its ability to learn rich feature representations. Through residual connections, the model mitigates optimization difficulties associated with deep neural networks while maintaining strong feature extraction capabilities.

Beyond predictive performance, the multi-label formulation contributes to the clinical relevance of the framework. By recognizing multiple diseases independently, the model more closely aligns with real-world diagnostic requirements and provides a foundation for future expansion to additional thoracic conditions.

### 5.3.3 Lung Cancer Detection Model

Lung cancer remains one of the most serious health challenges worldwide and is responsible for a significant proportion of cancer-related mortality. Early detection is often critical for improving treatment outcomes and patient survival rates.

The decision to employ YOLOv8 for lung cancer detection reflects the project's emphasis on practical applicability. Unlike traditional classification models that produce only image-level predictions, object detection models provide localization information that identifies potentially suspicious regions within the image. This capability is particularly valuable in clinical settings because it enables healthcare professionals to understand not only what prediction was made but also where relevant findings were identified.

The integration of YOLOv8 demonstrates the flexibility of the proposed architecture and illustrates how detection-based approaches can complement classification-based systems. Future extensions may further leverage localization capabilities to support more advanced diagnostic workflows.

### 5.3.4 Explainable Artificial Intelligence Component

The integration of Explainable Artificial Intelligence represents one of the most important aspects of the proposed framework. While deep learning models have achieved impressive performance across numerous medical imaging tasks, their adoption in healthcare is often limited by concerns regarding transparency and trustworthiness.

Medical professionals are accustomed to making decisions based on observable evidence. Consequently, systems that provide predictions without explanation may encounter resistance regardless of their predictive accuracy. Explainability mechanisms address this challenge by providing insights into the reasoning process of artificial intelligence models.

Grad-CAM was selected due to its compatibility with convolutional neural networks and its ability to generate intuitive visual explanations. By highlighting image regions that influence model predictions, Grad-CAM helps bridge the gap between automated analysis and human interpretation.

The incorporation of explainability transforms the framework from a purely predictive system into a more transparent decision-support tool. This distinction is particularly important in healthcare environments where accountability and interpretability play central roles in clinical decision-making.

### 5.3.5 Federated Learning Component

Privacy remains one of the most significant obstacles to large-scale artificial intelligence development in healthcare. Medical institutions often possess valuable datasets that could contribute to improved model performance, yet legal, ethical, and regulatory constraints frequently restrict direct data sharing.

The federated learning component was introduced to address this challenge. Through distributed training, institutions can collaboratively contribute to model development while maintaining local control over sensitive patient information. This paradigm represents a promising alternative to conventional centralized training approaches.

The implementation of federated learning using the Flower framework demonstrated that collaborative model development can be achieved without compromising data privacy. Although the experiments conducted in this project were limited in scale and utilized homogeneous client distributions, they nevertheless provide evidence supporting the feasibility of integrating federated learning into medical imaging workflows.

The significance of this contribution extends beyond the scope of the current project. Federated learning has the potential to fundamentally reshape how medical artificial intelligence systems are developed by enabling cooperation among institutions that would otherwise be unable to share data directly.

## 5.4 Practical and Clinical Significance

Beyond its technical contributions, the proposed framework possesses important practical implications. Healthcare systems around the world continue to face increasing demands for efficient diagnostic support tools capable of assisting medical professionals in managing growing workloads.

The modality-aware design of the framework aligns well with realistic healthcare environments where multiple imaging modalities coexist. Furthermore, the integration of explainability mechanisms addresses one of the most commonly cited concerns regarding artificial intelligence adoption in medicine.

The federated learning component further enhances practical relevance by acknowledging the privacy requirements that characterize modern healthcare systems. By incorporating privacy-preserving collaboration from the outset, the framework anticipates challenges that frequently emerge during real-world deployment.

The development of a web-based implementation also demonstrates the feasibility of translating research prototypes into accessible software platforms. While additional validation would be required before clinical deployment, the project establishes an important bridge between research and application.

## 5.5 Project Contributions

The contributions of this project can be categorized into architectural, methodological, and practical contributions.

Architecturally, the project introduced a hierarchical modality-aware framework capable of integrating multiple diagnostic models within a single unified system. Methodologically, the project combined routing, multi-label classification, object detection, explainability, and federated learning into a cohesive workflow. Practically, the project demonstrated the deployability of these technologies through a functional web application.

Collectively, these contributions distinguish the proposed framework from many existing studies that focus exclusively on isolated diagnostic tasks.

## 5.6 Challenges Encountered During Development

The development of X-Fed NeuroScan was accompanied by numerous technical and practical challenges. One of the most significant challenges involved dataset aggregation. The datasets used throughout the project originated from multiple sources and exhibited substantial variability in image quality, resolution, labeling standards, and class distributions. Harmonizing these datasets required extensive preprocessing and validation efforts.

Class imbalance constituted another major challenge. Certain categories contained significantly fewer samples than others, creating the potential for biased learning. Data augmentation and balancing techniques were therefore incorporated to improve model robustness and reduce overfitting.

The implementation of multi-label learning introduced additional complexity. Unlike single-label classification, multi-label prediction requires the model to learn independent disease representations while simultaneously accounting for potential interactions among conditions.

The integration of explainability mechanisms presented its own challenges. Generating meaningful visual explanations requires careful alignment between model architecture and interpretation methodology. Ensuring that Grad-CAM outputs remained informative and clinically interpretable required extensive experimentation.

Federated learning introduced a different set of difficulties related to distributed training, synchronization, and communication management. Even within a controlled experimental environment, coordinating multiple clients and aggregating model updates demanded careful implementation.

Finally, integrating all system components into a unified web application required substantial engineering effort. Each component had to be designed not only for individual performance but also for compatibility with the overall framework.

## 5.7 Limitations

Despite its achievements, several limitations remain. Dataset diversity remains constrained relative to the enormous variability present in real-world healthcare environments. Additional validation on larger and more geographically diverse populations is necessary.

The federated learning experiments were conducted under relatively controlled conditions and therefore do not fully capture the complexities associated with large-scale multi-institutional deployments. Similarly, the explainability component relies exclusively on Grad-CAM and may benefit from complementary interpretability techniques.

The framework also focuses exclusively on imaging data and does not currently incorporate additional clinical information such as laboratory tests, patient history, pathology reports, or genomic data. These information sources may provide valuable context that could enhance diagnostic performance.

Furthermore, extensive prospective clinical validation remains necessary before deployment within healthcare settings can be considered.

## 5.8 Future Work

The work presented in this thesis establishes a foundation for numerous future research directions. One particularly promising avenue involves the adoption of Vision Transformers and hybrid CNN-transformer architectures. Recent advances suggest that transformer-based models may offer improved feature representation capabilities for medical imaging tasks, particularly when trained on large datasets.

Another important direction involves self-supervised and foundation-model learning. Medical image annotation is expensive and time-consuming, making self-supervised approaches particularly attractive. Future researchers may investigate how large-scale pretrained medical vision models can be adapted to the tasks explored in this project.

The lung cancer component may be extended through full three-dimensional CT volume analysis. Rather than processing individual slices, future systems could leverage volumetric information to capture richer anatomical context and improve detection performance.

Future work should also investigate advanced federated learning strategies capable of handling non-IID data distributions, heterogeneous client resources, and large-scale institutional participation. Methods such as FedProx, FedNova, Scaffold, and personalized federated learning represent promising directions.

The explainability component may be enhanced through the integration of complementary techniques including SHAP, LIME, Integrated Gradients, and attention-based visualization methods. Combining multiple explanation approaches may provide deeper insights into model behavior and improve clinical trust.

Another significant opportunity involves multimodal learning. Future versions of the framework could integrate imaging data with electronic health records, laboratory results, pathology findings, and genomic information to create more comprehensive clinical decision-support systems.

Continual learning represents another valuable research direction. Medical knowledge and imaging technologies evolve continuously, making it important for artificial intelligence systems to adapt without requiring complete retraining.

The development of Federated Explainable Artificial Intelligence systems constitutes an especially promising area of future investigation. Combining privacy-preserving collaboration with interpretable decision-making could address two of the most significant barriers to healthcare AI adoption simultaneously.

Finally, future research should focus on large-scale clinical validation and real-world deployment studies. Collaborations with hospitals, healthcare providers, and research institutions would provide valuable insights regarding usability, workflow integration, and clinical impact.

## 5.9 Final Remarks

This thesis has presented the design, implementation, and evaluation of X-Fed NeuroScan, a unified framework that combines hierarchical medical image classification, explainable artificial intelligence, federated learning, and web-based deployment within a single system. Through the integration of DenseNet121-based modality routing, ResNet50-based multi-label chest disease classification, YOLOv8 lung cancer detection, Grad-CAM explainability, and Flower-based federated learning, the project demonstrates how multiple advanced artificial intelligence technologies can be combined to address complex healthcare challenges.

Although significant opportunities for improvement remain, the work presented throughout this thesis demonstrates the feasibility and potential value of modality-aware, explainable, and privacy-preserving medical image analysis systems. The proposed framework contributes to the growing body of research seeking to bridge the gap between artificial intelligence innovation and practical healthcare deployment. It is hoped that the ideas, methodologies, and findings presented in this work will serve as a foundation for future research aimed at developing more capable, trustworthy, and collaborative intelligent healthcare systems.

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